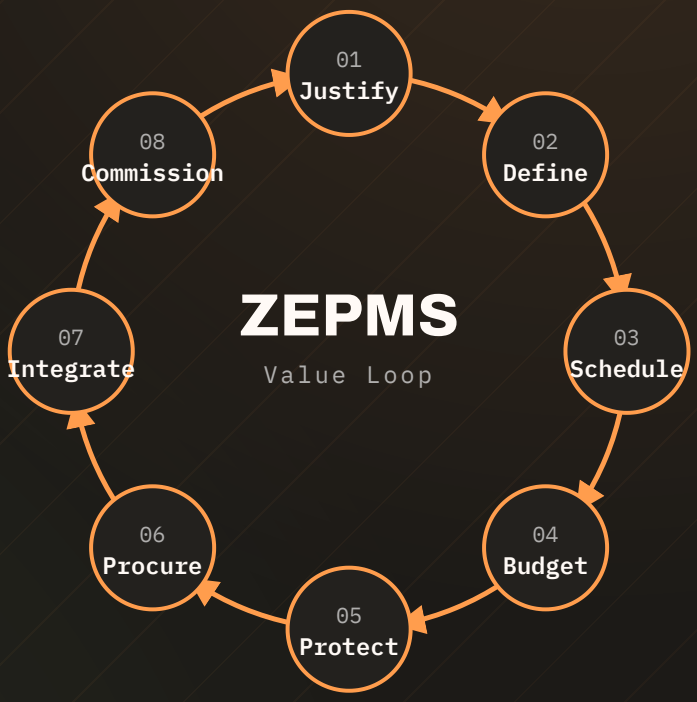
ZEPMS

ZyeemZia™ Engineering Project Management Specialization

Build it right, build it safe, hand it over ready.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 39 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

A professional specialization for people who deliver engineering projects in construction, infrastructure, manufacturing, energy and process plant: business cases and NPV, scope and WBS, critical path scheduling, cost and earned value, quantitative risk, contracts and claims, site safety and leadership, carbon and ESG, BIM information management and digital twins, and commissioning through to closeout, organised around one engineering delivery loop.

01
Learn
Work through 39 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 40-question official exam at zepms.zyeemzia.com, graded on the ZyeemZia server. Pass mark 80%.

04
Get certified
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Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Week 1	m1 Justify	Feasibility, NPV/IRR practice and a draft business case	☐
Week 2	m2 Define	Scope statement, battery limits and WBS for your capstone	☐
Weeks 3–4	m3 Schedule	CPM by hand, PERT estimates, levelling and a crashing exercise	☐
Week 5	m4 Budget	Estimate, contingency, S-curve and an earned value report	☐
Week 6	m5 Protect	Risk register, Monte Carlo exercise and HSE plan	☐
Week 7	m6 Procure	Contract strategy, tender evaluation and a variation log	☐
Week 8	m7 Integrate	Stakeholder plan, meeting rhythm and BIM/digital-twin use plan	☐
Weeks 9–10	m8 Commission	ITP, commissioning and handover plans, practice exam, capstone and certification exam	☐

Learner name .....

Start date ..... Target exam date .....

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4. The Business Case & Investment Decision · 35 min
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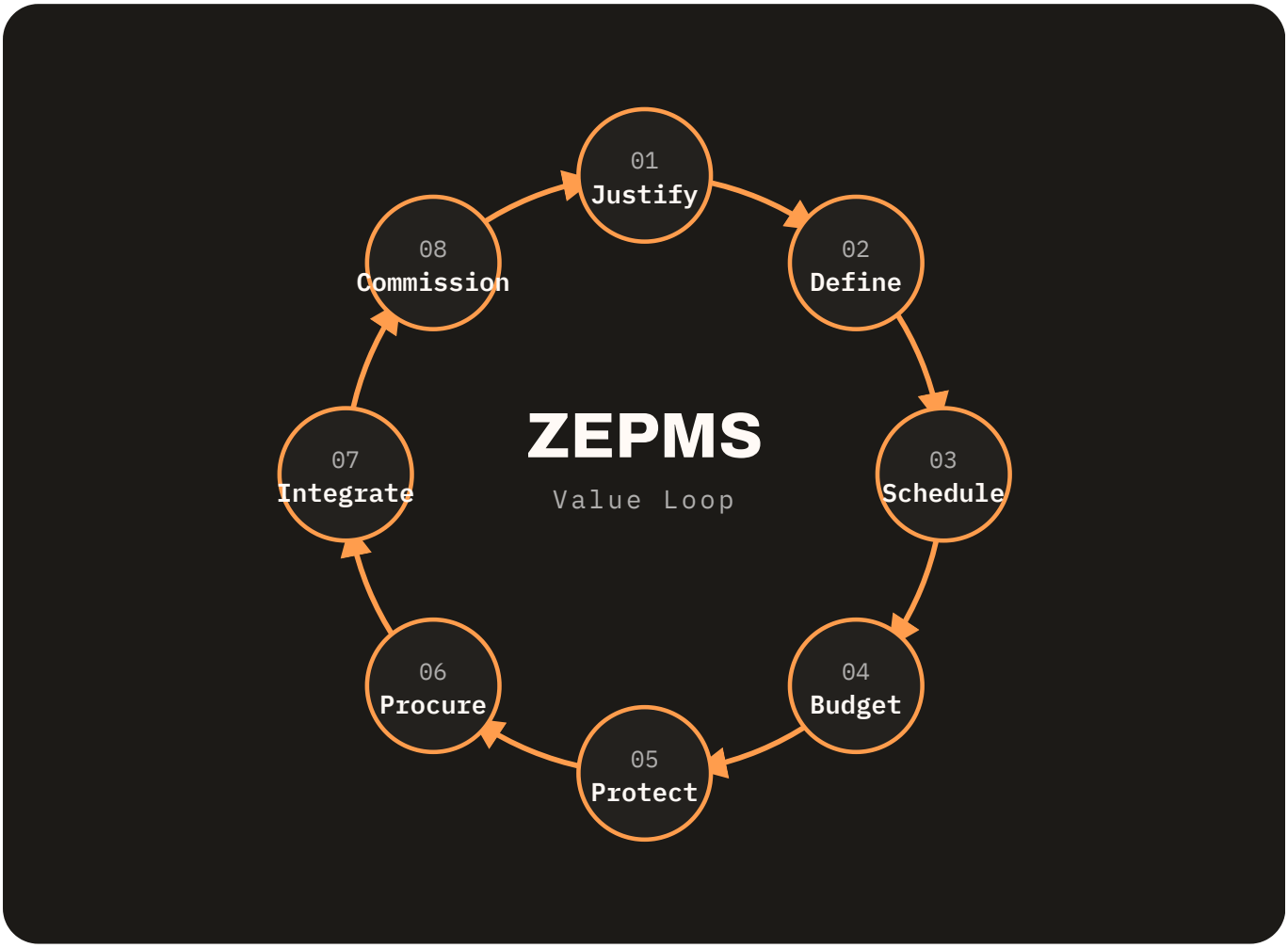
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THE FRAMEWORK

The ZEPMS Delivery Loop

Eight stages every engineering project moves through: justify the investment, define the scope, schedule the work, budget and measure it, protect people and value, procure and contract, integrate teams and data, then commission and hand over. Lessons from each handover feed the next business case.



01 Justify
Decide whether an engineering project deserves to exist, and set it up to succeed

02 Define
Turn an approved investment into a defined, buildable and controlled scope

03 Schedule
Turn the WBS into a logic-driven, resourced and defensible programme that shows where time really matters

04 Budget
Estimate with honest ranges, set a cost baseline that matches the schedule, and use earned value to see trouble early

05 Protect
Manage uncertainty, protect people and respect the places and communities where engineering happens

06 Procure
Buy the right capability, allocate risk deliberately and manage change and claims without losing the project to disputes

07 Integrate

08 Commission

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Bring people, information and data together so the site builds the right thing, safely, from the latest information

Prove the asset works, hand it over ready to operate, close the project cleanly and feed the lessons back

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01

MODULE 1 · JUSTIFY

Initiation, Feasibility & the Business Case

Decide whether an engineering project deserves to exist, and set it up to succeed

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain engineering project life cycles, stage gates and the roles of owner, consultant, contractor and operator
- Conduct a feasibility study and options analysis with an honest do-nothing baseline and sound site selection
- Calculate and interpret NPV, IRR, payback, discounted payback and benefit-cost ratio
- Build a business case with sensitivity and scenario analysis, whole-life carbon and an appropriate funding route
- Draft a project charter with governance, decision rights, tolerances and stage-gate approvals

LESSONS

1. Engineering Projects, Life Cycles & Stage Gates · 30 min
2. Feasibility Studies & Options Analysis · 35 min
3. Time Value of Money: NPV, IRR, Payback & Benefit-Cost Ratio · 40 min
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MODULE 1 · JUSTIFY · LESSON 1 · 30 MIN

Engineering Projects, Life Cycles & Stage Gates

How capital projects move from idea to operating asset, and who does what along the way

- Distinguish the project life cycle from the asset life cycle and explain why owners must care about both
- Describe the roles of owner, owner's engineer, designer, EPC contractor, vendors and O&M operator
- Explain how stage gates and front-end loading protect capital investment decisions

Engineering projects create physical assets that must work for decades: a substation, a viaduct, a dyeing house, an LNG jetty, a hospital wing. That changes the nature of project management. Most of the money is committed through **irreversible decisions**: once piles are driven, a turbine is ordered or a raft foundation is poured, changing your mind becomes very expensive. Many organisations are involved at once, each with its own contract, incentives and appetite for risk. And the real test of success is not the ribbon-cutting but whether the asset delivers the output, safety and reliability it was built for, year after year.

It helps to separate two life cycles. The **project life cycle** runs from the first idea to handover: identify, select, define, execute and commission. The **asset life cycle** is far longer: it contains the project, then years of operation and maintenance, upgrades and finally decommissioning or demolition. Owners pay for the whole asset life cycle, so decisions made during the project, such as material choices, equipment redundancy, access for maintenance and energy efficiency, should be judged on whole-life cost and performance, not on capital cost alone. Contractors are mostly paid for the project life cycle, which is one reason owner and contractor priorities can quietly diverge. In the ZEPMS Delivery Loop, **Justify** covers the Identify and Select phases below, **Define** covers Define, the middle stages run across Define and Execute, and **Commission** closes the project and opens operations. Asset management standards (ISO 55000 and ISO 55001, both revised in 2024) expect project decisions to trace back to the organisation's asset management objectives, while ISO 21502:2020 gives general guidance on governing and managing the project itself.

A typical capital project life cycle (owner's view)

01 Identify — Business need or opportunity, strategic fit and a first rough cost range

→

02 Select — Pre-feasibility and feasibility; compare options and choose the preferred concept

→

03 Define — Basis of design, FEED, execution plan, estimate and schedule good enough to sanction

→

04 Execute — Detailed design, procurement, fabrication and construction

→

05 Commission — Test systems, prove performance and hand over to operations

→

06 Operate — Run, maintain and measure benefits; feed lessons into the next project

Who does what on an engineering project

ROLE	MAIN INTEREST	TYPICAL RESPONSIBILITIES	RISK USUALLY CARRIED
Owner (employer, client)	Return on investment, public service or production capacity	Sets objectives, funds the project, approves gates, accepts the asset	Business risk: demand, financing, permits, owner-supplied information

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ROLE	MAIN INTEREST	TYPICAL RESPONSIBILITIES	RISK USUALLY CARRIED
Owner's engineer or consultant	Professional reputation and fees	Advises the owner, prepares feasibility and FEED, reviews contractor designs, supervises quality	Liability for the professional care of its own advice and design
Designer (in-house or appointed)	A buildable, compliant design delivered on time	Calculations, drawings, specifications, models and design reviews	Design errors within its own scope
EPC or main contractor	Margin, cash flow and repeat work	Detailed design (if EPC), procurement, construction, subcontractor management, site safety	Price, productivity and schedule risk as far as the contract allocates them
Vendors and subcontractors	Selling equipment or specialist services	Supplying switchgear, turbines, chillers, piling or steel erection, with performance guarantees	Performance and delivery of their own package
O&M operator	Safe, reliable and efficient operation	Operations input to design, participation in commissioning, running the asset	Availability and performance after handover

Same discipline, different emphasis by sector

SECTOR	ILLUSTRATIVE PROJECTS	DEFINING TECHNICAL CHALLENGE	WHAT TENDS TO DOMINATE THE BUSINESS CASE
Buildings	Hospital blocks, university campuses, high-rise offices	MEP coordination in tight ceiling voids, fire safety, fit-out interfaces	Functional brief, whole-life running cost, occupancy date
Transport infrastructure	Bridges, expressways, metro lines, ports	Geotechnics, temporary works, traffic and utility diversions, systems integration	Economic benefits to society, such as time savings, and public funding
Manufacturing	Garment and textile factories, semiconductor fabs, food plants	Process layout, utilities, clean or controlled environments, buyer compliance	Production capacity, time to market, product quality
Energy	Gas-fired and solar plants, substations, offshore wind	Grid connection, heavy lifts, equipment lead times, performance tests	Tariff or power purchase terms, availability, fuel supply
Process plant	Effluent treatment, LNG terminals, refineries, chemical plants	Process safety, hazardous areas, complex commissioning	Throughput, regulatory compliance, safety integrity

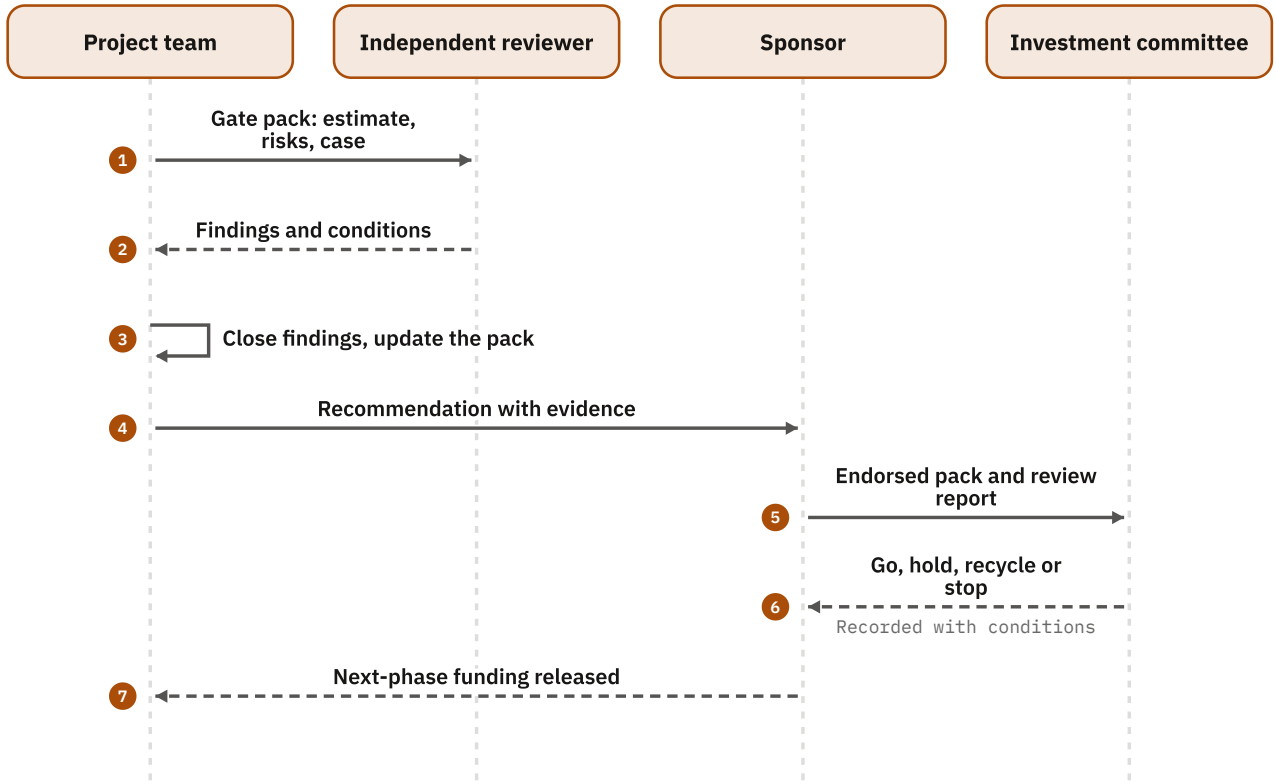
A **stage gate** is a planned decision point between life-cycle phases where the owner decides whether to proceed, recycle the phase, put the project on hold or stop it. Each gate has defined entry deliverables, for example a feasibility report, an estimate of a stated maturity, a risk register and an updated business case, and a named decision-maker. The value of a gate lies in the **option to stop**: money spent before the gate buys information, and the gate is where that information is used. Gates that only ever say yes are rubber stamps; they give boards false comfort and teams no reason to surface bad news.

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What a good gate review checks

- Is the need still valid and still aligned with strategy?
- Are the gate deliverables complete and at the agreed level of maturity?
- Have the estimate and schedule been independently challenged?
- Are the main risks understood, owned and allowed for?
- Is funding secured for the next phase, and is the plan for that phase credible?
- Is the decision recorded, with any conditions, owners and dates?

How a stage-gate decision is made

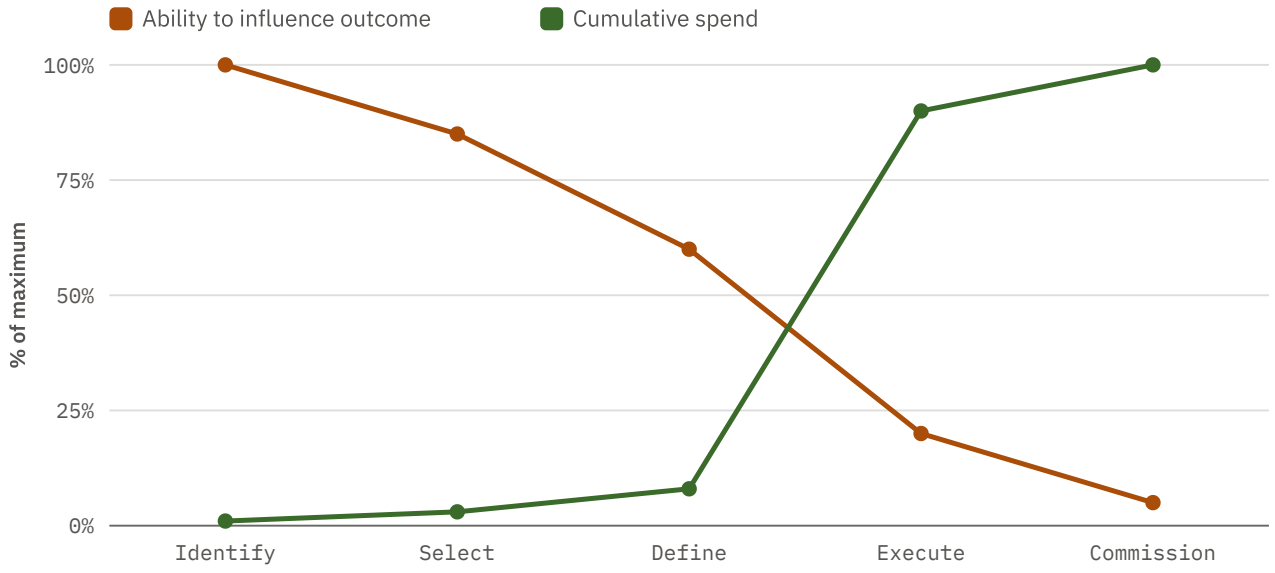


The independent challenge comes before the sponsor endorses the pack, and stop or recycle are real outcomes. Money for the next phase is released only after the decision and its conditions are recorded.

Front-end loading means investing effort early, in definition, design basis, site investigation, permits and execution planning, before committing to full-scale execution. The logic is simple: the ability to influence the outcome is highest at the start, when only a small share of the budget has been spent, and it falls steeply once contracts are signed and construction begins. A geotechnical investigation that costs a tiny fraction of the foundations can avoid redesigning pile caps after the piling rigs have arrived. Owners who rush the front end to start construction sooner often finish later, because unresolved scope and design questions surface during execution, when they are most expensive to fix.

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Influence falls as spending rises



Illustrative shape, not data from a real project: most of the ability to shape cost, carbon and performance is used before execution, while most of the money is spent during it. That gap is the case for front-end loading.

Rushed front end vs well-loaded front end

Rushed front end

- Construction starts with design 'to follow'
- Site investigation skipped or minimal
- Estimate based on a rough benchmark rate
- Operations staff not consulted on layouts
- Changes and claims dominate the execution phase

Well-loaded front end

- Design basis frozen before major contracts are signed
- Ground, utilities and access investigated
- Estimate built on defined quantities and assessed risk
- Operators review layouts and maintenance access
- Execution effort goes into productivity and safety

INSIGHT · CARBON AND CONTRACTORS FOLLOW THE SAME LOGIC

Whole-life carbon behaves like cost: carbon management guidance such as **PAS 2080:2023** stresses that the largest reductions come from early choices, such as questioning the need, reusing existing assets or building less, long before materials are bought. Buildability knowledge is also most valuable early, which is why many owners now use **early contractor involvement**, often under a two-stage arrangement, so that a contractor helps shape the design and plan before the price is fixed.

EXAMPLE · A MEGAPROJECT'S FRONT END

Bangladesh's **Padma Multipurpose Bridge**, which opened to road traffic in 2022 and carries both road and rail, shows how much of a megaproject happens before construction: studies of a very large and mobile river, decisions on alignment and river training, funding arrangements that changed during the front end, and land acquisition and resettlement. For any project of this kind, the quality of the front-end work largely determines how predictable execution can be.

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SCENARIO · DECIDE · SKIP THE FRONT END?

You are the owner's project manager for a new dyeing and finishing unit at a garment group's site in Gazipur. A major buyer has signalled larger orders from next year. The managing director wants piling to start next month using the layout of a similar plant the group built elsewhere, and says the soil report and utility capacity study can be done 'in parallel'. The structural consultant warns that part of the site is filled land. What should you do FIRST?

(A) Mobilise the piling contractor as instructed and ask the consultant to adjust the pile design once the soil report arrives.

(B) Present the managing director with a short fast-track front-end plan: commission the soil investigation and utility check now, show the cost and delay risk of starting without them, and propose early works that do not depend on the results.

(C) Refuse to proceed until a full feasibility study, FEED and detailed design are complete.

(D) Reuse the other plant's pile design but add a generous margin to pile lengths as a safety factor.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a project you have worked on. Where were the stage gates, who made the decision at each one, and was there ever a real possibility of stopping? What front-end work, if it had been done earlier, would have saved the most money or time?

Key terms

Capital project
A project that creates or substantially upgrades a long-lived physical asset, such as a plant, building or piece of infrastructure, funded from capital rather than operating budgets.

Asset life cycle
The full life of a physical asset from concept through design, construction, operation, maintenance and renewal to decommissioning; the project is only its first part.

Front-end loading
Deliberately investing effort in definition, design basis, investigations and execution planning before sanctioning execution, when influence over outcomes is highest and changes are cheapest.

O&M operator
The organisation, in-house or contracted, that will operate and maintain the asset after handover and should influence its design and commissioning.

Project life cycle
The sequence of phases a project passes through from identifying the need to handing over the finished asset, typically identify, select, define, execute and commission.

Stage gate
A planned decision point at the end of a life-cycle phase where a named authority reviews defined deliverables and decides to proceed, recycle, hold or stop.

Owner's engineer
An engineering consultant appointed by the owner to advise on technical matters, prepare or review designs and monitor contractors on the owner's behalf.

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Key takeaways

- Judge project decisions on whole-life cost, carbon and performance: the owner pays for the asset life cycle, not only the project.
- Each party is paid for different things and carries different risks, so owner and contractor priorities must be actively aligned.
- A stage gate is worth having only if stop, hold and recycle are real outcomes, decided on defined deliverables by a named decision-maker.
- Front-end loading spends a little early, when influence is highest, to avoid expensive change during execution.

KNOWLEDGE CHECK

1. A hospital owner chooses chillers with a higher purchase price but much lower energy and maintenance costs over 25 years. Which view is the owner applying?

   (A) The project life cycle, because chillers are bought during execution
   (B) The procurement cycle of the mechanical package
   (C) The asset life cycle, judging whole-life cost and performance
   (D) The contractor's view, which ends at handover

2. What gives a stage gate its real value to an owner?

   (A) The genuine option to stop, hold or recycle based on new information
   (B) A formal occasion to recognise the team's work at the end of each phase
   (C) The chance to transfer all remaining risk to the contractor
   (D) The requirement to publish the budget to all bidders

3. Why is front-end loading usually worth its extra early cost?

   (A) Because early studies are normally paid for by contractors
   (B) Because regulators require FEED for every building
   (C) Because it guarantees that no changes occur in execution
   (D) Because influence is highest and changes cheapest early on

4. On an EPC-delivered substation, who normally prepares the detailed design?

   (A) The owner's operations team
   (B) The EPC contractor or its design subconsultant
   (C) The lender's technical adviser acting for the banks
   (D) The O&M operator after handover

5. Which party is best placed to advise on maintenance access to transformers and switchgear during design?

   (A) The piling subcontractor
   (B) The lender's legal counsel
   (C) The future O&M operator
   (D) The architect's visualisation team

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · JUSTIFY · LESSON 2 · 35 MIN

Feasibility Studies & Options Analysis

Testing whether a project can and should be done, and comparing real alternatives including doing nothing

- Explain the purpose and depth of pre-feasibility and feasibility studies
- Assess a project across technical, market, economic, environmental, social and legal dimensions
- Run a structured options analysis with an honest do-nothing or do-minimum baseline
- Apply practical site selection criteria to an engineering facility

A feasibility study answers two questions that are easy to confuse: **can** this project be done, and **should** it be done? A design can be technically excellent and still fail because demand is too low, the tariff does not cover costs, the land cannot be acquired or a permit will never be granted. A good study is therefore multidisciplinary and candid. Its job is to find the reasons the project might not work while it is still cheap to change direction, not to dress up a decision that has already been made.

Studies are staged so that effort grows with confidence. An **identification note** frames the need and a rough range of costs. A **pre-feasibility study** screens a long list of options and eliminates those with fatal flaws, using desk data, benchmarks and reconnaissance visits. A **feasibility study** develops one to three short-listed options in enough depth, with site investigations, preliminary layouts, demand studies, cost estimates and environmental and social screening, to support an investment decision or to raise finance. Lenders often ask for a bankable feasibility study, meaning one robust enough for their technical adviser to rely on.

The dimensions of feasibility

DIMENSION	KEY QUESTION	TYPICAL EVIDENCE	RED FLAG
Technical	Can it be designed, built and operated with proven technology on this site?	Surveys, geotechnical data, preliminary design, technology references	Unproven process, poor ground, no route for heavy equipment
Market and demand	Will there be enough demand, throughput or users?	Demand forecasts, buyer commitments, traffic counts, offtake agreements	Forecasts resting on optimistic growth with no evidence
Economic and financial	Do benefits exceed costs, and can the owner fund it?	Capital and operating cost estimates, NPV, IRR, funding plan	Positive only under the most favourable assumptions
Environmental	Can impacts be avoided, reduced or managed acceptably?	Screening, baseline surveys, climate-risk and whole-life carbon screening, initial impact assessment (developed in Protect)	Wetland, protected habitat or flood-prone land in the footprint; flood levels that ignore climate projections
Social	Are affected people, land and livelihoods respected?	Land records, stakeholder mapping, resettlement screening	Large involuntary resettlement or strong community opposition
Legal and regulatory	Can permits, land rights and approvals be obtained in time?	Permit register, zoning checks, legal opinions	Disputed land title or a permit path longer than the schedule

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Pre-feasibility vs feasibility

Pre-feasibility study

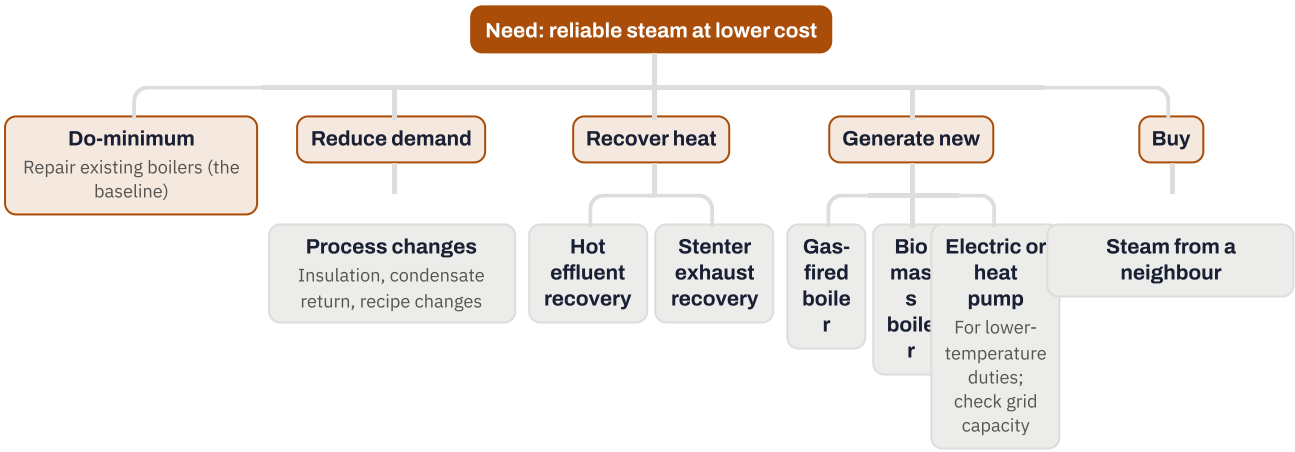
- Screens a long list of options
- Uses desk data, reconnaissance visits and benchmarks
- Gives cost estimates as broad ranges
- Identifies fatal flaws and data gaps
- Recommends which options deserve further study

Feasibility study

- Develops one to three short-listed options
- Uses site surveys, investigations and preliminary design
- Gives estimates with defined scope and contingency
- Quantifies risks, impacts and benefits
- Recommends a preferred option for the investment decision

Options analysis starts with the **need**, not the solution. If a textile mill's need is 'reliable steam at lower cost', the long list might include repairing the existing boilers, a new gas-fired boiler, a biomass boiler, waste-heat recovery from stenters and hot effluent, buying steam from a neighbouring plant, or cutting steam demand through process changes. Framing the need as 'build a new boiler house' would have silently removed half of those choices. Every options analysis must include a **do-nothing** (or do-minimum) option as the baseline: it shows what happens without new investment and is the reference against which the benefits of every other option are measured.

A long list built from the need, not the solution



Stating the need as an outcome opens up whole families of options, with do-minimum kept as the baseline. Each branch is screened for fatal flaws before the short list is developed in depth.

WATCH OUT · DO-NOTHING IS RARELY FREE

Doing nothing usually has costs: rising maintenance on an ageing transformer, penalties or loss of buyer certification when an effluent treatment plant fails discharge limits, lost production from frequent breakdowns, or growing safety risk in a deteriorating structure. Cost the do-nothing case honestly. If it is understated, every alternative looks better than it really is; if it is ignored, the analysis has no baseline at all.

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PRO TIP · CARBON AND CLIMATE BELONG IN THE OPTIONS APPRAISAL

Owners, lenders and regulators increasingly expect each short-listed option to show **whole-life carbon** alongside whole-life cost: embodied carbon from materials and construction (EN 15978 stages A1 to A5), carbon in use (stage B) and at end of life (stage C). The biggest savings come from options-stage choices such as reuse, building less or a different technology. Test every option against climate projections as well: a design flood level based only on past records may be too low for a 40-year asset.

Running an options analysis

1. **Define the need and objectives** State the problem in outcome terms, with measurable objectives and constraints such as a budget ceiling, a completion date and compliance requirements.
2. **Generate a long list** Include different technologies, locations, scales, phasing and non-build options. Involve operations, engineering and finance so the list is not limited to one discipline's preferences.
3. **Screen for fatal flaws** Remove options that fail a mandatory criterion, such as unproven technology, no viable site, an unlawful discharge or an unaffordable capital cost. Record why each was dropped.
4. **Assess the short list** Develop the remaining options far enough to compare capital and operating cost, NPV, whole-life carbon, risks, schedule, environmental and social effects and operability.
5. **Compare and score** Use discounted cash flow for the effects that can be valued in money and transparent multi-criteria scoring for the rest. Test whether the ranking changes when the weights change.
6. **Recommend and record** Recommend a preferred option with the reasons, the main uncertainties and what must be true for it to succeed, and keep the audit trail for the gate review.

TEMPLATE ### Options appraisal matrix

Compare short-listed options side by side, including the do-nothing baseline, so that the recommendation and its reasons are visible to gate reviewers.

OPTION	DESCRIPTION	CAPEX / ANNUAL OPEX (RELATIVE)	NPV VS DO-NOTHING	MAIN RISKS	NON-FINANCIAL SCORE (OUT OF 100)	STATUS
0 Do-minimum	*Repair the existing physico-chemical ETP and increase chemical dosing*	*Low / high*	*Baseline*	*Discharge limits still exceeded at peak load; failed buyer audits*	*35*	*Retained as baseline*
2 New biological ETP	*Equalisation, biological treatment, tertiary filtration and sludge dewatering*	*Medium / medium*	*Positive*	*Space on site, start-up of the biology, power interruptions*	*78*	*Recommended for feasibility*

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Site selection checklist for an industrial or energy facility

☐ Land: clear title, enough area for future expansion, no unresolved acquisition or encroachment

☐ Flood and drainage: floor and equipment levels above an agreed design flood level, with a drainage outfall

☐ Ground: geotechnical conditions known well enough to choose a foundation type, including filled or soft ground

☐ Access: road, river or rail access for heavy loads such as transformers, turbines and boilers, including bridges and turning radii

☐ Utilities: capacity and connection route for power, gas, water and telecoms, confirmed in writing by the utility where possible

☐ Effluent and waste: a lawful discharge point or treatment route, and space for waste storage

☐ Neighbours: distance from homes, schools and sensitive sites, and the noise, dust and traffic implications

☐ Permits and zoning: land-use classification and approvals required, with realistic timelines

☐ Labour and logistics: availability of skilled workers, accommodation and material supply routes

CASE STUDY · CHOOSING AN EFFLUENT SOLUTION FOR A DENIM WASHING PLANT

A hypothetical denim washing and dyeing plant near Narayanganj has an ageing physico-chemical effluent treatment plant that struggles at peak production. A major international buyer has warned that its next environmental audit must show consistent compliance with discharge limits, and the plant wants to expand washing capacity by about a third within two years. The owner's engineer has prepared a long list: do-minimum (repairs and extra chemical dosing), a new biological ETP on the existing site, a biological ETP with partial water recycling, connection to a planned central ETP for the industrial area, and zero liquid discharge. Land inside the factory boundary is tight, grid power has occasional interruptions, and the central ETP has no confirmed completion date. All figures in the study are illustrative.

1. Which options would you screen out at pre-feasibility, and on what grounds?

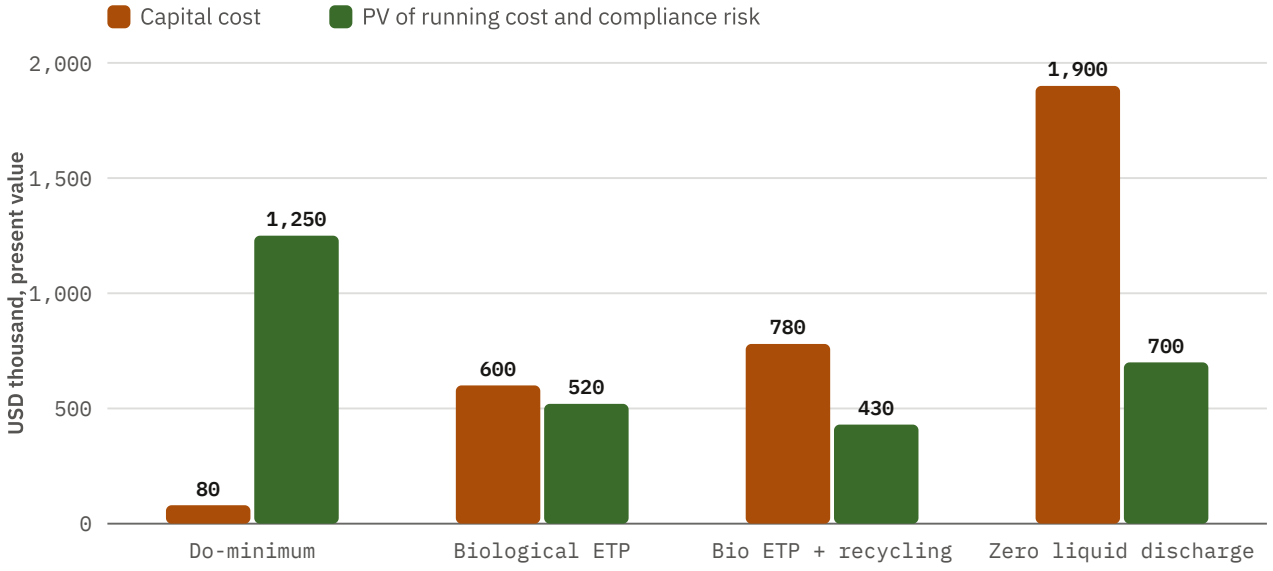
2. How should the uncertain completion date of the central ETP be treated?

3. Which non-financial criteria should be scored, and who should take part in the scoring?

Model answer: module answer key.

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How the effluent options might compare



Illustrative figures for the hypothetical denim plant: do-minimum is cheapest to build but carries the highest running cost and compliance risk, and zero liquid discharge fails on capital cost. The two biological options are close, so both go to feasibility.

SCENARIO · DECIDE · THE FEASIBILITY STUDY THAT MUST SAY YES

You lead a feasibility study for a hypothetical 50 MW gas-fired peaking plant. The draft results show a positive NPV only if gas is supplied at the full contracted volume every year; with the kind of supply curtailment the region has experienced, the NPV turns negative. The director who championed the project asks you to present only the full-supply case to the board because 'curtailment is temporary'. What should you do FIRST?

(A) Present only the full-supply case as requested, since the director owns the recommendation.

(B) Rework the analysis with a lower discount rate so that the curtailed case also shows a positive NPV.

(C) Discuss the finding with the director first, then present both cases with the gas-availability switching point and mitigation options such as dual-fuel capability.

(D) Stop the study and recommend cancelling the project immediately.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Pick a project in your organisation that is still at idea stage. Write its need as an outcome rather than a solution, then list at least five options including do-nothing. Which option would have been overlooked if the need had been framed as a solution?

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Key terms

Pre-feasibility study
An early, lower-cost study that screens a long list of options using desk data and limited site work, eliminating those with fatal flaws and selecting a short list for detailed study.

Feasibility study
A multidisciplinary study that develops short-listed options in enough technical, market, economic, environmental, social and legal depth to support an investment decision or financing.

Options analysis
The structured generation, screening and comparison of alternative ways to meet a defined need, leading to a recommended preferred option with recorded reasons.

Do-nothing option
The baseline option of making no new investment, or only the minimum needed to keep operating, against which the costs and benefits of all other options are measured.

Fatal flaw
A characteristic that makes an option unacceptable regardless of its other merits, such as an unobtainable permit, unproven technology or unaffordable cost, used to screen options early.

Multi-criteria analysis
A transparent method of scoring options against weighted financial and non-financial criteria, used alongside discounted cash flow when not all effects can be valued in money.

Site selection study
An assessment of candidate locations against criteria such as land, flood risk, ground conditions, access, utilities, neighbours and permits to choose the most suitable site.

Key takeaways

- Feasibility asks whether a project can be done and whether it should be; its job is to find fatal flaws while change is cheap.
- Start from the need in outcome terms and include an honestly costed do-nothing or do-minimum baseline.
- Screen the long list for fatal flaws, record why options were dropped and develop only one to three in depth.
- Compare the short list on whole-life cost, whole-life carbon, risk and non-monetised effects, and test the ranking.
- Site factors such as land title, flood level, ground, heavy-load access and utilities can each stop a project.

KNOWLEDGE CHECK

6. A pre-feasibility study is primarily intended to:

(A) Produce construction drawings for the preferred option
(B) Screen a long list and remove options with fatal flaws
(C) Fix the contract price with the EPC contractor
(D) Obtain final environmental clearance from regulators

7. A university needs more teaching space. Which option list best reflects good options analysis?

(A) Three architectural designs for the same new tower on the same plot
(B) A new tower with three alternative facade materials
(C) Three contractors' prices for an identical new tower
(D) Do-minimum, timetabling, refurbishment, a new block or leasing

8. Why must a do-nothing option be costed honestly?

(A) It is the baseline against which every other option's benefits are measured
(B) Regulators always require the do-nothing option to be shown as the cheapest one
(C) Do-nothing is normally the option that boards and investment committees choose
(D) Lenders will only finance projects that beat do-nothing on capital cost alone

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9. Which finding is most clearly a social feasibility red flag?

   (A) The preferred boiler technology has few regional installations

   (B) Demand forecasts rely on one buyer's verbal assurance

   (C) Large-scale involuntary resettlement is needed for the site

   (D) The discount rate has been set below the cost of capital

10. When candidate substation sites are compared, which item belongs in the site selection checklist rather than the tariff study?

    (A) The forecast electricity tariff in the tenth year of operation

    (B) Transformer delivery access and the design flood level

    (C) The parent company's dividend policy

    (D) The exchange rate assumed for imported spares

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · JUSTIFY · LESSON 3 · 40 MIN

Time Value of Money: NPV, IRR, Payback & Benefit-Cost Ratio

The investment arithmetic every engineering manager must be able to do by hand

- Explain discounting and choose a discount rate consistent with the cash flows
- Calculate NPV, IRR, simple and discounted payback and benefit-cost ratio from a cash flow table
- Interpret the measures together and recognise where each one can mislead

A sum of money today is worth more than the same sum in five years, for three reasons: money available now can be invested to earn a return, inflation erodes purchasing power, and future cash is uncertain. **Discounting** converts future cash flows into their equivalent value today, so a project's costs, which come mostly at the start, can be compared fairly with its benefits, which arrive over many years. Engineering investments are especially sensitive to discounting because their capital outlay is concentrated at the front and their benefits are long and spread out.

The **discount rate** carries much of the decision. A private owner usually starts from its weighted average cost of capital, the blended cost of its debt and equity, often adds a premium for project risk, and may set a minimum acceptable return called a **hurdle rate**. Public bodies use a social discount rate set by the finance ministry or funding agency for economic appraisal. Be consistent: if cash flows are in **nominal** terms (including expected inflation), use a nominal rate; if they are in **real** terms (constant prices), use a real rate. Mixing the two is one of the most common errors in engineering business cases. When inflation is volatile, as many economies experienced from 2021 onwards, state the inflation assumption explicitly and test it, because nominal and real figures drift apart quickly.

Discounting and net present value

MEASURE	FORMULA	WHAT IT TELLS YOU
Discount factor	`DF_t = 1 / (1 + r)^t`	Multiplier that converts a cash flow at the end of year t into today's value at discount rate r (as a decimal). At 10%, DF_3 = 1 / 1.1^3 = 0.7513.
Present value	`PV_t = CF_t / (1 + r)^t = CF_t x DF_t`	Today's value of a single cash flow CF_t received at the end of year t.
Net present value	`NPV = sum of CF_t / (1 + r)^t, for t = 0 to n`	Sum of all discounted cash flows, with the initial investment entered as a negative CF_0. NPV > 0 means the project earns more than the discount rate and adds value.
Present value of a level annuity	`PV = A x [1 - (1 + r)^(-n)] / r`	Shortcut for a constant amount A received at the end of each year for n years, useful for level savings, tariffs or availability payments.
Real and nominal rates	`(1 + nominal) = (1 + real) x (1 + inflation)`	Converts between rates. Real is roughly nominal minus inflation when rates are small. Use nominal rates with nominal cash flows and real rates with real cash flows.

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Investment decision measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Internal rate of return	`IRR = the rate r at which sum of CF_t / (1 + r)^t = 0`	The discount rate that makes NPV exactly zero. For conventional cash flows, accept if IRR exceeds the hurdle rate. Found by iteration or a spreadsheet function.
IRR by linear interpolation	`IRR ≈ r1 + NPV1 x (r2 - r1) / (NPV1 - NPV2)`	Hand estimate between a rate r1 giving a positive NPV1 and a higher rate r2 giving a negative NPV2. A narrow bracket, such as one percentage point, gives a close answer.
Simple payback	`Payback = Y + U / CF_(Y+1)`	Y is the last whole year at whose end cumulative cash flow is still negative, U is the unrecovered amount at the end of year Y, and CF_(Y+1) is the next year's cash flow. Ignores time value and all cash after payback.
Discounted payback	`Discounted payback = Y + U_d / PV_(Y+1)`	The same method applied to discounted cash flows: Y is the last year with negative cumulative PV, U_d the unrecovered PV and PV_(Y+1) the next year's present value. Always longer than simple payback when r > 0.
Benefit-cost ratio	`BCR = PV of benefits / PV of costs`	Present value returned per unit of present-value cost. BCR > 1 corresponds to NPV > 0 when the same cash flows and discount rate are used.

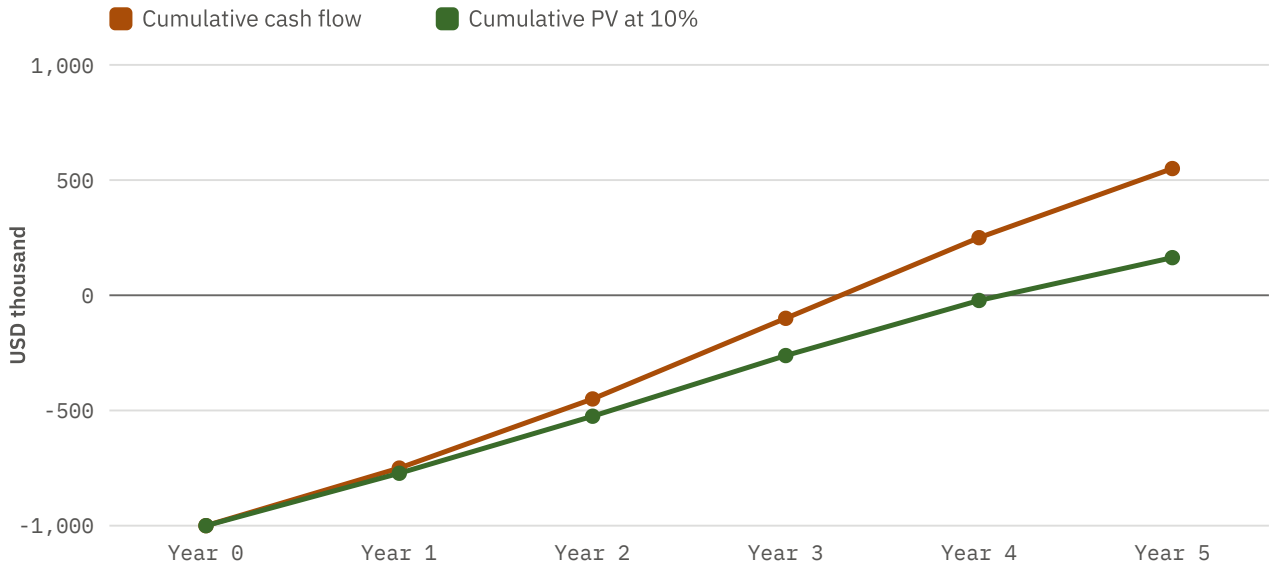
Worked example. A hypothetical dyeing and finishing plant in Narayanganj is considering a waste-heat recovery system that captures heat from hot effluent and stenter exhaust to preheat process water, reducing gas use in its boilers. The installed cost is **1,000** (all figures in USD thousand and purely illustrative), paid at the start, year 0. Net savings after extra maintenance and pumping power are forecast at **250** in year 1 while the system is tuned, **300** in year 2, **350** in years 3 and 4, and **300** in year 5 as heat-exchanger fouling increases. The analysis period is five years with no residual value, and the company's discount rate is **10%**.

Discounted cash flow table at r = 10% (USD thousand, illustrative)

YEAR T	NET CASH FLOW CF_T	DISCOUNT FACTOR 1/1.1^T	PRESENT VALUE	CUMULATIVE CASH FLOW	CUMULATIVE PV
0	-1,000	1.0000	-1,000.00	-1,000	-1,000.00
1	250	0.9091	227.27	-750	-772.73
2	300	0.8264	247.93	-450	-524.80
3	350	0.7513	262.96	-100	-261.84
4	350	0.6830	239.05	250	-22.79
5	300	0.6209	186.28	550	163.49

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Cumulative cash flow and cumulative present value



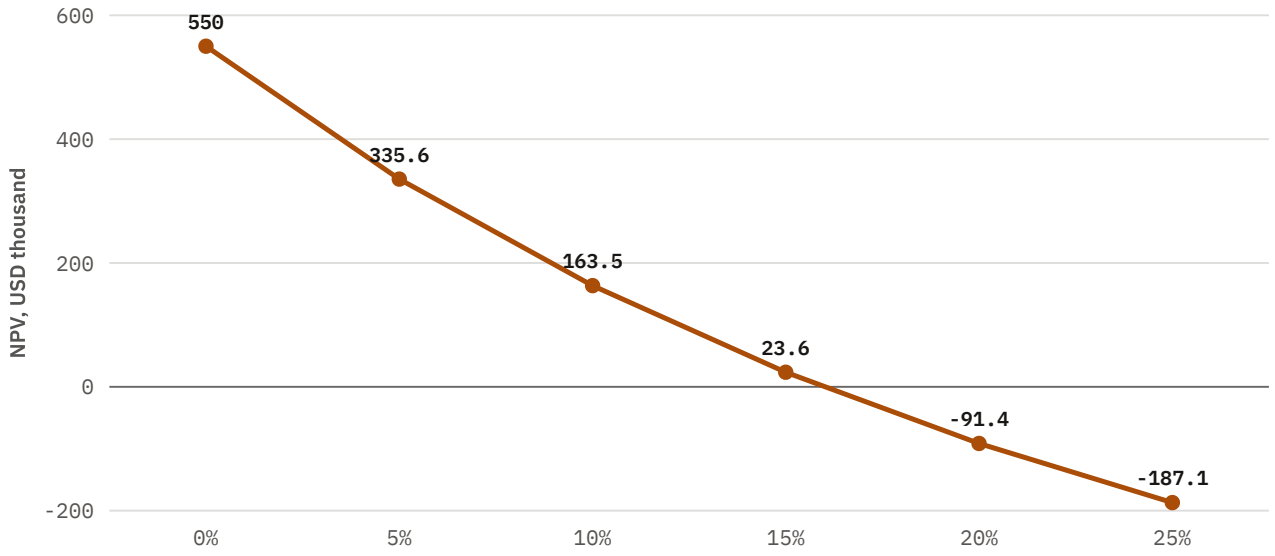
Illustrative figures from the table above. The undiscounted line crosses zero during year 4 (payback 3.29 years); the discounted line crosses only in year 5 (discounted payback 4.12 years), ending at the NPV of +163.5.

Reading the five measures from the table

1. **NPV** Add the present values: -1,000.00 + 227.27 + 247.93 + 262.96 + 239.05 + 186.28 = **+163.49**, which is also the final cumulative PV. The project creates about 163 of value today over and above a 10% return, so it passes the NPV test. (Present values are calculated with unrounded discount factors.)
2. **IRR** Try higher rates. At 15% the NPV is about +23.6; at 16% it is about -1.2. Interpolating: IRR ≈ 15% + 23.6 x (16% - 15%) / (23.6 - (-1.2)) = 15% + 0.95% = **about 15.95%**, well above the 10% hurdle. A wide bracket of 15% to 20% (NPV about -91.4 at 20%) would give about 16.03%, slightly overstated, which is why a narrow bracket is better.
3. **Simple payback** Cumulative cash flow is -100 at the end of year 3 and turns positive during year 4, when 350 arrives. Payback = 3 + 100 / 350 = **3.29 years**.
4. **Discounted payback** Cumulative PV is -22.79 at the end of year 4, and year 5 contributes 186.28. Discounted payback = 4 + 22.79 / 186.28 = **4.12 years**, almost a year longer than simple payback because later savings are worth less today.
5. **Benefit-cost ratio** The only cost here is the initial investment, because operating costs are already netted off the savings. PV of benefits = 1,163.49 and PV of costs = 1,000, so BCR = 1,163.49 / 1,000 = **1.16**: each unit of capital returns 1.16 units in present-value terms.

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NPV profile of the waste-heat recovery project



Illustrative figures from the worked example: NPV falls as the discount rate rises and crosses zero just below 16%, which is the IRR. At the company's 10% rate the NPV is +163.5, and at 0% it is simply the undiscounted total of 550.

INSIGHT · WHAT THE NUMBERS SAY TOGETHER

NPV says the investment creates value at the company's cost of capital. IRR says there is a margin of roughly six percentage points before the decision would flip. A discounted payback of just over four years in a five-year analysis warns that the case depends on the system still performing in year 5, so heat-exchanger fouling, maintenance and gas prices deserve sensitivity testing (see the next lesson). No single measure tells the whole story; investment committees should see all of them, with the assumptions behind them.

Strengths and blind spots of each measure

MEASURE	DECISION RULE	STRENGTH	BLIND SPOT
NPV	Accept if NPV > 0; among mutually exclusive options choose the highest NPV	Measures value created in money; additive across projects	Depends heavily on the discount rate; silent on how much capital is tied up
IRR	Accept if IRR > hurdle rate	Intuitive percentage, easy to compare with the cost of capital	Can rank mutually exclusive projects wrongly; multiple or no IRR with non-conventional cash flows
Simple payback	Accept if within a maximum period	Simple; a reasonable proxy for liquidity and exposure to change	Ignores time value and all cash after payback
Discounted payback	Accept if within a maximum period	Adds time value to the payback idea	Still ignores everything after the payback point
Benefit-cost ratio	Accept if BCR > 1	Useful for ranking under a fixed budget; common in public appraisal	Changes with how items are classified as costs or negative benefits; ignores scale

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WATCH OUT · TRAPS THAT CATCH EXPERIENCED ENGINEERS

Non-conventional cash flows: if a project has a large outflow late in life, such as decommissioning an offshore platform, a mid-life turbine overhaul or quarry restoration, the cash flow changes sign more than once and there may be two IRRs or none; rely on NPV. **Scale**: a small project can have a higher IRR but a much lower NPV than a large one; when options are mutually exclusive and capital is available, NPV is the better guide. **Classification**: treating operating costs as negative benefits rather than as costs changes the BCR, though not the NPV, so always state the convention. **Timing**: confirm whether cash flows fall at the start or end of each year. **Spreadsheets**: most spreadsheet NPV functions assume the first value in the range falls at the end of period 1, so enter the year-0 investment outside the function (NPV = CF_0 + NPV(rate, CF_1 to CF_n)), show the discount factor column explicitly and have a second person rebuild the headline figures by hand before any gate review. **Automated drafts**: when software or AI tools build a cash-flow model or extract figures from documents, treat the result as a draft, check every formula and input, and have a competent person sign off the numbers.

SCENARIO · DECIDE · HIGHER IRR OR HIGHER NPV?

A hypothetical spinning mill can install either Option A, a rooftop solar system costing 500 (USD thousand) with an IRR of 19% and an NPV of 180 at its 10% discount rate, or Option B, a larger 1,400 system covering the whole roof and a carport structure, with an IRR of 15% and an NPV of 390. The roof can host only one design, the company has funds for either, and the risks are similar. The finance manager prefers A 'because it has the better return'. What should the project manager recommend?

(A) Option A, because its higher IRR shows it is the more profitable investment.

(B) Option B, because the options are mutually exclusive, capital is available and B adds more value; show the board both measures and explain the difference.

(C) Option A, because it will have the shorter payback and therefore less risk.

(D) Neither option until both IRRs exceed 20%.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Find the last investment proposal you saw, or prepare one for your capstone project. Rebuild its NPV by hand with a discount factor column. Were the cash flows real or nominal, and did the discount rate match? What happens to the decision if the discount rate rises by two percentage points?

Key terms

Time value of money
The principle that money is worth more now than the same amount in the future, because of earning potential, inflation and risk; the basis of discounting.

Discount rate
The annual rate used to convert future cash flows into present values, typically the owner's cost of capital or hurdle rate for financial appraisal, or a social discount rate for public economic appraisal.

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Net present value
The sum of all of a project's cash flows discounted to today at the chosen discount rate; a positive NPV means the project creates value above the required return.

Payback period
The time needed for cumulative undiscounted cash inflows to recover the initial investment; simple to compute but ignores time value and cash after recovery.

Benefit-cost ratio
The present value of benefits divided by the present value of costs; a ratio above 1 indicates that benefits exceed costs at the chosen discount rate.

Internal rate of return
The discount rate at which a project's NPV equals zero; for conventional cash flows the project is acceptable when its IRR exceeds the hurdle rate.

Discounted payback period
The time needed for cumulative discounted cash flows to recover the initial investment; always longer than simple payback when the discount rate is positive.

Key takeaways

- Discount future cash flows so front-loaded costs and long-tail benefits compare fairly; match nominal flows with nominal rates and real with real.
- NPV measures value created in money and is the primary rule, especially for mutually exclusive options and non-conventional cash flows.
- IRR, payback, discounted payback and BCR add views of margin, exposure and ranking, but each can mislead on its own.
- Show the discount factors, enter year 0 outside spreadsheet NPV functions and have someone rebuild headline figures by hand.

KNOWLEDGE CHECK

11. At a discount rate of 8%, what is the present value of 1,000 received at the end of year 2?
    - (A) 1,166.40
    - (B) 925.93
    - (C) 857.34
    - (D) 840.00

12. A project costs 600 at year 0 and returns 200 a year for four years. What is its simple payback?
    - (A) 2.5 years
    - (B) 3.0 years
    - (C) 3.5 years
    - (D) 4.0 years

13. NPV is +40 at 12% and -20 at 14%. Using linear interpolation, the IRR is approximately:
    - (A) 12.7%
    - (B) 13.0%
    - (C) 13.5%
    - (D) 13.3%

14. Which situation most strongly suggests relying on NPV rather than IRR?
    - (A) A project with a large decommissioning outflow at the end of its life
    - (B) A project with one initial outflow followed by steady savings
    - (C) A project assessed in real rather than nominal terms
    - (D) A project whose payback period is shorter than three years

15. Cash flows have been forecast in constant (today's) prices. Which discount rate should be used?
    - (A) The nominal cost of capital including expected inflation
    - (B) The expected inflation rate on its own
    - (C) A real discount rate that excludes inflation
    - (D) The interest rate on short-term bank deposits

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · JUSTIFY · LESSON 4 · 35 MIN

The Business Case & Investment Decision

Turning feasibility evidence into a decision a board, a ministry or a lender can trust

- Structure a business case around strategic, economic, commercial, financial and management dimensions
- Use sensitivity analysis, switching values and scenarios to test the robustness of a case
- Compare funding routes and explain what each type of funder expects
- Recognise optimism bias and apply practical correctives

A **business case** is the reasoned argument for an investment: why it is needed, which options were considered, why the preferred option offers the best value, whether it is affordable and fundable, and how it will be delivered. It is not a one-off document written to win approval. It matures in stages, typically a strategic outline at identification, an outline case after options analysis and a full case before the final investment decision, and it is revisited at each gate and at closeout, when the promised benefits are checked against what actually happened.

How a business case matures

IDENTIFY	SELECT	DEFINE	EXECUTE	OPERATE
Strategic outline case	**Outline business case**	**Full business case**	**Revalidated at each gate**	**Benefits review**
Need, strategic fit, long list and a rough cost range	Preferred option from the options analysis, with sensitivities	Firm estimate, procurement route and funding for the investment decision	Updated for cost, schedule and risk; stop if the value is lost	Actual benefits compared with those promised; lessons fed back

The same argument is refined at each stage rather than rewritten. It does not end at approval: it is revalidated during execution and checked against reality once the asset is operating.

The five dimensions of a business case

S

Strategic

Why is it needed? The problem, the objectives, the fit with corporate, sector or national strategy and the consequences of not acting.

E

Economic

Which option gives best value? Options including do-nothing compared with NPV, BCR and non-monetised effects, from society's viewpoint for public projects or the owner's for private ones.

C

Commercial

Can it be procured and contracted? Supplier and contractor market, packaging, contract types and risk allocation (developed in Procure).

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F

Financial

Is it affordable and fundable? Capital and operating budgets, cash flow by year, funding sources and the effect on the balance sheet or tariffs.

M

Management

Can it be delivered? Governance, team, schedule, risk management, benefits realisation and the readiness of the operator.

Engineers often blur **economic** and **financial** appraisal. Financial appraisal looks at cash to and from the owner: tolls, tariffs, production margins, capital and operating costs. Economic appraisal, used for public infrastructure, looks at value to society: travel time savings and fewer accidents on an expressway, reduced flood damage behind an embankment, health benefits from clean water, including effects that never appear as cash. A metro line can have a negative financial NPV because fares do not cover its costs and still a strongly positive economic NPV because of time savings, reduced congestion and lower emissions. Both views are legitimate, but they answer different questions and must not be blended into one number. Many public appraisal frameworks now also value changes in greenhouse gas emissions, often using a carbon value set by the government or funder; use the official value where one exists and state it.

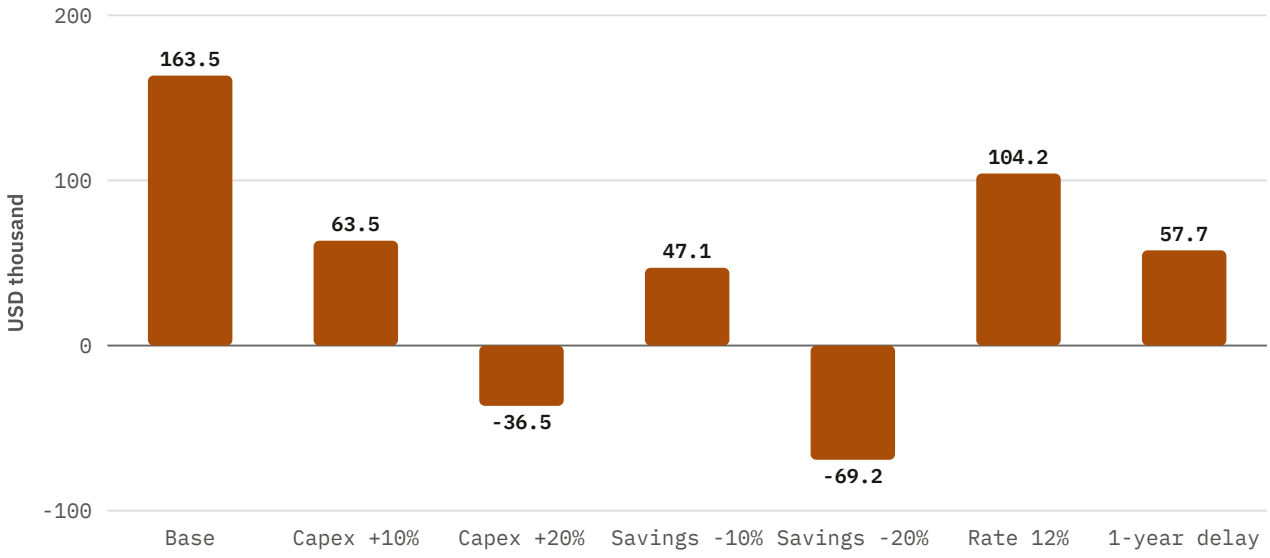
Every business case rests on forecasts that will be wrong to some degree. **Sensitivity analysis** varies one input at a time, such as capital cost, savings, discount rate or start date, and records the effect on NPV. The **switching value** is the change in an input that would make NPV exactly zero; it tells decision-makers how much room for error the case has. **Scenario analysis** changes several related inputs together to describe coherent futures, such as a low-demand, high-gas-price world. Quantitative risk analysis with Monte Carlo simulation goes further and is developed in Protect.

Sensitivity of the waste-heat recovery example (base NPV +163.5 at 10%, USD thousand, illustrative)

VARIABLE CHANGED	CHANGE	RESULTING NPV	COMMENT
Capital cost	+10%	+63.5	Still positive; the capital cost switching value is about +16%
Capital cost	+20%	-36.5	Negative: an overrun of this size destroys the case
Annual savings	-10%	+47.1	Savings switching value is about -14%: the most sensitive input
Annual savings	-20%	-69.2	Negative: vendor performance guarantees matter
Discount rate	12% instead of 10%	+104.2	Robust: the rate would have to exceed the IRR of about 16%
Start of savings	Delayed by one year	+57.7	A commissioning delay costs well over half the NPV

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NPV under each sensitivity case



Illustrative figures from the table: bars below zero are the cases that destroy value. Savings and capital cost dominate, so vendor performance guarantees and price risk deserve the most management attention.

Sensitivity analysis vs scenario analysis

Sensitivity analysis

- Changes one variable at a time
- Shows which inputs drive the result
- Produces switching values and ranked sensitivities
- Quick to compute and easy to explain
- Can mislead when variables move together

Scenario analysis

- Changes several linked variables together
- Describes coherent, plausible futures
- Produces a small set of results such as low, base and high
- Needs judgement about how variables relate
- Better for strategic and funding conversations

WATCH OUT · OPTIMISM BIAS AND HOW TO CORRECT IT

Research on large infrastructure programmes in many countries has repeatedly found that, at the decision stage, capital costs and durations tend to be underestimated and demand or benefits overestimated. Two causes are usually distinguished: genuine cognitive **optimism bias**, and **strategic misrepresentation**, where promoters understate costs to win approval. Practical correctives include reference class forecasting (adjusting an estimate using the actual outcomes of a class of similar completed projects), explicit uplifts where funding guidance requires them, independent estimate reviews at gates, and asking 'what would have to be true?' of every key assumption.

PRO TIP · PRICE VOLATILITY AND DELAY BELONG IN THE CASE

Since the early 2020s, prices and lead times for steel, cement, cable, transformers, switchgear and freight have swung sharply. A credible business case states which prices are fixed and when, how escalation is allowed for (for example a contract price-adjustment formula or an escalation allowance) and how long-lead equipment affects the date benefits start. Treat delay as a sensitivity in its own right: in the example above, one year of delay removes about two-thirds of the NPV.

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Funding routes for engineering projects

ROUTE	TYPICAL USE	WHAT THE FUNDER EXPECTS	IMPLICATIONS FOR THE PROJECT MANAGER
Public budget	Roads, bridges, hospitals and universities funded through the government's development programme	Fit with national plans, value for money, compliance with public procurement rules	Annual allocations can slow work; every decision is open to public audit
Public-private partnership	Toll roads, power plants, ports or hospitals financed, built and often operated by a private party under a long-term contract	A bankable revenue stream (tolls, availability payments, tariffs) and clear risk allocation	Long negotiations; the lenders' technical adviser reviews design, schedule and contracts
Development finance	Infrastructure and energy funded with multilateral or bilateral development banks or export credit agencies	Strong environmental and social safeguards (such as the World Bank Environmental and Social Framework or the IFC Performance Standards; many commercial lenders apply the Equator Principles), procurement under the lender's rules, independent monitoring	Safeguard compliance becomes a condition of disbursement; reporting is detailed and regular
Owner equity and corporate borrowing	Factory expansions, utilities and process plant funded from the company's own resources	A return above the hurdle rate, strategic fit, a manageable effect on cash flow and, increasingly, consistency with the company's climate targets and sustainability disclosures (for example under IFRS S1 and S2)	Faster decisions but tight capital rationing; projects compete with other uses of cash
Non-recourse project finance	Power plants and terminals where lenders are repaid only from the project's own cash flows	Firm offtake contracts, fixed-price construction, completion support, conservative financial ratios	Heavy due diligence and strong pressure for fixed-price, fixed-date contracts (see Procure)

TEMPLATE

Sensitivity and switching value table

Test how robust a business case is by recording the effect of each key variable on NPV and the change that would make NPV zero.

VARIABLE	BASE VALUE	LOW CASE	HIGH CASE	NPV AT LOW CASE	NPV AT HIGH CASE	SWITCHING VALUE
Annual energy savings	*250-350 per year (see cash flow)*	*-20%*	*+10%*	*-69.2*	*+279.8*	*About -14% (NPV = 0)*
Capital cost	*1,000*	*-10%*	*+20%*	*+263.5*	*-36.5*	*About +16% (NPV = 0)*

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VARIABLE	BASE VALUE	LOW CASE	HIGH CASE	NPV AT LOW CASE	NPV AT HIGH CASE	SWITCHING VALUE

CASE STUDY · A SOLAR PARK SEEKING DEVELOPMENT FINANCE

A hypothetical developer plans a 100 MW solar park on low-lying agricultural land in northern Bangladesh, selling power to the national utility under a long-term purchase agreement. The draft business case shows an attractive equity return. A multilateral development bank is considering a loan and has appointed a technical adviser. The adviser's first comments are that the energy yield estimate uses a single year of irradiance data, the capital cost assumes module prices at a recent low point, the land plan does not address sharecroppers who farm the land, and there is no scenario for grid curtailment. Parties and figures are illustrative.

1. Which dimensions of the business case do the adviser's comments touch, and how serious is each?

2. How would you restructure the sensitivity and scenario analysis?

3. What will the lender's safeguard requirements mean for the project manager's plan?

Model answer: module answer key.

SCENARIO · DECIDE · TAKE OUT THE DOWNSIDE?

You are preparing the full business case for a hypothetical expressway interchange to be funded from the public budget. The economic NPV is positive in the base case, but the low-traffic scenario, based on recent counts that show slower growth than the original forecast, gives a BCR of 0.9. The sponsoring agency's director asks you to drop the low-traffic scenario from the paper going to the planning ministry because 'it will only confuse them'. What should you do FIRST?

(A) Remove the scenario but keep it in your own files in case questions are asked later.

(B) Explain to the director why the scenario must stay, and propose presenting it clearly with the traffic-growth switching value and measures that could protect the benefits.

(C) Replace the recent traffic counts with the original forecast so every scenario looks positive.

(D) Send the full analysis directly to the planning ministry without telling the director.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

For a business case you know, which single assumption has the smallest switching value? Who owns that assumption, and what evidence would make you more confident in it before the next gate?

Key terms

Business case
The documented justification for an investment, setting out need, options, value for money, affordability, deliverability and expected benefits; developed in stages and revisited at each gate.

Five-case model
A widely used business case structure that tests a proposal against strategic, economic, commercial, financial and management dimensions.

Sensitivity analysis
Testing how a result such as NPV changes when one input is varied at a time, to identify the variables that matter most.

Switching value
The change in a single input that would make NPV zero or reverse the preferred option; a direct measure of how much error a business case can tolerate.

Scenario analysis
Assessing a project under a small number of internally consistent futures in which several related inputs change together, such as low demand combined with high fuel prices.

Optimism bias
The systematic tendency to underestimate costs and durations and overestimate benefits at the decision stage, corrected by reference class forecasting, uplifts and independent review.

Public-private partnership
A long-term contract in which a private party finances, builds and often operates a public asset or service, repaid through user charges or government payments, with risks shared by agreement.

Key takeaways

- A business case is a living argument across five dimensions: strategic, economic, commercial, financial and management.
- Keep financial appraisal (cash to the owner) and economic appraisal (value to society) separate; they answer different questions.
- Switching values and scenarios show how much room for error a case has; present downsides openly rather than removing them.
- Counter optimism bias with reference class forecasting, independent review and explicit allowances for price volatility and delay.
- Each funding route brings its own expectations, from bankable revenue to environmental and social safeguards.

KNOWLEDGE CHECK

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16. A business case section explains contractor market capacity, packaging and contract type. Which dimension is this?

(A) Strategic
(B) Management
(C) Financial
(D) Commercial

17. NPV falls to zero if annual savings drop by 14%. This 14% is best described as:

(A) The switching value for savings
(B) The internal rate of return
(C) The optimism bias uplift on the estimate
(D) The benefit-cost ratio

18. A new metro line shows a negative financial NPV but a positive economic NPV. The most likely explanation is that:

(A) The economic analysis used a discount rate that was set too high
(B) Construction costs were left out of the economic analysis
(C) Fares do not cover costs, but time savings and wider benefits do
(D) The two analyses mixed real and nominal cash flows

19. Which practice most directly counters optimism bias in a cost estimate?

(A) Asking the promoter to sign the estimate personally
(B) Reference class forecasting against completed projects
(C) Reducing contingency to make the project more competitive
(D) Using a lower discount rate for the most uncertain costs

20. Which funding route most typically makes environmental and social safeguards a condition of disbursement?

(A) Owner equity from retained profits
(B) A short-term overdraft from a commercial bank
(C) Payments from the annual maintenance budget
(D) A loan from a multilateral development bank

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · JUSTIFY · LESSON 5 · 30 MIN

Charter, Governance & Initiation

Authorising the project, setting decision rights and launching the team well

- Draft a project charter that authorises an engineering project and delegates authority
- Design a governance structure with clear decision rights, tolerances and stage-gate approvals
- Plan the initiation sequence and kickoff from investment decision to mobilisation

Once the investment decision is taken, the project needs formal authority to exist. The **project charter** provides it: it names the project, its purpose and its sponsor, appoints the project manager, sets high-level scope, budget, milestones and constraints, and states what the project manager may decide without asking. On engineering projects the charter is also where the owner records the outputs of the front end, the approved business case, the basis of design, the estimate and the target dates, so that everyone starts from the same baseline rather than from their own memory of earlier discussions. International guidance such as ISO 21502:2020 treats this formal authorisation, with defined governance and a business case that stays under review, as a basic condition for managing any project.

What a good engineering project charter contains

- Purpose, business need and a link to the approved business case
- Objectives and success criteria: capacity, performance, safety, cost, completion, carbon and benefits
- High-level scope with key inclusions, exclusions and battery limits (detailed in Define)
- Summary budget with contingency, and key milestones including gate dates
- Sponsor, project manager and governance bodies, with decision rights and tolerances
- Major assumptions, constraints, risks and dependencies such as permits, land and grid connection
- Information requirements and the common data environment to be used, following the ISO 19650 approach
- Approval signatures and date

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Governance layers on a typical owner-led capital project (top first)

Board or investment committee	Takes the final investment decision Sets strategic priorities and capital limits Receives exceptions beyond steering committee tolerance
Steering committee	Usually chaired by the sponsor Approves gates and major changes within delegated limits Resolves cross-functional conflicts
Project sponsor	Owns the business case and benefits Champions the project and secures resources Escalates decisions beyond own authority
Project manager and PMO	Plans and controls delivery within tolerances Integrates design, procurement, construction and commissioning Reports status, risks and forecasts
Package and contractor teams	Deliver work packages under contracts Manage subcontractors, safety and quality on site Raise issues and early warnings

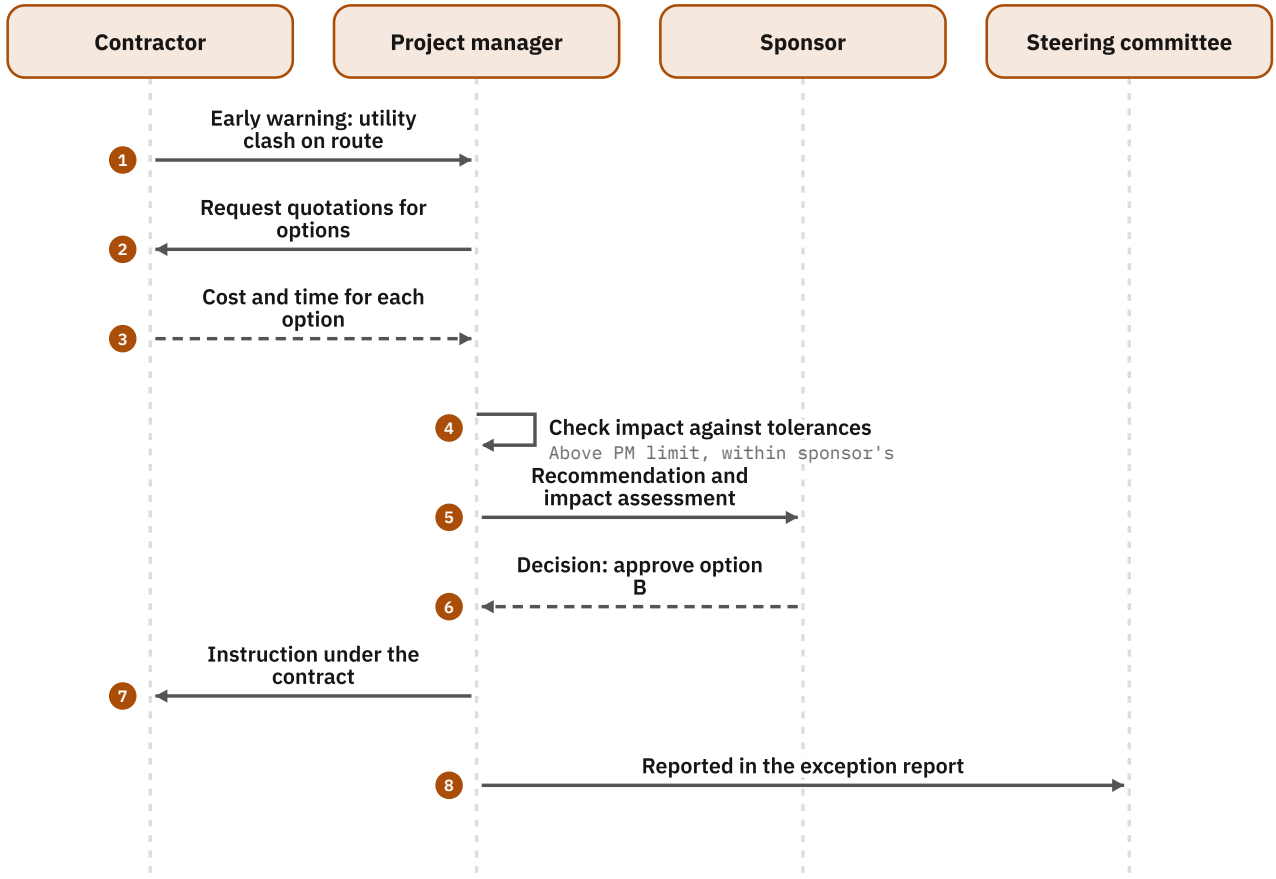
Governance works when everyone knows who decides what. **Decision rights** assign each class of decision, such as approving a gate, awarding a contract, releasing contingency or changing the design basis, to a specific person or body. **Tolerances** set the limits within which each level may act before escalating: for example, the project manager may approve changes up to a stated value and a few days of schedule impact, the sponsor up to a higher limit, and anything above goes to the steering committee. Tolerances should cover cost, time, scope, quality and safety, and they must be meaningful: too tight and the steering committee drowns in trivia; too loose and real problems stay hidden until it is too late to act. On projects that manage a carbon budget, carbon needs a tolerance too.

Illustrative decision rights matrix (thresholds are examples only)

DECISION	PROJECT MANAGER	SPONSOR	STEERING COMMITTEE	BOARD
Proceed through the gate into execution	Recommends	Endorses	Reviews	Decides (final investment decision)
Approve a change within the PM's tolerance	Decides	Informed	Informed in the monthly report	Not involved
Release contingency above the PM's tolerance	Requests with justification	Decides within own limit	Decides above the sponsor's limit	Not involved unless budget is exceeded
Award a major EPC or equipment contract	Recommends after evaluation	Endorses	Decides within delegated limit	Decides above that limit
Change the basis of design or a performance requirement	Assesses impact	Recommends	Decides	Informed
Stop work for a serious safety concern	Decides immediately (as can anyone on site)	Informed at once	Informed at once	Informed

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Escalating a change beyond the PM's tolerance



The decision goes to the lowest level whose tolerance covers it, here the sponsor, and the committee is informed rather than asked. The contractor is instructed only after the decision, through the contract's own procedure.

A **project management office** supports governance with standards and data. On a single large project it may be the project controls team that runs the schedule, cost reports, risk register and document control. Across an owner's portfolio it may set procedures, templates and gate criteria, provide planners and cost engineers, and assure the quality of reporting. PMOs range from **supportive** (offering tools and coaching) through **controlling** (requiring compliance with standards) to **directive** (managing projects directly). The test of a useful PMO is whether its data helps decision-makers act earlier, not how many reports it produces.

EXAMPLE · MANY ORGANISATIONS, ONE PROGRAMME

Urban rail projects such as Dhaka's first metro line, which began passenger service in late 2022, involve an implementing company, ministries, city authorities, utility owners, traffic police, lenders and several contractors for civil works, rolling stock and systems. Governance on programmes like this needs interface forums as well as a steering committee, because many critical decisions, such as utility diversions, road closures, power supply and station access, sit outside the project's direct control.

PRO TIP · SAFETY LEADERSHIP STARTS IN THE CHARTER

Since 2022 the ILO has counted a safe and healthy working environment among its fundamental principles and rights at work. Owners give that weight from the start: safety objectives sit beside cost and time in the charter, the owner's own duties in design and contractor selection are assigned, and the sponsor states that anyone may stop unsafe work. Site rules and contracts follow that lead.

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From investment decision to mobilisation

1. **Confirm the mandate** Obtain the signed investment decision and any conditions set by the board or lenders, and carry them into the charter.
2. **Issue the charter** The sponsor signs the charter, appointing the project manager and confirming decision rights, tolerances and governance bodies.
3. **Form the core team** Appoint leads for engineering, procurement, construction, project controls, HSE, quality and commissioning, and engage the owner's engineer.
4. **Set up controls** Establish the baseline schedule and budget, risk register, document control, reporting calendar and change procedure before major contracts are signed.
5. **Run the kickoff** Bring owner, consultants and early contractors together (under two-stage or alliance arrangements the contractor joins before the price is fixed) to align on objectives, safety leadership, roles, interfaces, information management and the first 90 days.
6. **Mobilise** Start site establishment, permits, early works and long-lead orders, and hold the first progress review against the plan.

Kickoff meeting readiness

☐ Charter signed and circulated, with objectives and success criteria in measurable terms
☐ Organisation chart showing owner, consultant and contractor roles with named contacts
☐ Decision rights and tolerances agreed and ready to explain
☐ Safety expectations and site rules prepared, including emergency arrangements
☐ Baseline milestones and gate dates ready to present
☐ Top risks and open issues listed with owners
☐ Communication plan: meeting rhythm, reporting dates and document control procedure
☐ Action log ready to capture decisions, owners and dates during the meeting

SCENARIO · DECIDE · A COMMITTEE MEMBER INSTRUCTS THE CONTRACTOR

Three months into construction of a hypothetical four-storey hospital block in Chattogram, a steering committee member visits site and verbally asks the main contractor's site manager to add an extra operating theatre in space designated for storage. The contractor starts setting out partitions and services and tells your site engineer the cost will follow later. The charter states that design changes go through the change procedure, with decisions above the project manager's tolerance taken by the steering committee as a body. What should the project manager do FIRST?

(A) Let the work continue, because a steering committee member outranks the project manager.

(B) Ask the contractor to pause the unapproved work, then promptly contact the member and the sponsor so the request goes through the change procedure with an impact assessment.

(C) Write to the contractor rejecting the request and forbidding any instruction from the member in future.

(D) Note the change in the next monthly report to the steering committee.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Look at a project you are involved in. Could every team member say who decides on contingency release, design changes and stopping work for safety? Where are the tolerances unclear, and what would you change in the charter?

Key terms

Project charter
The document, signed by the sponsor, that formally authorises a project, appoints the project manager and records purpose, objectives, high-level scope, budget, milestones and delegated authority.

Steering committee
A group of senior stakeholders, usually chaired by the sponsor, that gives direction, approves gates and major changes within delegated limits and resolves conflicts the project cannot.

Tolerance
The permitted deviation in cost, time, scope, quality or other measures within which a management level may act without escalating to the level above.

Kickoff meeting
The meeting at the start of execution or of a phase that aligns owner, consultants and contractors on objectives, roles, safety, interfaces, communication and immediate priorities.

Project sponsor
The senior individual accountable for the business case and benefits, who champions the project, secures resources and takes or escalates decisions beyond the project manager's authority.

Decision rights
The explicit assignment of each class of project decision to a named role or body, so that everyone knows who recommends, who decides and who is informed.

Project management office
A function that defines standards, provides project controls services and assures reporting for a project or portfolio; it may be supportive, controlling or directive.

Key takeaways

- The charter authorises the project, appoints the project manager and records the front-end baseline everyone starts from.
- Decision rights and tolerances say who decides what; a change goes to the lowest level whose tolerance covers it.
- Committees decide as bodies: individual members, however senior, should not instruct contractors directly.
- Put controls, safety leadership and information management in place before mobilisation, not after.

KNOWLEDGE CHECK

21. Which document formally appoints the project manager and delegates authority at the start of the project?

(A) The project charter
(B) The feasibility study
(C) The basis of design
(D) The contractor's method statement

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22. The project manager may approve changes up to a set cost and schedule impact without escalating. This limit is a:

(A) Hurdle rate
(B) Switching value
(C) Tolerance
(D) Stage gate

23. A PMO that offers templates and coaching but does not require compliance is best described as:

(A) Directive
(B) Supportive
(C) Controlling
(D) Executive

24. Who is normally accountable for the business case and the realisation of benefits?

(A) The EPC contractor's project director
(B) The owner's engineer
(C) The project management office
(D) The project sponsor

25. Which action belongs before the kickoff meeting rather than after it?

(A) Agreeing decision rights and tolerances to present to the team
(B) Holding the first monthly progress review against the baseline
(C) Starting inspections of long-lead equipment fabrication
(D) Walking down the first systems for punch list items

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · JUSTIFY

Answer key

Knowledge checks

1. C Judging a choice on running and maintenance costs over decades is an asset life cycle view. The project life cycle and contractor view both stop at handover and would favour the lower purchase price.

2. A A gate is valuable because information bought in the phase can change the decision. Celebrations, risk transfer or publishing budgets are not the purpose of a gate.

3. D Early definition is cheap relative to the value it protects, because changes made before contracts and construction cost far less. It reduces, but never guarantees against, later change.

4. B Under EPC delivery the contractor takes on engineering, procurement and construction, so detailed design sits with the contractor or its designers. The owner and its advisers review rather than produce it.

5. C The people who will maintain the equipment know what access, lifting and isolation they need. Bringing the operator in early is a core front-end loading practice.

6. B Pre-feasibility is a screening stage. Drawings, contract prices and final clearances come much later, after a preferred option has been developed.

7. D Good options analysis starts from the need and includes genuinely different solutions, including non-build and do-minimum options. The other lists are variations of one predetermined solution.

8. A Benefits of each option are measured relative to the do-nothing baseline, so an understated baseline inflates every option. The other statements are not general rules.

9. C Involuntary resettlement directly affects people and livelihoods. The other findings are technical, market and financial red flags respectively.

10. B Access for heavy loads and flood level are physical site criteria. Tariffs, dividends and exchange rates belong to the financial analysis.

11. C PV = 1,000 / 1.08^2 = 1,000 / 1.1664 = 857.34. 925.93 discounts only one year, 1,166.40 compounds instead of discounting, and 840.00 deducts simple interest.

12. B Cumulative cash flow reaches exactly zero at the end of year 3 (200 x 3 = 600), so payback is 3.0 years. The other values misread the cumulative position.

13. D IRR ≈ 12% + 40 x (14% - 12%) / (40 - (-20)) = 12% + 80/60 % = 13.3%. The other values place the zero point in the wrong part of the bracket.

14. A A late large outflow makes the cash flow change sign twice, which can produce multiple IRRs or none. The other situations do not undermine IRR in that way.

15. C Constant-price cash flows are real, so they must be discounted at a real rate. A nominal rate would double-count inflation and understate NPV.

16. D Procurement route, packaging and contract strategy belong to the commercial dimension. Financial covers affordability and funding, and management covers governance and delivery capability.

17. A A switching value is the change in one input that makes NPV zero. The IRR is a discount rate, not a change in savings.

18. C Financial appraisal counts only cash to the owner, while economic appraisal also counts benefits to society such as time savings. The other options describe errors, not the normal reason for the difference.

19. B Reference class forecasting anchors the estimate in the actual outcomes of comparable completed projects. Signatures, lower contingency or rate changes do not remove bias and may worsen it.

20. D Development banks attach safeguard requirements to their loans and monitor compliance before releasing funds. The other routes do not normally impose such conditions.

21. A The charter authorises the project and appoints the project manager. Feasibility studies and the basis of design are technical inputs, and method statements belong to contractors.

22. C A tolerance defines how far a level may act before escalating. A hurdle rate and switching value are financial measures, and a stage gate is a decision point between phases.

23. B A supportive PMO provides help without enforcing standards. A controlling PMO requires compliance and a directive PMO manages projects itself.

24. D The sponsor owns the business case and benefits on the owner's side. Contractors, consultants and the PMO contribute, but are not accountable for the investment outcome.

25. A Decision rights and tolerances should be settled so they can be explained at kickoff. Progress reviews, fabrication inspections and punch lists come later in execution.

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Scenarios & case studies

Scenario · Skip the front end?
B is best. Best. This respects the commercial urgency while applying front-end loading logic. A ground investigation takes weeks, not months, and protects the most irreversible work. Site clearance, boundary walls, temporary power and access roads can start meanwhile, so the decision-maker gets speed and evidence.

Case study · Choosing an effluent solution for a denim washing plant
Zero liquid discharge may be screened out at this stage if its capital and energy costs are clearly beyond the owner's means and it is unproven at this scale, but the reason must be recorded, not assumed. The central ETP should not be relied on as the only solution while its completion date is unconfirmed; it can be kept as a future option, with the on-site design checked so that a later connection remains possible. Do-minimum is the baseline and must be costed honestly, including the commercial risk of losing the buyer. The likely short list is a new biological ETP with and without water recycling, compared on NPV, footprint, resilience to power interruptions (for example standby generation for aeration), sludge handling, ease of operation by the plant's staff and capacity for the planned expansion. Operations, environment and compliance staff, finance and the owner's engineer should all score, and the ranking should be tested against changes in the weights.

Scenario · The feasibility study that must say yes
C is best. Best. You respect the director by raising it privately first, keep the board fully informed, and turn a problem into a decision with options. The board can then judge whether mitigation makes the project worthwhile.

Scenario · Higher IRR or higher NPV?
B is best. Best. When options are mutually exclusive and capital is not constrained, the higher NPV creates more value for the owner. Presenting both measures openly helps the finance manager and board understand why IRR alone misleads here.

Case study · A solar park seeking development finance
The comments touch the economic and financial dimensions (yield, capital cost and curtailment), the management and strategic dimensions (deliverability and land) and the commercial dimension through module procurement risk. The single year of irradiance data and the missing curtailment scenario go to the heart of revenue, so they need long-term irradiance datasets, a probabilistic yield estimate and a curtailment scenario based on the utility's grid data. Module price should be tested with sensitivities and a switching value, and the procurement plan should state when prices will be fixed. Sharecroppers depend on the land for their livelihoods even without formal title; development lenders' safeguards require them to be identified, consulted and compensated or supported, which affects the schedule, budget and stakeholder plan (developed in Protect and Integrate). The project manager should plan all of this as front-end work before financial close, not treat it as paperwork.

Scenario · Take out the downside?
B is best. Best. You address the director's concern about clarity while keeping the evidence. A switching value and mitigation options make the downside understandable and actionable rather than confusing.

Scenario · A committee member instructs the contractor
B is best. Best. Pausing stops unassessed, unpriced work and possible rework, while engaging the member and sponsor quickly treats the request as potentially valuable. The change procedure then gives the committee what it needs to decide.

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02

MODULE 2 · DEFINE

Scope, Requirements & Design Definition

Turn an approved investment into a defined, buildable and controlled scope

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Translate owner's requirements into a basis of design, specifications and applicable codes and standards
- Write a scope statement with battery limits, exclusions, assumptions and interfaces
- Build a deliverable-oriented WBS with a WBS dictionary, responsibility matrix and control accounts
- Manage design development through FEED, detailed design, reviews, value engineering, HAZOP and a common data environment to design freeze
- Establish a scope baseline and run disciplined change control that prevents scope creep

LESSONS

1. Requirements & Basis of Design · 35 min
2. Scope Statement & Work Breakdown Structure · 40 min
3. Design Development & Design Management · 40 min
4. Scope Baseline & Change Control · 35 min

MODULE 2 · DEFINE · LESSON 1 · 35 MIN

Requirements & Basis of Design

From what the owner needs to what the designers must design, written down, agreed and traceable

- Trace the cascade from business need to owner's requirements, basis of design and specifications
- Choose between performance and prescriptive specifications and apply codes and standards correctly
- Build constructability, operability and maintainability into requirements from the start

Many of the costliest problems on engineering projects begin as requirements nobody wrote down. If the designer of a dyeing house is not told that the owner plans to add machines in three years, the steam header, effluent channels and floor loading will be sized for today. If the operator's need to remove a transformer without dismantling the switch room is never stated, the room will be built tightly around it. Requirements management is the discipline of capturing what the owner needs, translating it into technical terms designers can work to, and keeping every requirement traceable through design, construction and testing to handover.

The requirements cascade

01 Business need
The outcome the investment must deliver, taken from the business case

→

02 Owner's project requirements
What owner and users need in their own terms: capacity, quality, reliability, comfort, budget

→

03 Basis of design
The designers' technical response: criteria, site data, codes, design life, key choices

→

04 Specifications and data sheets
Detailed requirements for materials, equipment, workmanship and testing

→

05 Drawings and models
The design deliverables that contractors build from

→

06 Verification
Tests and inspections proving each requirement is met (developed in Commission)

The **owner's project requirements** are written in the language of the business and its users: the factory must finish a stated number of tonnes of fabric per day, the ward must be kept at a comfortable temperature and pressure for infection control, the substation must keep supplying the export processing zone if one transformer fails. The **basis of design** is the engineers' answer: the design criteria, site data, codes, assumptions and main technical choices that will satisfy those requirements. It is the single reference every discipline designs to, and it should be controlled like a contract document, because once it is approved, changing it is a change to the project.

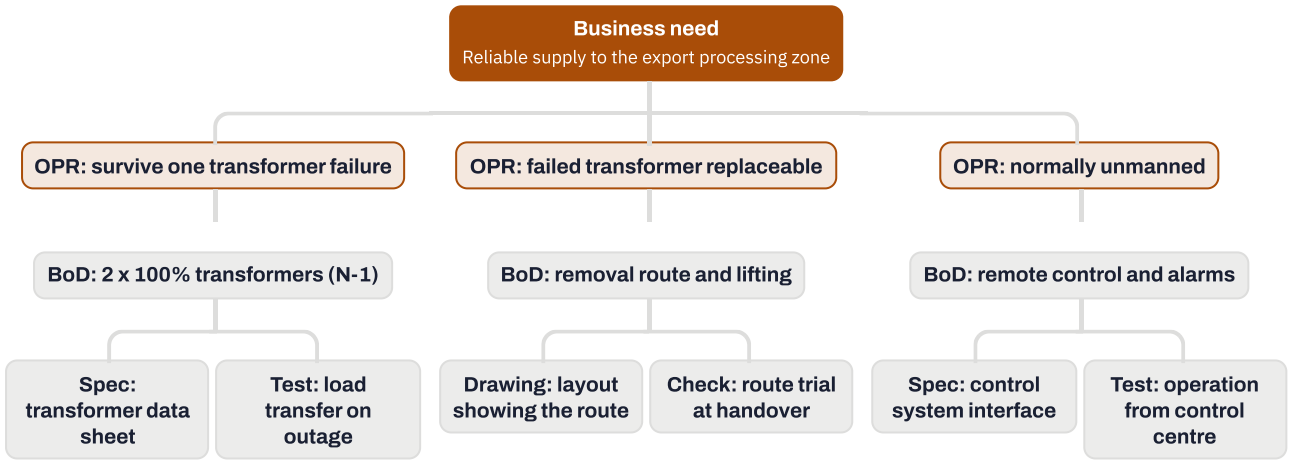
What a basis of design records: example for a hypothetical 33/11 kV substation

SECTION	CONTENT	ILLUSTRATIVE ENTRY
Design criteria	Ratings, capacities, redundancy and performance targets	Two 33/11 kV transformers, each able to carry the full forecast load if the other fails (N-1)
Site data	Climate, design flood level, seismic zone, ground conditions, pollution level	Floor and equipment plinths above the agreed design flood level; insulators chosen for a coastal pollution class
Codes and standards	Applicable national codes and international standards, with editions	National building and electrical codes plus the relevant IEC equipment standards, each edition listed

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SECTION	CONTENT	ILLUSTRATIVE ENTRY
Design life and durability	Required service life of structures and equipment	Long design life for civil structures; concrete cover and coatings matched to exposure
Operating philosophy	How the asset will be run and maintained	Normally unmanned with remote control; one transformer out for maintenance at a time
Carbon and resilience	Whole-life carbon target and climate assumptions	Embodied carbon budget for civil works; design flood level checked against climate projections, not only past records
Information requirements	Asset data and models the owner needs to operate and maintain	Data for every transformer and breaker delivered in the structure of the owner's maintenance system
Assumptions and exclusions	What the design takes as given and what it does not cover	Incoming 33 kV lines by the utility up to the gantry; telecom link by others

Tracing requirements for the substation



Every owner requirement should trace down to a design criterion, a deliverable and a test. The removal-route branch is exactly what is missing in the scenario later in this lesson, so nothing in design or testing would have caught the gap.

Performance vs prescriptive specification

Performance specification

- States the result: 'supply chilled water at 6 °C at peak load'
- Leaves the solution to the designer, contractor or supplier
- Encourages innovation and transfers design responsibility
- Needs clear, measurable acceptance tests
- Common in EPC and design-build packages

Prescriptive specification

- States the means: chiller type, materials and configuration
- Owner or its designer keeps responsibility for the outcome
- Gives control and standardisation, for example common spares
- Acceptance is by checking conformity to the description
- Common where the owner has strong standards or operating preferences

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Choosing a specification style

SUPPLY MARKET'S DESIGN CAPABILITY (LOW TO HIGH) →

Performance specification

- State outputs and tests
- Let suppliers innovate
- Typical of EPC packages

Performance plus owner standards

- Outputs plus a preferred equipment list
- Common spares and controls
- A hybrid specification

Design by the owner's engineer

- Owner's designer fixes the solution
- Simple, build-only package
- Owner keeps design risk

Prescriptive specification

- Owner's standard details and equipment
- Acceptance by conformity checks
- Owner keeps outcome risk

OWNER'S NEED FOR STANDARDISATION (LOW TO HIGH) →

A rule of thumb, not a law: performance specifications suit capable markets that can take design responsibility, while strong owner standards or a weak supply market push towards prescription. Many packages end up as hybrids.

Codes and standards turn decades of experience into minimum rules. A typical project draws on the national building code and fire regulations, electrical and grid codes, environmental discharge standards, and international standards for equipment and materials, such as IEC standards for electrical equipment, ISO standards for many management systems and test methods, and ASME or European standards for pressure equipment. The basis of design should list each applicable code with its edition, set an **order of precedence** for conflicts (normally statutory requirements first, then project-specific requirements, then referenced standards) and say who rules when two documents disagree. Codes set minimums; the owner's requirements may legitimately go further. In Bangladesh, for example, the Bangladesh National Building Code (BNBC) 2020 is the national building code for structural, fire and building services design.

WATCH OUT · EDITIONS, MIXING AND SILENT ASSUMPTIONS

Standards are revised, and designers and suppliers may work to different editions unless the edition is fixed. Mixing design methods from different code families, for example loads from one and resistance factors from another, can produce unsafe or wasteful results. Equally dangerous are assumptions that live only in a designer's head: allowable bearing pressure, fault level at the point of connection, the strength and flow pattern of incoming effluent. If it matters to the design, it belongs in the basis of design.

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Design-for lenses every requirement set should cover

C

Constructability

Can it be built safely and efficiently? Offsite and modular units designed for manufacture and assembly, standard reinforcement details, crane access, temporary works, sequence and lay-down space.

O

Operability

Can operators run it well? A clear control philosophy, sensible alarms, safe access to valves and instruments, flexibility for start-up and upsets.

M

Maintainability

Can it be maintained and replaced? Lifting beams over pumps, removal routes for transformers and chillers, isolation points, standardised spares.

S

Safety in design

Are hazards removed at source? Less work at height, segregated pedestrian routes, safe chemical unloading points, fire separation.

E

Energy and environment

Does it minimise whole-life energy, water and carbon? Efficient motors, heat recovery, daylight, water reuse, lower-carbon materials and less material overall.

EXAMPLE · CONSTRUCTABILITY ON A VIADUCT

On elevated metro and expressway viaducts, designers who standardise pier heads and girder lengths allow formwork and precast moulds to be reused many times and simplify lifting or launching sequences. A constructability review in which the contractor's temporary works and lifting engineers comment on early layouts can expose crane outreach limits, lay-down needs over live traffic and pile cap clashes with buried utilities long before they become site problems.

Basis of design review checklist

- ☐ Every owner's requirement is traceable to a design criterion or is explicitly excluded
- ☐ Site data (flood level, ground conditions, climate, seismicity) is sourced and dated
- ☐ Codes and standards are listed with editions and an order of precedence
- ☐ Design life, redundancy and availability targets are stated
- ☐ Operating and maintenance philosophy has been reviewed by the future operator
- ☐ Constructability input has been obtained from construction and temporary works specialists
- ☐ Assumptions, exclusions and interfaces with other parties are listed with owners
- ☐ Allowances for future expansion are stated or deliberately excluded
- ☐ Whole-life carbon targets, climate assumptions and the owner's information requirements are stated
- ☐ The document is under revision control with approval signatures

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SCENARIO · DECIDE · BUILT AROUND THE TRANSFORMER

During detailed design of a hypothetical 33/11 kV substation for an export processing zone, the owner's maintenance manager sees the layout for the first time and points out that once the switchgear building is complete, a failed power transformer could only be removed by demolishing a wall. The design consultant says the layout meets all codes and that the owner's requirements said nothing about removal routes. Civil works start in six weeks. What should the project manager do FIRST?

(A) Proceed with the current layout, because it complies with the codes and the written requirements.

(B) Ask the consultant to assess layout options for a transformer removal route, with cost and time impact, and take the change through change control before civil works start.

(C) Ask the contractor to leave one wall unfinished so it can be completed later if needed.

(D) Instruct the consultant to redesign at its own cost because it should have foreseen the need.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Take one system you know well: a boiler, a chiller plant, a pump station or a switch room. List three requirements an operator or maintainer would want but a designer working only from codes might miss. Where in your project's documents would they need to be written so that they are designed, built and tested?

Key terms

Owner's project requirements
A document that sets out, in the owner's and users' terms, what the asset must achieve, including functions, capacities, performance, quality, budget and operational expectations, against which the design is checked.

Performance specification
A specification that states the required outcome and acceptance tests, leaving the means of achieving it to the designer, contractor or supplier.

Constructability
The extent to which a design allows safe and efficient construction, considering sequence, access, temporary works, standardisation and site conditions.

Maintainability
The ease with which an asset can be inspected, maintained, repaired and replaced, including access, lifting provisions, isolation and spares.

Basis of design
The designers' controlled record of the technical criteria, site data, codes and standards, assumptions and key decisions used to meet the owner's requirements; the reference for all subsequent design.

Prescriptive specification
A specification that defines the materials, equipment types, methods or workmanship to be used, so that compliance is checked by conformity to the description.

Operability
The ease with which operators can run an asset safely and effectively across its operating modes, including start-up, normal operation, upsets and shutdown.

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Key takeaways

- Requirements flow from business need to owner's requirements, basis of design, specifications and tests, and must stay traceable.
- The basis of design is the single technical reference: control it like a contract document, with code editions and an order of precedence.
- Performance specifications transfer design responsibility and need measurable tests; prescriptive ones keep it with the owner.
- Build constructability, operability, maintainability, safety and whole-life carbon into requirements before layouts are fixed.

KNOWLEDGE CHECK

1. Which document is written mainly in the owner's and users' terms rather than designers' technical terms?
   - (A) The basis of design
   - (B) The owner's project requirements
   - (C) The equipment data sheets
   - (D) The issued-for-construction drawings
2. A chiller specification states only the required chilled water temperature, capacity and test method. This is a:
   - (A) Prescriptive specification
   - (B) Method statement
   - (C) Basis of design
   - (D) Performance specification
3. Two referenced standards give different requirements for cable derating. Where should the project already have defined how to resolve this?
   - (A) In the order of precedence in the basis of design
   - (B) In the contractor's monthly progress report
   - (C) In the terms of reference of the steering committee
   - (D) In the site safety induction materials
4. Providing a lifting beam above large pumps and a clear route to remove them mainly improves:
   - (A) Constructability
   - (B) Operability
   - (C) Maintainability
   - (D) Code compliance
5. Why should the edition of each code be fixed in the basis of design?
   - (A) Because older editions are always cheaper to design to
   - (B) Because designers and suppliers may use different editions
   - (C) Because regulators accept only the newest edition published
   - (D) Because the code edition determines the discount rate

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MODULE 2 · DEFINE · LESSON 2 · 40 MIN

Scope Statement & Work Breakdown Structure

Drawing the boundary of the work and breaking it into packages that can be planned, priced and controlled

- Write a scope statement with deliverables, battery limits, exclusions, assumptions and interfaces
- Build a deliverable-oriented WBS that obeys the 100% rule, supported by a WBS dictionary
- Link WBS, OBS and CBS through a responsibility assignment matrix, control accounts and work packages

The **scope statement** is the project's written boundary. It turns the charter's high-level scope into a precise description of what will be delivered, where the project starts and stops, what is excluded and what is assumed. On engineering projects vagueness here is rarely harmless: if the statement does not say who provides the 11 kV supply to a new pump house, or whether demolition of the old ETP tanks is included, someone will find out during construction, usually through a claim. Precision costs a few days of careful writing; ambiguity costs months of argument.

Elements of an engineering scope statement

- **Objectives and deliverables**: the facilities, systems and documents to be handed over, with capacities and performance
- **Battery limits**: the physical and contractual boundaries of the scope, such as tie-in flanges, cable terminations and plot boundaries
- **Exclusions**: items explicitly not in scope, to stop false assumptions
- **Assumptions**: conditions taken as true for planning, such as ground conditions or availability of grid power, each with an owner to verify it
- **Constraints**: fixed dates, budget ceilings, working-hour limits and live-plant shutdown windows
- **Interfaces**: points where this scope meets other contracts, existing plant or third parties
- **Acceptance criteria**: the tests and documents that prove each deliverable is complete

EXAMPLE · BATTERY LIMITS IN PRACTICE

For a new effluent treatment plant added to an operating textile factory, the battery limits might be: raw effluent from the inlet flange of the new equalisation tank; treated water to the outlet flange of the final discharge pit; power from the outgoing terminals of a named feeder in the existing LV panel; sludge to the loading point of the dewatered sludge store. Everything inside belongs to the ETP contractor. Everything outside, such as the factory drains, the LV panel and sludge disposal, belongs to others and must appear as an interface.

TEMPLATE

Battery limits and interface register

Record every point where the scope meets another party, contract or existing system, so that responsibilities, data and dates are agreed before they cause clashes.

INTERFACE ID	LOCATION / TIE-IN POINT	PARTY A (PROVIDES)	PARTY B (RECEIVES)	WHAT IS EXCHANGED	NEED DATE	STATUS
IF-07	*Outgoing terminals, feeder 4, existing LV panel*	*Factory electrical team*	*ETP contractor*	*400 V supply, cable termination and protection settings*	*Before pre-commissioning*	*Open: protection settings awaited*

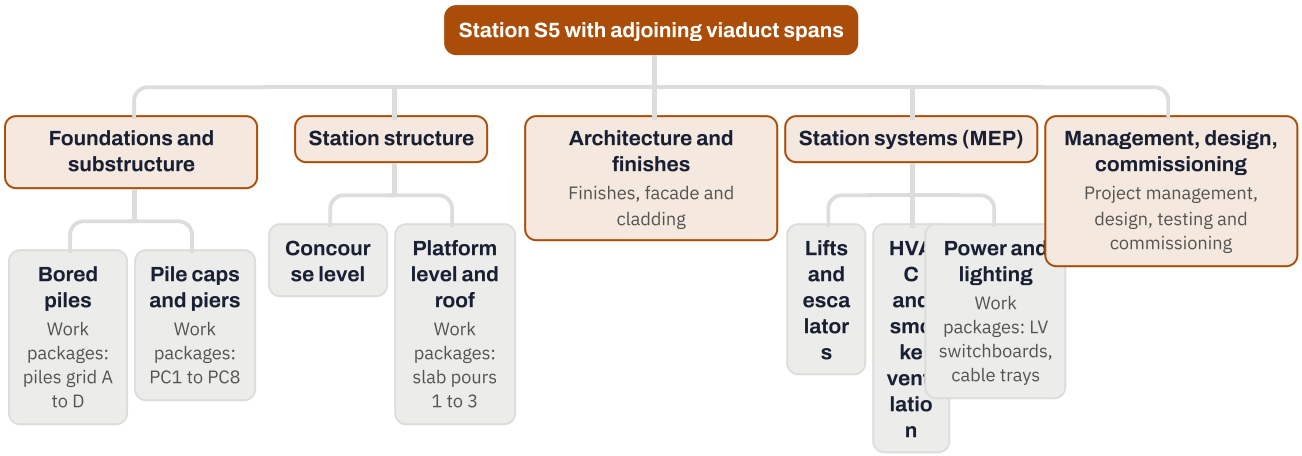
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INTERFACE ID	LOCATION / TIE-IN POINT	PARTY A (PROVIDES)	PARTY B (RECEIVES)	WHAT IS EXCHANGED	NEED DATE	STATUS
IF-12	*Outlet flange of final discharge pit*	*ETP contractor*	*Factory civil team*	*Treated effluent; pipe size, invert level and flow data*	*Before drain connection works*	*Agreed*

The **work breakdown structure** decomposes the total scope into progressively smaller, manageable parts. A good engineering WBS is **deliverable-oriented**, organised around what is produced (a station box, a turbine hall, a substation, an ETP) rather than who produces it, and it obeys the **100% rule**: each level contains all the work of its parent and nothing more, so nothing is missed and nothing is counted twice. Elements at the same level should not overlap. The lowest level is the **work package**, small enough to estimate, schedule, assign to one responsible party and measure; a common rule of thumb is that a work package should be completed within one or two reporting periods. Project management, design and commissioning are deliverables too and belong in the WBS. ISO 21511:2018 gives international guidance on work breakdown structures. Offsite and modular items, such as prefabricated MEP risers or bathroom pods, need work packages for factory production, transport and installation, so progress and interfaces at each stage stay visible. Coding model objects and documents to the WBS also lets quantities, 4D sequencing and cost be drawn from the model.

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WBS for a hypothetical elevated metro station



Each branch contains all of its parent's work and nothing else (the 100% rule), and management, design and commissioning appear as deliverables. Work packages sit at the next level down, small enough to estimate, schedule and assign to one party.

WBS, OBS and CBS: three ways to slice the same project

Work breakdown structure

- Breaks down the work and deliverables
- Answers: what must be produced?
- Basis for scope control and scheduling
- Lowest level: the work package

Organisational breakdown structure

- Breaks down the organisation: owner, consultants, contractors, departments
- Answers: who is responsible?
- Basis for assigning accountability
- Lowest level: a team or responsible manager

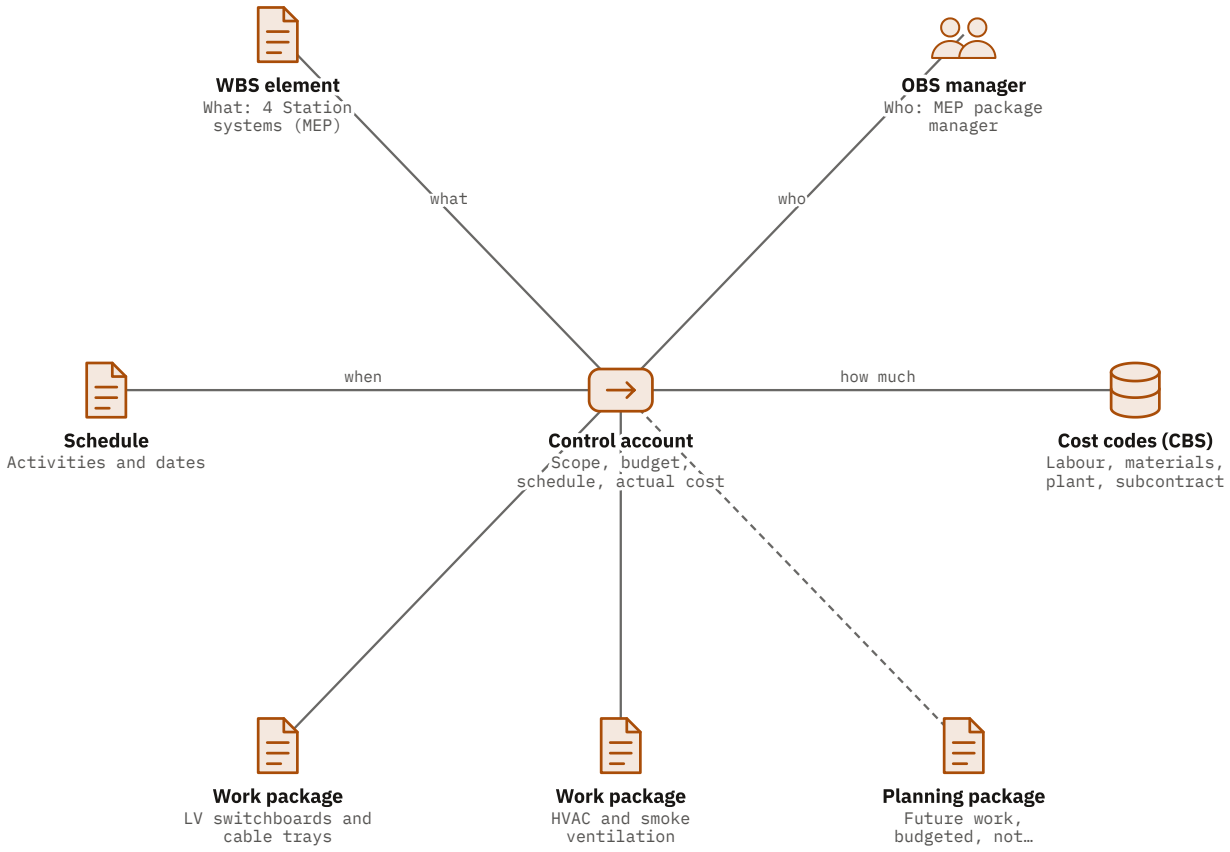
Cost breakdown structure

- Breaks down costs by account or cost type
- Answers: where does the money go?
- Basis for estimates, budgets and cost reports (see Budget)
- Lowest level: cost codes such as labour, materials and plant
- Where carbon is a managed baseline, a carbon budget can be held against the same codes

Where the WBS and the organisational breakdown structure (OBS) intersect, responsibility is assigned. A **responsibility assignment matrix** (RAM) lists WBS elements down the side and organisations or roles across the top; a common format marks who is Responsible, Accountable, Consulted and Informed (RACI), and each element should have exactly one accountable party. A **control account** is the management control point where scope, budget, schedule and actual cost are integrated for one WBS element under one responsible manager. It contains one or more work packages and is where earned value is measured and variances are explained (see Budget). Choose the level with care: too high and problems hide inside large accounts; too low and the reporting effort overwhelms the team.

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Where a control account sits



A control account sits where one WBS element meets one responsible manager, with its own budget, schedule and actual cost; earned value is measured here. Work packages are planned in detail now, while planning packages hold budget for later work.

Responsibility assignment matrix (RACI) extract for a hypothetical ETP project

WBS ELEMENT	OWNER'S PM	OWNER'S ENGINEER	ETP CONTRACTOR	FACTORY OPERATIONS
1.1 Process design and basis of design	A	R	C	C
1.3 Civil works: tanks and pump house	I	C	A/R	I
1.5 Electrical and controls	I	C	A/R	C
1.7 Tie-ins to existing drains and LV panel	A	C	R	R
1.8 Commissioning and performance test	A	C	R	C
1.9 Operator training and O&M manuals	I	C	R	A

TEMPLATE **WBS dictionary entry**

Describe each WBS element precisely, so that scope, deliverables, responsibility, estimate and acceptance are unambiguous to everyone who uses the WBS.

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WBS CODE	ELEMENT NAME	DESCRIPTION OF WORK	DELIVERABLES AND ACCEPTANCE CRITERIA	RESPONSIBLE (CONTROL ACCOUNT MANAGER)	ESTIMATE (COST / HOURS)	PLANNED START - FINISH
1.3.2	*Equalisation tank*	*Excavation, blinding, reinforced concrete raft and walls, waterproofing, inlet and outlet nozzles, walkway and handrails*	*Tank built to IFC drawings; water-tightness test passed; inspection records signed*	*ETP contractor's civil manager*	*Illustrative: 85 (USD thousand), 6,200 hours*	*Weeks 6 - 15*

SCENARIO · DECIDE · WHO OWNS THE CABLE TRAYS?

On a hypothetical gas-engine power plant extension, the WBS has separate elements for '3.2 Electrical installation' (main electrical contractor) and '3.5 Instrumentation and control' (I&C contractor). While reviewing the cost-loaded schedule, the planner notices that cable trays between the engine hall and the control building appear in both contractors' bills of quantities, while fire stopping of cable penetrations appears in neither. What should the project manager do FIRST?

(A) Leave both contracts as they are and let the site teams agree who installs the trays.

(B) Correct the WBS and WBS dictionary so each item sits in exactly one element, then align both contracts' scopes and the interface register.

(C) Remove the trays from both contracts and buy them directly as owner-supplied material.

(D) Add the fire stopping to the I&C contractor because its scope is smaller.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Sketch the top three levels of a WBS for your capstone project. Does every element pass the 100% rule? Which elements are organised by deliverable, and which, perhaps out of habit, by discipline or by company?

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Key terms

Scope statement
The documented description of a project's deliverables, boundaries, exclusions, assumptions, constraints, interfaces and acceptance criteria, used as the basis for detailed planning and control.

Work breakdown structure
A hierarchical, deliverable-oriented decomposition of the total project scope into progressively smaller elements, down to work packages.

WBS dictionary
A companion document giving, for each WBS element, its description of work, deliverables, acceptance criteria, responsible party, estimate and schedule information.

Control account
A management control point where scope, budget, schedule and actual cost are integrated for a WBS element under one responsible manager; it contains one or more work packages.

Battery limit
A defined physical or contractual boundary, such as a tie-in flange, cable termination or plot line, at which one party's scope ends and another's begins.

100% rule
The WBS principle that each level must include all of the work of its parent element and no work outside it, so that no scope is missed or double-counted.

Work package
The lowest level of the WBS: a defined piece of work small enough to be estimated, scheduled, assigned to one responsible party and measured for progress.

Key takeaways

- The scope statement fixes the boundary: deliverables, battery limits, exclusions, assumptions, constraints, interfaces and acceptance.
- A deliverable-oriented WBS obeys the 100% rule, with management, design, commissioning and offsite work included as deliverables.
- The WBS dictionary removes doubt about what each element contains and how its completion is accepted.
- Control accounts integrate scope, budget, schedule and actual cost under one manager; size them so that problems cannot hide.

KNOWLEDGE CHECK

6. Which statement best describes a battery limit?

(A) A cap on the number of batteries that may be installed in a UPS room
(B) The largest sum the PM may spend without approval
(C) A defined boundary between one party's scope and another's
(D) A limit on working hours inside a live plant area

7. A WBS element 'Substation' has children that together omit earthing and include a road outside the site. Which principle is breached?

(A) The 100% rule
(B) The rule of credit
(C) The critical path
(D) Decision rights

8. Which top-level breakdown is most deliverable-oriented for a hospital block?

(A) Civil engineers, mechanical engineers, electrical engineers and architects
(B) Labour, materials, plant, subcontracts and overheads
(C) January works, February works, March works
(D) Substructure, superstructure, envelope, MEP systems, fit-out

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9. In a RACI matrix, how many parties should be Accountable for each WBS element?

   (A) Every party involved, so the load is shared
   (B) Exactly one party for each element
   (C) One party per organisation involved
   (D) At least two parties, for resilience

10. Where are scope, budget, schedule and actual cost integrated under one responsible manager?

   (A) The control account
   (B) The scope statement
   (C) The interface register
   (D) The project charter

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MODULE 2 · DEFINE · LESSON 3 · 40 MIN

Design Development & Design Management

Maturing the design from concept to construction: reviewed, challenged, frozen and tracked

- Describe the purpose and deliverables of concept design, FEED or basic design, and detailed design
- Plan design reviews, value engineering and HAZOP studies at the right points
- Control design progress with a deliverables register and a disciplined design freeze

Design matures in steps, and each step should answer the questions the next decision needs. **Concept design** establishes the arrangement, the main technical choices and an order-of-magnitude cost. **Front-end engineering design (FEED)**, called basic or preliminary design in some sectors, fixes the process, layouts, main equipment sizes and specifications in enough detail to estimate reliably and to tender an EPC or construction contract. **Detailed design** produces everything needed to procure, fabricate, build and test: calculations, drawings, schedules, models and data sheets, issued for construction. Skimping on maturity at one stage simply moves the uncertainty into the next, where it costs more to resolve.

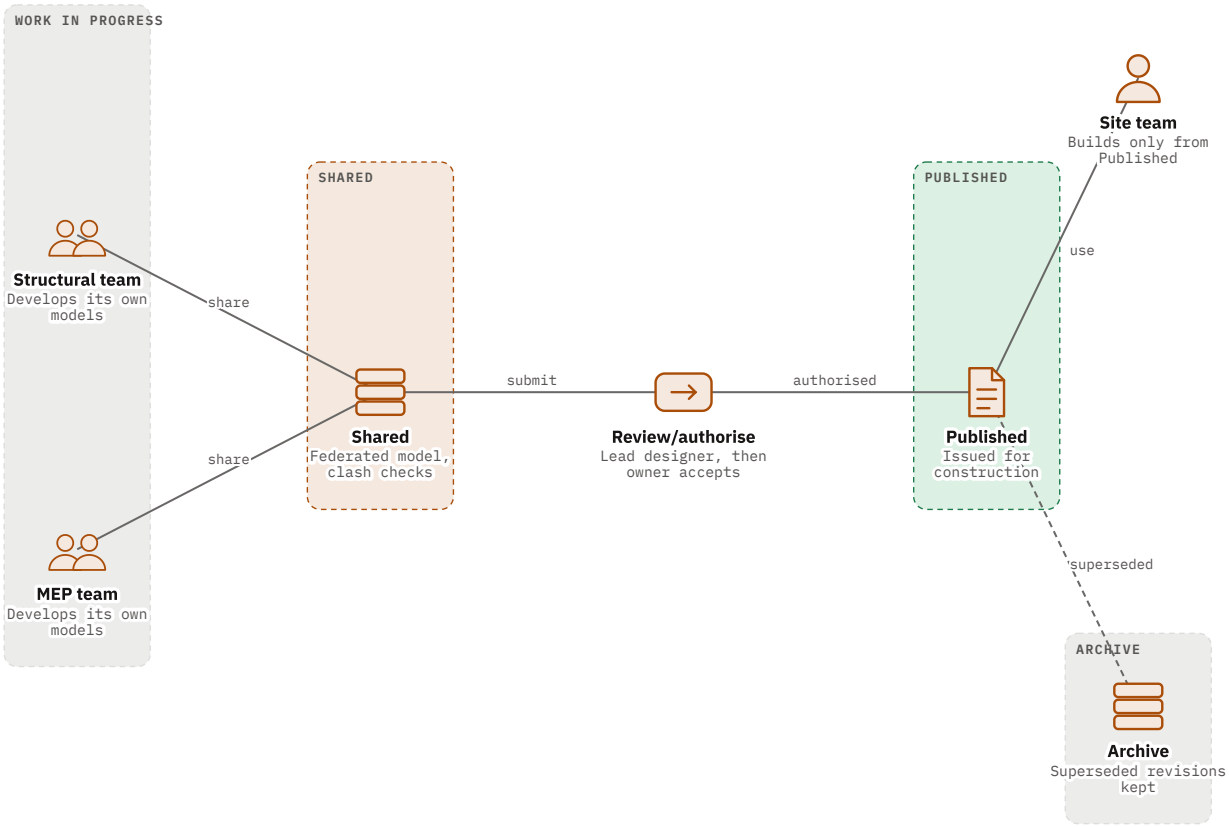
Design stages at a glance

STAGE	PURPOSE	TYPICAL DELIVERABLES	DECISION SUPPORTED
Concept	Choose the arrangement and technology	Block diagrams, site layout options, outline sizing, concept report	Selecting the preferred option at feasibility
FEED or basic design	Define the facility well enough to estimate and contract	Process flow diagrams, P&IDs, plot plans, equipment lists and data sheets, single-line diagrams, specifications	Final investment decision and EPC or construction tender
Detailed design	Complete the design for procurement and construction	Calculations, general arrangement and detail drawings, 3D model, cable schedules, bar bending schedules	Purchase orders, fabrication and construction
Issued for construction and as-built	Build to approved documents and record what was built	IFC drawings, revisions, field change records, as-built records	Construction, commissioning and handover (see Commission)

Design management is project management applied to design work. The design manager plans the deliverables, sequences the disciplines (process before piping, structural loads before foundations, equipment data before electrical sizing), chases vendor information and coordinates reviews. The core tool is the **deliverables register**: a list of every drawing, calculation, specification and model with its discipline, revision, planned and actual dates and status, such as issued for review, approved or issued for construction. It makes design progress measurable rather than a matter of opinion, and it shows when late vendor data or unanswered comments are blocking other disciplines downstream. On projects run to the ISO 19650 series, this register takes the form of master and task information delivery plans, and every drawing or model is an information container whose status is managed in a **common data environment** (CDE), as shown below.

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Information states in a common data environment



Information moves through states instead of being emailed around: teams work privately, share for coordination, and only authorised containers are published. Superseded revisions are archived, never deleted, so the history of every change survives.

Design reviews that add value

- **Staged reviews** at defined maturity, often around 30%, 60% and 90% of detailed design, plus a final review before issue for construction
- **Interdisciplinary checks** so structural, mechanical, electrical and architectural documents agree, increasingly supported by clash detection in a federated 3D model (see Integrate)
- **Automated checks**, such as rule-based or AI-assisted model checking and clash triage, speed reviews, but a competent engineer still verifies and signs off every finding
- **Operations and maintenance reviews** with the future operator walking through the 3D model
- **Constructability reviews** with contractors and temporary works engineers
- **Formal comment resolution**: every comment logged, answered and closed, with the reason recorded

Running a HAZOP study as a design review

1. **Prepare** Use reasonably mature P&IDs and a clear design intent. Appoint an independent, experienced chair and a scribe; the team includes process, instrument, mechanical and operations engineers.
2. **Divide into nodes** Break the system into manageable sections, such as a boiler feedwater line or a chemical dosing skid, each with a defined design intent.
3. **Apply guide words** Combine guide words with process parameters, such as no flow, more pressure, less temperature or reverse flow, to generate credible deviations.

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4. **Analyse each deviation** Identify causes, consequences and existing safeguards, and judge whether the safeguards are adequate.
5. **Record actions** Where safeguards are inadequate or uncertain, raise an action with a named owner, such as adding a non-return valve or a high-level trip.
6. **Close out** Track every action to closure and update the design before freeze; any open actions are reported at the design freeze gate.

HAZOP guide words applied to a hypothetical boiler feedwater line

GUIDE WORD + PARAMETER	DEVIATION	POSSIBLE CAUSE	POSSIBLE CONSEQUENCE	EXAMPLE ACTION
No + flow	No feedwater reaches the boiler	Feed pump trip, closed valve, blocked strainer	Low drum level, tube overheating	Confirm low-level alarm and burner trip; standby pump auto-start
More + pressure	Overpressure in the feed line	Discharge valve closed against a running pump	Pipe or gasket failure	Check pump shut-off head against line rating; add minimum-flow recirculation
Less + temperature	Feedwater colder than design	Deaerator heating failure	Thermal stress, oxygen corrosion	Alarm on deaerator temperature; review chemical dosing
Reverse + flow	Backflow from boiler towards the feed tank	Non-return valve failure when the pump stops	Hot water and steam in the feed system	Specify a second non-return valve; confirm valve test regime
Other than + composition	Contaminated feedwater	Condensate return contaminated by a process leak	Scaling, foaming, tube damage	Conductivity monitoring with automatic dumping of returned condensate

INSIGHT · VALUE ENGINEERING IS NOT COST CUTTING

Value engineering analyses the functions an element must perform and looks for ways to deliver them at lower whole-life cost without reducing performance, safety or quality. Replacing cast-in-situ crossheads with precast units to cut work at height, standardising pump sizes to reduce spares, or orienting a building to cut its cooling load are value engineering. Deleting a standby pump or thinning roof insulation to meet a budget is cost cutting with consequences. Value engineering pays best at concept and FEED, when change is cheap; late proposals during construction must be weighed against redesign, delay and approval costs. Switching to a lower-carbon concrete mix that meets the same strength and durability is value engineering too.

PRO TIP · DECIDE OFFSITE, CARBON AND DATA NEEDS EARLY

Precast elements, volumetric modules, prefabricated MEP risers and plant skids change grids, tolerances, crane and transport limits and the design programme, because factories need frozen information earlier than site trades. **Design for manufacture and assembly** (DfMA) decisions therefore belong at concept and FEED, together with a whole-life carbon assessment at each design stage. Likewise, if the owner wants an operational digital twin (concepts in ISO/IEC 30173:2023), define the asset data it needs during design, not at handover.

CASE STUDY · VALUE ENGINEERING ON AN ELEVATED EXPRESSWAY

A hypothetical elevated expressway in a South Asian city is at FEED. The reference design uses cast-in-situ box girders on falsework, which would occupy traffic lanes beneath for long periods. An independently facilitated value engineering workshop, attended by the designer, the owner's engineer, a contractor's adviser and the traffic

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police, proposes precast segmental girders erected by a launching gantry. The proposal needs a casting yard on land the owner does not yet hold, changes the pier head details and requires heavier transport routes. Design freeze for the viaduct is scheduled in four months. Figures and parties are illustrative.

1. What functions should the workshop have defined before comparing the two methods?

2. What information is needed to decide whether to adopt the proposal before design freeze?

3. How should the change be controlled if it is accepted?

Model answer: module answer key.

WATCH OUT · DESIGN FREEZE MUST MEAN SOMETHING

A **design freeze** is the point after which design changes are allowed only through formal change control with impact assessment. Freezes are usually staged, for example process design, then layouts, then detailed discipline design, and linked to procurement: equipment cannot be ordered until its data sheet is frozen, and foundations cannot be finalised until vendor loads are confirmed. A freeze that is announced but not enforced is worse than none, because procurement and construction proceed on documents that are still moving.

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Staged freezes on a typical process plant

CONCEPT
Technology and layout chosen

FEED
Process design frozen
Flow diagrams and balances fixed; P&IDs ready for HAZOP

END OF FEED
Plot plan and data sheets frozen
Long-lead equipment can be ordered

DETAILED 30-60%
Vendor data confirmed
Certified loads fix foundations and piping

DETAILED 90%
Discipline design frozen
Later changes only through change control

IFC
Issued for construction

HANDOVER
As-built records
Documents and models match the installed asset

Each freeze is tied to the work that depends on it: orders need frozen data sheets and foundations need certified vendor loads. After the last freeze, change is still possible, but only through change control.

SCENARIO · DECIDE · EQUIPMENT NOT YET CHOSEN

You are the design manager for a hypothetical 300-bed hospital in Dhaka. Detailed design is at about 60%, and the structural and MEP design freeze is in five weeks. The clinical team has not yet chosen the MRI and CT equipment, which affects floor loading, radiation and magnetic shielding, cooling and power. The architect proposes freezing the whole design on generic allowances and dealing with any differences later. What should you do FIRST?

(A) Freeze on generic allowances as proposed and handle any differences through site instructions later.

(B) Postpone the entire design freeze until the clinical team has chosen all the equipment.

(C) Agree an equipment decision date with the client, freeze the rest of the design, and hold the imaging suites on an envelope design that covers the likely options.

(D) Choose the equipment yourself, based on the most common models, so the design can proceed.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

For a design you have seen develop, at what point was it truly frozen, and how many significant changes came after that point? Which of them would a HAZOP, a constructability review or a value engineering workshop at FEED have caught?

Key terms

Front-end engineering design
The design stage, also called basic design, that defines process, layouts, main equipment and specifications in enough detail to support the investment decision, a reliable estimate and an EPC or construction tender.

Design review
A planned, structured examination of design deliverables at a defined maturity by designers, owner, operator or contractor, with comments logged, resolved and closed.

HAZOP
Hazard and operability study: a systematic, team-based design review that applies guide words to process parameters for each node of a design to identify deviations, their causes, consequences and safeguards.

Deliverables register
A controlled list of every design deliverable, such as drawings, calculations, specifications and models, with discipline, revision, planned and actual dates and status, used to plan and measure design progress.

Detailed design
The design stage that completes the calculations, drawings, models, schedules and data sheets needed to procure, fabricate, construct and test, ending with documents issued for construction.

Value engineering
A function-based, structured review that seeks to deliver the required functions at the lowest whole-life cost without compromising performance, safety or quality.

Design freeze
The agreed point after which a design, or a defined part of it, may be changed only through formal change control with impact assessment.

Key takeaways

- Design matures from concept to FEED to detailed design; skimping on one stage moves uncertainty into the next, where it costs more.
- A deliverables register or information delivery plan, with a common data environment, makes design progress and status visible.
- Reviews, HAZOP, value engineering, DfMA and carbon decisions add most value before freeze; value engineering keeps functions.
- Freezes are staged and tied to procurement; after a freeze, change goes through control rather than being forbidden.

KNOWLEDGE CHECK

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11. Which stage most typically produces P&IDs, equipment data sheets and specifications good enough for an EPC tender?

(A) Concept design
(B) As-built documentation
(C) Detailed design
(D) FEED or basic design

12. In a HAZOP, combining the guide word 'reverse' with the parameter 'flow' is used to:

(A) Calculate the pump's required suction head
(B) Generate a credible deviation for the team to analyse
(C) Set the order of precedence between conflicting standards
(D) Rank options in a multi-criteria analysis

13. Which proposal is genuine value engineering rather than cost cutting?

(A) Deleting the standby fire pump to meet the budget
(B) Reducing roof insulation below the design basis
(C) Standardising pump sizes so fewer spares are needed
(D) Removing the second transformer from an N-1 substation

14. What does the deliverables register mainly allow the design manager to do?

(A) Measure design progress and spot blocked deliverables
(B) Value variations submitted by the contractor each month
(C) Assign safety responsibilities on site
(D) Calculate the discount rate for the business case

15. Why are design freezes often staged and tied to procurement?

(A) Because regulators require one freeze per discipline
(B) Because value engineering is banned after the first freeze
(C) Because orders and foundations depend on frozen data
(D) Because the steering committee meets only at freeze dates

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · DEFINE · LESSON 4 · 35 MIN

Scope Baseline & Change Control

Protecting the agreed scope while allowing the right changes quickly, visibly and with their full impact understood

- Define the scope baseline and explain why it must be formally approved and controlled
- Run a change control process with multidisciplinary impact assessment and a change control board
- Prevent scope creep using trend notices, configuration management and site discipline

Once scope is defined and approved it becomes the **scope baseline**: the approved scope statement, WBS and WBS dictionary. Together with the schedule and cost baselines (developed in Schedule and Budget), it is the reference against which progress is measured and changes are judged. A baseline is not a prediction that nothing will change. Engineering projects always change: ground conditions differ, regulations evolve, operators learn and markets move. A baseline simply means that each change is a deliberate decision with known consequences, rather than something that just happens.

Change control process

01 Identify
Anyone may raise a change request or an early trend notice

→

02 Log
Register with a unique number, description, originator and reason

→

03 Assess impact
Scope, cost, schedule, risk, safety, quality, interfaces and operations

→

04 Decide
Approve, reject or defer at the level set by decision rights; the CCB for larger changes

→

05 Update baselines
Revise scope documents, schedule, budget, drawings and, where affected, contracts

→

06 Implement and communicate
Instruct the work, inform affected parties and update the change log

→

07 Close
Verify implementation and capture the actual cost and time

Sources of scope creep on engineering projects and how to control them

SOURCE	EXAMPLE ON SITE	CONTROL
Informal client requests	An owner's director asks the site team to add a mezzanine in the warehouse during a walk-round	Instructions only through designated channels; site teams trained to log requests, not act on them
Design development beyond intent	Designers add features or higher grades than the basis of design requires	Design reviews check against the basis of design; deviations raised as change requests
Gold-plating by the team	Engineers specify stainless steel where coated carbon steel meets the requirement	Specification reviews and value engineering against stated requirements
Unclear battery limits	The contractor assumes the owner will build access roads, and the owner assumes the opposite	Interface register and explicit exclusions in the scope statement

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SOURCE	EXAMPLE ON SITE	CONTROL
Accumulated small changes	Dozens of minor layout changes, each below the approval limit, add up to a large overrun	Cumulative change log reviewed monthly, with tolerances applied to the total as well as each item
Regulatory or stakeholder demands	New fire or drainage requirements raised during the permit process	Track permit conditions and treat them as changes with impact assessment rather than absorbing them silently

An **impact assessment** is the heart of change control. A request to move a switchgear room on a factory extension looks like an architectural change, yet it may affect cable lengths and voltage drop, fire separation, ventilation, structural openings, cable already on order and the commissioning sequence. A good assessment is done by every affected discipline, considers direct and indirect effects (including disruption to other work and safety during construction), estimates cost and time with a stated confidence and identifies any new risks. Speed matters too: a change that waits weeks for assessment while work continues on the old design creates rework whichever way the decision goes. Where the project manages a carbon budget, include the effect on whole-life carbon. In volatile markets, price changes at current rates and state how long each quotation remains valid.

Trend notice, change request and contract variation

Trend notice

- Early warning that a change may be coming
- Raised as soon as a potential deviation is spotted, often by engineers or planners
- Prompts early assessment and forecast adjustment
- Does not approve any work

Change request

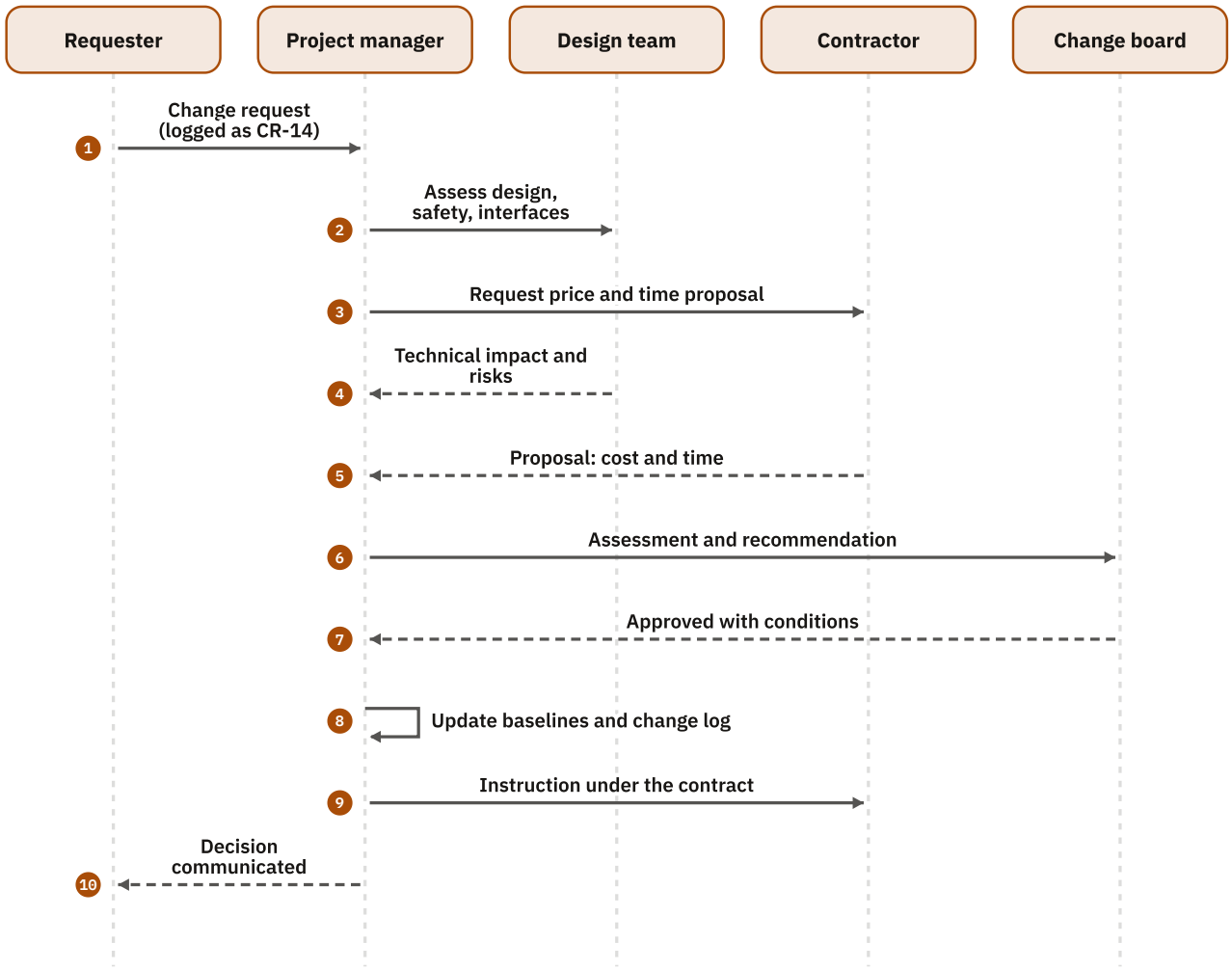
- Formal proposal to change an approved baseline
- Raised by owner, designer, contractor or project team
- Needs impact assessment and a decision at the right level
- If approved, baselines are updated

Contract variation

- A change to the work a contractor must perform under its contract
- Instructed through the contract's procedure by the authorised party
- Valued and paid under the contract's rules (developed in Procure)
- Follows an approved change wherever a contract is affected
- Called a variation under FIDIC 2017 and, for most owner changes, a compensation event under NEC4

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From change request to contract instruction



The board decides only once both the technical impact and the contractor's price and time are known. The contractor is instructed through the contract's own procedure after the decision, and the baselines are updated so records stay consistent.

Making a change control board work

- Chaired by someone with authority for the decisions in its remit: the project manager or the sponsor, depending on value
- Includes engineering, project controls, construction, HSE, operations and commercial representatives
- Meets on a regular rhythm, with an urgent route for safety-critical and time-critical changes
- Decides only on changes with a completed impact assessment and a clear recommendation
- Records decisions, reasons and conditions in the change log and reviews cumulative impact
- Escalates changes above its tolerance to the steering committee

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DEFINITION · CONFIGURATION MANAGEMENT

Configuration management keeps the description of the asset, including requirements, design documents, models, vendor data and the physical installation, consistent and under control as changes occur. It answers questions such as: which revision of this P&ID is current, which approved changes have been incorporated, and does the installed valve match the approved data sheet? It relies on unique identification of items, revision control, a change history linked to approved changes, and audits that compare documents with the built asset. Without it, as-built records drift away from reality, which becomes a safety problem during commissioning and maintenance. Reality capture, such as laser scanning and drone surveys, now makes it practical to compare what was built with the model and the approved revisions, and an operational digital twin is only as trustworthy as this discipline.

CASE STUDY · DEATH BY A THOUSAND CHANGES

A hypothetical four-storey hospital block in Chattogram is halfway through construction under a remeasurement contract. The owner's clinical staff have requested more than sixty changes: extra power outlets, relocated hand-wash basins, a revised nurse station, additional medical gas outlets, and conversion of two rooms into isolation rooms with negative pressure. Most were approved one by one by the site engineer because each fell below his limit. The cost report now shows a significant forecast overrun, the MEP subcontractor has asked for more time, and the HVAC designer warns that the isolation rooms need a dedicated exhaust system that has not been designed.

1. What went wrong with the change control process?

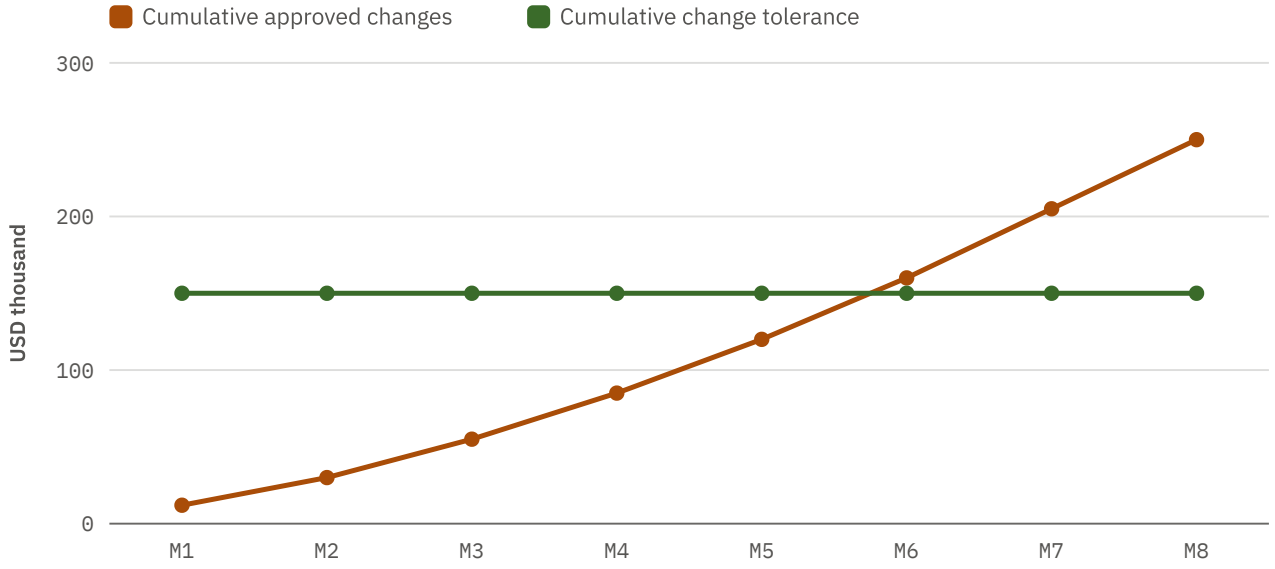
2. How should the isolation room change have been handled?

3. What should the project manager put in place now?

Model answer: module answer key.

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Small approvals, large total



Illustrative figures for a case like the hospital block: every change was below the site engineer's limit, yet the total passes the cumulative tolerance in month 6. Only a cumulative review would have triggered escalation in time.

SCENARIO · DECIDE · A VERBAL REQUEST ON SITE

On a hypothetical garment factory expansion near Chattogram, the owner's production manager asks your site engineer to shift a row of columns by one bay to suit a new sewing-line layout. The steel fabricator has already started cutting the affected beams, and the foundation bolts are cast. The site engineer thinks the change is small and wants to agree it on the spot to keep good relations. What should the project manager do FIRST?

(A) Let the site engineer agree the change to keep the relationship strong, and record it in the change log later.

(B) Raise a change request with an urgent impact assessment by the structural, fabrication and planning leads, and ask the fabricator to hold cutting of the affected members until the decision.

(C) Refuse the request because the steel design is frozen and changes are not allowed.

(D) Ask the fabricator to make both the original and revised beams so either can be used.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Review the change log of a project you know, or imagine one for your capstone. How many changes were raised informally first? What was the largest change, and was its full impact, including interfaces and operations, assessed before it was approved?

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Key terms

Scope baseline
The approved version of the scope statement, WBS and WBS dictionary, which can be changed only through formal change control and is the reference for measuring scope performance.

Scope creep
The uncontrolled growth of project scope through additions that are not assessed, approved or funded, often through many small informal requests.

Change control board
A formally constituted group with authority to review, approve, reject or defer change requests within defined limits, and to record its decisions and reasons.

Trend notice
An early-warning record raised when a potential change or deviation is first identified, prompting assessment and forecast updates before a formal change request is needed.

Change request
A formal proposal to modify an approved baseline, deliverable or document, recorded and assessed before a decision is taken at the authorised level.

Impact assessment
The multidisciplinary evaluation of a proposed change's effects on scope, cost, schedule, risk, safety, quality, interfaces and operations, carried out before a decision is taken.

Configuration management
The discipline of identifying, controlling and auditing the requirements, design documents and physical configuration of an asset so they remain consistent as changes are made.

Key takeaways

- The scope baseline (scope statement, WBS and WBS dictionary) makes every change a deliberate decision with known consequences.
- Raise trend notices early and assess change requests across every affected discipline, including safety, interfaces and carbon.
- Decide at the level set by decision rights, update the baselines and instruct contractors only through the contract's procedure.
- Track cumulative change against a tolerance, and keep documents, models and the built asset consistent through configuration management.

KNOWLEDGE CHECK

16. Which set of documents makes up the scope baseline?

(A) The approved scope statement, WBS and WBS dictionary
(B) The charter, the business case and the feasibility study
(C) The contract, drawings and site instructions
(D) The risk register, issue log and lessons register

17. Many small changes, each below the approval limit, have together caused a large overrun. Which control would have helped most?

(A) Giving the site engineer a higher individual approval limit
(B) Raising the steering committee's own tolerances
(C) Reviewing cumulative change impact against a tolerance
(D) Holding the kickoff meeting again with the contractor

18. A planner notices that revised vendor pump data may enlarge a foundation, but nothing is confirmed yet. What should be raised now?

(A) A contract variation to the civil contractor
(B) A trend notice for early assessment
(C) A formal waiver of the design freeze
(D) A claim in the contractor's final account

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19. Which question is primarily answered by configuration management?

(A) Which contractor offered the lowest tender price for the works?

(B) What is the critical path of the project schedule?

(C) Who is accountable for each WBS element?

(D) Does the installed valve match the approved revision?

20. Why should a change control board decide only on changes with a completed impact assessment?

(A) Because the board cannot meet more than once a month

(B) Because unassessed decisions undermine the baseline

(C) Because contractors are paid only after the board meets

(D) Because assessments transfer liability to the designer

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · DEFINE

Answer key

Knowledge checks

1. B Owner's project requirements describe what the business and users need. The basis of design, data sheets and drawings translate those needs into technical terms.

2. D It defines the result and how it will be tested, not the equipment or method, so it is a performance specification.

3. A The basis of design should state an order of precedence for conflicting documents. Progress reports, committee terms and inductions are not where technical precedence is set.

4. C Lifting provisions and removal routes make repair and replacement easier, which is maintainability. They do little for construction efficiency or day-to-day operation.

5. B Fixing editions keeps all designers and suppliers working to the same rules. The other statements are not true in general.

6. C A battery limit is a scope boundary, such as a tie-in flange or cable termination. The other options describe equipment, tolerances and site constraints.

7. A The children must contain exactly the parent's work: missing earthing and an extra road breach the 100% rule. The other concepts relate to progress measurement, scheduling and governance.

8. D Physical deliverables make a deliverable-oriented WBS. The others are organisational, cost-type and time-based breakdowns.

9. B Single-point accountability avoids gaps and finger-pointing. Several parties may be responsible, consulted or informed, but only one is accountable.

10. A A control account is the integration point for scope, cost and schedule performance. The other documents define or record scope and authority but do not integrate performance data.

11. D FEED defines the facility well enough to estimate and tender an EPC contract. Concept is too early, detailed design follows the award, and as-built records come at the end.

12. B Guide word and parameter combinations generate deviations that the team analyses. They are not a calculation, a precedence rule or a scoring method.

13. C Standardisation keeps every function while cutting whole-life cost. The other proposals remove required functions or performance and are cost cutting.

14. A The register tracks status, revisions and dates for every deliverable, making progress and blockages visible. It is not a commercial, safety or financial tool.

15. C Orders need frozen data sheets and foundations need confirmed vendor loads, so freezes follow the logic of procurement. The other reasons are not general rules.

16. A The scope baseline is the approved scope statement with the WBS and WBS dictionary. The other sets are governance, contractual or risk records.

17. C Tolerances on cumulative impact reveal the total effect of many small changes. Higher limits would make the problem worse, and a second kickoff does not control change.

18. B A trend notice gives early warning so the effect can be assessed and forecasts adjusted. A variation, waiver or claim would be premature before the change is confirmed and approved.

19. D Configuration management keeps documents and the physical asset consistent. Tender prices, critical paths and accountability are handled by other processes.

20. B The baseline's value is that every change is a deliberate decision with known effects. The other reasons are not why impact assessment is required.

Scenarios & case studies

> **Scenario · Built around the transformer**
> **B** is best. Best. The missing requirement is real and is cheapest to fix now, before concrete is poured. Assessing options and using change control keeps the decision visible and properly authorised.

> **Scenario · Who owns the cable trays?**
> **B** is best. Best. Overlap and omission are both breaches of the 100% rule. Fixing the WBS at the root and then aligning the contracts removes the duplicate cost and closes the fire-safety gap.

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Case study · Value engineering on an elevated expressway

The workshop should start from functions such as carrying traffic loads safely over the design life, spanning between piers, being buildable with minimal disruption to traffic below, and being inspectable and maintainable, rather than from 'provide a box girder'. Deciding on precast segments needs evidence on whole-life cost, schedule, traffic disruption, quality control of segments, casting yard land and permits, transport routes, launching gantry availability and the redesign effort for pier heads. The decision should be taken during FEED, before design freeze, with an impact assessment covering cost, time, risk and interfaces. If accepted, the basis of design, scope statement, WBS and deliverables register must all be updated and the change formally approved, so that the tender documents describe one consistent design.

Scenario · Equipment not yet chosen

C is best. Best. A partial freeze with a defined hold point protects the programme while keeping the imaging rooms flexible. The envelope and the decision date make the remaining uncertainty explicit and manageable.

Case study · Death by a thousand changes

Approving changes one by one below a site-level limit hid their cumulative effect: nobody reviewed the total impact on cost, time and building systems, and there is no sign of impact assessment by all affected disciplines. The isolation room conversion was never a minor change; it alters the HVAC concept, room pressure regimes, controls and infection-control requirements and should have gone to the change control board with a full design and cost assessment before any work began. The project manager should now pause further non-essential changes, consolidate approved and pending changes into one log showing cumulative cost and time, commission the HVAC design for the isolation rooms, and set tolerances that apply to cumulative as well as individual changes. A clinical user group with one authorised representative and agreed cut-off dates should channel future requests, and the forecast with options should go to the steering committee.

Scenario · A verbal request on site

B is best. Best. Holding the affected fabrication limits waste while the change is assessed quickly. The production manager's need is taken seriously and the decision is made with its real consequences known.

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03

MODULE 3 · SCHEDULE

Scheduling: CPM, PERT, Levelling & Crashing

Turn the WBS into a logic-driven, resourced and defensible programme that shows where time really matters

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Build a logic-linked activity network with correct relationship types, lags, calendars, milestones and a minimum of constraints
- Calculate early and late dates, total float and free float by hand and identify critical and near-critical paths
- Apply three-point estimating and PERT to state the probability of meeting a target date, and explain merge bias
- Load, smooth and level resources such as cranes and specialist crews and judge the effect on float and the critical path
- Compress, update and health-check a schedule using crashing, fast-tracking, look-ahead planning and progress updates

LESSONS

1. Activities, Logic & Networks · 32 min
2. The Critical Path Method · 38 min
3. PERT & Three-Point Estimating · 36 min
4. Resource Loading, Levelling & Smoothing · 34 min
5. Schedule Compression, Updating & Look-Ahead Planning · 40 min

MODULE 3 · SCHEDULE · LESSON 1 · 32 MIN

Activities, Logic & Networks

From work packages to a logic-linked network that behaves like the real site

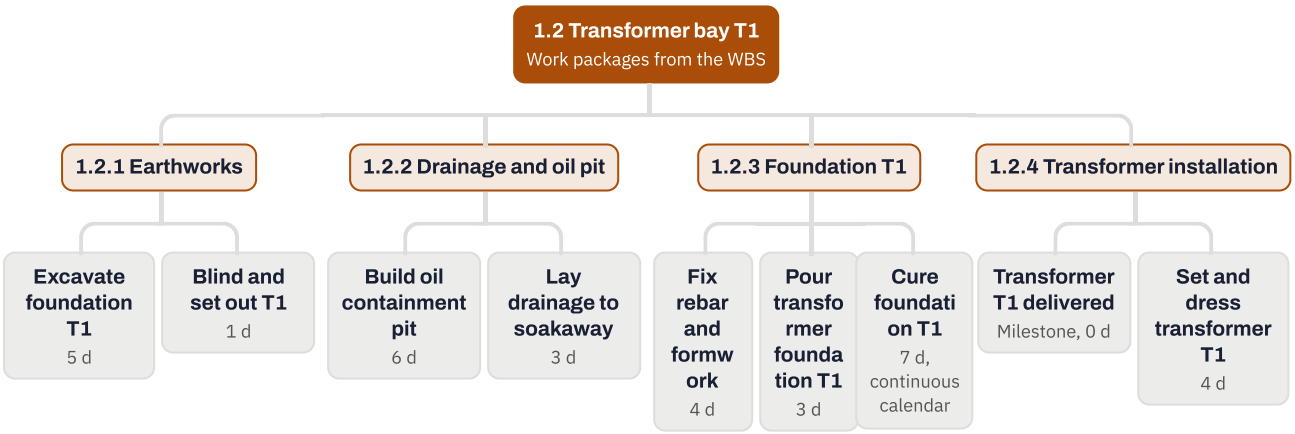
- Define activities from work packages that crews can perform, measure and report
- Choose the right relationship type (FS, SS, FF, SF) and use leads and lags responsibly
- Apply calendars, constraints, milestones and schedule levels without breaking the logic

A schedule is a model of how the project team intends to build the works. It is not a bar chart drawn to match a contract date; it is a **network** of activities joined by logic, so that when one piece of work slips the model shows what else moves. The starting point is the WBS developed in Define: each work package is broken down into **activities** that a crew, subcontractor or engineer can actually perform and report on. A well-built network lets the project manager answer the three questions that matter on site every week: what must start next, what is driving completion, and how much room there is before a delay bites.

DEFINITION · WHAT MAKES A GOOD ACTIVITY

A good activity has a **verb-noun name** a foreman understands ('Pour pile cap PC-07', not 'Concrete'), a **single responsible party**, a start and finish that can be observed on site, and a duration short enough to be measured between updates. A practical rule for construction work is that most activities should last no longer than about two update periods. Each activity belongs to exactly one work package, so that time, cost and progress roll up cleanly to the control accounts used in Budget.

From work package to activities (illustrative)



Each work package breaks into a few verb-noun activities that one crew can perform and report. Notice that curing runs on its own continuous calendar and the delivery is a zero-duration milestone linked into the logic.

The four logic relationships

RELATIONSHIP	MEANING	SITE EXAMPLE	USE WITH CARE
Finish-to-start (FS)	Successor cannot start until the predecessor finishes	Complete transformer plinth, then set the transformer on it	The default and clearest link; most links in a good network are FS

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RELATIONSHIP	MEANING	SITE EXAMPLE	USE WITH CARE
Start-to-start (SS)	Successor cannot start until the predecessor has started, often with a lag	Start laying ducts 2 days after cable-trench excavation starts (SS + 2)	Pair with an FF link so the follower cannot overtake the leader
Finish-to-finish (FF)	Successor cannot finish until the predecessor finishes	Cable termination finishes 1 day after cable pulling finishes (FF + 1)	Common with SS for overlapping linear work such as pipelines and trenches
Start-to-finish (SF)	Successor cannot finish until the predecessor starts	Supply from the old feeder cannot end until the new temporary supply has started	Rare and easily misread; many planning standards discourage it

A **lag** is a delay imposed on a relationship, such as FS + 7 days so that a raft slab gains strength before props are removed. A **lead** is a negative lag that lets a successor start before its predecessor finishes, for example starting cable tray installation in a zone three days before ceiling framing there is complete. Both are legitimate, but both hide work and risk. If the seven days are really curing that needs protection, cube tests and an engineer's release, model them as an activity with its own duration and calendar. Check which calendar a lag runs on: a seven-day curing lag counted on a six-day working calendar silently becomes more than a week of real time.

Calendars that reflect the real site

- **Working-week calendars**: six-day weeks are common on South Asian construction sites, while design offices and overseas factory acceptance tests may run five-day weeks.
- **Continuous seven-day calendars** for curing, ground consolidation, cooling, slip-forming and other processes that do not stop at weekends.
- **Holiday calendars** for national and religious holidays, such as Eid periods when much of the workforce travels home and productivity drops before and after.
- **Weather windows**: monsoon months that limit earthworks, piling platforms and roofing; wind limits for crane lifts and offshore installation.
- **Heat and climate**: hot-season work-rest regimes and shifted working hours cut productive time; build them into calendars or productivity rates instead of hoping crews will catch up.
- **Shutdown and outage windows** for brownfield work, such as a boiler tie-in permitted only during a planned plant shutdown.
- **Resource calendars** for scarce plant, such as a heavy-lift crane available only between two other contracts.

WATCH OUT · CONSTRAINTS ARE NOT A SUBSTITUTE FOR LOGIC

A **date constraint** fixes when an activity may start or finish regardless of logic. Soft constraints such as 'start no earlier than' a confirmed delivery date are reasonable. **Hard constraints** such as 'must finish on' override the network: the software shows the date you typed even when predecessors are late, so float becomes negative or meaningless and the critical path can vanish. Use constraints only for genuine external dates such as a utility connection window, a permit date or a contract key date, and record why each one exists.

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Levels of schedule (top first)

Level 1: Master summary	Key milestones and major phases on one page For sponsors, boards and lenders
Level 2: Summary programme	Phases by area, facility or discipline Management reporting and contract key dates
Level 3: Project control schedule	Logic-linked network across engineering, procurement, construction and commissioning Basis of the critical path, float and baseline
Level 4: Execution schedule	Detailed activities per area or subcontract Often owned by contractors and rolled up into Level 3
Level 5: Task-level plans	Day-by-day crew work for the next few weeks Look-ahead and weekly work plans on site

Five rules for a sound network

N1

Every activity has logic

Every activity except the first has at least one predecessor and every activity except the last has at least one successor. Open ends make float meaningless.

N2

Logic drives dates

Dates come from relationships and durations, not from typing. Constraints are the documented exception, never the rule.

N3

Right link, right lag

Prefer FS links; use SS/FF pairs for overlapping linear work; replace long or unexplained lags with real activities.

N4

Durations from quantities

Duration = quantity / (output per crew per day x number of crews). Write the basis down so it can be challenged and updated.

N5

Milestones mark decisions

Use zero-duration milestones for contract key dates, permits, major deliveries and handovers, and link them into the logic.

Durations should be calculated, not guessed. A hypothetical blockwork package of 1,800 m2 with two gangs, each laying 15 m2 per day, takes 1,800 / (2 x 15) = 60 working days on the site calendar. The same discipline applies to cable pulling in metres per day, pipe welding in inch-diameters per day or steel erection in tonnes per day. Writing the quantity and productivity next to each duration turns arguments about whether 60 days is 'too long' into a useful discussion about crew size, productivity and access. **Milestones**, by contrast, have zero duration: they mark events such as 'transformer delivered to site' or 'sectional completion of the station box' and are linked into the logic so that their dates move when the work driving them moves.

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PRO TIP · DIGITAL AND OFFSITE INPUTS TO THE NETWORK

Three changes shape how networks are built today. **4D planning** links model elements to activities so the team can rehearse the sequence and spot clashes in space and time before they happen on site; under the ISO 19650 series the schedule and the model are shared as managed information containers in the common data environment (CDE), so everyone works from the current revision. **Offsite and modular construction** moves work into factories, so production slots, factory acceptance, transport and delivery windows become real activities and milestones. **AI-assisted tools** can propose activities, logic and durations from past schedules in minutes: treat the output as a first draft and have a planner check every link, duration basis and calendar before it is used for control. ISO 21502:2020, the international guidance on project management, treats this planning as part of integrated project control, not a drawing exercise.

SCENARIO · DECIDE · REVIEWING A CONTRACTOR'S LEVEL 3 SCHEDULE

You are the owner's project manager for a hypothetical extension to a dyeing and finishing unit in Narayanganj. The contractor's planner submits a Level 3 schedule in which 40 of 220 activities carry 'must finish on' constraints matching the dates in the tender programme, and several links have unexplained finish-to-start lags of 10 to 15 days. The schedule shows a tidy critical path through mechanical installation and completion exactly on the contract date. What should the project manager do FIRST?

(A) Accept the schedule as the baseline, because completion matches the contract date, the critical path through mechanical installation is clearly shown and the planner has used the tender dates.

(B) Return it with a request to remove unjustified hard constraints, explain or replace the long lags with real activities, and re-run the network so the true critical path and float appear.

(C) Ask the planner to add more detail until the schedule reaches Level 4 for every area, and review the logic only once the extra detail is in place.

(D) Instruct the contractor to accelerate mechanical installation now, because the submitted schedule shows that it is the critical path for the whole project.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Activity and logic register**

Records each activity with its duration basis, logic and calendar so that the network can be reviewed and challenged before it is baselined.

ACTIVITY ID	ACTIVITY (VERB-NOUN)	WBS CODE	DURATION AND BASIS	PREDECESSORS (TYPE, LAG)	CALENDAR	RESPONSIBLE PARTY
SS-C-110	*Pour transformer foundation T1*	*1.2.3*	*3 d: 45 m3 at 15 m3 per day*	*SS-C-100 FS*	*6-day site*	*Civil subcontractor*
SS-C-120	*Cure transformer foundation T1*	*1.2.3*	*7 d: specified curing period*	*SS-C-110 FS*	*7-day continuous*	*Civil subcontractor*

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ACTIVITY ID	ACTIVITY (VERB-NOUN)	WBS CODE	DURATION AND BASIS	PREDECESSORS (TYPE, LAG)	CALENDAR	RESPONSIBLE PARTY

REFLECT

Open a schedule from a current or recent project. How many activities carry hard constraints, have open ends or rely on lags longer than five days? For each one, would a real activity or a written reason explain it better?

Key terms

Precedence diagramming method (PDM)
A way of drawing a schedule network in which activities are boxes (nodes) and arrows show the logic between them, allowing FS, SS, FF and SF relationships with lags. It is the method used by virtually all modern scheduling software.

Lag
A time delay applied to a logic relationship, such as FS + 7 days for curing. It consumes time but carries no resources or cost, so long lags should be justified or replaced by real activities.

Hard constraint
An imposed date, such as 'must finish on', that overrides network logic. It can hide the true critical path and should be used only for genuine fixed dates with a recorded reason.

Schedule level
The degree of detail in a schedule, from Level 1 (one-page milestone summary) to Level 5 (day-by-day crew tasks), with each level serving a different audience while staying consistent with the others.

Finish-to-start (FS) relationship
The default logic link: the successor cannot start until the predecessor has finished, for example setting a transformer only after its plinth is complete.

Lead (negative lag)
A negative lag that allows a successor to start before its predecessor finishes, such as starting cable tray installation in a zone before ceiling framing there is complete.

Milestone
A zero-duration event marking a significant point such as a contract key date, permit approval, equipment delivery or sectional completion, linked into the logic rather than left floating.

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Key takeaways

- Build the schedule as a logic-linked network of verb-noun activities, each tied to one work package and one responsible party.
- Prefer FS links, pair SS with FF for overlapping work, and replace long or unexplained lags with real activities.
- Calculate durations from quantities and crew output, and put weather, heat, holidays and outage windows into calendars.
- Use hard constraints only for genuine external dates, with a written reason: logic, not typed dates, drives the programme.
- Treat AI-generated or model-linked schedules as drafts until a planner has checked the logic, durations and calendars.

KNOWLEDGE CHECK

1. Duct laying may begin two days after cable-trench excavation begins, but must not finish before the excavation finishes. Which relationship pair models this best?

   (A) SS + 2 days, paired with an FF link
   (B) FS + 2 days, paired with an SF link
   (C) SF + 2 days, paired with an SS link
   (D) FF + 2 days, paired with an FS link

2. Why is a 'must finish on' constraint on an internal activity a concern in a Level 3 control schedule?

   (A) It forces the software to add resources so that the constrained date is always achieved
   (B) It converts the activity into a milestone with a duration of zero working days
   (C) It removes the activity from the critical path calculation for the whole project
   (D) It overrides the logic, so float and the critical path may no longer reflect reality

3. A cable-pulling package of 9,600 m will be done by three crews, each pulling 400 m per day. What duration should be entered?

   (A) 6 working days
   (B) 8 working days
   (C) 12 working days
   (D) 24 working days

4. Which level of schedule is normally the basis of the critical path and the baseline used for project control?

   (A) Level 1 master summary for the board
   (B) Level 2 summary programme by facility
   (C) Level 3 project control schedule
   (D) Level 5 task plans for crews on site

5. A 7-day curing lag is placed between two activities that both use a 6-day site calendar. What is the likely problem?

   (A) None, because every scheduling tool counts all lags in calendar days
   (B) The lag may count working days, delaying the successor beyond the real curing need
   (C) The predecessor's own duration will be reduced by one day to compensate for the lag
   (D) The successor will automatically be converted to a continuous 7-day calendar

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · SCHEDULE · LESSON 2 · 38 MIN

The Critical Path Method

Forward pass, backward pass, total and free float, worked by hand on a substation job

- Perform forward and backward passes to calculate ES, EF, LS and LF
- Calculate total float and free float and explain the difference on site
- Identify the critical path and manage near-critical paths and negative float

The **critical path method (CPM)** turns a logic network into dates. A **forward pass** from the start calculates the earliest each activity can start and finish; a **backward pass** from the finish calculates the latest each can start and finish without delaying completion. The difference between the two is **float**, the time an activity can slip. The longest path through the network has the least float and sets the project duration: that is the **critical path**. Engineers who can do this by hand understand what the scheduling software is doing, spot nonsense quickly and can explain the effect of a delay to a client with confidence.

CPM by hand

1. **List activities and logic** Record each activity, its duration and its predecessors. Check that the network has a single start and a single finish and no open ends.
2. **Forward pass** Give the first activity ES = 0. For every activity EF = ES + duration. Where several predecessors merge, ES is the largest of their EFs.
3. **Set the project finish** The largest EF in the network is the project duration. Unless a contract date is imposed, the last activity's LF equals that EF.
4. **Backward pass** For every activity LS = LF - duration. Where an activity has several successors, its LF is the smallest of their LSs.
5. **Calculate float** Total float = LS - ES (which equals LF - EF). Free float = earliest ES among the successors - EF.
6. **Identify critical and near-critical paths** Activities with the lowest total float (zero when the finish is not constrained) form the critical path. Paths with small float are near-critical and need watching.

CPM formulas (zero-based day numbers, FS links)

MEASURE	FORMULA	WHAT IT TELLS YOU
Early finish	`EF = ES + D`	D is the duration in working days; the first activity starts at ES = 0, so EF is the end of the last working day.
Early start	`ES = max(EF of all predecessors)`	An activity can start only when its last predecessor has finished.
Late finish	`LF = min(LS of all successors)`	The final activity's LF equals the project finish: its own EF, or an imposed completion date.
Late start	`LS = LF - D`	The latest start that still allows the activity to finish by its late finish.
Total float	`TF = LS - ES = LF - EF`	How long the activity can slip without delaying project completion.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Free float	`FF = min(ES of all successors) - EF`	How long the activity can slip without delaying the early start of any successor.
Links with lags	`FS: ES(succ) = EF(pred) + lag; SS: ES(succ) = ES(pred) + lag; FF: EF(succ) = EF(pred) + lag`	A lead is entered as a negative lag. Apply the largest result where several links drive one activity.

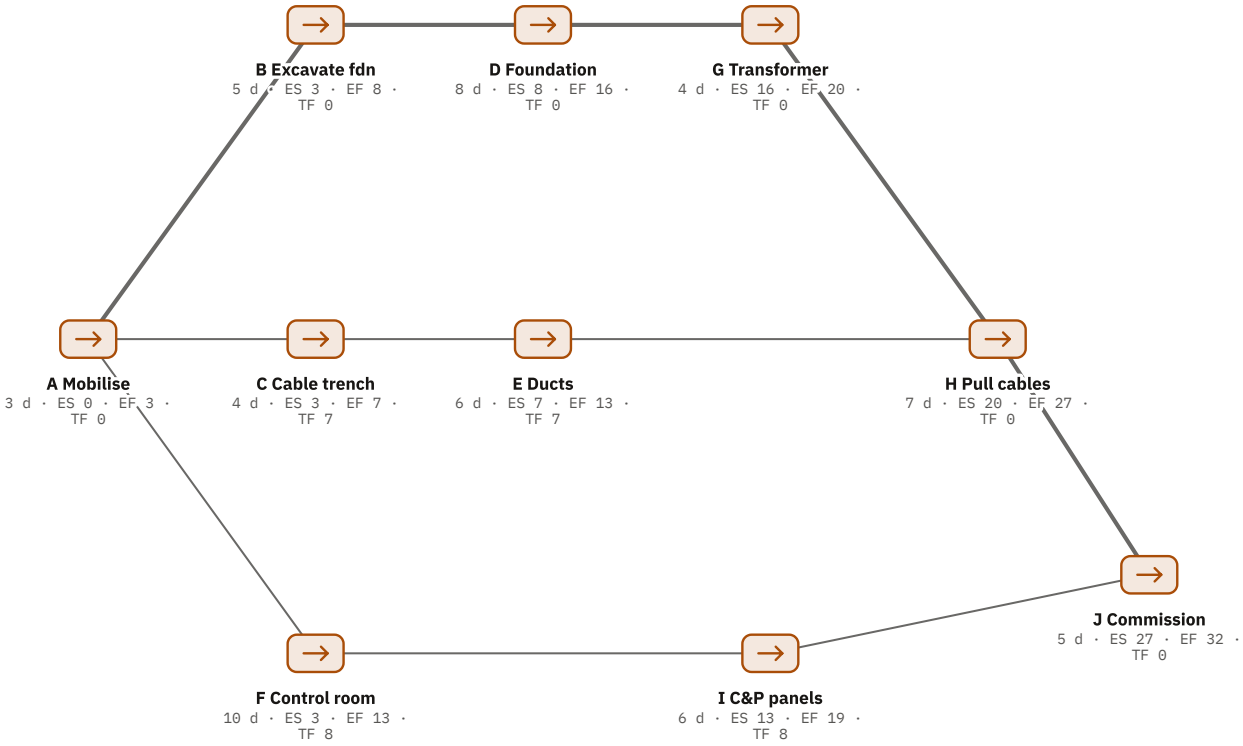
Consider a hypothetical extension of a 33/11 kV substation serving an industrial park near Gazipur. Durations are in working days and every link is finish-to-start with no lag. The forward pass gives a project duration of 32 days. Look at activity H, pulling and terminating cables: it has two predecessors, so it cannot start until both the duct work (E, early finish day 13) and the transformer setting (G, early finish day 20) are complete, and its ES is the larger value, 20. On the backward pass, activity A has three successors whose late starts are 3, 10 and 11, so its LF is the smallest of them, 3. Day numbers are zero-based: 'ES = 3' means the activity starts at the end of day 3, that is, on the morning of working day 4. Many software packages display the same dates as day 4.

Worked example: substation extension (working days, zero-based)

ACTIVITY	DUR	PRED.	ES	EF	LS	LF	TF / FF
A · Mobilise and set out site	3	-	0	3	0	3	0 / 0
B · Excavate transformer foundation	5	A	3	8	3	8	0 / 0
C · Excavate cable trench	4	A	3	7	10	14	7 / 0
D · Construct and cure transformer foundation	8	B	8	16	8	16	0 / 0
E · Lay ducts and backfill trench	6	C	7	13	14	20	7 / 7
F · Build control room	10	A	3	13	11	21	8 / 0
G · Deliver and set transformer	4	D	16	20	16	20	0 / 0
H · Pull and terminate cables	7	E, G	20	27	20	27	0 / 0
I · Install control and protection panels	6	F	13	19	21	27	8 / 8
J · Test and commission	5	H, I	27	32	27	32	0 / 0

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The substation network as an activity-on-node diagram



Logic runs left to right and every link is finish-to-start. Bold links trace the critical path A-B-D-G-H-J (32 days, zero float); H waits for both E and G, and J for both H and I.

EXAMPLE · READING THE RESULT

Three paths run from A to J. **A-B-D-G-H-J** = 3 + 5 + 8 + 4 + 7 + 5 = 32 days: the critical path, zero float. **A-C-E-H-J** = 3 + 4 + 6 + 7 + 5 = 25 days: 7 days of float. **A-F-I-J** = 3 + 10 + 6 + 5 = 24 days: 8 days of float. Notice that C has 7 days of total float but **zero free float**: any slip in C immediately delays E, although completion is safe until the shared 7 days are used up. C and E share the same float, so if C uses 4 days, E has only 3 left. Free float sits at the end of a path, just before the merge point: E (7) and I (8).

Total float vs free float

Total float

- Slip possible without delaying project completion
- Shared along a path: whatever one activity uses, the others lose
- The basis for identifying critical and near-critical paths
- Becomes negative when an imposed date is earlier than the date the network can achieve

Free float

- Slip possible without delaying the early start of any successor
- Belongs to the activity: using it harms no other activity
- Usually found on the last activity of a path before a merge point
- Never greater than total float for the same activity

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WATCH OUT · NEGATIVE FLOAT AND WHO OWNS FLOAT

If a contract requires completion on day 30 but the network calculates day 32, setting the final LF to 30 gives every activity on the critical path **negative float** of -2: the plan cannot meet the date and time must be recovered, as shown in the compression lesson of this module. Float ownership is a contract question, developed in Procure: some contracts treat float as a shared project resource, others reserve it to one party. Agree the rule before the first delay, not after it.

Managing near-critical paths

- Set a threshold, for example total float of 10 working days or less, and report those paths alongside the critical path every month.
- A near-critical path that carries long equipment deliveries, utility connections or permits can become critical overnight.
- Watch **float erosion**: a path whose float falls from 15 to 5 days over three updates is a warning even before it becomes critical.
- When several paths are close to critical, the chance of finishing on time falls sharply: the merge bias explained in the PERT lesson.
- Longest path and lowest float can differ when calendars or constraints vary, so check both views in the scheduling software.

SCENARIO · DECIDE · A LATE CONTROL ROOM ON THE SUBSTATION JOB

On the substation extension in the table above, the builder of the control room (activity F) reports that blockwork will finish 6 working days late because of a brick shortage. The client's engineer is worried and asks whether commissioning and completion will now slip. What should the project manager conclude FIRST?

(A) Completion slips by 6 days, because the control room is on a path that leads directly to testing and commissioning.

(B) Completion holds, because F has 8 days of total float; but panel installation (I) is pushed back and the path A-F-I-J now has only 2 days of float, so it must be watched closely.

(C) Completion holds and no further action is needed, because the control room has 8 days of free float of its own that it can use without affecting any other activity or the commissioning date.

(D) Completion slips by 2 days, because the 6-day delay exceeds the free float of the control room activity by exactly that amount.

Circle your choice. Answer and feedback: module answer key.

REFLECT

In your current schedule, which three near-critical paths would you report to the sponsor, and what single event on each could turn it critical?

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Key terms

Forward pass
The calculation from project start to finish that gives each activity's early start and early finish, taking the largest predecessor early finish wherever paths merge.

Backward pass
The calculation from project finish back to the start that gives each activity's late finish and late start, taking the smallest successor late start wherever paths branch.

Total float
The time an activity can be delayed without delaying project completion, calculated as LS - ES or LF - EF. It is shared by the activities on the same path.

Free float
The time an activity can be delayed without delaying the early start of any immediate successor, calculated as the earliest successor ES minus the activity's EF.

Critical path
The longest sequence of logically linked activities through the network, which sets the shortest possible project duration. Its activities have the lowest total float.

Near-critical path
A path whose total float is below a threshold set by the project, so that a modest delay or a realised risk could make it critical.

Negative float
Float below zero, which appears when an imposed completion or milestone date is earlier than the date the network can achieve. It shows how much time must be recovered.

Key takeaways

- Forward pass: ES is the largest predecessor EF at a merge. Backward pass: LF is the smallest successor LS at a burst.
- Total float (LS - ES) is shared along a path; free float (earliest successor ES - EF) belongs to the activity alone.
- The critical path is the longest path with the lowest float; report near-critical paths and watch float erosion between updates.
- Negative float means the logic cannot meet an imposed date: recover the time or report the impact.
- Agree who owns float under the contract before the first delay, not after it.

KNOWLEDGE CHECK

6. An activity has ES = 12, a duration of 5 days and LF = 22. What is its total float?

(A) 3 days
(B) 5 days
(C) 10 days
(D) 7 days

7. Activity K has an early finish of day 14. Its two successors have early starts of day 16 and day 19. What is K's free float?

(A) 5 days
(B) 0 days
(C) 2 days
(D) 3 days

8. In a forward pass, an activity has three FS predecessors with early finishes of day 9, day 14 and day 11. What is its early start?

(A) Day 9
(B) Day 11
(C) Day 34
(D) Day 14

9. Why can using up the total float of one activity harm another activity?

(A) Total float is shared along a path, so other activities on that path lose it
(B) Total float is converted into a lag on the successor's logic relationship
(C) Scheduling software deletes all remaining float once an activity has started
(D) Total float always equals free float for every activity in a network

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10. A contract requires completion by day 60, but the forward pass gives day 64. If the final LF is set to day 60, the critical activities will show:

   (A) Total float of +4 days
   (B) Total float of -4 days
   (C) Free float of -4 days on the last activity only
   (D) Zero float, because critical paths always show zero

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · SCHEDULE · LESSON 3 · 36 MIN

PERT & Three-Point Estimating

Turn uncertain durations into a probability of meeting the date, and know where PERT misleads

- Calculate beta (PERT) and triangular means and the PERT standard deviation from three-point estimates
- Combine activity variances along a path and use z to find the probability of meeting a target date
- Explain merge bias and the other limitations of PERT and when a fuller risk analysis is needed

Single-point durations hide uncertainty. Bored piling may take 15 days if the ground matches the site investigation, or 24 if the rig meets boulders and casing problems. **Three-point estimating** asks for an optimistic (O), most likely (M) and pessimistic (P) duration, which gives both a better mean and a measure of spread. The **Program Evaluation and Review Technique (PERT)** treats each activity as following a beta distribution, then adds means and variances along the critical path to estimate the probability of finishing by a target date. It is a quick, hand-calculable cousin of the full schedule risk analysis developed in Protect.

PRO TIP · GETTING HONEST THREE-POINT ESTIMATES

Ask for the pessimistic value first ('what could realistically go wrong?'), then the optimistic, and only then the most likely, so that M is not anchored on a hopeful first number. O and P should be rare but credible outcomes, not physical impossibilities or catastrophes. Ask the people who will do the work, and record the drivers behind P, such as rock at pile toe level, monsoon rain, crane breakdown or late shop drawings, because they belong in the risk register too.

PERT and three-point formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
PERT (beta) mean	`te = (O + 4M + P) / 6`	Weighted mean duration: the most likely value carries four times the weight of each extreme.
Triangular mean	`tm = (O + M + P) / 3`	Simple average of the three points; it gives more weight to the tails, so with right-skewed estimates it is higher than the PERT mean.
Activity standard deviation (PERT)	`sigma = (P - O) / 6`	Approximate spread: the range from O to P is treated as about six standard deviations.
Activity variance	`variance = sigma^2 = ((P - O) / 6)^2`	Variances, not standard deviations, are what can be added along a path.
Path mean	`Te = sum of te for the activities on the path`	Expected duration of the critical path.
Path standard deviation	`sigma_path = square root of (sum of activity variances on the path)`	Assumes the activity durations are independent of each other.
Probability of meeting a target	`z = (T - Te) / sigma_path`	Look up z in a standard normal table: P(finish by T) is the area to the left of z.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Date for a chosen confidence	`T = Te + z x sigma_path`	For example z = 1.28 gives about 90% confidence and z = 1.645 about 95%.

A hypothetical contractor is building one four-span segment of an elevated metro viaduct. The critical path for the segment runs through bored piling, pile caps, pier columns and girder erection. The team's three-point estimates in working days are shown below. Girder erection is strongly skewed to the right: launching can go well, but crane availability, night-time possession windows and wind limits can stretch it a long way. Notice that the sum of the most likely durations is 15 + 8 + 13 + 15 = 51 days, while the PERT mean of the path is 56 days: the long pessimistic tails add five days of expected time that a single-point plan would miss.

Worked example: three-point estimates on the critical path (working days)

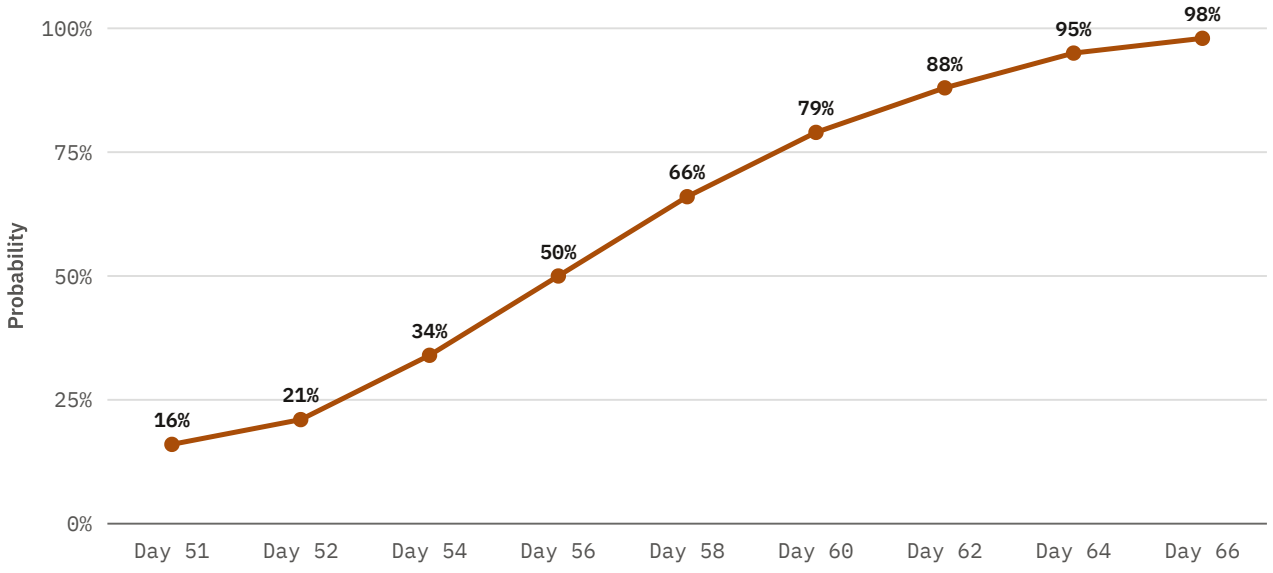
ACTIVITY	O	M	P	TE = (O + 4M + P) / 6	SIGMA = (P - O) / 6	VARIANCE
Bored piling	12	15	24	16.0	2.0	4.0
Pile caps	5	8	11	8.0	1.0	1.0
Pier columns	10	13	22	14.0	2.0	4.0
Girder erection	12	15	36	18.0	4.0	16.0
Path total	-	51 (sum of M)	-	56.0	square root of 25 = 5.0	25.0

EXAMPLE · PROBABILITY OF FINISHING BY DAY 60

Path mean Te = 56 days and path standard deviation = 5 days (the square root of the summed variance of 25; adding the four sigmas instead would wrongly give 9). For a target of 60 days, z = (60 - 56) / 5 = 0.8 and the standard normal table gives about 0.79, roughly a **79%** chance. For 52 days, z = -0.8 and the chance is about 21%. A plan built on the most likely values, 51 days, has z = (51 - 56) / 5 = -1.0, only about a **16%** chance. To be about 95% confident, T = 56 + 1.645 x 5 = 64.2 days, so the team would commit to 65 working days.

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Confidence of finishing the viaduct segment by day T



Normal approximation for the hypothetical worked example (Te = 56, sigma = 5 days). The most likely plan of 51 days has only about a 16% chance, and moving from about 80% to 95% confidence costs another four days.

Useful standard normal values

Z	PROBABILITY OF FINISHING BY T	HOW IT IS TYPICALLY READ
-1.0	about 16%	A date one standard deviation earlier than the mean: very unlikely to be met
0	50%	The mean date: a coin toss
0.8	about 79%	The 60-day target in the worked example
1.0	about 84%	A reasonably confident internal target
1.28	about 90%	A high-confidence (P90) commitment
1.645	about 95%	Very high confidence, at the price of a large time buffer

Beta (PERT) mean vs triangular mean

Beta (PERT)

- Mean = (O + 4M + P) / 6
- Weights the most likely value heavily
- Mean sits closer to M and is less sensitive to an extreme P
- Traditional for hand calculations and quick checks
- Girder erection: (12 + 60 + 36) / 6 = 18 days

Triangular

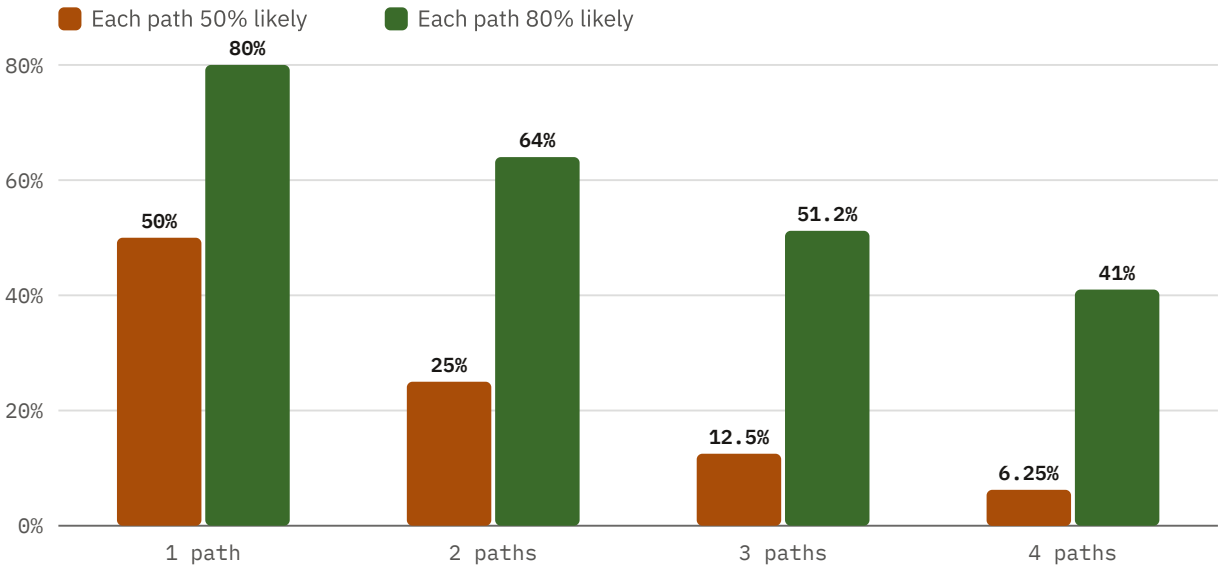
- Mean = (O + M + P) / 3
- Weights all three points equally
- Mean is pulled further towards a long tail
- A common input shape in simulation-based risk tools
- Girder erection: (12 + 15 + 36) / 3 = 21 days

PERT's biggest weakness is that it looks only at the critical path. Where two or more paths merge, the successor can start only when **all** of them have finished, so for independent paths the probability of an on-time start is the product of the individual probabilities. If two independent paths each have a 50% chance of finishing by day 40, the chance that both are done by day 40 is 0.5 x 0.5 = 25%; with three such paths it falls to

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12.5%. This is **merge bias**: projects with many near-critical paths finish later than a critical-path PERT calculation suggests. A Monte Carlo schedule risk analysis, covered in Protect, captures the effect by simulating every path together.

Merge bias: chance that every parallel path is finished on time



Illustrative, assuming independent paths: the joint chance is the product of the individual chances. Even paths that are each 80% likely give only about a 41% chance when four of them feed one milestone.

WATCH OUT · LIMITATIONS OF PERT

PERT assumes activity durations are independent, but monsoon rain, a late design package or a shared crane affects many activities at once, so real variance is larger than PERT shows. It assumes the critical path stays critical, ignores merge bias, and relies on a normal approximation that is shaky when a path has only a handful of activities. Its answers are also only as good as the O and P values people are willing to state. Treat PERT probabilities as a disciplined estimate for discussion, not a guarantee.

INSIGHT · BEYOND HAND PERT

On major projects the hand calculation is a sense check for a **Monte Carlo schedule risk analysis** of the whole network, which models every path, correlation between activities and discrete risk events together. Owners and lenders commonly ask for P50 and P80 (sometimes P90) dates, so always state which confidence level a committed date represents. Some tools now suggest duration ranges from historical project data: use them as a reference class, adjust them for this site's ground, method and supply chain, and keep a named planner accountable for the inputs. Ranges for long-lead equipment such as power transformers and switchgear should reflect current supplier lead times, which have been volatile in recent years, not figures copied from an old programme.

CASE STUDY · COMMITTING TO A GIRDER LAUNCHING DATE

A hypothetical contractor on an elevated metro extension must give the client a committed date for completing deck erection on a ten-span section, because a station fit-out contractor depends on it. The planner's critical path through piling, pile caps, piers, pier heads and girder launching gives a PERT mean of 140 working days with a path standard deviation of 8 days. A parallel path for a river-crossing span, with its own temporary works, has a

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mean of 132 days and a standard deviation of 10 days, and both paths must finish before the deck is complete. The commercial manager wants to commit to 140 days to look competitive.

1. What is the probability of meeting 140 days on the critical path alone?

2. What does the parallel river-crossing path do to that probability?

3. What date would you recommend committing to, and how would you justify it?

Model answer: module answer key.

SCENARIO · DECIDE · A SUBCONTRACTOR WHO WILL NOT GIVE A RANGE

A marine piling subcontractor on a hypothetical LNG terminal jetty gives the planner a single duration of 30 days and refuses to discuss ranges, saying 'that is our number'. The geotechnical report shows variable rock levels along the jetty line, and the piling sits on the critical path. The sponsor wants a stated confidence level for the jetty completion milestone. What should the project manager do FIRST?

(A) Treat 30 days as the most likely value and add and subtract a standard 10% to create an optimistic and a pessimistic value for the PERT calculation.

(B) Hold a short estimating session with the subcontractor, the geotechnical engineer and the planner to agree O, M and P with their drivers, then calculate te and sigma for the path.

(C) Replace the subcontractor's figure with a triangular mean based on a range that the planner chooses alone from previous jetty projects in other countries, without involving the subcontractor.

(D) Report a 50% confidence level for the milestone, because a single-point estimate always represents the median of the duration distribution.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Choose one critical activity on your project and write down O, M and P with a reason for each. How far apart are the PERT and triangular means, and what does that gap tell you about the risk in the activity?

Key terms

Program Evaluation and Review Technique (PERT)
A network scheduling technique that uses optimistic, most likely and pessimistic durations, a beta-distribution approximation and path variances to estimate the probability of meeting a target date.

Three-point estimate
A duration or cost estimate expressed as optimistic, most likely and pessimistic values, so that both the expected value and the uncertainty can be calculated.

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PERT (beta) mean
The expected activity duration (O + 4M + P) / 6, which gives the most likely value four times the weight of each extreme.

PERT standard deviation
The approximate spread of an activity duration, (P - O) / 6, treating the optimistic-to-pessimistic range as about six standard deviations.

Merge bias
The tendency for a milestone fed by several parallel paths to finish later than any single path suggests, because every incoming path must finish before the milestone is reached.

Triangular mean
The expected duration (O + M + P) / 3 under a triangular distribution; it weights the extremes more heavily than the PERT mean.

Path variance
The sum of the variances of the activities on a path, from which the path standard deviation is the square root; it assumes the activities are independent.

Key takeaways

- PERT mean = (O + 4M + P) / 6 and sigma = (P - O) / 6; add variances, not standard deviations, along a path.
- z = (T - Te) / sigma_path turns a target date into a probability; commit to a stated confidence such as P80 or P90.
- A plan built on most likely durations is usually optimistic when risks are skewed to the right.
- Merge bias and correlated risks make the real chance lower than a single-path PERT result; use a Monte Carlo analysis for commitments that matter.

KNOWLEDGE CHECK

11. A critical activity has O = 4, M = 7 and P = 16 days. What is its PERT mean duration?
    - (A) 7 days
    - (B) 9 days
    - (C) 8 days
    - (D) 9.5 days

12. For the same activity (O = 4, M = 7, P = 16), what is the PERT standard deviation?
    - (A) 2 days
    - (B) 4 days
    - (C) 12 days
    - (D) 1.5 days

13. Three critical activities have variances of 4, 9 and 3 (days squared). What is the path standard deviation?
    - (A) 16 days
    - (B) about 6.7 days
    - (C) 8 days
    - (D) 4 days

14. A path has a PERT mean of 90 days and a standard deviation of 6 days. Roughly what is the chance of finishing within 96 days?
    - (A) About 50%
    - (B) About 84%
    - (C) About 69%
    - (D) About 98%

15. Why do projects with several parallel near-critical paths tend to finish later than a critical-path PERT calculation predicts?
    - (A) Parallel paths always have larger variances than the critical path does
    - (B) Scheduling software ignores all parallel paths when calculating the critical path
    - (C) Merge points need every incoming path to finish, so the probabilities multiply
    - (D) The beta distribution cannot be applied to activities that are not critical

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · SCHEDULE · LESSON 4 · 34 MIN

Resource Loading, Levelling & Smoothing

Cranes, specialist crews and work-face space: making the logic network buildable

- Load resources onto a schedule and read a resource histogram
- Distinguish resource levelling from resource smoothing and apply each appropriately
- Judge how resource constraints change float and the critical path

A logic network quietly assumes unlimited resources. Real sites have one tower crane, two crews of coded welders, a single team of protection-relay testers and a laydown area that holds steel for only one zone at a time. **Resource loading** assigns the labour, plant and specialist crews each activity needs; when the loaded schedule demands more than is available on some days, the programme is not achievable as drawn. Resource planning turns a theoretical CPM schedule into one the site can actually work to, and it often changes which activities are critical.

Resources that constrain engineering projects

RESOURCE	WHY IT CONSTRAINS	PLANNING RESPONSE
Cranes and heavy-lift plant	Few units, long mobilisation, wind and ground-bearing limits; one hook serves many trades	Plan by hook time, prepare lift plans, give heavy-lift windows their own resource calendar
Specialist crews (coded welders, relay testers, refractory installers, commissioning engineers)	Small pool, often shared across projects or brought from abroad; visas and qualifications take time	Book early, give the crew continuous work, avoid stop-start sequences
General labour and trades	Seasonal availability, festival periods, productivity loss in heat and congestion	Histogram by trade, realistic ramp-up and ramp-down, welfare facilities sized for the peak
Work-face space and access	Too many crews in one area cut productivity and raise safety risk	Zone-based sequencing, limits on trade density, access and laydown plans
Formwork, falsework and temporary works	Limited sets of formwork, props or launching gantries cycle through the job	Model as resources; plan cycle times and reuse between zones
Owner resources and outages	Shutdown windows, permit issuers and owner's operators needed for tie-ins	Agree calendars with operations and protect the windows in the schedule

A **resource histogram** plots the quantity of a resource needed in each period, usually first from the early-start schedule. Peaks show where demand exceeds availability; deep valleys show crews standing idle or being demobilised and remobilised, which costs money and wastes the learning a crew has built up. A good labour histogram for a building or plant job rises steadily, holds a reasonably flat plateau through the main construction phase and falls away through commissioning. Sharp spikes are a sign that the schedule needs work even before availability is checked.

Worked example: one tower crane, three lifting

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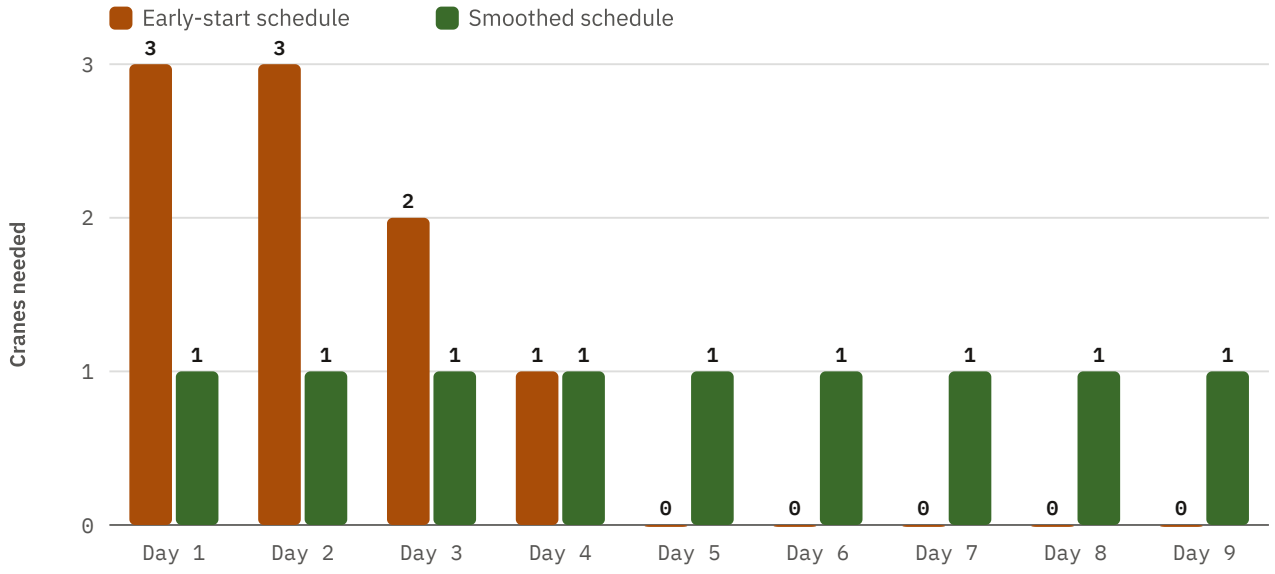
activities (hypothetical garment factory expansion)

ACTIVITY	DURATION (DAYS)	CRANES NEEDED	TOTAL FLOAT (DAYS)	EARLY-START DAYS	SMOOTHED DAYS	FLOAT LEFT
P: Erect roof trusses, zone 1	4	1	0	1-4	1-4	0
Q: Install precast stair cores	3	1	5	1-3	5-7	1
R: Lift rooftop chiller units	2	1	8	1-2	8-9	1

EXAMPLE · SMOOTHING, THEN LEVELLING

On early starts the site would need 3 cranes on days 1-2, 2 on day 3 and 1 on day 4, then none. Moving Q by 4 days and R by 7 days, both within their float, gives exactly 1 crane every day from day 1 to day 9 and leaves completion unchanged: that is **smoothing**, and it has used most of the float. Now suppose R had only 4 days of total float. Delaying R by 7 days would then push completion back by 3 days. Re-sequencing helps: lift R on days 5-6 (using all 4 of its float) and install Q on days 7-9 (6 days late against 5 days of float), and completion slips by only 1 day. Accepting a later finish because the crane limit is fixed is **levelling**, and choosing the best sequence is a planning judgement, not something to leave to a software rule.

Tower crane demand before and after smoothing



From the hypothetical worked example: early starts need three cranes on days 1-2 against one available. Moving Q and R within their float holds demand at one crane every day without moving completion.

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Resource levelling vs resource smoothing

Resource levelling

- The resource limit is fixed; the end date may move
- Used when the resource genuinely cannot be increased, such as one heavy-lift crane or one shutdown crew
- Can lengthen the project and create a resource-driven critical path
- Software delays lower-priority activities using priority rules
- Results must be reviewed, because rules can make odd choices

Resource smoothing

- The end date is fixed; activities move only within their float
- Used to cut peaks and valleys for efficiency, cost and safety
- Does not change the project duration, but consumes float
- May not remove every over-allocation
- Leaves less float to absorb future delays

INSIGHT · WHAT LEVELLING DOES TO THE CRITICAL PATH

After levelling, an activity can become critical not because of logic but because it is waiting for a resource. Scheduling software may represent this with resource-driven links that do not appear in the logic diagram, so the critical path can look broken. Make the reason visible: either add a documented resource relationship or report a **resource-constrained critical path** that names the crane, crew or outage window driving it. Remember that smoothing consumes float: the plan becomes more efficient but also more fragile.

Choosing a response to a resource clash

ABILITY TO ADD MORE OF THE RESOURCE →

Add resource or crash

- Second crane or crew if cheaper than the delay
- Check work-face space and supervision first

Smooth first, add only if cheap

- Move work within its float
- Add resource only if the float is needed for risk

Level and re-sequence

- Accept a resource-driven date or change the method
- Night lifts, offsite assembly or a new sequence
- Report the resource-constrained critical path

Smooth within float

- Delay non-critical work to fit the limit
- Record the float consumed

FLOAT ON THE CLASHING WORK →

Ask two questions: how much float does the clashing work have, and can the resource really be increased? Clashes in the right-hand column are usually solved by smoothing; the bottom-left quadrant is true levelling.

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From logic to a resource-feasible programme

01 Load — Assign labour, plant and specialist crews to each activity →

02 Plot — Produce a histogram per key resource from the early-start schedule →

03 Compare — Overlay availability and find over-allocations and idle periods →

04 Smooth — Move non-critical work within its float to cut peaks →

05 Level — Where limits remain, re-sequence or delay work and accept the date effect →

06 Review — Check the new critical path, float and risks, then agree and baseline

PRO TIP · PLANNING AROUND THE CRANE

On congested sites the tower crane is often the real critical resource. Plan by **hook time**: list every lift by zone and day, give priority to lifts that release other trades (formwork panels, reinforcement cages, steel members), move small-item handling to hoists, telehandlers or permitted night shifts, and keep heavy-lift days clear of competing work. Wind-speed limits, lift plans and exclusion zones, developed in Protect, belong in the same plan as the crane schedule.

INSIGHT · OFFSITE WORK, HEAT AND CROWDED WORK FRONTS

Modern methods of construction such as precast frames, volumetric modules and pre-assembled pipe racks move labour off site. The site labour histogram flattens, but demand concentrates on crane hook time, delivery slots and laydown space, and factory capacity and transport become resources to plan. **Heat stress** in the hot season brings work-rest regimes and earlier shifts that cut productive hours, so histograms based on cool-season output overstate capacity. Limiting trade density at a work face is a **safety control** as well as a productivity measure, and leaders who set and enforce those limits prevent the congestion behind many incidents. Optimisation and AI-assisted levelling tools can test many sequences quickly, but their priority rules still need a planner's review.

TEMPLATE

Weekly resource loading sheet

Compares weekly demand for each constrained resource with what is available and records the action taken to close any gap.

WEEK	RESOURCE (TRADE, CREW OR PLANT)	ACTIVITIES USING IT	REQUIRED	AVAILABLE	OVER (+) / UNDER (-)	ACTION AND OWNER
Wk 14	*Tower crane TC-1 (hook days)*	*Roof trusses zone 1; precast stairs; chiller lift*	*9*	*6*	*+3*	*Move chiller lift to a permitted night shift; planner*
Wk 14	*Coded pipe welders*	*Boiler feed-water piping; steam header tie-ins*	*10*	*8*	*+2*	*Release 2 welders from utility rack work; mechanical lead*

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WEEK	RESOURCE (TRADE, CREW OR PLANT)	ACTIVITIES USING IT	REQUIRED	AVAILABLE	OVER (+) / UNDER (-)	ACTION AND OWNER

SCENARIO · DECIDE · ONE CRANE, TWO LIFTS

On a hypothetical 100 MW solar park, a single mobile crane is on site. Setting the main power transformer in the project substation is critical with zero float; lifting the containerised inverter skids across the array has 15 days of total float. Both are planned for the same three days, and the crane supplier offers a second crane at short notice for a premium. What should the project manager do FIRST?

(A) Hire the second crane immediately, so that both the transformer setting and the inverter lifts can keep their early-start dates.

(B) Give the crane to the inverter lifts first, because they involve more units, more lifts and a larger share of the array's value.

(C) Split each day's crane time between the two activities, so that neither the transformer setting nor the inverter lifting is completely stopped while the crane is shared.

(D) Allocate the crane to the transformer setting, move the inverter lifts within their float, and confirm that the revised schedule keeps the completion date.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Sketch a quick histogram for the scarcest resource on your project. Where are the peaks, which activities cause them, and how much float do those activities have to move?

Key terms

Resource loading
Assigning the quantities of labour, plant, equipment and specialist crews required by each activity in the schedule so that demand can be totalled over time.

Resource over-allocation
A period in which the schedule requires more of a resource than is available, meaning the programme cannot be worked as drawn.

Resource smoothing
Adjusting activity dates within their float to reduce peaks and valleys in resource demand without changing the project end date.

Resource-constrained critical path
The sequence of activities that drives completion once resource limits are applied, including links created by waiting for a scarce resource rather than by logic alone.

Resource histogram
A bar chart showing the quantity of a resource required in each time period, used to spot peaks, valleys and over-allocation.

Resource levelling
Adjusting activity dates so that resource limits are never exceeded, accepting that the project end date may move later.

Resource calendar
A calendar showing when a specific resource, such as a heavy-lift crane or a commissioning team, is available to the project.

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Key takeaways

- A logic network assumes unlimited resources: load the scarce ones, such as cranes, specialist crews and work-face space, and compare demand with availability.
- Smoothing keeps the end date and consumes float; levelling respects a fixed limit and may move the end date.
- Choose the sequence deliberately and review it, because software priority rules can make poor choices.
- After levelling, report the resource-constrained critical path and name the resource that drives it.

KNOWLEDGE CHECK

16. A project has one heavy-lift crane, so three lifts planned for the same day must be spread out even if completion moves later. This is:

    (A) Resource smoothing
    (B) Fast-tracking
    (C) Resource levelling
    (D) Crashing

17. When non-critical activities are moved to reduce a resource peak without changing the end date, what is consumed?

    (A) The float of the activities that are moved
    (B) The management reserve held by the sponsor
    (C) The lags between critical activities
    (D) The contract time allowed for completion

18. Two activities need the only tower crane. X lasts 5 days with zero total float; Y lasts 3 days with 2 days of total float. Both have an early start of day 1. If Y waits until X finishes, how much does completion move?

    (A) 0 days
    (B) 2 days
    (C) 5 days
    (D) 3 days

19. After levelling, an activity that was not on the logic critical path now shows zero float. What is the most likely reason?

    (A) Levelling has added a hard constraint to the activity's start date in the software
    (B) It now waits for a scarce resource, so a resource-driven path controls completion
    (C) Levelling converts every non-critical activity into a milestone
    (D) Its duration has been doubled to match the available resource level

20. Which histogram shape usually signals a workable labour plan on a plant construction job?

    (A) A steady build-up, a broad plateau and a gradual run-down
    (B) Sharp weekly spikes followed by weeks with idle crews
    (C) A flat line at peak strength from mobilisation to closeout
    (D) A large peak in the first month, then a long thin tail

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · SCHEDULE · LESSON 5 · 40 MIN

Schedule Compression, Updating & Look-Ahead Planning

Crash with cost slopes, fast-track with care, update honestly and plan the next six weeks with the crews

- Crash a schedule economically using cost slopes and indirect costs
- Compare crashing with fast-tracking and judge the risks of each
- Update a schedule correctly against a data date and baseline
- Run look-ahead planning and schedule health checks that keep the programme credible

When the forecast finish is later than the required date, the project manager has two main compression tools. **Crashing** adds resources, such as extra shifts, more crews, a second crane or faster-curing concrete, to shorten critical activities at extra direct cost. **Fast-tracking** overlaps activities that were planned in sequence, such as starting foundations on a released design package before the superstructure design is frozen; it saves time with little direct cost but raises the risk of rework. Both help only when applied to the critical path: shortening a non-critical activity buys nothing except a bill.

Crashing formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Crash cost slope	`Cost slope = (Crash cost - Normal cost) / (Normal duration - Crash duration)`	Extra direct cost per day saved, assuming cost rises roughly in a straight line between the normal and crash points.
Maximum crash	`Maximum days saved = Normal duration - Crash duration`	The activity cannot be shortened beyond its crash duration, however much is spent.
Net effect of one day saved	`Change in total cost = Cost slope - (Daily indirect cost + daily delay damages avoided)`	If the result is negative, crashing that day lowers total cost. Stop when the cheapest available slope exceeds the daily saving.

Return to the substation extension from the CPM lesson. The critical path A-B-D-G-H-J takes 32 days; the parallel paths A-C-E-H-J and A-F-I-J take 25 and 24 days. Site indirect costs (supervision, site facilities, standby plant and security) run at 7k per day, where k means thousands of a currency unit and all figures are illustrative. The client wants the earliest economical finish. Crash data for the critical activities is below; mobilisation (A) cannot be shortened.

Crash data for the critical activities (illustrative, costs in k)

ACTIVITY	NORMAL DURATION	CRASH DURATION	NORMAL COST	CRASH COST	COST SLOPE PER DAY	MAX DAYS SAVED
B: Excavate transformer foundation	5	4	20	26	6	1
D: Construct and cure foundation	8	6	60	70	5	2
G: Deliver and set transformer	4	3	15	27	12	1
H: Pull and terminate cables	7	5	28	36	4	2

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ACTIVITY	NORMAL DURATION	CRASH DURATION	NORMAL COST	CRASH COST	COST SLOPE PER DAY	MAX DAYS SAVED
J: Test and commission	5	4	30	45	15	1

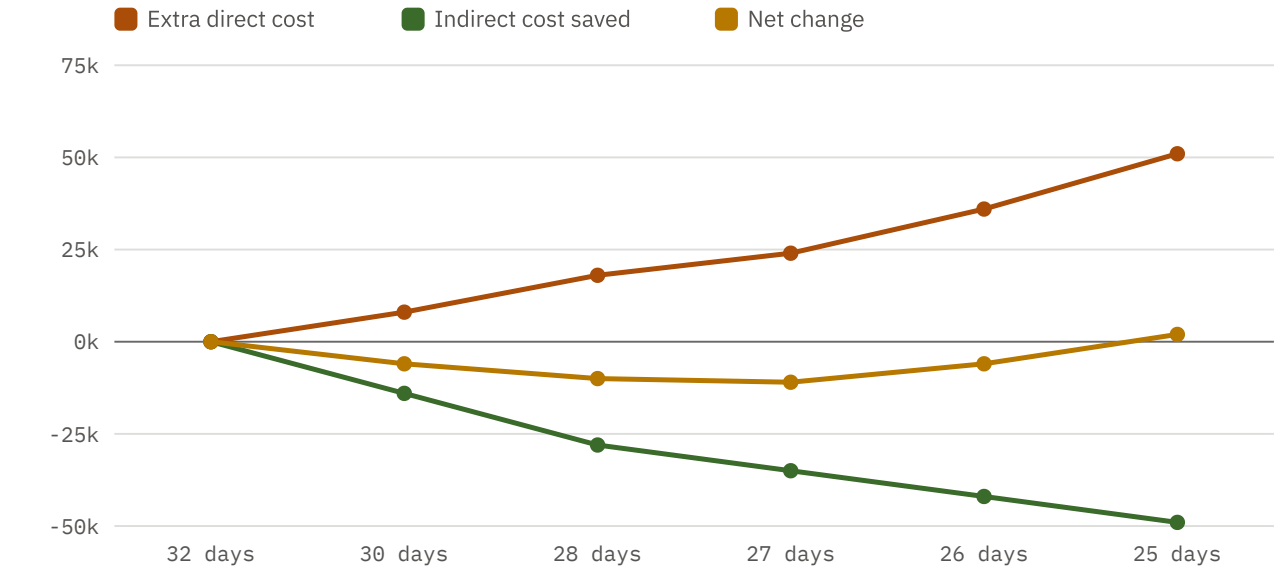
Crashing the substation step by step

1. **Crash the cheapest critical activity** H has the lowest slope, 4k per day, against 7k of indirect cost saved per day. Crash H by 2 days: duration 32 to 30, extra direct cost 8k, indirect saving 14k.
2. **Re-check every path** The critical path is now 30 days. H also sits on A-C-E-H-J, which falls to 23 days; A-F-I-J stays at 24. A single critical path remains.
3. **Crash the next cheapest** D at 5k per day: crash 2 days, 30 to 28, extra 10k, saving 14k.
4. **Continue while the slope is below the daily saving** B at 6k per day: crash 1 day, 28 to 27, extra 6k, saving 7k.
5. **Stop at the economical point** The remaining options, G at 12k and J at 15k per day, cost more than the 7k saved per day, so 27 days is the economical finish. The parallel paths (24 and 23 days) are still shorter, so they have not become critical.

EXAMPLE · THE RESULT, AND WHAT CHANGES IT

Duration falls from 32 to 27 days. Direct cost rises by 24k (8 + 10 + 6) while indirect cost falls by 35k (5 days x 7k), a net saving of 11k. If the contract also imposed delay damages of 10k per day for every day beyond day 25, each day saved down to day 25 would be worth 17k, so crashing G (12k) and then J (15k) would also pay and the economical finish would become 25 days. The method is the same; only the daily saving changes. In practice, check that crashed durations are achievable: a second cable crew needs drums, termination kits, supervision and work-face space.

Crashing the substation: change in cost against the 32-day plan



Illustrative figures from the worked example with no delay damages. The net change is lowest at 27 days (11k saved); crashing G and then J beyond that adds more direct cost than the indirect cost it saves.

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Crashing vs fast-tracking

Crashing

- Adds resources to shorten critical activities
- Raises direct cost, often steeply near the crash limit
- Logic unchanged; durations shorter
- Watch productivity losses from congestion, fatigue and long overtime
- Example: double-shift concrete pours or a second launching gantry

Fast-tracking

- Overlaps activities planned in sequence
- Little direct cost but more risk of rework, claims and safety conflicts
- Logic changed from FS to SS/FF links or leads
- Watch design maturity, interfaces and work on top of other work
- Example: start piling on released foundation drawings before superstructure design is frozen

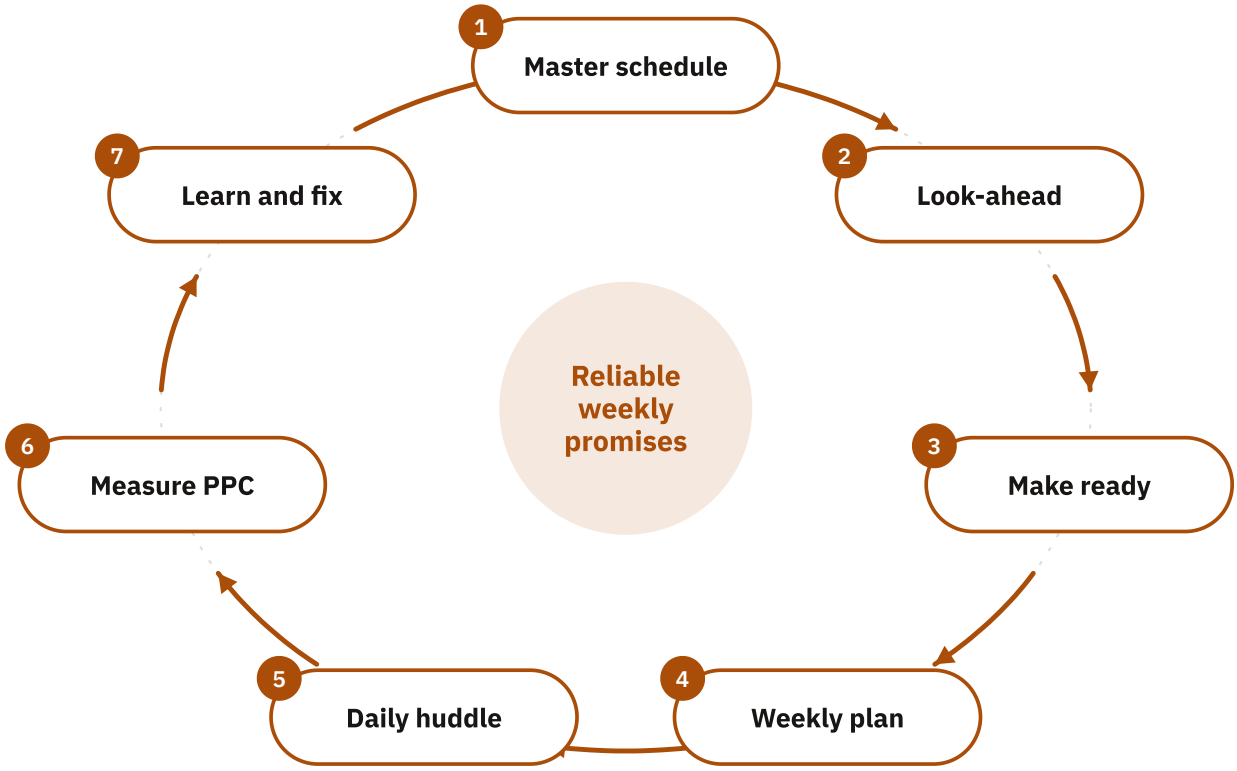
A schedule is useful only if it is updated on a fixed cycle, usually weekly on site and monthly for formal reporting. At each update the planner sets a new **data date**, records actual start and finish dates, enters a realistic **remaining duration** (not just a percent complete) for each activity in progress, and recalculates. Out-of-sequence progress, where work has started before its predecessors finished, must be handled deliberately: **retained logic** keeps the remaining work waiting for the predecessor, while **progress override** lets it continue as if the link no longer applied. Retained logic is usually the more honest choice. The update is compared with the **schedule baseline** to report variances and forecast completion; the baseline itself changes only through formal change control, as set out in Define. Progress evidence increasingly comes from **reality capture**: drone surveys, laser scans and 360-degree photos compared with the model show what is actually installed. Automated or AI-assisted progress recognition speeds this up, but site engineers must verify it before it becomes an actual date in the schedule. Modern contract forms such as FIDIC 2017 and NEC4 set out what a programme must contain and when revised programmes are due, so check what your contract requires at each update.

Look-ahead and last-planner style planning

- Roll a 4-6 week look-ahead from the Level 3 schedule and break it into crew-level tasks together with the foremen and subcontractor supervisors who will do the work.
- Run a **make-ready** process: list each task's constraints (drawings, RFI answers, materials, permits, access, crane time, inspections) and give each an owner and a clear-by date.
- Put into next week's work plan only tasks whose constraints are cleared, so crews are not sent to work that cannot be done.
- Measure **percent plan complete (PPC)** every week and record why tasks were missed; fix the recurring causes rather than blaming crews.
- Feed actual progress and newly discovered constraints back into the master schedule so the critical path stays realistic.

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The weekly planning loop



1. **Master schedule** — Level 3 milestones and the critical path set the targets
2. **Look-ahead** — The next 4-6 weeks broken into crew tasks with the foremen
3. **Make ready** — Constraints listed, owned and cleared by a date
4. **Weekly plan** — Crews commit only to tasks whose constraints are cleared
5. **Daily huddle** — Short coordination at the work face; early warnings raised
6. **Measure PPC** — Tasks done as planned divided by tasks planned; reasons logged
7. **Learn and fix** — Remove recurring causes and feed progress back to the schedule

The loop turns once a week. Notice that work enters the weekly plan only after make-ready, and that what is learned flows back to the master schedule so the critical path stays realistic.

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Schedule health check before baselining and at each major update

☐ No open ends: every activity except the project start and finish has a predecessor and a successor.

☐ Hard constraints are few, justified and documented.

☐ Lags are short and explained, and no leads hide overlapping work without a reason.

☐ Most links are finish-to-start; SS/FF pairs are used correctly for overlapping work.

☐ No unexplained very high float that suggests missing logic.

☐ Most activity durations are within about two update periods, apart from procurement and level-of-effort items.

☐ No actual dates after the data date and no forecast dates before it.

☐ A critical path test works: adding delay to a critical activity delays the finish by the same amount.

☐ Key resources are loaded and peaks have been checked against availability.

☐ Progress in the update is backed by verified evidence, and any logic or durations suggested by automated tools have been reviewed by a planner.

☐ Any negative float is reported with a recovery plan and an owner.

CASE STUDY · RECOVERING A HEAT RECOVERY STEAM GENERATOR PROGRAMME

On a hypothetical combined-cycle power plant extension, erection of the heat recovery steam generator (HRSG) is six weeks behind and lies on the critical path to first fire. The EPC contractor proposes three measures: double-shift welding on the high-pressure modules; starting insulation and cladding on completed sections before the hydrostatic test of the whole unit; and bringing the commissioning team forward to begin cold commissioning of auxiliary systems early. The owner's project manager must advise the steering committee.

1. Which proposals are crashing and which are fast-tracking?

2. What risks does each create, and how would you test whether they really recover time?

3. What should the project manager ask for before approving the recovery plan?

Model answer: module answer key.

TEMPLATE **Six-week look-ahead and constraint log**

Breaks the next six weeks of the control schedule into crew tasks and tracks every constraint that must be cleared before a task can enter a weekly work plan.

TASK (CREW LEVEL)	WEEK	CREW / FOREMAN	CONSTRAINTS TO CLEAR	CONSTRAINT OWNER	CLEAR BY	READY (Y/N)
Fix rebar cages, pier P12	*Wk 3*	*Rebar gang 2*	*Approved bar bending schedule; cages delivered; crane slot*	*Site engineer; stores; crane coordinator*	*Wk 2 Thu*	*N*
Pull 11 kV cables, feeder 4	*Wk 2*	*Cable crew A*	*Trench handed over; drums on site; permit to work*	*Civil lead; procurement; HSE*	*Wk 1 Sat*	*Y*

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TASK (CREW LEVEL)	WEEK	CREW / FOREMAN	CONSTRAINTS TO CLEAR	CONSTRAINT OWNER	CLEAR BY	READY (Y/N)

SCENARIO · DECIDE · THE CLIENT WANTS A MONTH BACK

On a hypothetical university laboratory building in Dhaka, the client asks for completion four weeks earlier to meet the start of the academic year. The planner proposes fast-tracking by starting laboratory fit-out on floors 1-3 while the roof slab is still being cast, and crashing MEP installation with a second shift. The laboratory design still has 30 open RFIs, and the fume-cupboard supplier has not confirmed exhaust duct sizes. What should the project manager do FIRST?

(A) Approve both measures straight away, so that the client sees a responsive team and the second shift can start on Monday.

(B) Refuse to accelerate at all, explaining that fast-tracking always causes rework and claims and that the contract date must stand.

(C) Analyse which activities are critical, test each measure's time gain, cost slope and rework risk, especially fit-out with open RFIs, and give the client costed options.

(D) Crash every MEP activity with a second shift, critical or not, so that no single trade can hold up the others during the fit-out and the client sees progress on every floor.

Circle your choice. Answer and feedback: module answer key.

Key terms

Crashing
Shortening the duration of critical activities by adding resources such as extra shifts, crews or equipment, accepting additional direct cost to save time.

Fast-tracking
Compressing a schedule by overlapping activities that were planned in sequence, which saves time but increases the risk of rework and interface problems.

Schedule baseline
The approved version of the schedule against which progress and variances are measured; it changes only through formal change control.

Crash cost slope
The extra direct cost per unit of time saved when an activity is crashed: (crash cost - normal cost) / (normal duration - crash duration).

Data date
The date up to which progress has been recorded in a schedule update; everything before it is actual, everything after it is forecast.

Look-ahead schedule
A short-term plan, typically covering the next four to six weeks, that breaks the control schedule into crew-level tasks and identifies constraints to remove.

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Percent plan complete (PPC)
The number of tasks completed as planned in a week divided by the number of tasks planned, expressed as a percentage; a measure of planning reliability.

Key takeaways

- Crash the cheapest critical activity first, re-check every path, and stop when the cost slope exceeds the daily saving.
- Fast-tracking buys time with rework risk: test design maturity and open RFIs before overlapping work.
- Update against a data date with actual dates and realistic remaining durations backed by verified progress evidence.
- Run a 4-6 week look-ahead with make-ready, commit only constraint-free work, and use PPC to improve reliability.

KNOWLEDGE CHECK

21. An activity's normal duration is 10 days at a cost of 50k; its crash duration is 7 days at 62k. What is the crash cost slope?
    - (A) 12k per day
    - (B) 4k per day
    - (C) 6.2k per day
    - (D) about 1.7k per day

22. Daily indirect cost is 5k and there are no delay damages. Which option should be crashed first?
    - (A) A critical activity with a slope of 7k per day, which can save 2 days
    - (B) A critical activity with a slope of 6k per day, which can save 3 days
    - (C) A critical activity with a slope of 3k per day, which can save 1 day
    - (D) A non-critical activity with a slope of 2k per day, saving 4 days

23. Starting piling on released foundation drawings before the superstructure design is frozen is an example of:
    - (A) Crashing
    - (B) Resource levelling
    - (C) Progress override
    - (D) Fast-tracking

24. During a weekly update, what should be entered for an activity that is in progress?
    - (A) Its actual start date and a realistic remaining duration
    - (B) Only the percent complete reported by the foreman that week
    - (C) A new baseline start date to match the actual start
    - (D) A hard constraint on its forecast finish date

25. A team planned 40 tasks for the week and completed 30 of them as planned. What is the percent plan complete?
    - (A) 133%
    - (B) 25%
    - (C) 75%
    - (D) 70%

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · SCHEDULE

Answer key

Knowledge checks

1. A An SS link with a 2-day lag lets ducting follow excavation closely, and the FF link stops the follower from finishing before the leader. The other pairs either force full sequencing or use SF, which does not describe this situation.

2. D Hard constraints fix the date regardless of predecessors, so the network no longer shows the real effect of delays. Scheduling software does not add resources, change durations or simply remove the activity.

3. B Duration = 9,600 / (3 x 400) = 9,600 / 1,200 = 8 working days. 24 days would be the figure for a single crew, and the other values do not follow from the quantities.

4. C The Level 3 schedule is the logic-linked network across all phases and is used for critical path analysis and baselining. Levels 1 and 2 are summaries; Level 5 is too detailed and short-term.

5. B Lags are often counted on a working calendar, so 7 working days on a 6-day calendar stretches beyond 7 real days. Tools differ, so the lag calendar must be checked; durations and successor calendars are not changed automatically.

6. B EF = 12 + 5 = 17, so TF = LF - EF = 22 - 17 = 5 days (equivalently LS = 17 and LS - ES = 5). 10 days is LF minus ES, which ignores the duration.

7. C Free float = earliest successor ES - EF = 16 - 14 = 2 days. Using the later successor (19) would overstate it, because K would then delay the first successor.

8. D An activity can start only when all predecessors are finished, so ES is the largest predecessor EF: day 14. Taking the smallest or adding them up does not reflect the logic.

9. A Total float belongs to the path, not the activity. Software does not convert or delete it, and total and free float are often different, as C and E in the worked example show.

10. B With LF = 60 and EF = 64 on the critical path, TF = LF - EF = -4 days, which signals that 4 days must be recovered. Critical paths show zero float only when the finish is unconstrained.

11. C te = (4 + 4 x 7 + 16) / 6 = 48 / 6 = 8 days. 9 days is the triangular mean (27 / 3) and 7 days is simply the most likely value.

12. A sigma = (P - O) / 6 = (16 - 4) / 6 = 2 days. 12 days is the full range, and the variance would be 4 days squared, not a standard deviation.

13. D Add the variances (4 + 9 + 3 = 16) and take the square root: 4 days. Adding the individual standard deviations (2 + 3 + 1.73 = about 6.7) is a common mistake.

14. B z = (96 - 90) / 6 = 1.0, and the standard normal table gives about 84%. 50% would apply at the mean, 69% at z = 0.5 and about 98% at z = 2.

15. C At a merge point every incoming path must be complete, so on-time probabilities multiply and the overall chance falls. The other statements are false: variances vary, software does calculate all paths, and PERT can be applied to any activity.

16. C Respecting a fixed resource limit and accepting a later end date is levelling. Smoothing keeps the end date fixed, fast-tracking overlaps work and crashing adds resources.

17. A Smoothing moves activities within their float, so float is used up and the plan becomes less tolerant of future delay. Reserves, lags and contract time are not affected.

18. D Y is delayed by 5 days (the length of X) but has only 2 days of float, so completion moves by 5 - 2 = 3 days. Ignoring float would wrongly suggest 5 days.

19. B Once resources are limited, waiting for a crane or crew can drive completion, creating a resource-constrained critical path. Levelling does not normally add constraints, create milestones or change durations.

20. A A steady rise, stable plateau and gradual decline lets crews mobilise, learn and demobilise efficiently. Spikes, constant peak staffing or an early peak waste money and cause congestion.

21. B Slope = (62 - 50) / (10 - 7) = 12 / 3 = 4k per day. 12k is the total extra cost, not the cost per day saved.

22. C Crash the cheapest critical activity whose slope is below the daily saving: 3k is less than 5k. The 6k and 7k options cost more than they save, and crashing a non-critical activity saves no project time.

23. D Overlapping design and construction that were planned in sequence is fast-tracking. Crashing adds resources, levelling respects resource limits and progress override is an update setting.

24. A The actual start and a realistic remaining duration let the software forecast the finish correctly. Percent complete alone may not reflect remaining time, and baselines and constraints should not be altered to fit progress.

25. C PPC = tasks completed as planned / tasks planned = 30 / 40 = 75%. 25% is the share missed, and 133% inverts the ratio.

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Scenarios & case studies

Scenario · Reviewing a contractor's Level 3 schedule
B is best. Best. Logic must drive the dates. Only after constraints and lags are justified can the owner trust the critical path, the float values and any later delay analysis.

Scenario · A late control room on the substation job
B is best. Best. The 6-day slip uses most of the shared float on the path, leaving 2 days. Completion is safe, but the path has become near-critical and needs active management.

Case study · Committing to a girder launching date
On the critical path alone, 140 days is the mean, so z = 0 and the chance is only about 50%. The river-crossing path has z = (140 - 132) / 10 = 0.8, about 79%, so if the paths are independent the chance that both finish by day 140 is roughly 0.50 x 0.79, about 39%: merge bias makes the commitment worse than it looks. At 152 days the critical path has z = 12 / 8 = 1.5 (about 93%) and the river path z = 2.0 (about 98%), giving a combined chance of roughly 91%. A sensible recommendation is to commit to about 152 days, explain the merge effect to the commercial manager, and use the gap to plan responses such as extended possession windows or a second launching crew. The commitment should then be confirmed with a full schedule risk analysis that models correlation between the paths.

Scenario · A subcontractor who will not give a range
B is best. Best. Three-point values based on the real drivers (rock levels, weather, rig performance) give an honest mean and spread, and the session builds shared ownership of the range.

Scenario · One crane, two lifts
D is best. Best. This is smoothing: the critical activity keeps its dates, non-critical work moves within its float, and the change is checked rather than assumed.

Case study · Recovering a heat recovery steam generator programme
Double-shift welding is crashing: it adds resources and cost to a critical activity, but productivity and weld quality can fall at night, so the welder pool, non-destructive testing capacity, lighting and supervision must be checked. Insulating sections before the full hydrotest is fast-tracking: it overlaps sequential work and risks costly removal if leaks are found, so it should be limited to sections that have already passed their own pressure tests. Bringing the commissioning team forward helps only if that work is on the critical path and its prerequisites, such as permanent power, instrument air and water, will really be available. Before approval the project manager should ask for a revised logic-linked schedule showing the recovered critical path, the cost slope of each measure, the added risks with their mitigations, and evidence that near-critical paths do not become critical.

Scenario · The client wants a month back
C is best. Best. Compression must target the critical path and be justified by time gained against cost and risk; the open RFIs make fit-out fast-tracking particularly risky.

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04

MODULE 4 · BUDGET

Cost Estimation, Budgeting & Earned Value

Estimate with honest ranges, set a cost baseline that matches the schedule, and use earned value to see trouble early

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Select and apply estimating methods suited to the maturity of design, including parametric, factored and bottom-up estimates with escalation, location and currency adjustments
- Build a time-phased cost baseline with contingency and management reserve, and turn it into an S-curve and a cash-flow forecast
- Calculate and interpret PV, EV, AC, CV, SV, CPI and SPI for a project and its control accounts
- Forecast final cost and completion using EAC variants, ETC, VAC, TCPI and earned schedule, and apply reporting thresholds
- Measure physical progress with objective rules of credit and control committed, accrued and forecast costs on site

LESSONS

1. Cost Estimating Classes & Methods · 38 min
2. Budgeting & the Cost Baseline · 34 min
3. Earned Value Fundamentals · 36 min
4. Forecasting with Earned Value · 38 min
5. Progress Measurement & Cost Control on Site · 36 min

MODULE 4 · BUDGET · LESSON 1 · 38 MIN

Cost Estimating Classes & Methods

Match the method to the maturity of the design, and state the range honestly

- Relate estimate classes and indicative accuracy ranges to the maturity of design
- Apply analogous, parametric, factored, bottom-up and BOQ unit-rate methods
- Adjust estimates for capacity, escalation, location and currency, and document a basis of estimate

Every engineering estimate is a forecast made with incomplete information. At concept stage the team may know only the plant's capacity and the site; at tender stage it has drawings, a bill of quantities and supplier quotations. Good estimators match the **method** to the **maturity** of the design, state the **accuracy range** honestly, and write a **basis of estimate** so that anyone can see what is included, what is excluded and where each number came from. The estimate feeds the business case developed in Justify and becomes the starting point for the cost baseline built in the next lesson.

Estimate classes by design maturity (AACE-style ranges for process plant; indicative)

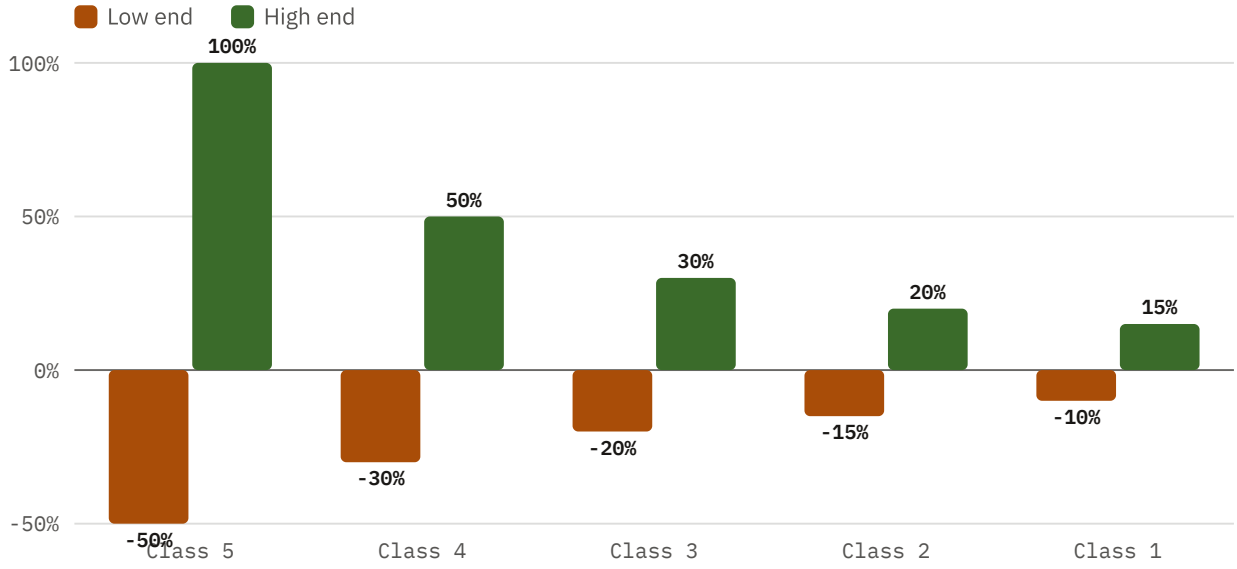
CLASS	DESIGN MATURITY	TYPICAL PURPOSE	TYPICAL METHODS	INDICATIVE ACCURACY RANGE (LOW / HIGH)
Class 5: screening	Concept only: capacity, location, broad scope (about 0-2% of full definition)	Screening options, strategic planning	Analogous, capacity scaling	Low -20% to -50%; high +30% to +100%
Class 4: feasibility	Early design: main equipment list, block layouts (about 1-15% of full definition)	Feasibility study, option selection	Parametric, equipment-factored	Low -15% to -30%; high +20% to +50%
Class 3: budget authorisation	Basic design or FEED largely complete (about 10-40% of full definition)	Funding decision, budget approval	Unit costs for major items, factors for the rest	Low -10% to -20%; high +10% to +30%
Class 2: control	Detailed design substantially advanced (about 30-75% of full definition)	Cost baseline, checking bids	Detailed quantities and unit rates	Low -5% to -15%; high +5% to +20%
Class 1: definitive	Design complete, quantities measured, prices quoted (about 65-100% of full definition)	Check estimate, tender pricing	Full bottom-up with firm quotations	Low -3% to -10%; high +3% to +15%

INSIGHT · RANGES ARE SKEWED, AND THAT IS DELIBERATE

Notice that the high side of each range is wider than the low side. Engineering estimates are far more likely to grow than to shrink, because scope gaps, ground conditions and market prices tend to surprise upwards. The pattern follows AACE International's recommended practice for the process industries; AACE publishes companion practices for other sectors, such as building and general construction, and organisations calibrate the ranges to their own technology, complexity and data. Each bound is itself a range: a first-of-a-kind process plant sits at the wide end, a repeat warehouse at the narrow end. The class describes how mature the definition is, not how much effort went into the estimate.

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The accuracy funnel: widest typical range by class



Indicative, using the widest bounds in the table. The range narrows as the design matures, and the high side always stays wider than the low side.

Five estimating methods

E1

Analogous

Scale the known cost of a similar completed project, adjusting for size, date, location and scope differences. Fast and useful at screening stage, but only as good as the comparability of the reference project.

E2

Parametric

Apply a rate derived from historical data to a key parameter: cost per MW, per m2 of floor area, per km of 132 kV line, per m3/day of treatment capacity. Capacity scaling with an exponent is a parametric technique.

E3

Factored

Price the major equipment, then multiply by factors for installation, piping, electrical, instrumentation, civil works and indirect costs to reach total installed cost. Common for process plant at feasibility stage.

E4

Bottom-up

Estimate each work package from quantities, productivity, labour, plant, materials and subcontract prices, then roll up. The most accurate and the most time-consuming method.

E5

BOQ and unit rates

Measure quantities from drawings into a bill of quantities and price each item with a built-up unit rate. The standard method for civil and building work and the basis of most tender comparisons.

Estimating formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Capacity scaling	`C2 = C1 x (Q2 / Q1)^x`	C is cost and Q capacity. The exponent x is often around 0.6 for process plant (the 'six-tenths rule'), reflecting economies of scale; use data for the specific plant type where it exists.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Parametric estimate	`Cost = Parameter quantity x Rate per unit`	For example floor area x cost per m2, adjusted for specification and site conditions.
Factored estimate	`Total installed cost = Major equipment cost x Installation factor`	Overall factors for process plant commonly lie between about 3 and 5; discipline-by-discipline factors give better accuracy.
Escalation	`C_future = C_base x (1 + e)^n`	e is the annual escalation rate for the cost category; n is the number of years from the estimate base date to the mid-point of spend.
Location adjustment	`C_site = C_reference x (LF_site / LF_reference)`	LF is a location factor or index reflecting labour productivity, wages, materials and logistics at each location.
Currency conversion	`C_local = C_foreign x exchange rate (local units per foreign unit)`	Apply only to the imported content, and record the rate and the date used.
Unit rate	`Unit rate = (Labour + Plant + Materials + Subcontract cost per unit) x (1 + overheads and profit %)`	The building block of BOQ pricing.

EXAMPLE · SCALING AND ESCALATING A TREATMENT PLANT ESTIMATE

A reference effluent treatment plant with a capacity of 2,000 m3/day cost BDT 80 million at its estimate base date. A garment group now needs a 3,000 m3/day plant at a similar site. Capacity scaling with x = 0.6 gives 80 x (3,000 / 2,000)^0.6 = 80 x 1.275 = **BDT 102.0 million**. Escalating at an assumed 7% per year for two years to the mid-point of spend gives 102.0 x 1.07^2 = 102.0 x 1.1449 = **BDT 116.8 million**. The plant is 50% larger but, before escalation, costs only about 28% more. All figures are illustrative, and the result is still a screening-class number to be presented with a wide range.

A **bill of quantities (BOQ)** lists every measurable item of work, such as m3 of concrete in pile caps, tonnes of reinforcement, m2 of formwork, metres of cable tray or numbers of light fittings, measured from the drawings according to an agreed method of measurement. Each item is priced with a **unit rate** that bundles labour, plant, materials, wastage, overheads and profit. Because quantities and rates are kept separate, the client can compare tenders line by line and value variations later; the commercial use of remeasurement contracts is developed in Procure. For estimators the discipline lies in the rate build-up: productivity assumptions, wastage allowances and crew mix are where most estimating errors hide.

Unit rate build-up: 1 m3 of reinforced concrete in pile caps (illustrative)

ELEMENT	BASIS	COST PER M3 (BDT)
Ready-mix concrete	1.05 m3 (5% wastage) x 9,000	9,450

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ELEMENT	BASIS	COST PER M3 (BDT)
Reinforcement, supplied and fixed	120 kg x 95	11,400
Formwork	4.0 m2 x 450	1,800
Labour: place, vibrate, cure	3.0 labour-hours x 250	750
Plant: pump and vibrators	Allowance per m3	600
Direct cost subtotal	Sum of the above	24,000
Overheads and profit	15% of direct cost	3,600
Unit rate	Direct cost + overheads and profit	27,600

PRO TIP · ESTIMATING WITH MODELS, MACHINES AND VOLATILE MARKETS

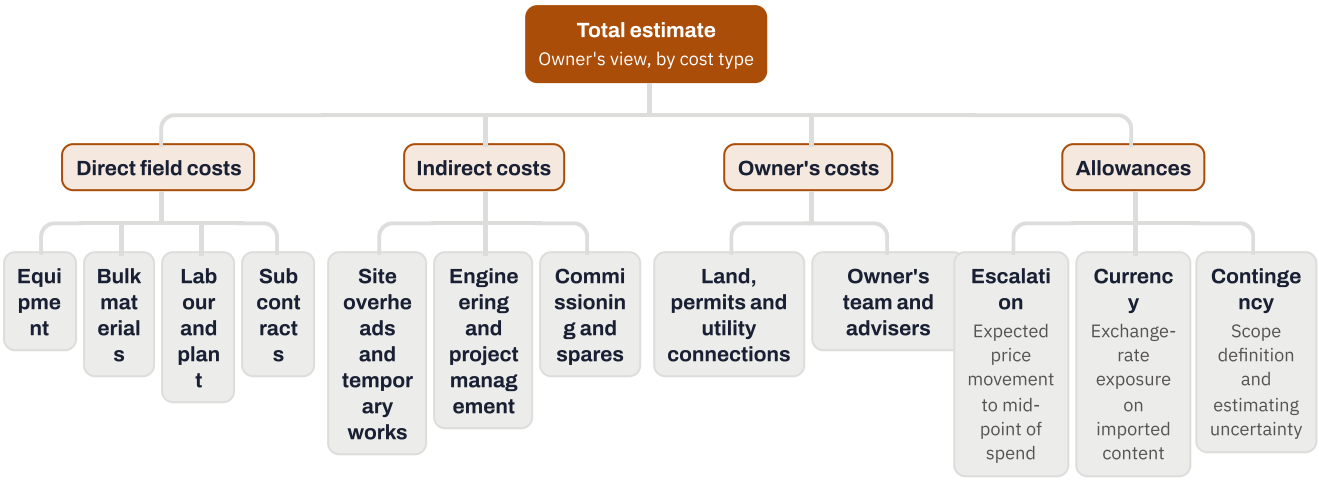
Model-based quantities (often called 5D) let estimators take quantities straight from a BIM model, but check them against the agreed method of measurement and the model's maturity: a concept-stage model omits much of what a BOQ must price. **AI-assisted estimating tools** can suggest rates and benchmarks from past projects in seconds; use them to challenge the estimate, and have an estimator verify the basis, date and location of every rate before it is relied on. **Supply-chain volatility** means long-lead items such as transformers, switchgear and structural steel need current quotations with stated validity, escalation by cost-specific indices and, where the contract allows, price-adjustment provisions (FIDIC 2017 and NEC4 both offer optional mechanisms). The same quantities also drive a first **embodied-carbon** estimate for the product and construction stages (A1-A5), so options can be compared on cost and carbon together.

WATCH OUT · ESCALATION, CURRENCY AND CONTINGENCY ARE DIFFERENT THINGS

Escalation covers expected price movement over time. Currency allowances cover exchange-rate exposure on imported items such as transformers, turbines, switchgear or dyeing machines. Contingency covers uncertainty in scope definition and estimating. Rolling them into a single 'add 15%' hides the reasoning and makes later control impossible. Show each separately, state the base date, the exchange rate and its source, and keep imported and local content apart.

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Structure of a project estimate



Escalation, currency and contingency sit as separate, visible allowances rather than being buried in rates. Management reserve, held by the sponsor, sits above this estimate, as the next lesson shows.

SCENARIO · DECIDE · ONE NUMBER FOR THE BOARD

A hypothetical textile group in Chattogram asks for a cost figure for a new 30,000 m2 knit-dyeing factory within one week, to decide whether to fund a feasibility study. The only data available are the site, the capacity target and the final cost of a similar factory the group built in Gazipur four years ago. The managing director wants 'one number we can put in the board paper'. What should the estimator provide FIRST?

(A) A detailed bottom-up estimate with a full bill of quantities, asking the board to postpone its meeting until the drawings needed for measurement have been prepared.

(B) An analogous estimate scaled from the Gazipur factory and adjusted for escalation, location and scope, presented as a range with its class, assumptions and exclusions.

(C) The Gazipur factory's final cost unchanged, on the basis that both factories are knit-dyeing plants of a similar type and the board only needs an approximate figure for now.

(D) A single figure with a 30% contingency added, so that the board has a safe number to approve and the project is unlikely to need more money later.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE

Basis of estimate register

Documents where every major figure in the estimate comes from, so that reviewers can trace and challenge it and later estimators can reuse it.

ITEM / WBS	QUANTITY AND SOURCE	METHOD	RATE SOURCE AND BASE DATE	ESCALATION APPLIED	CURRENCY	EXCLUSIONS AND ASSUMPTIONS
1.3 Pile caps	*620 m3 from FEED drawings rev B*	*BOQ and unit rate*	*Built-up rate, local prices Q1 2026*	*5% per year to mid-point of spend*	*BDT*	*Excludes dewatering below 2 m depth*
2.1 Clarifier mechanisms	*2 units from vendor datasheet*	*Vendor budget quotation*	*Quotation dated Feb 2026*	*None: price firm to Dec 2026*	*EUR; rate and date recorded*	*Installation priced separately under 2.4*

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ITEM / WBS	QUANTITY AND SOURCE	METHOD	RATE SOURCE AND BASE DATE	ESCALATION APPLIED	CURRENCY	EXCLUSIONS AND ASSUMPTIONS

REFLECT

Take one estimate you have prepared or reviewed recently. Could you state its class and range, and could someone else trace every major figure to a quantity, a rate and a date?

Key terms

Analogous estimate
An estimate based on the actual cost of a similar past project, adjusted for differences in size, date, location and scope; quick but only as reliable as the comparison.

Factored estimate
An estimate that prices the major equipment and applies factors for installation, bulk materials and indirect costs to reach total installed cost, typical of process plant at feasibility stage.

Bill of quantities (BOQ)
A list of measured quantities of every work item, prepared to an agreed method of measurement and priced with unit rates to produce an estimate or tender price.

Cost escalation
The expected increase in costs over time due to inflation and market conditions, applied from the estimate base date to when the money is spent.

Parametric estimate
An estimate that multiplies a key project parameter, such as MW, m2 or km, by a rate derived from historical data, sometimes with a scaling exponent.

Bottom-up estimate
An estimate built from the detailed cost of each work package or activity, using quantities, productivity and prices, then rolled up to the project total.

Estimate class
A category describing an estimate's maturity, purpose, method and expected accuracy range, from early screening estimates to definitive estimates based on complete design.

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Key takeaways

- Match the estimating method to the maturity of design, and always state the estimate class and its range.
- Ranges are skewed upwards: early estimates are far more likely to grow than to shrink.
- Show escalation, currency and contingency as separate allowances, with the base date, exchange rate and sources.
- Model quantities and AI-suggested rates speed estimating, but a named estimator verifies them against the method of measurement.
- Write a basis of estimate so that every major figure traces to a quantity, a rate and a date.

KNOWLEDGE CHECK

1. Capacity-scaling a plant to twice the size of a reference plant with an exponent of 0.6 multiplies its cost by about:

   (A) 1.20 (B) 1.52
   (C) 2.00 (D) 1.60

2. An item costs 100 at the estimate base date. With escalation of 6% per year, what should be carried for spend two years later?

   (A) 112.00 (B) 106.00
   (C) 112.36 (D) 113.40

3. Which method best suits a process plant at feasibility stage when only a priced list of major equipment exists?

   (A) A bottom-up estimate built from measured quantities (B) An analogous estimate from an unrelated industry
   (C) A bill of quantities priced with built-up unit rates (D) A factored estimate built from the equipment cost

4. A unit rate has direct costs of 2,000 per tonne and overheads and profit of 12%. What is the unit rate?

   (A) 2,240 per tonne (B) 2,012 per tonne
   (C) 2,120 per tonne (D) 2,400 per tonne

5. Why should escalation not be hidden inside the contingency allowance?

   (A) Contingency must always be returned to the client at the end of the project (B) They address different causes and are calculated and controlled differently
   (C) Escalation may not legally be included in public-sector cost estimates (D) Contingency can only be calculated after the main contract has been awarded

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BUDGET · LESSON 2 · 34 MIN

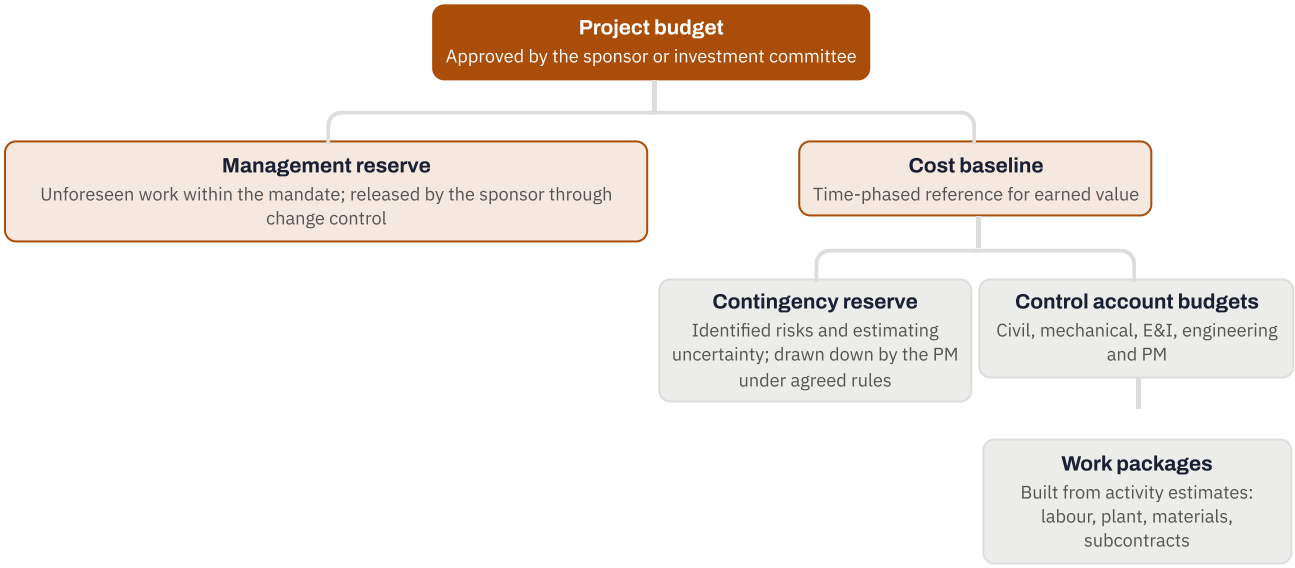
Budgeting & the Cost Baseline

Reserves, cost-loaded schedules, S-curves and the difference between cost and cash

- Build a time-phased cost baseline from the estimate, the CBS and the schedule
- Distinguish contingency reserve from management reserve and control each correctly
- Convert a cost curve into a cash-flow forecast allowing for retention, advance payments and commitments

An approved estimate is not yet a budget that can be controlled. The **cost baseline** is the estimate broken down into control accounts and work packages, spread over time according to the schedule, and approved as the reference against which performance will be measured. It contains the planned cost of the work plus **contingency reserve** for identified risks; **management reserve** for the unforeseen sits above it under the sponsor's control. Because the baseline is time-phased, it also tells the finance team when money will be needed, which for a lender-funded power plant or a factory expansion paid from operating cash can matter as much as the total.

From activity estimates to project budget



Contingency sits inside the cost baseline and is drawn down by the project manager; management reserve sits outside it with the sponsor and enters the baseline only through change control.

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Contingency reserve vs management reserve

Contingency reserve

- Covers identified risks and estimating uncertainty (known unknowns)
- Part of the cost baseline
- Sized from the risk register and quantitative analysis, developed in Protect
- Drawn down by the project manager under documented rules
- Example: an allowance for poor ground at some pile locations

Management reserve

- Covers unforeseen events within the approved scope (unknown unknowns)
- Outside the cost baseline but inside the project budget
- Often set by policy or from portfolio experience
- Released by the sponsor through change control, then added to the baseline
- Example: a new regulatory requirement introduced mid-construction

A **cost-loaded schedule** assigns each activity's budget to its dates, so that adding up costs period by period gives the planned spend. Plotting the cumulative total gives the familiar **S-curve**: slow at the start through design and mobilisation, steep during main construction, flattening through commissioning. The S-curve becomes the planned value line used in earned value. But cost is not cash. Work is certified some weeks after it is done, payment follows certification after an agreed period, **retention** is withheld, **advance payments** are recovered, and major equipment may need milestone payments long before delivery. A cash-flow forecast adjusts the cost curve for these timings.

Illustrative S-curve data: hypothetical ETP for a garment factory (BDT million; control account budgets total 40.0, contingency held separately)

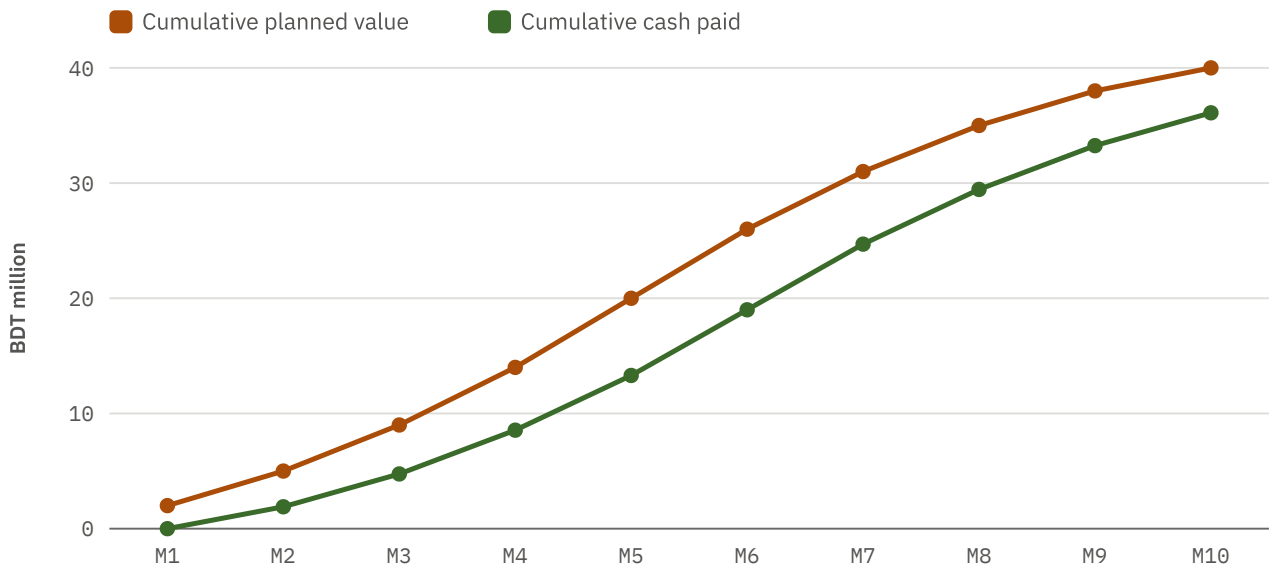
MONTH	PLANNED VALUE IN MONTH	CUMULATIVE PLANNED VALUE	CUMULATIVE % OF BUDGET
1	2.0	2.0	5%
2	3.0	5.0	12.5%
3	4.0	9.0	22.5%
4	5.0	14.0	35%
5	6.0	20.0	50%
6	6.0	26.0	65%
7	5.0	31.0	77.5%
8	4.0	35.0	87.5%
9	3.0	38.0	95%
10	2.0	40.0	100%

EXAMPLE · FROM COST TO CASH ON ONE CERTIFICATE

In month 5 the contractor's work for the month, valued at BDT 6.0 million, is certified. The contract withholds 5% retention and recovers a 10% advance payment by deducting 10% from each interim certificate. Net payment = 6.0 - 0.30 (retention) - 0.60 (advance recovery) = **5.10**, paid within the period stated in the contract, for example 28 days after certification. Under many contracts half the retention is released at taking-over and the rest at the end of the defects period, so the owner's cash curve lags its cost curve and has a tail after completion.

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Cost curve and cash curve for the ETP



Illustrative: each month's work is paid one month later with 5% retention held back and no advance payment. By month 10 the owner has paid 36.1 of 40.0; the last month's work and the retention follow after completion.

Retention, advances and commitments

- **Retention** protects the employer against defects and incomplete work. It is money the contractor has earned but not yet received, so contractors price its financing cost.
- **Advance payments** help contractors mobilise and order long-lead equipment. They are usually secured by an advance payment guarantee and recovered by deductions from interim certificates.
- **Commitments** are orders and subcontracts placed but not yet invoiced. They matter because they usually cannot be cancelled cheaply; tracking them is covered in the final lesson of this module.
- Milestone payments for major equipment such as transformers, turbines and boilers can be large and early: put them in the cash-flow forecast at their contractual dates.
- Foreign-currency payments need treasury planning, such as letters of credit or forward cover, and the timing of those payments belongs in the cash-flow forecast.
- **Price adjustment**: where the contract links payments to published cost indices, certificate values move with inflation, so forecast the adjustment in the cash flow instead of discovering it in each certificate.
- **Target-cost, cost-reimbursable and alliance** contracts pay actual cost through open-book accounts, so the owner's cash curve follows the contractor's real spend and any pain/gain share must be forecast too.

Building the cost baseline

1. **Freeze the estimate and the CBS** Confirm the approved estimate, its class and its mapping to WBS control accounts.
2. **Assign budgets to work packages** Allocate every cost item to one control account and work package, so that nothing is double-counted or left orphaned.
3. **Cost-load the schedule** Spread each budget over its activities' dates using sensible profiles: linear for steady work, front- or back-loaded only where justified.

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4. **Add contingency reserve** Hold contingency at project or control-account level according to policy, visibly, not buried in activity rates.
5. **Produce the S-curve and cash flow** Cumulate planned value and convert it to a cash-flow forecast using payment terms, retention, advances and equipment milestones.
6. **Approve and baseline** Obtain formal approval. From then on, changes to the baseline go through change control and are logged.

INSIGHT · A CARBON BASELINE BESIDE THE COST BASELINE

Many owners now set a **carbon budget** alongside the cost budget. PAS 2080:2023 sets out carbon management for buildings and infrastructure across the value chain, and whole-life carbon is reported by life-cycle stage following EN 15978: A1-A5 product and construction, B use, C end of life and D beyond the system boundary. In practice the estimator uses the same quantities to calculate embodied carbon for A1-A5, the baseline records cost and carbon for each control account, and every change is assessed for both. State the carbon factors and their source and version, just as you state a price base date. Where an owner reports under IFRS S1/S2 or a similar framework, project carbon data may feed corporate disclosures, so keep it auditable.

WATCH OUT · FRONT-LOADING AND HIDDEN PADDING

Contractors sometimes load early activities with more value than their true cost to improve cash flow, or bury allowances inside rates. Both distort earned value: early progress looks profitable, later work looks loss-making, and the owner overpays early, weakening its position if the contractor fails. Owners should test activity values against their own estimate, look for unusually rich early items such as mobilisation and preliminaries, and require contingency to be visible.

SCENARIO · DECIDE · A STORM ON THE SOLAR PARK

A hypothetical 100 MW solar park is 40% complete. A storm damages a section of mounting structures already installed. The repair cost is within the range assessed for 'extreme weather damage', which is listed in the risk register and covered by the contingency reserve. The site manager asks the sponsor to release management reserve to pay for the repair. What should the project manager do FIRST?

(A) Agree with the site manager and ask the sponsor for management reserve, since management reserve exists to cover any unplanned cost that arises during construction.

(B) Increase the cost baseline by the full repair cost through a change request, without drawing on either reserve, so that contingency is preserved for later risks.

(C) Check any insurance recovery, fund the net repair cost from contingency under the agreed drawdown procedure, record it against the risk and update the forecast.

(D) Instruct the contractor to absorb the full repair cost on the grounds that the approved cost baseline cannot be changed once construction has started on site.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Look at the budget on your current project. Can you see where contingency sits, who can release it and how much remains? How does your cost curve differ from your cash curve?

Key terms

Cost baseline
The approved, time-phased budget, made up of control account budgets plus contingency reserve, against which cost and earned value performance are measured.

Management reserve
Budget outside the cost baseline but within the project budget, held by the sponsor for unforeseen work within scope and released through change control.

S-curve
A graph of cumulative planned (or actual) cost or progress over time, typically shaped like an S because spending is slow at the start, fast in the middle and slow at the end.

Advance payment
A payment made to the contractor early in the contract to fund mobilisation or long-lead purchases, usually secured by a guarantee and recovered through deductions from later certificates.

Contingency reserve
Budget within the cost baseline set aside for identified risks and estimating uncertainty, drawn down by the project manager under documented rules.

Cost-loaded schedule
A schedule in which each activity carries its budgeted cost, so that planned spend can be summed by period and plotted over time.

Retention money
A percentage withheld from interim payments to the contractor as security for completion and defects, usually released in stages at taking-over and the end of the defects period.

Key takeaways

- The cost baseline is the time-phased estimate by control account plus contingency; management reserve sits outside it with the sponsor.
- Contingency funds identified risks under agreed drawdown rules; unforeseen work needs change control and management reserve.
- Cost is not cash: certification, payment periods, retention and advance recovery make the cash curve lag the cost curve.
- Test contractor activity values for front-loading and keep allowances visible, not buried in rates.
- Where a carbon budget is set, baseline it and control changes to it alongside cost.

KNOWLEDGE CHECK

6. Where does management reserve normally sit in the budget structure?

(A) Inside each work package budget as a hidden allowance

(B) Inside the cost baseline, alongside contingency reserve

(C) Outside the cost baseline but within the project budget

(D) Outside the project budget, within the contractor's price

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7. An interim certificate is for 8.0 million with 5% retention and no other deductions. How much retention is withheld?

   (A) 0.40 million
   (B) 0.80 million
   (C) 0.04 million
   (D) 4.00 million

8. Why is a typical project cost curve shaped like an S?

   (A) Contractors are paid a higher rate per unit of work at the start and end of a project
   (B) Contingency reserve is released at the start and the end of the project only
   (C) Scheduling software applies a standard S profile to every activity budget
   (D) Spending is slow in mobilisation, fast in main construction and slow in commissioning

9. Which item belongs in contingency reserve rather than management reserve?

   (A) An unforeseeable change in national environmental legislation
   (B) A quantified risk of poor ground at some pile locations
   (C) A sponsor's decision to add a new production line
   (D) A general allowance for events no one has identified

10. A certificate for 10.0 million has 5% retention and a 10% advance-payment recovery. What is the net payment?

    (A) 9.50 million
    (B) 9.00 million
    (C) 8.50 million
    (D) 8.00 million

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BUDGET · LESSON 3 · 36 MIN

Earned Value Fundamentals

PV, EV and AC: separating cost performance from schedule performance

- Define planned value, earned value and actual cost and obtain each from project data
- Calculate CV, SV, CPI and SPI for a project and for its control accounts
- Interpret earned value results and recognise the classic traps

Comparing actual spend with the budget tells you very little on its own. If a project has spent 22.5 against a plan of 20.0 by now, is that because more work has been done than planned, or because the work is costing more than it should? **Earned value management (EVM)** answers the question by adding a third measure: the budgeted value of the work actually completed. With three numbers, **planned value**, **earned value** and **actual cost**, the project manager can separate schedule performance from cost performance and spot trends months before the final account.

Earned value fundamentals

MEASURE	FORMULA	WHAT IT TELLS YOU
Planned value (PV)	`PV = budgeted cost of the work scheduled to be done by the status date`	Read from the cost-loaded baseline (the S-curve) at the status date.
Earned value (EV)	`EV = sum of (budget of each work item x its % complete)`	Budgeted value of the work actually done, measured with objective rules of credit.
Actual cost (AC)	`AC = actual cost incurred for the work performed to the status date`	Booked costs plus accruals for work done but not yet invoiced, on the same basis as the budget.
Cost variance	`CV = EV - AC`	Negative means the work done has cost more than its budget.
Schedule variance	`SV = EV - PV`	Negative means less work has been done than planned, expressed in money.
Cost performance index	`CPI = EV / AC`	Budgeted value earned per unit of money spent; below 1.0 is unfavourable.
Schedule performance index	`SPI = EV / PV`	Rate of progress against the plan; below 1.0 means behind.
Variance percentages	`CV% = CV / EV x 100; SV% = SV / PV x 100`	Allow fair comparison and thresholds across control accounts of different sizes.
Percent complete and percent spent	`% complete = EV / BAC x 100; % spent = AC / BAC x 100`	BAC is the budget at completion. A large gap between the two is an early warning.

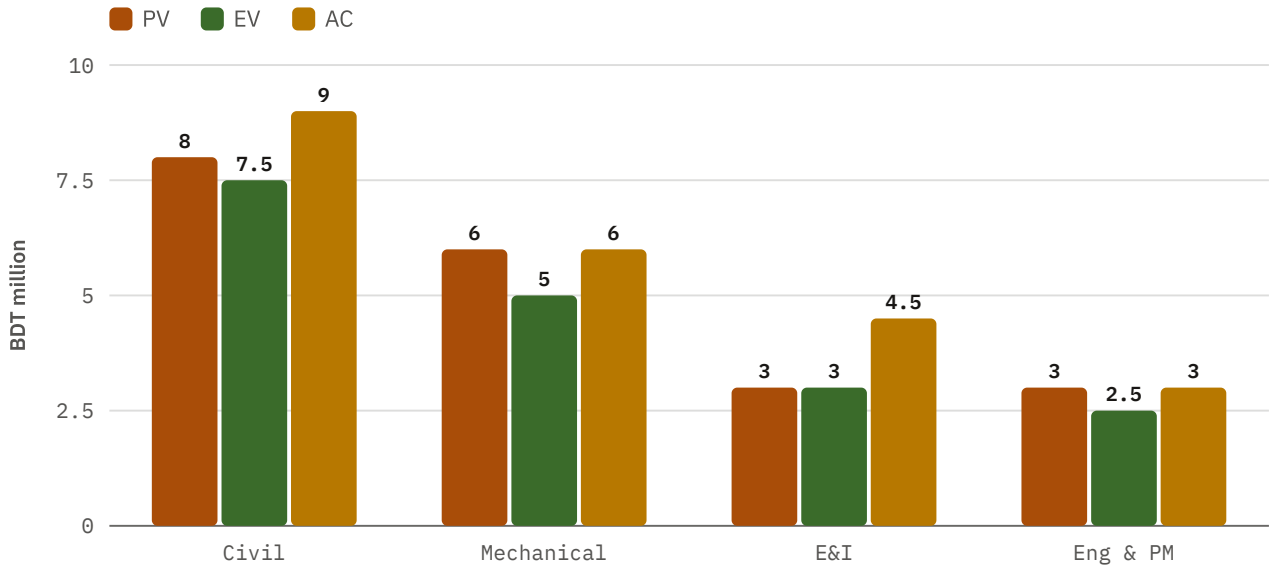
Return to the hypothetical effluent treatment plant (ETP) from the budgeting lesson, with a budget at completion (BAC) of BDT 40.0 million over 10 months. At the end of month 5 the cumulative planned value read from the S-curve is 20.0. The site team measures physical progress by control account using the rules of credit described in the last lesson of this module, and the cost team collects actual costs including accruals. The status by control account is shown below in BDT million, with indices rounded to two decimals.

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Status at the end of month 5 (BDT million, illustrative)

CONTROL ACCOUNT	PV	EV	AC	CV	SV	CPI	SPI
Civil works	8.0	7.5	9.0	-1.5	-0.5	0.83	0.94
Mechanical equipment and installation	6.0	5.0	6.0	-1.0	-1.0	0.83	0.83
Electrical and instrumentation	3.0	3.0	4.5	-1.5	0.0	0.67	1.00
Engineering and project management	3.0	2.5	3.0	-0.5	-0.5	0.83	0.83
Project total	20.0	18.0	22.5	-4.5	-2.0	0.80	0.90

PV, EV and AC by control account at month 5



Illustrative figures from the table. E&I is on schedule (EV equals PV) yet has the worst cost efficiency, while civil works carries the largest absolute overspend.

EXAMPLE · PROJECT-LEVEL RESULTS

CV = 18.0 - 22.5 = **-4.5**: the work done so far has cost 4.5 more than its budget. SV = 18.0 - 20.0 = **-2.0**: work worth 2.0 that should have been done by now is not yet done. CPI = 18.0 / 22.5 = **0.80**: every BDT 1.00 spent has earned only BDT 0.80 of budgeted work. SPI = 18.0 / 20.0 = **0.90**: progress is running at 90% of the planned rate. The project is 45% complete (18.0 / 40.0) but has spent about 56% of its budget (22.5 / 40.0). CV% = -4.5 / 18.0 = -25% and SV% = -2.0 / 20.0 = -10%.

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Reading CPI and SPI together

CPI (COST PERFORMANCE; 1.0 AT THE CENTRE LINE) →

Behind but economical

- Is the work under-resourced?
- Is the critical path slipping?
- Are costs deferred or not yet booked?

Ahead and within budget

- Is progress over-claimed?
- Is the baseline padded?
- Can resources move to other work?

Behind and over cost

- Root causes: productivity, design, supply or rework?
- A recovery plan is needed

On or ahead of plan but expensive

- Is progress bought with overtime or extra crews?
- Was that acceleration agreed and funded?

SPI (SCHEDULE PERFORMANCE; 1.0 AT THE CENTRE LINE) →

Each quadrant suggests the first questions to ask, not the answer. The ETP at month 5 (CPI 0.80, SPI 0.90) sits bottom-left.

Drilling down matters more than the headline. Electrical and instrumentation is exactly on schedule (SPI 1.00) but has spent 1.5 more than it has earned (CPI 0.67): the team may be buying progress with overtime or a more expensive cabling subcontractor. Mechanical is behind and over cost in equal measure (CPI and SPI both 0.83), which on a treatment plant often points to late or incomplete equipment deliveries causing stop-start installation. Civil works shows a large overspend in absolute terms on the biggest account. The project manager now asks why each variance has occurred, not just how big it is, and agrees actions with each control account manager.

WATCH OUT · THREE CLASSIC TRAPS

First, SV is measured in money, not time: a project can show a large negative SV while the critical path is fine, or an SV near zero while a critical activity is weeks late, so always read EVM together with the schedule. **Second**, SPI drifts back towards 1.0 at the end of every project, because when all work is done EV equals PV even if the project finished late. **Third**, EVM is only as good as the EV figure: subjective percent-complete claims inflate EV and hide trouble, and missing accruals understate AC.

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The monthly earned value cycle

01 Baseline — The cost-loaded schedule gives PV for the period →

02 Measure — Rules of credit give EV for each work package →

03 Collect — The cost system gives AC, including accruals →

04 Calculate — CV, SV, CPI and SPI by control account and in total →

05 Analyse — Explain variances beyond agreed thresholds →

06 Act — Corrective actions, updated forecasts and change requests

INSIGHT · EVM STANDARDS AND CONTRACT FORMS

ISO 21508:2018 gives international guidance on earned value management, and ISO 21502:2020 places performance measurement within overall project management, so the vocabulary in this lesson is widely shared. EVM works under any contract form, but the source of AC differs: on lump-sum and remeasurement contracts the owner's AC is usually certified value plus accruals, while on target-cost, cost-reimbursable and alliance contracts open-book records give the actual cost of the work directly. Some owners also apply the same planned, earned and actual logic to a carbon budget, tracking embodied carbon against progress.

SCENARIO · DECIDE · 'WE ARE UNDER BUDGET'

On a hypothetical hospital block in Chattogram, the monthly report shows actual cost of 14.0 against planned value of 15.0 (BDT million). The contractor's commercial manager says, 'We are 1.0 under budget, everything is fine.' The owner's site engineer, measuring with the agreed rules of credit, puts earned value at 12.0. What should the project manager conclude FIRST?

(A) The project is 1.0 under budget, as the commercial manager says, because actual cost is below the planned value for the period, so no corrective action is needed.

(B) The project is behind schedule but its cost performance is good, because actual cost is below the planned value and the contractor has therefore spent less than expected.

(C) The figures cannot yet be interpreted, because earned value results are only meaningful once a project has passed the half-way point of its schedule.

(D) The project is behind schedule (SPI 0.80) and over cost for the work done (CPI about 0.86), so variance explanations and recovery actions are needed.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Pick one control account you know well. Could you produce PV, EV and AC for it today, and would you trust the EV figure? What would make it more objective?

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Key terms

Planned value (PV)
The authorised budget for the work scheduled to be completed by a given date, read from the time-phased cost baseline.

Earned value (EV)
The budgeted value of the work actually completed by a given date, measured using agreed rules of credit.

Actual cost (AC)
The cost actually incurred, including accruals, for the work performed up to a given date, recorded on the same basis as the budget.

Cost variance (CV)
The difference between earned value and actual cost (EV - AC); a negative value means the completed work has cost more than its budget.

Schedule variance (SV)
The difference between earned value and planned value (EV - PV), measured in money; a negative value means less work has been done than planned.

Cost performance index (CPI)
The ratio EV / AC, showing the budgeted value earned for each unit of money spent; values below 1.0 indicate cost overrun.

Schedule performance index (SPI)
The ratio EV / PV, showing the rate of progress against plan; values below 1.0 indicate work behind schedule.

Key takeaways

- PV comes from the time-phased baseline, EV from objective rules of credit, and AC from the cost system including accruals.
- CV = EV - AC and CPI = EV / AC measure cost; SV = EV - PV and SPI = EV / PV measure schedule in money terms.
- Drill down by control account: a project total can hide very different problems.
- Read EVM with the critical path, because SV is money, not time, and SPI drifts back to 1.0 at the end.

KNOWLEDGE CHECK

11. A control account has EV = 300 and AC = 375. What is its CPI?

(A) 1.25
(B) 0.80
(C) 0.75
(D) -75

12. A control account has PV = 450 and EV = 360. What is its schedule variance?

(A) +90
(B) 0.80
(C) -0.20
(D) -90

13. A control account shows CPI 1.08 and SPI 0.85. What is the most useful first question?

(A) Is the team under-resourced, so that critical work is slipping?
(B) Is the team buying progress with expensive overtime at weekends?
(C) Has the client paid all of the interim certificates on time?
(D) Is the contingency reserve too large for the remaining risks?

14. Why might SPI be close to 1.0 on a project that finished three months late?

(A) Because SPI is measured in time units and is reset for late projects
(B) Because AC rises in the final months and offsets the schedule delay
(C) Because at completion EV equals PV, so SPI returns to 1.0 regardless
(D) Because management reserve is added to PV in the final months of work

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15. A project has BAC = 800, EV = 200 and AC = 260. What percentage of the work is complete?

(A) 25%
(B) 32.5%
(C) 77%
(D) 30%

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BUDGET · LESSON 4 · 38 MIN

Forecasting with Earned Value

EAC variants, ETC, VAC, TCPI and earned schedule: telling the sponsor what it will really cost

- Calculate EAC with several formulas and explain the assumption behind each
- Use ETC, VAC and TCPI to judge whether a forecast or a recovery target is credible
- Apply the earned schedule idea and produce an independent EAC with reporting thresholds

Earned value earns its keep when it is used to forecast. The sponsor's real question is not 'what is the CPI?' but 'what will this cost, and when will it finish?'. EVM offers several **estimate at completion (EAC)** formulas, each embodying a different assumption about future performance. A competent project manager calculates more than one, explains which assumption is most credible and why, and compares the result with a bottom-up re-estimate by the people doing the work.

Forecasting formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
EAC: current cost performance continues	`EAC = BAC / CPI`	Assumes the cumulative CPI to date is typical of the remaining work; widely used and often realistic.
EAC: remaining work at budget rate	`EAC = AC + (BAC - EV)`	Assumes past variances were one-off (atypical) and the rest of the work will be done exactly to budget.
EAC: cost and schedule both influence	`EAC = AC + (BAC - EV) / (CPI x SPI)`	Assumes schedule pressure will also drive cost, for example paying for acceleration on a late project; usually the most pessimistic.
EAC: bottom-up	`EAC = AC + bottom-up ETC`	The remaining work is re-estimated from current quantities, rates and the latest schedule.
Estimate to complete	`ETC = EAC - AC`	The cost still to be spent from the status date.
Variance at completion	`VAC = BAC - EAC`	Negative means a forecast overrun.
TCPI to meet BAC	`TCPI = (BAC - EV) / (BAC - AC)`	The cost efficiency needed on all remaining work to finish within the original budget.
TCPI to meet a revised EAC	`TCPI = (BAC - EV) / (EAC - AC)`	The efficiency needed to meet an approved revised forecast.
Earned schedule (ES)	`ES = C + (EV - PV_C) / (PV_C+1 - PV_C)`	C is the number of whole periods for which cumulative PV is less than or equal to EV. ES is the time by which the plan expected today's EV (not the CPM early start).
Time-based schedule performance	`SPI(t) = ES / AT; forecast duration = planned duration / SPI(t)`	AT is the actual time elapsed. Gives a schedule forecast in time units rather than money.

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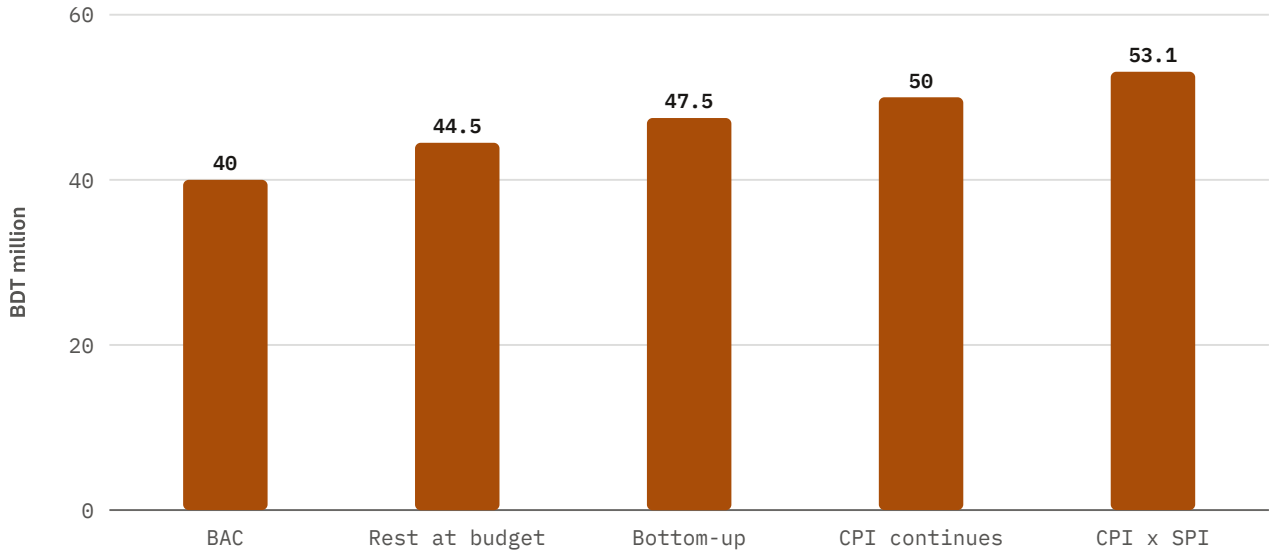
Worked forecasts for the ETP at month 5 (BAC 40.0, EV 18.0, AC 22.5, CPI 0.80, SPI 0.90; BDT million)

MEASURE	CALCULATION	RESULT	WHEN IT IS CREDIBLE
EAC: CPI continues	40.0 / 0.80	50.0	Cost problems are systemic, such as low productivity across trades
EAC: rest at budget	22.5 + (40.0 - 18.0)	44.5	The overrun came from a one-off cause that has been fixed
EAC: CPI x SPI	22.5 + 22.0 / (0.80 x 0.90) = 22.5 + 22.0 / 0.72	53.1	Late work will need acceleration or extended site overheads
EAC: bottom-up	22.5 + 25.0 (re-estimated ETC)	47.5	Fresh quantities and rates from the control account managers
VAC (using EAC 50.0)	40.0 - 50.0	-10.0	The forecast overrun to report against the budget
TCPI to meet BAC	(40.0 - 18.0) / (40.0 - 22.5) = 22.0 / 17.5	1.26	The efficiency needed to hit the original budget

EXAMPLE · WHAT THE NUMBERS SAY

The formula forecasts range from 44.5 to 53.1 against a budget of 40.0. To finish on budget, the remaining work would need a CPI of 1.26 when the project has achieved only 0.80 so far, a jump that experience suggests is very unlikely. Using EAC = 50.0, ETC = 50.0 - 22.5 = **27.5**, VAC = **-10.0**, and TCPI to that EAC = 22.0 / 27.5 = **0.80**, which simply matches current performance. The honest message to the sponsor: expect a final cost of roughly 47.5 to 50.0 unless the causes in civil and E&I works are fixed, and decide now whether contingency covers it or whether management reserve and a formal change are needed.

Range of forecasts for the ETP at month 5

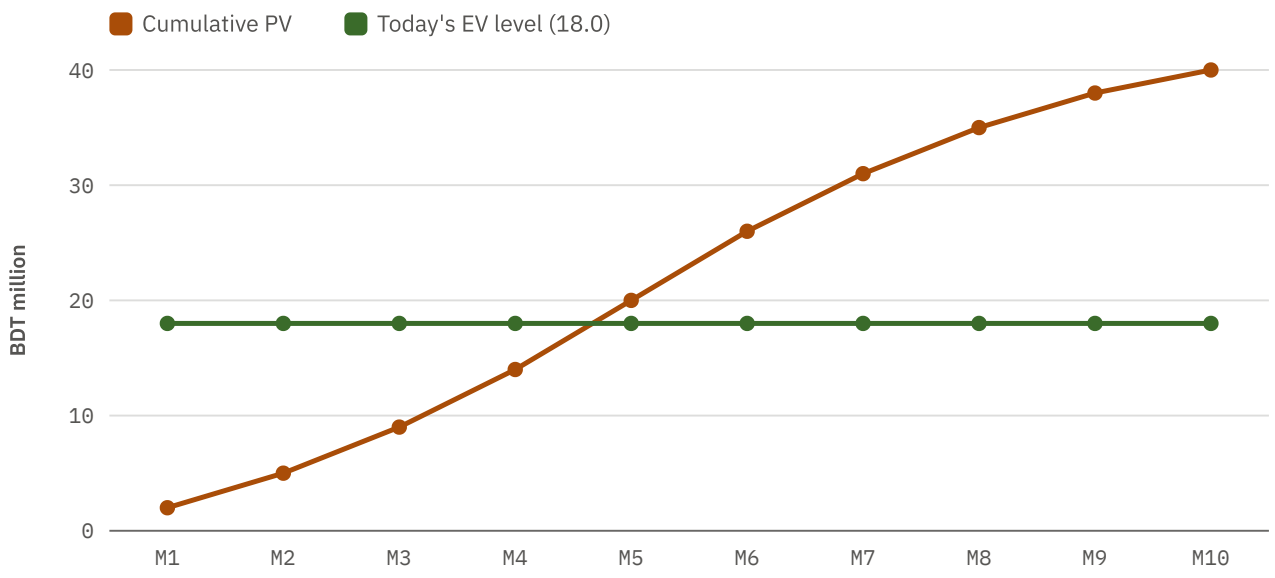


Illustrative values from the worked table. Every forecast sits well above the 40.0 budget, and the bottom-up re-estimate lands between the optimistic 'rest at budget' figure and the CPI-based ones.

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Schedule variance in money becomes misleading late in a project, so many practitioners add **earned schedule**. Using the S-curve from the budgeting lesson, the EV of 18.0 at month 5 lies between the cumulative PV at month 4 (14.0) and month 5 (20.0). Earned schedule = 4 + (18.0 - 14.0) / (20.0 - 14.0) = 4 + 4 / 6 = **4.67 months**: the project has done the work it planned to have done about two-thirds of the way through month 5. SPI(t) = 4.67 / 5 = **0.93**, and a simple time forecast is 10 / 0.933 = about **10.7 months**. Like every EVM schedule measure, check this against the critical path, which may tell a different story.

Reading earned schedule from the S-curve



Illustrative: the level line at today's EV crosses the planned-value curve between month 4 (14.0) and month 5 (20.0), at about 4.67 months. That crossing is the earned schedule; the project is about a third of a month behind.

Illustrative reporting thresholds (set the real values in the project controls plan)

STATUS	CPI OR SPI	CV% OR SV%	REQUIRED ACTION
Green	0.95 to 1.10	Within -5% to +10%	Report only
Amber	0.90 to below 0.95	Between -5% and -10%	Written variance explanation and recovery actions from the control account manager
Red	Below 0.90	Worse than -10%	Escalate to the project manager and sponsor; formal recovery plan and revised EAC
Favourable outlier	Above 1.10	Better than +10%	Check for over-claimed progress, padded budgets or costs not yet booked

Producing an independent EAC

1. **Check the data** Confirm that EV uses objective rules of credit and that AC includes accruals for work done and goods received.
2. **Calculate the formula EACs** Run BAC / CPI, AC + (BAC - EV) and the CPI x SPI version, by control account and in total.

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3. **Re-estimate the remaining work** Ask control account managers for a bottom-up ETC from remaining quantities, current rates and the latest schedule.
4. **Add known changes and risks** Include approved and pending variations, likely claims, the remaining risk exposure from the register, and escalation or contract price adjustment on the remaining work at current indices.
5. **Reconcile** Compare bottom-up and formula EACs; challenge any bottom-up figure much better than past performance unless a specific change explains it. Treat automated or AI-generated forecasts as one more input to challenge, not as the answer.
6. **Report and decide** Present the range and the recommended EAC with VAC, TCPI, the actions proposed and the decisions needed from the sponsor.

INSIGHT · WHY CUMULATIVE CPI DESERVES RESPECT

Studies of large defence and engineering programmes have long reported that cumulative CPI tends to change little once a project is roughly 20% complete, and that it seldom improves dramatically afterwards. Treat any forecast that assumes a sudden recovery with suspicion unless a specific, funded change in method, resources or scope explains it. A TCPI well above the CPI achieved to date is a warning sign, not a target.

CASE STUDY · FORECASTING A HOSPITAL BLOCK

A hypothetical six-storey hospital block in Chattogram has a BAC of BDT 600 million and a planned duration of 20 months. At month 9, PV = 270, EV = 240 and AC = 300 (BDT million). The contractor's project manager reports an EAC of 610, explaining that the overrun was caused by a one-off piling problem that has now been resolved. The owner's cost engineer notes that the MEP subcontracts, which start next month, were let at prices about 8% above the estimate.

1. Calculate CPI, SPI and the EAC using BAC / CPI.

2. Is the contractor's EAC of 610 credible? Use TCPI to support your view.

3. What EAC would you report to the sponsor, and why?

Model answer: module answer key.

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SCENARIO · DECIDE · WHICH FORECAST GOES TO THE BOARD?

A hypothetical 50 MW gas-engine power plant is 35% complete with a CPI of 0.92 and an SPI of 0.85. The contract carries liquidated damages for late completion, and the contractor has just proposed weekend working on the critical engine-hall works to recover time. The sponsor asks the project manager which EAC to report to the board. What should the project manager do FIRST?

(A) Report EAC = AC + (BAC - EV), because the overrun to date is modest and the remaining work can reasonably be assumed to proceed at the budgeted rate.

(B) Report EAC = BAC / CPI only, because it is the most commonly used formula and boards are familiar with it from other projects across the portfolio.

(C) Compare the formula EACs, including CPI x SPI, with a bottom-up ETC that prices the weekend working, and report a reasoned range and recommendation.

(D) Report the original BAC until the contractor's recovery plan has been approved, to avoid alarming the board with forecasts that may yet change.

Circle your choice. Answer and feedback: module answer key.

REFLECT

For a project you know, which EAC formula best reflects how the remaining work will really go? What evidence would you need to defend that choice to a sceptical sponsor?

Key terms

Budget at completion (BAC)
The total approved budget for the work in the performance measurement baseline; the value that EV will reach when all work is complete.

Estimate to complete (ETC)
The forecast cost of the remaining work from the status date to completion, equal to EAC minus AC.

To-complete performance index (TCPI)
The cost efficiency that must be achieved on the remaining work to meet a target, either BAC or an approved EAC: remaining work divided by remaining funds.

Variance threshold
A pre-agreed limit on cost or schedule variance, or on CPI and SPI, beyond which a variance must be explained, escalated or acted on.

Estimate at completion (EAC)
The forecast total cost of the project or control account at completion, based on actual cost to date plus the expected cost of the remaining work.

Variance at completion (VAC)
The difference between the budget at completion and the estimate at completion (BAC - EAC); a negative value is a forecast overrun.

Earned schedule
An extension of earned value that finds the time at which the planned value equalled the current earned value, giving schedule variance and performance in time units.

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Key takeaways

- Calculate several EACs and state the assumption behind each: BAC / CPI, AC + (BAC - EV) and the CPI x SPI version.
- Reconcile formula EACs with a bottom-up ETC that includes changes, risks and price escalation on the remaining work.
- A TCPI well above the CPI achieved so far is a warning sign, not a target.
- Use earned schedule to express schedule performance in time units, and check it against the critical path.
- Report a reasoned range and a recommended EAC early, with the decisions needed from the sponsor.

KNOWLEDGE CHECK

16. A project has BAC = 1,000 and a cumulative CPI of 0.80. What is the EAC if current cost performance continues?

(A) 800
(B) 1,200
(C) 1,250
(D) 1,800

17. A project has BAC = 1,000, EV = 400 and AC = 500. What TCPI is needed to finish within the BAC?

(A) 1.20
(B) 0.83
(C) 1.25
(D) 0.80

18. Using BAC = 1,000, EV = 400 and AC = 500, what is the EAC if the remaining work is done at the budgeted rate?

(A) 1,250
(B) 1,000
(C) 1,500
(D) 1,100

19. A project has EAC = 1,250 and BAC = 1,000. What is the variance at completion?

(A) +250
(B) -250
(C) 1.25
(D) -0.25

20. When is EAC = AC + (BAC - EV) / (CPI x SPI) most appropriate?

(A) When past overruns came from one event that will not recur
(B) When schedule slippage is expected to drive cost on remaining work
(C) When the project is ahead of plan and under budget everywhere
(D) When no actual costs have yet been recorded against the cost baseline

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BUDGET · LESSON 5 · 36 MIN

Progress Measurement & Cost Control on Site

Rules of credit, joint measurement, commitments, accruals and capturing the cost of change

- Select objective rules of credit for different kinds of site work
- Calculate earned value with units-completed, weighted-milestone and fixed-formula methods
- Run a monthly cost cycle that tracks commitments, accruals, forecasts and the cost of change

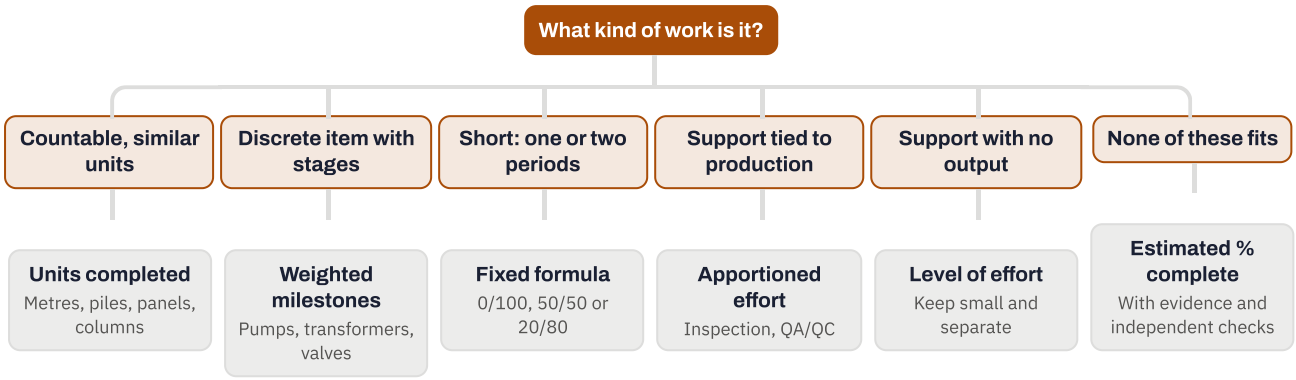
Earned value depends entirely on how progress is measured. If a foreman's 'about 70% done' becomes EV, then CPI and SPI are opinions. **Rules of credit** fix in advance how each type of work earns value: by units installed, by weighted steps, by fixed formulas or, as a last resort, by estimated percentage. Combined with disciplined cost capture of commitments, accruals and the cost of changes, they give a cost report that the project manager, the sponsor, the lenders and the auditors can all trust.

Rules of credit

METHOD	HOW VALUE IS EARNED	BEST FOR	WATCH-OUT
Units completed	EV = budget x (units done / total units)	Repetitive measurable work: metres of cable pulled, piles installed, panels fixed	Units must take similar effort; define 'done' (installed, tested, accepted?)
Weighted milestones	A pre-agreed % is credited as each observable step is reached	Discrete items with distinct stages: pumps, transformers, valves, steel frames	Steps must be objectively verifiable and weights agreed before work starts
Fixed formula (0/100, 50/50, 20/80)	A fixed % at start, the remainder at finish	Short activities lasting one or two reporting periods	On long activities it distorts EV; 0/100 is the most conservative
Percent complete (estimated)	The responsible engineer estimates the % complete	Work with no clear units, such as design studies	Subjective: needs supporting evidence and independent checks
Level of effort	EV equals PV as time passes	Support work: site management, security, document control	Earns value without output; keep it separate and small
Apportioned effort	EV is a fixed % of a related discrete task's EV	Inspection, QA/QC and supervision tied to production	Only as good as the measurement of the base task

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Choosing a rule of credit



Work down from the nature of the work to the most objective rule that fits it. Estimated percent complete is the last resort, not the default.

Weighted milestones: installing a dyeing machine (hypothetical budget BDT 2.0 million)

STEP	WEIGHT	CUMULATIVE	EV WHEN STEP REACHED (BDT MILLION)
Received on site and inspected	10%	10%	0.20
Set on foundation and levelled	20%	30%	0.60
Aligned and grouted	20%	50%	1.00
Piping, utilities and electrical connected	30%	80%	1.60
Tested and released for commissioning	20%	100%	2.00

EXAMPLE · UNITS COMPLETED: A CABLE PACKAGE

An 11 kV and LV cable package has a budget of BDT 6.0 million for 12,000 m of cable. At month end, 4,500 m have been pulled, tested and tagged. EV = 6.0 x 4,500 / 12,000 = **2.25**, which is 37.5% complete. If the subcontractor claims 50% because 'the hard runs are done', the project manager can either agree, before work starts, a weighted rule that reflects difficulty, or hold to the measured 37.5%. Arguing about the rules after the work is done is how EV loses its credibility.

Physical progress is backed by **quantity surveys**: joint measurement of concrete poured, steel erected, pipe welded and tested, or cable pulled, usually agreed each month between the contractor's and the owner's engineers and reconciled with drawings, delivery records and test certificates. On the cost side, the controller tracks several figures for every cost code: **commitments** (orders and subcontracts placed), **actuals** (invoices received and booked) and **accruals** (work done or goods received but not yet invoiced). Without accruals, AC is understated and CPI looks better than it really is. Without commitments, the forecast misses money that is already spoken for.

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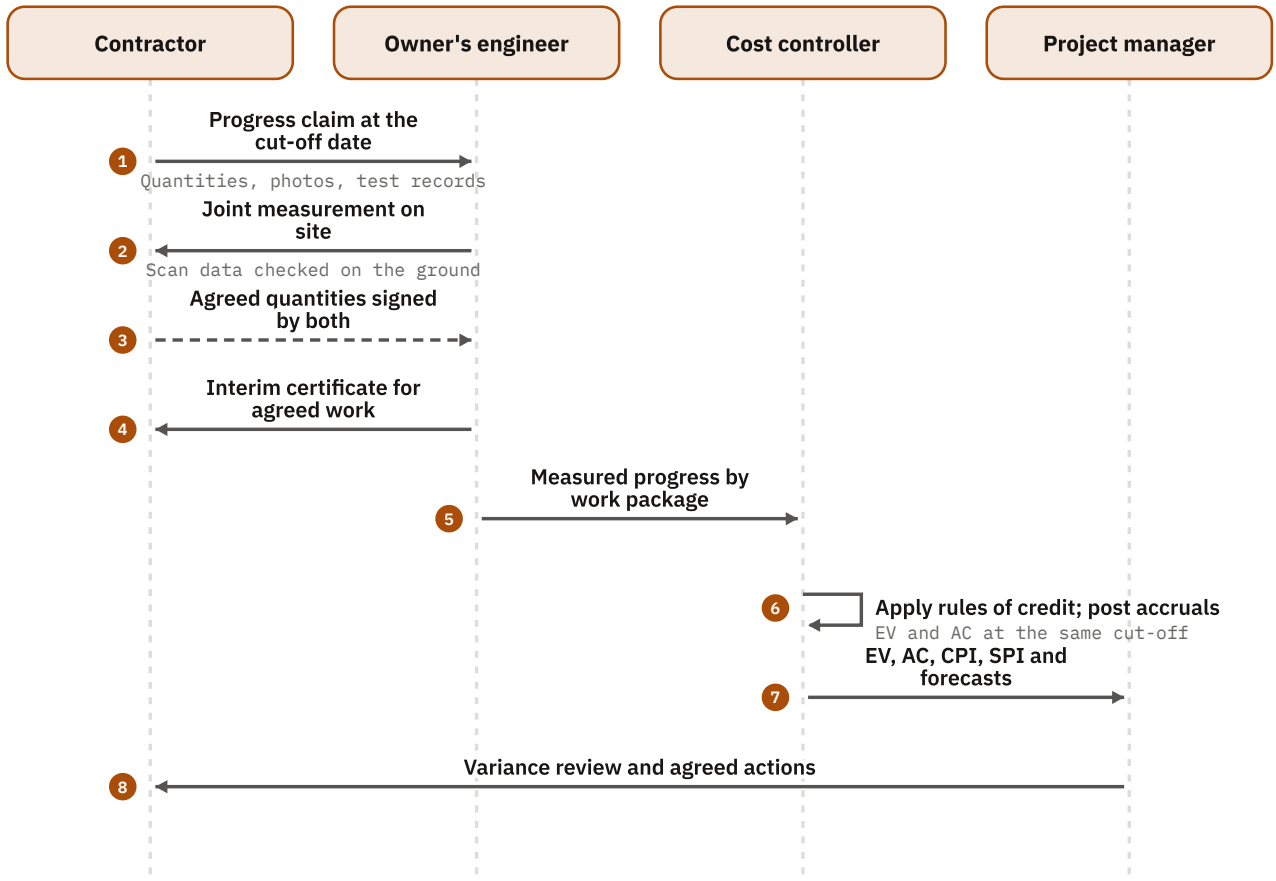
PRO TIP · REALITY CAPTURE AS PROGRESS EVIDENCE

Drone surveys, laser scans and 360-degree photo walks, compared with the BIM model and stored in the CDE, give dated, objective evidence for joint measurement: earthwork volumes, steel erected, pipe runs installed. Automated or AI-based recognition of installed elements can speed up the counting, but a named engineer checks and signs off the quantities, especially where a rule of credit defines 'done' as tested or accepted, which a camera cannot see. In volatile markets also watch commitments closely: quotations with short validity, price-adjustment provisions and expediting costs can move the forecast before a single invoice arrives.

The monthly site cost cycle

1. **Cut-off** Fix the reporting cut-off date and freeze progress and cost data at that date.
2. **Measure** Jointly measure quantities and milestone steps; keep evidence such as photographs, test certificates and inspection records.
3. **Capture cost** Book invoices, accrue for work done and goods received, and update commitments for new orders and variations.
4. **Calculate** Produce EV, AC, CPI and SPI by control account and update the forecast at completion for each cost code.
5. **Explain** Control account managers explain variances beyond the agreed thresholds and propose actions.
6. **Report and act** Issue the cost report, agree actions in the monthly review and track them to closure.

From monthly measurement to certified value and earned value



Nothing is credited before it is measured, and the same agreed quantities drive both the payment certificate and earned value, so the two reports reconcile.

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TEMPLATE

Cost report by cost code

Shows, for each cost code, how much is budgeted, committed, spent and forecast, so that overruns are visible while there is still time to act.

COST CODE / DESCRIPTION	CURRENT BUDGET	COMMITTED TO DATE	ACTUAL + ACCRUED TO DATE	FORECAST AT COMPLETION	VARIANCE (BUDGET - FORECAST)	COMMENT / ACTION
3.2 Pumps and blowers	*6.0*	*5.6*	*3.1*	*6.3*	*-0.3*	*Blower price increase; seek value engineering on spares*
4.1 11 kV and LV cabling	*6.0*	*4.8*	*2.3*	*6.1*	*-0.1*	*Slow productivity on first runs; monitor monthly*

WATCH OUT · CAPTURE THE COST OF CHANGE AS IT HAPPENS

Every variation instruction, site instruction or RFI response that changes the work should receive its own change reference and cost code on the day it is issued. Labour, plant and materials used on the changed work are then recorded separately, with daywork sheets signed daily and delivery tickets tagged, so that the cost can be valued, claimed or defended later. Changes valued months afterwards from memory are where both owners and contractors lose money; the rules for valuing variations are developed in Procure.

Monthly cost control checklist

☐ Every work package has an agreed rule of credit, recorded before work started.

☐ Progress is backed by joint measurement or other objective evidence.

☐ Level-of-effort budgets are separated from discrete work and kept small.

☐ Accruals are posted for work done and materials received but not yet invoiced.

☐ Commitments are updated for new orders, subcontracts and approved variations.

☐ Each change has its own reference and cost capture from the day it was instructed.

☐ Forecasts at completion are updated for each cost code, not copied from the budget.

☐ Variances beyond thresholds have written explanations and actions with named owners.

☐ Contingency drawdowns and the remaining balance are reported.

☐ The cost report reconciles with the finance ledger at the cut-off date.

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CASE STUDY · HEALTHY FIGURES ON A FACTORY UTILITIES PACKAGE

A hypothetical garment factory expansion in Gazipur reports a CPI of 1.05 and an SPI of 1.02 for its utilities package at month 6. The owner's newly appointed cost engineer walks the site and finds that the boiler feed pumps are credited at 80% although they have only been set on their foundations; that steam piping is credited using a percent complete estimated by the subcontractor; and that invoices for BDT 1.2 million of pipe fittings delivered last month have not yet been booked or accrued.

1. What is wrong with how EV and AC are being measured?

2. In which direction are the errors pushing CPI and SPI?

3. What should the project manager change before the next report?

Model answer: module answer key.

SCENARIO · DECIDE · WHOSE 60%?

On a hypothetical metro station box, the structural steel subcontractor's monthly claim credits 60% of the roof steel package, while the owner's engineer measures 45% of the tonnage erected and bolted. The subcontractor argues that its off-site fabrication progress should count. The subcontract and the project controls plan both state that steel earns value on erection. What should the project manager do FIRST?

(A) Accept the 60% claim to preserve a good working relationship, and settle any difference with the subcontractor later when the final account is agreed.

(B) Report 45% under the agreed rule of credit and, separately, discuss whether off-site fabrication should earn value under a formally agreed future rule change.

(C) Average the two figures and report 52.5% as a fair compromise between the subcontractor's claim and the owner's measurement, to avoid a dispute this month.

(D) Stop reporting steel progress altogether until the contract administrator has formally decided the dispute between the subcontractor and the owner's engineer.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Choose two work packages on your project, one repetitive and one made up of discrete equipment items. Which rule of credit is used for each today, and would an auditor accept the evidence behind this month's figure?

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Key terms

Rule of credit
A pre-agreed method stating how a type of work earns value as it progresses, such as units completed, weighted milestones or a fixed formula.

Weighted-milestone method
A rule of credit that divides a discrete item of work into observable steps, each carrying an agreed percentage of the budget that is earned when the step is reached.

Level of effort
Support work, such as site management or security, that has no measurable output and earns value in step with the passage of time, so its EV always equals its PV.

Accrual
A cost recognised for work done or goods received before the invoice is booked, included in actual cost so that cost performance is not overstated.

Units-completed method
A rule of credit in which earned value is the budget multiplied by the proportion of units completed, such as metres of cable pulled or piles installed.

Fixed-formula method
A rule of credit that credits a fixed percentage of budget when an activity starts and the remainder when it finishes, such as 0/100, 50/50 or 20/80; suitable for short activities.

Commitment
An obligation to spend money created by a placed order or subcontract that has not yet been invoiced, tracked so that forecasts include money already spoken for.

Key takeaways

- Agree a rule of credit for every work package before work starts, choosing the most objective rule that fits.
- Back progress with joint measurement and evidence; reality-capture data helps, but an engineer verifies it.
- Post accruals and update commitments every month, or CPI and forecasts will look better than they are.
- Give every change its own reference and cost capture from the day it is instructed.

KNOWLEDGE CHECK

21. A pump with step weights of 10/20/20/30/20 has reached 'aligned and grouted', the third step. Its budget is 500k. What is its EV?
    - (A) 150k
    - (B) 250k
    - (C) 300k
    - (D) 400k

22. Under a 0/100 fixed-formula rule, a three-week activity that has started but is not yet finished earns:
    - (A) 50% of its budget, half at start
    - (B) The foreman's estimated percentage
    - (C) 100% of its budget, once started
    - (D) 0% of its budget until it finishes

23. Why must accruals be included in actual cost?
    - (A) Accruals replace the need to track commitments in the cost report
    - (B) Accruals are required only by contractors and never by owners
    - (C) Without them AC is understated and CPI looks better than it is
    - (D) Accruals convert the planned value of the period into earned value

24. Which rule of credit best suits 40 identical lighting columns being installed along an expressway?
    - (A) Units completed, crediting each installed column
    - (B) Level of effort, crediting value as time passes
    - (C) Apportioned effort, tied to the road surfacing
    - (D) Percent complete as estimated by the site supervisor

25. A cable package has a budget of 6.0 million for 12,000 m. 3,000 m have been pulled and tested. What is its EV?
    - (A) 1.0 million
    - (B) 2.0 million
    - (C) 1.5 million
    - (D) 3.0 million

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☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BUDGET

Answer key

Knowledge checks

1. B 2^0.6 = about 1.52, so doubling capacity raises cost by roughly half. 2.00 ignores economies of scale, and 1.20 and 1.60 do not follow from the formula.

2. C 100 x 1.06^2 = 112.36. 112.00 adds simple rather than compound escalation, and 106.00 covers only one year.

3. D With a priced equipment list and little else, factoring up to total installed cost is the established approach. Bottom-up and BOQ methods need much more design, and an unrelated analogy is unreliable.

4. A 2,000 x 1.12 = 2,240 per tonne. The other values misapply the percentage.

5. B Escalation reflects expected price movement over time; contingency reflects scope and estimating uncertainty. Keeping them separate makes both visible and controllable. The other statements are not general rules.

6. C Management reserve is part of the project budget but not of the baseline; it enters the baseline only when released through change control. Contingency, by contrast, sits inside the baseline.

7. A 8.0 x 5% = 0.40 million. 0.80 million would be 10%, and the other values misplace the decimal point.

8. D The S shape comes from the rate of work: few resources at the start, peak resources mid-project and a tail through commissioning. It is not caused by pricing, reserves or software defaults.

9. B Poor ground at pile locations is an identified, quantified risk, which is what contingency covers. Unforeseeable events belong to management reserve, and new scope requires a change and new funding.

10. C 10.0 - 0.5 (retention) - 1.0 (advance recovery) = 8.50 million. 9.50 and 9.00 apply only one of the two deductions.

11. B CPI = EV / AC = 300 / 375 = 0.80. 1.25 inverts the ratio and -75 is the cost variance, not an index.

12. D SV = EV - PV = 360 - 450 = -90, meaning work worth 90 is behind plan. 0.80 is the SPI, and +90 has the sign reversed.

13. A Good cost efficiency with slow progress often means too few resources, or costs not yet booked. Buying progress with overtime would show the opposite pattern, and payment or reserve questions do not explain the indices.

14. C When all the work is done, EV equals the full budget, which equals PV, so SPI = 1.0 however late the finish. SPI is a money-based ratio, AC does not enter it, and reserves do not alter PV in this way.

15. A % complete = EV / BAC = 200 / 800 = 25%. 32.5% is the percent spent (AC / BAC), and 77% is the CPI expressed as a percentage.

16. C EAC = BAC / CPI = 1,000 / 0.80 = 1,250. 800 multiplies instead of dividing, and 1,200 adds 20% rather than dividing by the index.

17. A TCPI = (BAC - EV) / (BAC - AC) = 600 / 500 = 1.20. 0.80 is the CPI so far and 1.25 is its inverse, neither of which is the required future efficiency.

18. D EAC = AC + (BAC - EV) = 500 + 600 = 1,100. 1,250 assumes the CPI continues, and 1,000 ignores the overrun already incurred.

19. B VAC = BAC - EAC = 1,000 - 1,250 = -250, a forecast overrun. +250 has the sign reversed and 1.25 is a ratio, not a variance.

20. B Dividing by CPI x SPI assumes both cost and schedule performance will affect future cost, as when a late project must accelerate. A one-off overrun suits AC + (BAC - EV), and the other conditions make the formula meaningless.

21. B Cumulative weight after the third step is 10 + 20 + 20 = 50%, so EV = 0.50 x 500k = 250k. 150k counts only the first two steps, and 400k assumes the next step is already done.

22. D A 0/100 rule credits nothing at the start and the full budget only on completion. 50% would apply under a 50/50 rule, and estimated percentages are a different rule of credit.

23. C Work done or goods received but not yet invoiced are real costs; omitting them understates AC and overstates CPI. Accruals do not replace commitments, apply to both parties and have nothing to do with converting PV to EV.

24. A Identical, countable items are ideal for the units-completed method. Level of effort ignores output, apportioned effort ties value to unrelated work, and estimated percentages are needlessly subjective.

25. C EV = 6.0 x 3,000 / 12,000 = 1.5 million (25% complete). 3.0 million would be half the package, and the other figures do not follow from the units.

Scenarios & case studies

> **Scenario · One number for the board**
> **B** is best. Best. The method fits the information available, the adjustments are explicit, and the range and class stop the board treating a screening figure as a budget.

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Scenario · A storm on the solar park
C is best. Best. This is an identified risk that has occurred: contingency is the right source, insurance may reduce the net cost, and the drawdown must be recorded and reported.

Scenario · 'We are under budget'
D is best. Best. SPI = 12.0 / 15.0 = 0.80 and CPI = 12.0 / 14.0 = about 0.86. Both are unfavourable, so the project manager needs explanations and a recovery plan.

Case study · Forecasting a hospital block
CPI = 240 / 300 = 0.80, SPI = 240 / 270 = about 0.89, and EAC = 600 / 0.80 = 750. Even if the piling overrun really was one-off, the 'rest at budget' formula gives 300 + (600 - 240) = 660, not 610. Reaching 610 would need a TCPI of (600 - 240) / (610 - 300) = 360 / 310 = about 1.16 on the remaining work, when the project has achieved 0.80 and the MEP packages are already priced above estimate, so 610 is not credible. A reasonable reported EAC lies between about 660 and 750, refined by a bottom-up ETC that includes the known MEP price increase. The project manager should report a figure in that range with its assumptions and trigger a review of contingency and funding now rather than later.

Scenario · Which forecast goes to the board?
C is best. Best. Several formulas show the spread of outcomes, the bottom-up ETC captures the specific acceleration cost, and the board gets a recommendation with its assumptions.

Case study · Healthy figures on a factory utilities package
EV is inflated because the pumps are credited far beyond the observable step reached: under a weighted-milestone rule like the dyeing machine example in this lesson, 'set on foundation' would earn about 30%. The steam piping uses subjective estimates rather than measured welds, metres or test packs. AC is understated because delivered fittings worth 1.2 million have not been accrued. Both errors push CPI upwards and the EV error also pushes SPI upwards, so the package probably looks healthier than it is. The project manager should agree weighted-milestone rules for equipment and units-completed rules for piping, re-measure jointly, post the missing accruals, restate the month-6 figures and explain the restatement openly in the next report.

Scenario · Whose 60%?
B is best. Best. EV must follow the agreed rule. If fabrication progress deserves recognition, that is a separate, documented change to the rule, not a monthly negotiation.

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05

MODULE 5 · PROTECT

Risk, Health, Safety & Environment

Manage uncertainty, protect people and respect the places and communities where engineering happens

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Identify threats and opportunities on engineering projects and write them as actionable cause-risk-effect statements in a living register
- Prioritise risks with calibrated probability-impact scales and choose owned, triggered responses within the organisation's risk appetite
- Quantify risk with EMV, decision trees and Monte Carlo results, and explain P50 and P80 confidence to decision-makers
- Lead site health and safety using the hierarchy of controls, RAMS, permits-to-work and leading indicators
- Manage environmental and social impacts through an ESIA, an ESMP, a grievance mechanism and carbon-aware design choices

LESSONS

1. Risk Management Process & Identification · 32 min
2. Qualitative Analysis & Risk Responses · 34 min
3. Quantitative Risk Analysis & Monte Carlo Intuition · 40 min
4. Health & Safety Management on Site · 38 min
5. Environmental & Social Management · 36 min

MODULE 5 · PROTECT · LESSON 1 · 32 MIN

Risk Management Process & Identification

Find the uncertainties that matter early, and write them so someone can act on them

- Distinguish risks from issues, and threats from opportunities, on an engineering project
- Use a risk breakdown structure and an assumptions review to identify risks systematically
- Write cause-risk-effect statements and build a risk register that drives action

Every engineering project is a bet placed on incomplete information. The geotechnical report rests on a handful of boreholes, the transformer arrives from a factory you have never visited, the monsoon may start early, and the exchange rate on imported switchgear will move between award and delivery. **Risk management** is the disciplined way a project team finds those uncertainties, decides which ones matter, and acts on them while acting is still cheap. It is not a register that is filled in for the lender and then forgotten; it is a weekly management habit that changes what people do on site and in the design office.

DEFINITION · RISK, ISSUE AND THE DIFFERENCE THAT MATTERS

A **risk** is an uncertain event or condition that, if it occurs, would affect at least one project objective: safety, cost, time, quality, scope or reputation. An **issue** is something that has already happened or is certain to happen and now needs a decision. 'Piles may meet a buried obstruction' is a risk; 'pile P-17 has refused at 6 metres on an old foundation' is an issue. Risks are managed with responses planned in advance; issues are managed with decisions and actions now. A risk that occurs becomes an issue, and the register should record that transfer rather than quietly deleting the line.

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The risk management process is a loop, not a line



1. **Plan** — Agree scales, appetite, roles, tools and review rhythm
2. **Identify** — Find threats and opportunities across the whole RBS
3. **Analyse** — Prioritise qualitatively; quantify the significant few
4. **Respond** — Choose strategies, owners, actions and triggers
5. **Monitor** — Track triggers and exposure; control contingency drawdown
6. **Learn** — Feed what actually happened into the next estimate and register

Learn feeds straight back into Identify: every review, stage gate and incident restarts the loop. The steps are broadly consistent with ISO 31000:2018 and the risk guidance in ISO 21502:2020.

Uncertainty cuts both ways. A **threat** would harm an objective if it occurred; an **opportunity** would help. Engineering teams are trained to spot threats and often ignore opportunities, yet opportunities are real money: a supplier may offer an earlier factory slot for the gas turbine, a value-engineered pile cap design may let you use a smaller crane, or the grid company may energise the substation early so you can stop renting diesel generators. Treat opportunities with the same discipline: write them, own them and plan actions that make them more likely. A register with no opportunities usually means nobody looked, not that none exist.

A risk breakdown structure (RBS) for engineering projects

RBS LEVEL 1	LEVEL 2 PROMPTS	TYPICAL ENGINEERING EXAMPLE
Technical and design	Design maturity, interfaces, new technology, codes	Late vendor data for dyeing machines delays foundation and drainage design

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RBS LEVEL 1	LEVEL 2 PROMPTS	TYPICAL ENGINEERING EXAMPLE
Ground and site	Geology, groundwater, contamination, existing services, access	Unrecorded high-voltage cable found while excavating for a metro station entrance
Supply chain and logistics	Long-lead items, factory capacity, shipping, customs, heavy haul, single-source suppliers	Power transformer delayed at the port; no lowbed trailer available for the river crossing
Commercial and contractual	Contract form, subcontractor solvency, inflation and price volatility, currency	Steel price rise on a remeasured viaduct contract with a price-adjustment formula
External	Permits, weather, flooding, politics, community, utilities	Monsoon flooding of the batching plant access road in Chattogram
Information and digital	Model quality, CDE access and status, data handover, cyber security	Clashes in the federated model are not resolved before steel shop drawings are released
Organisational and management	Decision speed, staffing, owner approvals, funding	Owner's design approvals take four weeks instead of the two assumed in the programme

PRO TIP · THE CAUSE-RISK-EFFECT STATEMENT

Write every risk in three parts: **Because of** [a fact or condition that exists today], **there is a risk that** [an uncertain event may occur], **which would** [affect named objectives]. Example: 'Because only three boreholes were drilled across the effluent treatment plant footprint, there is a risk that piles meet a soft clay lens below the design toe level, which would extend piling by three to four weeks and require longer piles.' The cause tells you where to act to reduce probability; the effect tells you how to reduce impact; the event is what you monitor.

Weak versus useful risk descriptions

Weak (cannot be acted on)

- 'Delay'
- 'Cost overrun on civil works'
- 'Weather'
- 'Contractor performance'
- 'Safety'

Useful (cause, event, effect)

- Because the turbine vendor's shop is fully booked, rotor delivery may slip, delaying mechanical completion by six weeks
- Because rebar quantities came from a concept design, measured quantities may exceed the estimate, increasing civil cost
- Because the pour is scheduled in late May, pre-monsoon storms may stop the raft pour, risking a cold joint
- Because the piling subcontractor has two other jobs starting, rig availability may fall, slowing the critical path
- Because the roof steel will be erected over a live production floor, a dropped object could seriously injure workers

Running a risk identification workshop

1. **Prepare the inputs** Circulate the scope statement, summary schedule, estimate basis, site investigation report and lessons from similar projects. Invite design, construction, procurement, commissioning, HSE and an operations representative.
2. **Set the frame** State the objectives at stake, the time horizon and the RBS headings. Explain that opportunities are wanted, and that naming a risk is not blaming anyone.

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3. **Review assumptions** Walk through the key assumptions in the schedule and estimate, for example 'two-week design approvals' or 'grid connection available by month 14'. Every fragile assumption is a candidate risk.
4. **Generate, then cluster** Let people write risks silently first to avoid anchoring on the loudest voice, then cluster duplicates by RBS heading.
5. **Write proper statements** Convert each cluster into a cause-risk-effect statement. Separate out anything that is already an issue and route it to the issues log.
6. **Assign provisional owners** Give every risk a named owner who will analyse it and propose responses before the next review. Close by agreeing the review date.

Workshops are only one route. Good teams also run a **pre-mortem** ('it is two years from now and the project has gone badly; what happened?'), interview specialists one-to-one, walk the site with the construction team, review the schedule logic for single points of failure, and read the contract for risks it allocates to you. Watch for bias: optimism ('our crews are faster than average'), anchoring on the last project, and groupthink when the sponsor is in the room. Identification is never finished; new risks appear at every stage gate, every design review and every major procurement award. Digital help is welcome: searching past registers and lessons-learned records, or using AI tools to suggest candidate risks from the schedule, contract and site reports, widens the net quickly. Treat every suggestion as a candidate that someone who knows the project must verify before it enters the register.

Identification continues across the life cycle

CONCEPT
Strategic risks
Site, technology, funding and permits; first register for the business case

FEASIBILITY
Option risks
Ground, routes and utility connections compared between options

INVESTMENT DECISION
Quantified risks
Top risks quantified; contingency and confidence level agreed

DESIGN REVIEWS
Design risks
Interfaces, buildability, safety in design, carbon hotspots

CONTRACT AWARD
Allocated risks
Read the contract for risks it gives you; supply-chain and price risks

CONSTRUCTION
Site risks
Weekly reviews: ground, weather, productivity and safety

COMMISSIONING
Handover risks
Testing, energisation, operator readiness and defects

Each stage gate and major award should trigger a fresh identification pass; risks retire and new ones appear as the project moves from paper to site.

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TEMPLATE **Risk register (core columns)**

Record each risk so that its cause, owner, priority and next action are visible at the weekly review.

ID	CAUSE-RISK-EFFECT STATEMENT	RBS CATEGORY	OWNER	P X I SCORE	RESPONSE AND ACTIONS	STATUS AND TRIGGER
R-014	*Because only three boreholes cover the ETP footprint, piles may meet a soft clay lens below design toe level, extending piling by 3-4 weeks and adding cost*	*Ground and site*	*Civil package manager*	*3 x 4 = 12*	*Reduce: four extra boreholes before ordering piles; designer to prepare longer-pile alternative*	*Open. Trigger: test pile fails to reach set at design level*
O-003	*Because the grid company is commissioning a nearby feeder, early energisation may be offered, letting us stop diesel generator rental two months early*	*External*	*Electrical lead*	*2 x 3 = 6*	*Enhance: submit protection settings and test reports early; agree a joint inspection date*	*Open. Trigger: utility confirms feeder test date*

SCENARIO · DECIDE · A REGISTER WRITTEN FOR THE LENDER, NOT FOR THE SITE

You have just taken over as project manager for a denim factory expansion in Gazipur: a new dyeing hall, eight dyeing machines and an upgraded effluent treatment plant. The register prepared for the lender has 46 lines such as 'delay', 'rain', 'price increase' and 'machine late'. Three lines describe things already happening, including a dyeing machine shipment that has missed its vessel. The sponsor wants to send the register to the lender tomorrow unchanged. What should the project manager do FIRST?

(A) Send it as requested; the lender only needs to see that a register exists.

(B) Hold a short session with the package leads to move current problems to the issues log, rewrite the top risks as cause-risk-effect statements with owners, and agree with the sponsor to send that version.

(C) Delete every vague line so the register looks shorter and more professional.

(D) Buy a dedicated risk software package before revising anything.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Pick one entry from a risk register you have seen recently. Rewrite it as a cause-risk-effect statement. What does the cause tell you that the original wording hid, and who should own it?

Key terms

Project risk
An uncertain event or condition that, if it occurs, would have a positive or negative effect on at least one project objective such as safety, cost, time, quality or scope.

Threat
A risk whose occurrence would harm one or more project objectives, for example late delivery of a power transformer.

Risk breakdown structure (RBS)
A hierarchy of risk categories, such as technical, ground, supply chain, commercial, external and organisational, used to prompt identification and to group and report risks.

Cause-risk-effect statement
A three-part way of writing a risk: the existing cause or condition, the uncertain event that may occur, and its effect on named objectives.

Issue
A problem or condition that has already occurred or is certain to occur and now requires a decision or action; a risk becomes an issue when it materialises.

Opportunity
A risk whose occurrence would benefit one or more project objectives, for example an earlier factory slot for a turbine or early grid energisation.

Risk register
The living record of identified risks with their statements, owners, scores, responses, triggers and status, reviewed regularly to drive action.

Key takeaways

- A risk is uncertain and gets a planned response; an issue has happened and needs a decision now. Move occurred risks to the issues log with a cross-reference.
- Write every risk as cause, event and effect so the team knows where to act and what to watch.
- Use the RBS, an assumptions review, a pre-mortem and site walks to find threats and opportunities; generate silently before discussing.
- Treat tool-generated or database suggestions as candidates that someone who knows the project must verify.
- Repeat identification at every stage gate, design review and major contract award.

KNOWLEDGE CHECK

1. During excavation for a substation cable trench, the team exposes an unrecorded water main. In risk management terms this is best described as:

   (A) An opportunity that the team should exploit
   (B) A secondary risk created by a planned response
   (C) An issue that now needs a decision and action
   (D) A threat that should be scored and monitored

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2. Which statement is written in proper cause-risk-effect form?

(A) Because the vendor's shop is fully booked, rotor delivery may slip, delaying mechanical completion

(B) Turbine delivery is a major concern for the project and should be watched closely by the whole team

(C) There may be delays and cost overruns on the turbine and generator package

(D) The turbine vendor is unreliable and will probably deliver the rotor late

3. What is the main purpose of a risk breakdown structure in an identification workshop?

(A) To calculate the expected monetary value of every risk on the register in one pass

(B) To allocate each risk to the contractor best able to manage it

(C) To replace the work breakdown structure for cost reporting purposes

(D) To prompt systematic thinking across all categories so that blind spots are reduced

4. A grid company may energise a new feeder early, letting the project stop renting diesel generators. How should this be recorded?

(A) As an issue, because it involves an external utility

(B) As an opportunity with an owner and enhancing actions

(C) As an assumption, because it is outside the team's control

(D) It should not be recorded, as it cannot harm the project

5. Why do experienced facilitators ask participants to write risks silently before group discussion?

(A) To make sure every participant's list is the same length

(B) To keep the workshop as short as possible for senior staff

(C) To reduce anchoring and groupthink around the most senior or loudest voice

(D) To avoid recording any opportunities until all of the threats have been listed

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · PROTECT · LESSON 2 · 34 MIN

Qualitative Analysis & Risk Responses

Prioritise with a calibrated matrix, then choose owned responses that change the odds

- Score risks consistently with defined probability and impact scales and a probability-impact matrix
- Choose response strategies for threats and opportunities within the organisation's risk appetite
- Assign owners, actions and triggers, and control contingency drawdown

A register with sixty risks cannot get sixty people's full attention. **Qualitative analysis** ranks risks quickly so that management effort goes where it changes outcomes. It works only if the scales are defined in project terms before scoring starts: 'high impact' must mean the same thing to the piling engineer, the planner and the sponsor. Scores are a prioritisation tool, not a measurement; they tell you which risks to respond to first and which deserve a closer, quantitative look (developed in the next lesson).

Example probability and impact scales (illustrative, for a hypothetical USD 60 million plant)

LEVEL	PROBABILITY	COST IMPACT	SCHEDULE IMPACT	SAFETY, ENVIRONMENT, REPUTATION
1 Very low	Below 10%	Under USD 50k	Under 1 week	First aid case; no external attention
2 Low	10-30%	USD 50k-250k	1-3 weeks	Medical treatment case; minor local complaint
3 Medium	30-50%	USD 250k-1m	3-6 weeks	Lost-time injury; reportable environmental incident
4 High	50-70%	USD 1m-3m	6-12 weeks	Permanent injury; regulator notice; national media
5 Very high	Above 70%	Over USD 3m	Over 12 weeks	Fatality or multiple serious injuries; licence at risk

Scoring and exposure

MEASURE	FORMULA	WHAT IT TELLS YOU
Risk score (ordinal)	`Score = P score x I score, each on a 1-5 scale, giving 1 to 25`	Ranks risks for attention. The numbers are labels on a scale, so a score of 12 is not 'twice as risky' as 6; use it for ordering, not arithmetic.
Cost exposure of a threat	`Exposure = probability x cost impact`	A 40% chance of a USD 600k impact gives an exposure of USD 240k. Opportunities use the same formula with a saving, recorded as a positive value to the project.
Schedule exposure	`Schedule exposure = probability x delay to the key milestone`	A 25% chance of an 8-week delay to first power is an exposure of 2 weeks. Only delay that reaches the milestone counts, so check the risk sits on or near the critical path.
Net value of a response	`Net value = (exposure before - exposure after) - cost of the response`	If positive, the response is worth funding on expected value alone; safety and legal risks may justify a response even when this is negative.

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EXAMPLE · IS EXPEDITING WORTH IT?

A 132/33 kV transformer has a 40% chance of arriving eight weeks late, with an estimated cost impact of USD 600k (generator rental, idle crews, delayed revenue). Exposure = 0.40 x 600k = **USD 240k**. Sending an expediter to the factory and witnessing the factory acceptance test would cost USD 60k and is judged to cut the probability to 15%. Exposure after = 0.15 x 600k = USD 90k. Net value = (240k - 90k) - 60k = **USD 90k**, so the response pays for itself on expected value and also cuts the worst case. Figures are illustrative.

INSIGHT · APPETITE DECIDES WHAT 'ACCEPTABLE' MEANS

Risk appetite is how much risk the organisation is willing to take in pursuit of its objectives; **thresholds** turn that into rules, such as 'any risk with a safety impact of 5 is escalated to the steering committee whatever its probability' or 'no single risk may carry more than USD 1m exposure without a funded response'. Never average a catastrophic safety consequence away: a crane lift over a live 33 kV bay scored 1 x 5 = 5 looks 'low' on the heat map but is unacceptable to almost every owner and contractor.

Reading the probability-impact grid as four zones

IMPACT (LOW TO HIGH) →

Rare but severe
- Never dismiss as 'green'
- Severity thresholds apply
- Specific controls and a fallback
- e.g. a lift beside a live 33 kV bay

Likely and severe
- Act now and escalate
- Funded, owned responses
- Model it quantitatively
- e.g. late transformer on the critical path

Rare and minor
- Accept and monitor
- Review at the monthly cycle
- e.g. late delivery of site cabins

Likely but minor
- Reduce with routine controls
- Allow for it in contingency
- Watch for clusters
- e.g. frequent short rain stoppages

PROBABILITY (LOW TO HIGH) →

A cell's colour is only a starting point: the top-left zone holds the rare but catastrophic risks that appetite thresholds pull out for specific treatment, whatever their score.

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Response strategies for threats and for opportunities

Threats (reduce the downside)

- **Avoid**: remove the cause, e.g. reroute a pipeline away from a known landslip slope
- **Transfer**: move the financial consequence to a party better placed to bear it, e.g. insurance, or a fixed-price heavy-lift subcontract
- **Mitigate**: lower probability or impact, e.g. extra boreholes, a second vendor for cable, pre-assembled pipe racks
- **Accept**: acknowledge and hold contingency, or simply monitor if exposure is small
- **Escalate**: pass to the programme or owner level when it is outside the project's authority

Opportunities (increase the upside)

- **Exploit**: make it certain, e.g. lock in an early factory slot with a reservation fee
- **Share**: partner with a party better able to capture it, e.g. a gain-share with the contractor for a faster construction method
- **Enhance**: raise probability or benefit, e.g. submit grid-connection documents early
- **Accept**: take the benefit if it arrives without active pursuit
- **Escalate**: offer it to the owner when the benefit lies outside the project, e.g. a wider site master plan

From strategy to owned action

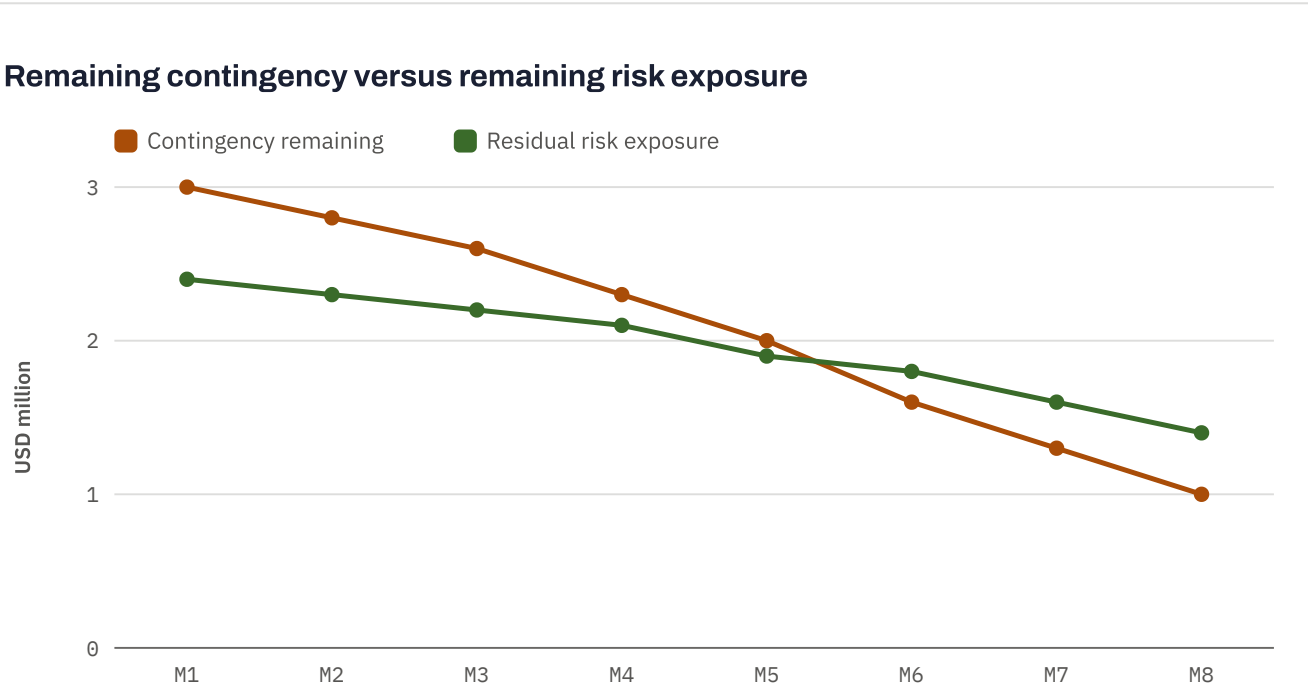
RISK	STRATEGY	SPECIFIC ACTION	OWNER	TRIGGER OR EARLY WARNING
Soft clay lens below design pile toe	Mitigate	Four additional boreholes and two preliminary test piles before ordering piles	Civil package manager	Test pile set not achieved at design level
Pre-monsoon storm during raft pour	Mitigate	Pour plan with weather window check, covers on standby and a pre-agreed construction joint location	Site manager	Forecast rainfall above the agreed limit within 24 hours
Currency movement on imported switchgear	Transfer	Forward contract for the foreign-currency payments on the purchase order	Finance lead	Purchase order signed
Steel and cement price volatility on a three-year package	Accept (funded)	Price-adjustment formula on published indices instead of a large bidder premium; escalation allowance held in the budget	Cost manager	Index movement beyond the allowance assumed in the estimate
Insolvency of the facade subcontractor	Mitigate	Monthly financial check, direct payment of key suppliers, identify a replacement firm	Commercial manager	Late payment to suppliers reported
Early energisation of the grid feeder	Enhance	Submit protection settings and test certificates two months early	Electrical lead	Utility confirms feeder test date

Every significant risk needs a **risk owner**: one named person with the authority and knowledge to manage it, who may delegate individual actions to action owners. A **trigger** is the early warning that the risk is about to occur or that a response must be activated; good triggers are observable on site or in reports, such as 'vendor misses the casting milestone'. Plan a **fallback** for high-impact risks in case the main response fails, and check each response for **secondary risks** it creates: adding a night shift to recover concrete works reduces delay but adds fatigue, lighting and noise risks near homes. Finally, record **residual risk**, the exposure that remains after responses, because that is what contingency must cover.

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Controlling contingency drawdown

1. **Confirm the event** The risk owner shows that a risk in the register has occurred or a trigger has been reached, with evidence such as a test pile record or vendor delay notice.
2. **Quantify the need** Estimate the cost and time required to deal with it, separating risk-related cost from scope change, which goes through change control (see Define).
3. **Approve within authority** Approve the drawdown at the level set in the delegation of authority; larger draws go to the sponsor or steering committee.
4. **Transfer the budget** Move the approved amount from contingency into the relevant control account so cost reports stay honest (see Budget).
5. **Update the register and forecast** Mark the risk as occurred or closed, update residual exposure and re-forecast the outturn.
6. **Compare remaining contingency with remaining risk** Report each month whether the contingency left is enough for the risk exposure left; a gap is an early warning to the sponsor.



Illustrative figures. From month 6 the exposure still ahead exceeds the contingency left; that crossover is the early warning the monthly report must give the sponsor before the money runs out.

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SCENARIO · DECIDE · A LOW SCORE ON A LETHAL RISK

On a combined-cycle power plant extension, the register scores 'dropped load during lift of a 40-tonne HRSG module next to a live 33 kV switchyard' as probability 1, impact 5, total 5, which shows green on the heat map. The construction manager says it therefore needs no specific response and the lift can proceed under the general lifting procedure next week. What should the project manager do FIRST?

(A) Accept the green rating; the matrix exists to stop the team over-reacting to unlikely events.

(B) Apply the safety threshold: escalate the risk, and require a specific lift plan that removes or reduces the hazard, such as an outage of the adjacent bay or a different crane position.

(C) Buy additional insurance to transfer the consequences of the lift to an insurer.

(D) Raise the probability score to 3 so that the risk turns red and gets attention.

Circle your choice. Answer and feedback: module answer key.

Is your risk register healthy?

☐ Every top-ten risk is written as cause, event and effect

☐ Every significant risk has one named owner with authority to act

☐ Each response is a specific action with a date, not a general intention

☐ Triggers are observable events on site or in reports

☐ Opportunities appear on the register and have owners

☐ Residual exposure is recorded and compared with remaining contingency

☐ Occurred risks have been moved to the issues log with a cross-reference

☐ Scores were reviewed in the last month and changes explained

REFLECT

Think of a risk on your current or last project that did occur. Was there a trigger that could have warned you earlier, and who was (or should have been) its owner?

Key terms

Probability-impact matrix
A grid that combines defined probability and impact scales to rank risks into priority bands, often shown as a red-amber-green heat map.

Risk appetite
The amount and type of risk an organisation is willing to accept in pursuit of its objectives, expressed in practice through thresholds and escalation rules.

Risk exposure
The expected effect of a risk, calculated as its probability multiplied by its impact on cost or time; used to compare risks and size responses.

Risk owner
The one named person accountable for monitoring a risk and making sure its agreed responses are carried out, even when actions are delegated.

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Risk trigger
An observable event or condition that warns that a risk is about to occur or that a planned response or fallback must now be activated.

Contingency drawdown
The controlled release of risk contingency into a control account when an identified risk occurs, approved within delegated authority and recorded in the register and forecast.

Residual risk
The risk that remains after responses have been implemented; residual exposure is what contingency and management attention must still cover.

Key takeaways

- Define probability and impact scales in project terms before scoring; scores rank risks, they do not measure them.
- Severity thresholds from the risk appetite override the heat-map colour for catastrophic safety and public risks.
- Fund a response when the exposure it removes exceeds its cost, and check it for secondary risks.
- Every significant risk needs one owner, specific dated actions and an observable trigger.
- Draw down contingency only for occurred risks, within delegated authority, and compare what is left with residual exposure every month.

KNOWLEDGE CHECK

6. A threat has a 25% probability of occurring and would cost USD 800k if it did. What is its cost exposure?
   - (A) USD 160k
   - (B) USD 200k
   - (C) USD 250k
   - (D) USD 600k

7. Why should a risk score of 12 not be described as 'twice as risky' as a score of 6?
   - (A) Because ordinal scores rank risks but are not true measurements that support ratio arithmetic
   - (B) Because the probability-impact matrix should only ever be used to rank opportunities, not threats
   - (C) Because scores above 10 are always escalated and so cannot be compared
   - (D) Because the impact scale must always be expressed in weeks, not money

8. A contractor takes out a forward currency contract to fix the cost of imported switchgear. Which threat response is this?
   - (A) Avoid
   - (B) Accept
   - (C) Escalate
   - (D) Transfer

9. A response costs USD 50k and reduces a risk's exposure from USD 300k to USD 120k. What is the net value of the response?
   - (A) USD 180k
   - (B) USD 70k
   - (C) USD 130k
   - (D) USD 250k

10. Adding a night shift to recover lost concrete progress introduces fatigue and noise risks near homes. What are these new risks called?
    - (A) Residual risks
    - (B) Consequential issues
    - (C) Triggers
    - (D) Secondary risks

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · PROTECT · LESSON 3 · 40 MIN

Quantitative Risk Analysis & Monte Carlo Intuition

EMV, decision trees and simulation: turning a register into a confidence level a board can use

- Calculate expected monetary value for risks and use a decision tree to compare engineering options
- Explain in plain language how Monte Carlo simulation produces P50 and P80 dates and costs
- Explain why a deterministic schedule is usually optimistic, including skew, merge bias, risk events and correlation
- Read tornado charts and use quantitative results to set contingency and communicate confidence

Qualitative scoring tells you which risks to work on; it cannot tell a board how much contingency to hold or how likely the commissioning date is. **Quantitative risk analysis** puts numbers on the combined effect of uncertainty and risk events on cost and time. It is worth the effort on large or complex projects, at major investment decisions, before setting a contractual completion date, and whenever a sponsor asks 'how confident are we?'. The techniques range from a simple expected value calculation on a page to a full simulation of an integrated cost and schedule model.

Expected value and decision-tree formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Expected monetary value of one risk	`EMV = P x I, where I is negative for a threat (cost) and positive for an opportunity (saving)`	A 30% chance of a USD 400k overrun has EMV = 0.30 x (-400k) = -USD 120k. It is the long-run average effect, not the amount you will see if the risk occurs.
EMV of a set of risks	`EMV_total = sum of (P_i x I_i) for i = 1 to n`	Adds threats and opportunities together. Useful for a first sense of contingency on a portfolio of many independent risks, where averages are meaningful.
Value of a chance node	`Chance node value = sum of (p_j x outcome_j), where the branch probabilities p_j add up to 1`	The probability-weighted outcome of everything that might follow at that point in the tree.
Value of a decision node	`Decision node value = best of (branch cost or value + value of the node that follows), e.g. the lowest expected cost`	Roll the tree back from right to left: evaluate chance nodes, then choose at each decision node, carrying only the best branch forward.
Triangular mean of a three-point range	`Mean = (a + m + b) / 3, where a = optimistic, m = most likely, b = pessimistic`	For a right-skewed range the mean sits above the most likely value, which is one reason single-point plans are optimistic (the beta-PERT mean is covered in Schedule).
Risk-based contingency	`Contingency = cost (or date) at the chosen confidence level, e.g. P80 - base estimate (or deterministic date)`	Sizes the allowance from the simulation result rather than a flat percentage. The chosen P-level is a governance decision that reflects risk appetite.

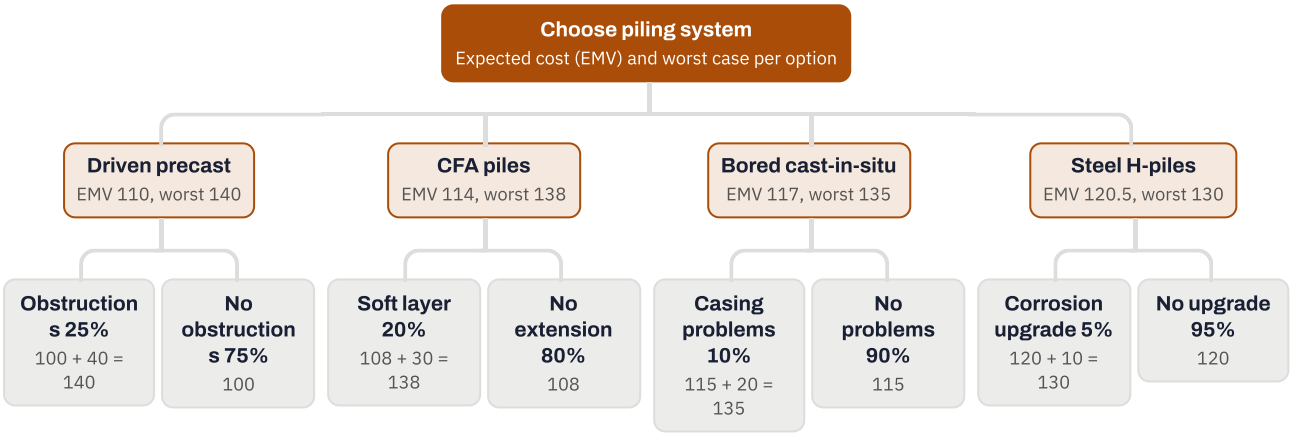
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EXAMPLE · EMV OF A SMALL REGISTER (FIGURES ILLUSTRATIVE)

Hypothetical 50 MW solar plant: (1) inverter delivery late, P 30%, cost USD 400k, EMV -120k; (2) rock in cable trenching, P 20%, cost USD 250k, EMV -50k; (3) currency movement on modules, P 50%, cost USD 180k, EMV -90k; (4) opportunity: early grid connection saves generator and site costs, P 25%, saving USD 160k, EMV +40k. Total EMV = -120 - 50 - 90 + 40 = **-USD 220k**. This is a sensible first estimate of the average risk cost, but notice that no single outcome equals it: if the inverters are late the project loses the full USD 400k. EMV supports comparison and contingency sizing; it does not describe the worst case.

A **decision tree** lays out a choice, the uncertain events that follow each option and their consequences, so that options can be compared on expected value and on spread. Suppose the foundation designer for an effluent treatment plant must choose a piling system before ground conditions are fully known. Each option has a base cost and a chance of an adverse outcome. Rolling back the tree, you compute each option's expected cost and also note its worst case. The tree below shows four options; drawing it makes the logic easy for a steering committee to follow.

Piling decision tree (BDT million, illustrative)



Roll back from the outcomes: each option's EMV is its base cost plus probability times the extra cost. Driven piles have the lowest EMV but the widest spread; bored piles cap the worst case at 135.

On expected cost, driven precast piles win at 110. But the decision is not purely arithmetic. Driven piles carry the widest spread and generate noise and vibration that may be unacceptable next to a working dyeing hall or a hospital; a risk-averse owner with little contingency might pay 7 more in expectation for bored piles to cap the worst case at 135. The tree makes that trade-off explicit, which is its real value. Now consider many uncertainties at once: hundreds of activities with ranges, a dozen risk events, logic that merges paths. A tree becomes unmanageable, and **Monte Carlo simulation** takes over.

The intuition is simple. Imagine you could build the same substation ten thousand times in parallel universes. In each one, every uncertain activity takes a duration drawn from its range: piling might take 41 days in one universe and 52 in the next. Each risk event happens in its share of universes: if the transformer has a 30% chance of a six-week delay, it is late in about three thousand of them. The computer runs the schedule logic in each universe and records the finish date. Sort the ten thousand finish dates and you have a distribution. The date that 50% of universes meet is the **P50**; the date that 80% meet is the **P80**. Nothing magical happens: the method just does, very quickly, the arithmetic of combining many ranges and events that the human mind cannot do reliably.

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A Monte Carlo schedule or cost risk analysis

01
Prepare the model
A logic-sound summary schedule or cost model; fix open ends and hard constraints first

→

02
Add uncertainty ranges
Three-point ranges on durations and quantities, from evidence and experts

→

03
Add risk events
Each top risk with its probability and impact, mapped to the activities it affects

→

04
Set correlation
Link inputs driven by the same cause, such as ground conditions or productivity

→

05
Run iterations
Several thousand passes, each sampling every input and recording the result

→

06
Read and act
S-curve, P50 and P80, tornado chart and criticality; then change the plan

Why the deterministic date is usually optimistic

- **Skewed ranges.** Engineering activities can overrun far more than they can underrun, so the mean of each range sits above its most likely value. Adding most-likely values produces a total that falls below the middle of the true distribution.
- **Merge bias.** Where parallel paths join, the milestone waits for the latest one. If two independent paths each have a 50% chance of being on time, the chance that both are is only 25%; with five such paths it is about 3%. The schedule logic cannot show this; simulation does.
- **Risk events left out.** The deterministic plan quietly assumes that no identified risk occurs. The simulation lets each one occur in its share of iterations.
- **Correlation ignored.** If piling and excavation both depend on the same ground, they will tend to overrun together. Treating them as independent lets a bad draw on one cancel a good draw on the other, which makes the forecast look narrower and safer than it is.
- **Planning optimism.** Durations are often estimated as 'what we can do if things go well', which is closer to the optimistic end than to the most likely value.

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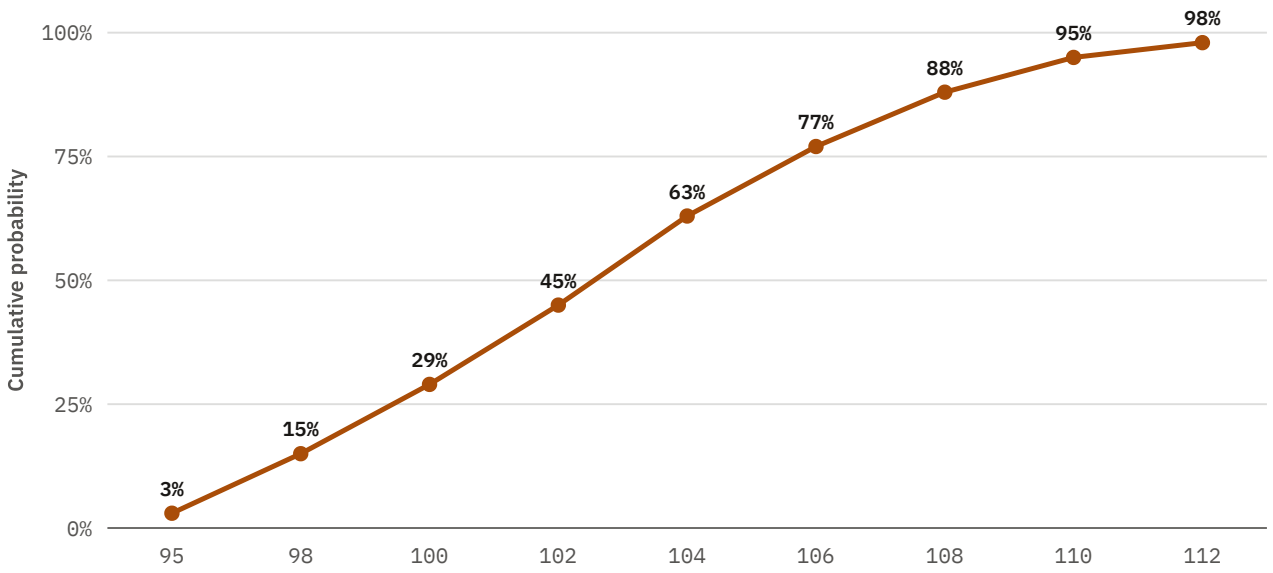
A tiny Monte Carlo of a substation finish date

Python notebook

```
import numpy as np
rng = np.random.default_rng(seed=7)
N = 20_000                                  # number of simulated projects
# Triangular(min, most likely, max) durations in days - illustrative
civil  = rng.triangular(42, 45, 56, N)      # foundations and plinths
steel  = rng.triangular(27, 30, 38, N)      # structure and cable trays
equip  = rng.triangular(66, 72, 92, N)      # transformer supply (parallel path)
hookup = rng.triangular(18, 20, 27, N)      # install, terminate, test
finish = np.maximum(civil + steel, equip) + hookup  # merge, then hook-up
print("Deterministic:", max(45 + 30, 72) + 20, "days")
print("P50:", np.percentile(finish, 50).round(1), "days")
print("P80:", np.percentile(finish, 80).round(1), "days")
print("P(finish <= 95 days):", (finish <= 95).mean().round(2))
# Output with this seed:
# Deterministic: 95 days
# P50: 102.5 days
# P80: 106.5 days
# P(finish <= 95 days): 0.03
```

The plan built from most-likely durations says 95 days, yet only about 3% of simulated projects finish by then, because every range is right-skewed and the transformer path sometimes overtakes the civil and steel path at the merge. This sketch samples each input independently; in a real model, durations sharing a cause (the same ground, the same crew) would be correlated, which widens the spread and pushes the P80 later still.

S-curve from the substation simulation



Computed from the sketch above (illustrative inputs, seed 7). The 95-day plan sits at about 3%, P50 falls near day 102.5 and P80 near day 106.5: read across from a confidence level to find the date.

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Reading the simulation output

RESULT	WHAT IT MEANS	TYPICAL USE
P10	Only 10% of iterations finish this early or cheaply	A stretch outcome; useful for spotting unrealistic targets
P50	Half the iterations finish by this value; the median	Internal target for the delivery team; often the basis of a contractor's bid
P80	80% of iterations finish by this value	Common basis for an owner's funded budget and publicly announced date
P90	90% of iterations finish by this value	Conservative governments and lenders; first-of-a-kind technology
Deterministic	The single-point plan from most-likely inputs	Shown on the S-curve to reveal how optimistic it is; it often sits below P20

Two outputs tell you what to do next. A **tornado chart** ranks the inputs by how strongly they drive the result, longest bar at the top, so management attention goes to the few risks and activities that move P80 the most, often transformer supply, ground conditions or design approvals rather than the issues the team argues about most. The **criticality index** shows how often each activity was on the critical path across the iterations, revealing near-critical paths that a single CPM run hides. On correlation, prefer to model the shared cause explicitly, for example a 'ground worse than expected' risk that lengthens both piling and excavation, rather than assigning arbitrary coefficients; either way, never leave obviously related inputs independent. Tools increasingly use AI to propose duration ranges, risk mappings and correlations from historical data. Treat that as a first draft: experienced planners must check every input against this project's evidence, and the analyst must be able to explain each one to the board. A model nobody can explain is not evidence.

SCENARIO · DECIDE · WHICH DATE GOES ON THE PLAQUE?

A 500-bed hospital in Dhaka is being built for a public owner. The schedule risk analysis shows the deterministic completion date has roughly a 15% chance of being achieved; the P50 is seven weeks later and the P80 is twelve weeks later. The ministry's communications office wants an opening date for the minister's speech next week and the owner's director suggests using the deterministic date 'to keep the contractor under pressure'. What should the project manager recommend?

(A) Announce the deterministic date; public pressure will motivate everyone to achieve it.

(B) Announce the P80 date or a window with a stated confidence, keep the earlier date as the internal delivery target, and focus the team on the top tornado drivers.

(C) Add three weeks of float to every activity so that the deterministic plan becomes safe.

(D) Ask the risk analyst to remove the three largest risks from the model so the P80 moves earlier.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · THE RISK ANALYSIS NOBODY BELIEVED

A hypothetical offshore wind project's owner commissioned a schedule risk analysis before final investment decision. The model was the contractor's full 4,000-activity schedule with many hard date constraints. Package managers supplied ranges of roughly plus or minus 5% on every activity, no discrete risk events were included, and all inputs were sampled independently. The result showed the P80 only two weeks after the deterministic date, on a programme with a novel installation vessel, a fixed weather window and a single export cable route.

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The investment committee was told the schedule was 'highly robust'. Eighteen months later, vessel availability and cable installation problems had pushed first power well beyond the reported P80.

1. Why was the reported distribution so narrow?

2. What should the analyst have changed in the model before running it?

3. How should the results have been presented to the investment committee?

Model answer: module answer key.

REFLECT

Look at the last schedule you worked on. Where do three or more paths merge into one milestone, and which activities share a common cause that would make them overrun together?

Key terms

Expected monetary value (EMV)
The probability-weighted value of an uncertain outcome, calculated as probability multiplied by impact and summed across outcomes; negative for threats and positive for opportunities.

Monte Carlo simulation
A method that recalculates a cost or schedule model thousands of times, each time sampling uncertain inputs and risk events, to build a distribution of possible outcomes.

Tornado chart
A bar chart that ranks model inputs by how strongly they influence the result, with the largest driver at the top, directing attention to the risks that matter most.

Schedule risk analysis (SRA)
A Monte Carlo analysis of a logic-linked schedule with duration ranges and risk events, producing finish-date confidence levels, drivers and criticality for each activity.

Decision tree
A diagram of decisions, chance events and outcomes that is rolled back from right to left to compare options on expected value while also showing the spread of outcomes.

Confidence level (P50, P80)
A point on the simulated outcome distribution; P80 is the value that 80% of iterations meet or beat, used to set funded budgets and committed dates.

Risk correlation
The tendency of uncertain inputs that share a common cause to move together; ignoring it narrows the simulated spread and understates high-confidence values such as P80.

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Key takeaways

- EMV = probability x impact; it suits comparison and contingency for many risks but never describes the worst case.
- Decision trees make the trade-off between expected cost and spread visible to decision-makers.
- Monte Carlo combines ranges, risk events and logic; P50 and P80 are cumulative confidence values, not predictions of a date.
- Single-point plans are optimistic because of skew, merge bias, omitted risks, ignored correlation and planning optimism.
- Use tornado and criticality results to direct action, and have experts verify any tool-generated inputs.

KNOWLEDGE CHECK

11. A register holds one threat (30% chance of a USD 400k overrun) and one opportunity (50% chance of a USD 100k saving). What is the combined EMV?

    (A) -USD 70k, an expected net cost
    (B) -USD 120k, an expected net cost
    (C) +USD 50k, an expected net saving
    (D) -USD 300k, an expected net cost

12. Option A costs 200 with a 20% chance of an extra 50. Option B costs 190 with a 40% chance of an extra 60. Which option has the lower expected cost?

    (A) Option B, at an expected cost of 190
    (B) Option B, at an expected cost of 214
    (C) Option A, at an expected cost of 210
    (D) Option A, at an expected cost of 250

13. A schedule risk analysis reports a P80 completion of 14 March. What does this mean?

    (A) The project will finish on 14 March in eight cases out of ten if nothing changes
    (B) 80% of the simulated outcomes finish on or before 14 March
    (C) The deterministic date has an 80% probability of being achieved
    (D) There is an 80% chance that the project finishes after 14 March

14. Three independent parallel paths each have a 60% chance of reaching a common milestone on time. What is the chance the milestone is met on time?

    (A) 60%, the same as each individual path
    (B) About 36%, two paths multiplied together
    (C) About 80%, because the paths support each other
    (D) About 22%, all three paths multiplied together

15. Piling and bulk excavation both depend on the same uncertain ground conditions, but the model samples them independently. What is the most likely effect?

    (A) The spread is understated, so the P80 is too early and contingency too small
    (B) The P50 is pushed later because every activity is treated as critical
    (C) The tornado chart will show both activities as having no influence
    (D) The deterministic date is shown as far more likely than it really is in practice

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · PROTECT · LESSON 4 · 38 MIN

Health & Safety Management on Site

Plan the hazard out, control what remains, and lead a culture where anyone can stop the job

- Apply the hierarchy of controls to high-risk site activities
- Explain how risk assessments, method statements, permits and toolbox talks fit together
- Use leading and lagging indicators and build a safety culture that surfaces problems early
- Draw structural and fire safety lessons from Bangladesh's garment sector for any change-of-use or expansion project

No schedule date or cost saving justifies a serious injury, and on a well-run project nobody is asked to make that trade. Construction, plant installation and commissioning combine heavy loads, heights, energy sources, moving plant and many companies working in the same space, often under time pressure and in heat. The project manager is not the safety officer, but sets the conditions in which safety succeeds or fails: realistic programmes, competent contractors, funded temporary works, safe designs and the visible expectation that unsafe work stops. Safety is managed as a line responsibility, with the HSE team advising, checking and coaching. Many organisations run their system to ISO 45001:2018, whose 2024 amendment asks them to consider climate change, which on site means heat stress, flooding and storms. Since 2022 the ILO has counted a safe and healthy working environment among the fundamental principles and rights at work, and its Convention C167 (1988) remains the international benchmark for construction; in Bangladesh, national labour law and the construction safety provisions of BNBC 2020 set the legal minimum.

The hierarchy of controls (most effective at the top)

Elimination	`Design out the hazard`
	`Assemble roof trusses at ground level and lift them complete`
	`Relocate a cable route away from a live substation`
	`Drone or laser-scan inspection instead of climbing`

Substitution	`Replace with something less hazardous`
	`Water-based coatings instead of solvent paint in an enclosed hall`
	`Mechanical splices instead of hot-work welding near fuel`

Engineering controls	`Isolate people from the hazard` `Edge protection and netting` `Trench boxes and shoring` `Interlocks and guarding on test rigs`
Administrative controls	`Change how people work` `Permits, exclusion zones, lift plans` `Training, supervision and toolbox talks`
Personal protective equipment	`Last line of defence` `Harness with rescue plan, hard hat, arc-rated clothing` `Works only if selected, fitted, maintained and worn`

Controls come from a structured **risk assessment** for each significant task, written into a **method statement** that tells the crew how the work will be done: sequence, plant, temporary works, competence, supervision, emergency and rescue arrangements, and the specific hazards and controls at each step. Together these are often called RAMS. They are only useful if they are task-specific, prepared with the people who will do the work, reviewed by a competent engineer, and briefed to the crew before work starts. A generic RAMS copied

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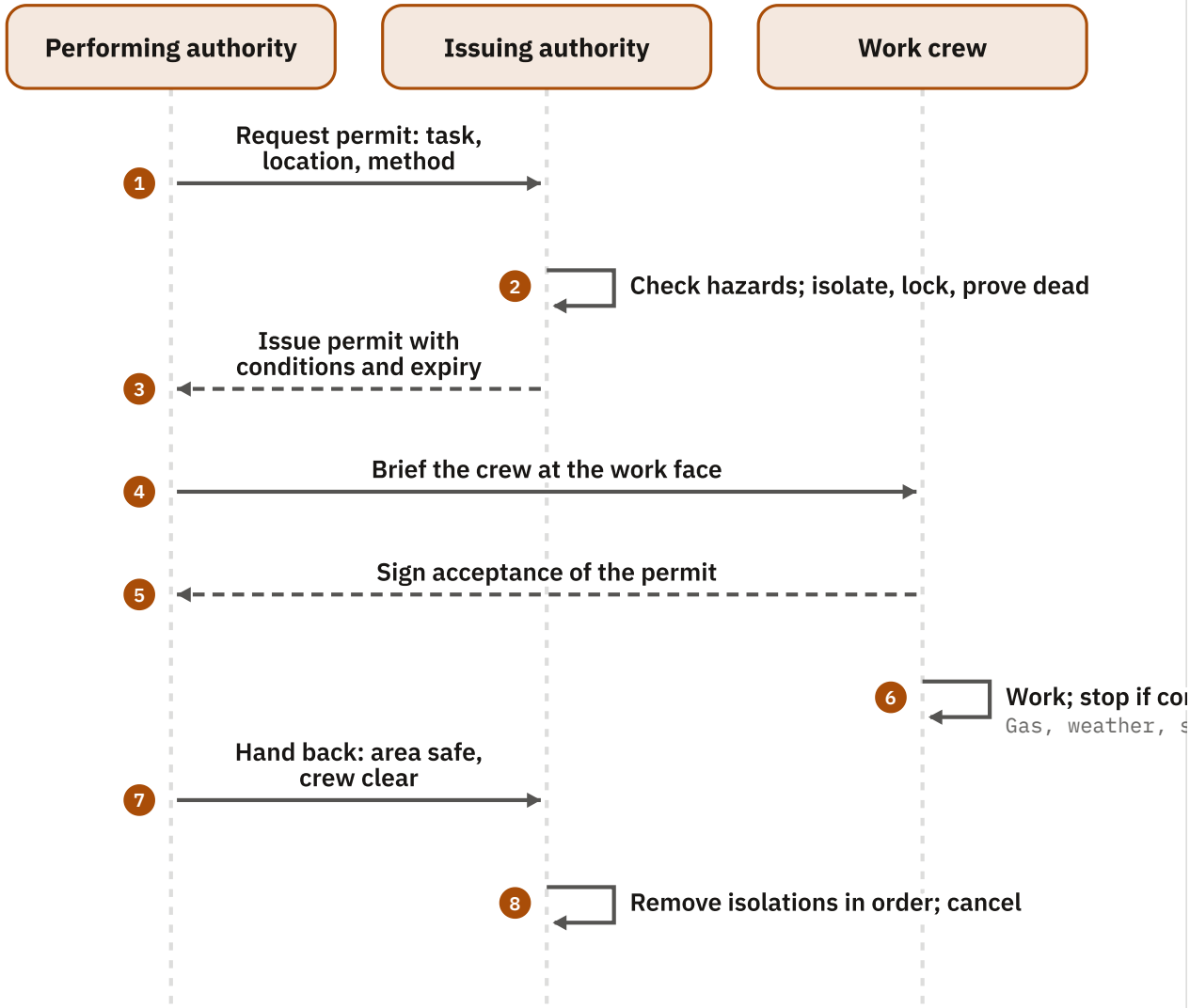
from the last project, signed by a foreman who never read it, gives no protection. When conditions change, such as a different crane, a wetter trench or a new crew, the RAMS is reviewed before work continues.

High-risk activities and their critical controls

ACTIVITY	TYPICAL HAZARDS	CRITICAL CONTROLS	PERMIT OR INSPECTION
Work at height (steel erection, roofing, scaffolds)	Falls from edges, through fragile roofs or openings; dropped objects	Collective protection first: guardrails, netting, mobile elevating platforms; harness with anchor and rescue plan; exclusion zones below	Scaffold inspected and tagged before use and after changes; work-at-height permit where required
Lifting operations (cranes, girder launches, transformer placement)	Load drop, crane overturning on soft ground, contact with power lines, wind	Lift plan by an appointed person; ground bearing check and outrigger mats; wind limits; certified rigging; banksman; critical-lift review for tandem or near-capacity lifts	Daily crane checks; thorough examination certificates; lift permit for critical lifts
Excavation (pile caps, cable trenches, station boxes)	Collapse, buried services, water ingress, falling into trench, plant at edge	Service detection and trial holes; shoring, trench boxes or safe batter; spoil and plant set back; safe access and barriers	Excavation permit; inspection at the start of each shift and after rain
Electrical isolation (switchgear, MCCs, energisation)	Electric shock, arc flash, unexpected re-energisation	Isolate, lock and tag; prove dead with a tested instrument; earthing; arc-rated PPE; energisation procedures agreed with commissioning	Electrical permit-to-work; sanction-to-test for commissioning
Confined spaces (tanks, ETP chambers, manholes)	Oxygen deficiency, toxic gas, engulfment, difficult rescue	Avoid entry where possible; ventilation; continuous gas monitoring; attendant outside; rescue equipment and trained team	Confined-space entry permit with gas test record
Hot work (welding, cutting, grinding)	Fire, burns, fumes, ignition of flammables	Remove or protect combustibles; fire watch during and after work; extinguishers; screens; fume extraction	Hot-work permit with area check and fire watch sign-off

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The permit-to-work cycle



Notice that the issuing authority both starts and ends the cycle: nobody removes an isolation until the work has been handed back and the crew is confirmed clear.

WATCH OUT · TEST BEFORE TOUCH, EVERY TIME

Many serious electrical incidents during construction and commissioning involve equipment that someone believed was dead. Isolate at the point of supply, apply a personal lock and tag for every person working, then **prove dead** with a voltage tester that has itself been proved on a known live source before and after use. Watch for back-feeds from generators, UPS systems, capacitor banks and parallel transformer circuits. On the day a new switchboard is energised, the site's whole permit regime changes; plan that transition as carefully as the energisation itself.

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Leading versus lagging safety indicators

Leading indicators (predict and prevent)

- Percentage of high-risk tasks with a reviewed, briefed RAMS
- Senior leader safety walks completed and actions closed
- Near-miss and hazard reports raised, and time to close them
- Toolbox talks delivered and attendance
- Permit audits passed; scaffold and crane inspection compliance
- Training and competence records current for critical roles

Lagging indicators (measure harm after the event)

- Lost-time injury frequency rate
- Total recordable injury rate
- Dangerous occurrences, such as a dropped load or scaffold collapse
- Environmental incidents and regulator notices
- Days lost to injury and occupational illness
- Fatalities, which must be zero

Indicators only work inside a healthy **safety culture**. Workers report near misses when they trust that reporting leads to fixes, not blame; this is the idea of a **just culture**, which separates honest error from reckless behaviour. Every worker, including the newest labourer from a subcontractor, must have and use **stop-work authority** without fear. A short **toolbox talk** at the start of a shift, given in the crew's own language and tied to that day's task, does more than a long induction slide deck. Welfare is part of safety: drinking water, shade and rest breaks in pre-monsoon heat, clean toilets, safe transport to site and reasonable working hours all reduce fatigue and error. And watch the leaders: if the project manager walks past an open edge, the crew learns what the rules really are.

CASE STUDY · LESSONS FROM BANGLADESH'S GARMENT SECTOR

In 2012 a fire at the Tazreen Fashions factory at Ashulia, near Dhaka, killed many workers; investigations and reporting highlighted stairways that led down into areas used to store yarn and fabric, inadequate emergency exits and workers being told to return to their machines after the alarm. In April 2013 the Rana Plaza building at Savar collapsed, causing one of the worst industrial disasters on record. It housed several garment factories; widely reported findings included upper floors added beyond what had been permitted, a building not designed for industrial loads or the vibration of heavy machinery and generators, and large cracks seen the day before, after which workers were still sent back inside. Afterwards, brand-union agreements such as the Accord on Fire and Building Safety, an industry alliance and a national inspection initiative inspected garment factories for structural, fire and electrical safety and tracked remediation such as structural strengthening, fire separation, fire doors, alarms and safe exit routes.

1. Which failures were about design and change of use, and which were about management decisions on the day?

2. You are managing the addition of a new floor and heavy dyeing machines to an existing factory in Narayanganj. What must happen before the design is fixed?

3. Which leading indicators would show that fire safety is being maintained once the factory is in operation?

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Model answer: module answer key.

From disaster to reform in Bangladesh's garment sector

2012
Tazreen fire
Exits and stairs fail as escape routes

APRIL 2013
Rana Plaza collapse
Illegal extra floors, industrial loads, warnings ignored

2013
Inspection programmes
Accord, industry Alliance and a national initiative inspect factories

2020
RMG Sustainability Council
A national body takes over the Accord's work in Bangladesh

2021
International Accord
Brands and unions extend the model beyond Bangladesh

2022
ILO fundamental right
Safe and healthy work becomes a fundamental principle

A decade of reform turned one-off inspections into permanent institutions. The engineering lesson is unchanged: structural and fire safety are designed and governed, not left to workers on the day.

SCENARIO · DECIDE · WIND RISING DURING A GIRDER LIFT

On an elevated metro viaduct, a night-time lift of a precast girder over a closed city road is under way. The road closure permit ends at 05:00 and the programme is already three weeks behind. At 02:30 the measured wind speed at boom height reaches the limit in the lift plan and is still rising. The crane supervisor suggests 'getting this last one in quickly while we still can'. What should the project manager, who is on site, do FIRST?

(A) Support the supervisor: one quick lift is less risky than leaving the work half finished.

(B) Stop the lift, make the load and crane safe in accordance with the lift plan, and arrange a new road closure through the agreed contingency procedure.

(C) Ask the crane operator to decide, since the operator has the best view of the load.

(D) Reduce the crane's radius slightly and continue, recording the decision in the site diary.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Task risk assessment summary**

Capture the hazards and controls for one task so it can be reviewed and briefed to the crew before work starts.

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TASK STEP	HAZARD	WHO COULD BE HARMED	CONTROLS (WITH HIERARCHY LEVEL)	RESIDUAL RISK	RESPONSIBLE
Lift 18 t transformer onto plinth	*Load drop; crane overturning on backfilled ground*	*Riggers; electricians in adjacent bay*	*Engineering: crane on mats over verified ground; Admin: lift plan by appointed person, exclusion zone, adjacent bay cleared*	*Low*	*Lifting supervisor*
Terminate 33 kV cables in switchgear	*Contact with live conductors via back-feed*	*Cable jointers*	*Elimination: outage of the incomer; Engineering: earths applied; Admin: electrical permit, lock and tag, prove dead*	*Low*	*Authorised electrical person*

REFLECT

Choose one high-risk task on your site. Which control in your current RAMS sits lowest in the hierarchy, and what would it take to move it one level higher?

Key terms

Hierarchy of controls
The ranking of risk controls from most to least effective: elimination, substitution, engineering controls, administrative controls and personal protective equipment.

Permit-to-work
A formal written system that authorises specific high-risk work at a set place and time, after hazards are checked and isolations made, and controls its hand-back.

Lock-out tag-out (LOTO)
An isolation practice in which energy sources are isolated, secured with personal locks and tags, and proved dead before work begins.

Risk assessment and method statement (RAMS)
A task-specific document set that identifies hazards and controls and describes the safe sequence, plant, competence, supervision and emergency arrangements for the work.

Toolbox talk
A short, task-focused safety briefing given to a crew at the work face, usually at the start of a shift, in the workers' own language.

Leading safety indicator
A measure of proactive safety activity or conditions, such as near-miss reports closed or RAMS briefed, that can predict and prevent harm.

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Lagging safety indicator
A measure of harm that has already happened, such as lost-time injuries or dangerous occurrences, useful for trends but too late to prevent those events.

Key takeaways

- Work down the hierarchy of controls from elimination; PPE is the last line of defence.
- RAMS must be task-specific, prepared with the crew and briefed; permits control high-risk work from isolation to hand-back.
- Prove dead with a proven tester every time, and plan energisation as a change to the whole permit regime.
- Lead with leading indicators, a just culture, stop-work authority, welfare and heat-stress planning.
- Structural and fire safety for any change of use are design and governance decisions made before work starts.

KNOWLEDGE CHECK

16. Steel erectors on a warehouse roof currently rely on harnesses. Which change moves the control highest up the hierarchy?

(A) Providing better harnesses with more frequent and thorough harness inspections
(B) Adding a second supervisor to watch the erectors at the edge
(C) Giving the crew a refresher toolbox talk on fall protection
(D) Assembling roof sections at ground level and lifting them complete

17. What is the main purpose of a permit-to-work system?

(A) To record who was on site for payroll and attendance purposes
(B) To authorise specific high-risk work once hazards are checked and isolations made
(C) To transfer legal responsibility for any accident to the subcontractor doing the work
(D) To replace the need for task-specific method statements

18. Which of these is a leading safety indicator?

(A) Lost-time injury frequency rate for the quarter
(B) Number of dangerous occurrences reported to the regulator this year
(C) Percentage of near-miss reports closed out within seven days
(D) Total days lost to occupational injury this year

19. An electrician has isolated and locked a motor control centre. What must happen before anyone touches the conductors?

(A) Prove the circuit dead with a tester that has been checked on a known live source
(B) Confirm verbally with the control room that the upstream feeder breaker has been opened
(C) Check the single-line diagram to confirm that no other supply exists
(D) Wait at least ten minutes for any stored energy to dissipate on its own

20. A factory owner wants to add a storey and install heavy, vibrating machines on an existing upper floor. What does the garment-sector experience say must come first?

(A) A full fire evacuation drill for all workers on the existing floors before any construction begins
(B) A structural assessment of the building and foundations for the new loads, with permits
(C) A new insurance policy that covers structural damage during the works
(D) A temporary reduction in working hours while the new storey is added

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · PROTECT · LESSON 5 · 36 MIN

Environmental & Social Management

Protect the air, water, land and communities around the project, and keep the social licence to build

- Explain the stages of an environmental and social impact assessment and how its commitments become an ESMP
- Plan practical site controls for dust, noise, water and waste, and for community health and safety
- Apply land acquisition, resettlement and grievance principles, and make carbon-aware project choices

An engineering project changes the place where it is built. Excavation, dewatering, batching plants, haul roads, labour camps and night work affect neighbours long before the asset delivers any benefit, and the finished asset may change flood patterns, traffic or livelihoods for decades. Regulators, lenders and communities therefore expect environmental and social risks to be assessed and managed with the same discipline as cost and schedule. In Bangladesh, the Department of Environment grants environmental clearance under the national Environment Conservation Rules, with the depth of assessment depending on the project's category (check the current rules and category list). Internationally financed projects also follow lender standards such as the IFC Performance Standards (2012), the World Bank Environmental and Social Framework (2018) or the Equator Principles (EP4, 2020) adopted by many banks. Failing here stops work as surely as a missing crane.

Environmental and social impact assessment (ESIA)

01 Screening — Decide whether and how deeply to assess, by project type, scale and location →

02 Scoping — Identify the key issues, study area, alternatives and stakeholders to consult →

03 Baseline — Measure existing air, noise, water, ecology, land use and social conditions →

04 Impacts — Predict impacts in construction and operation and rate their significance →

05 Mitigation and ESMP — Apply the mitigation hierarchy and set measures, owners and monitoring →

06 Consult and disclose — Share findings with affected people, respond to comments, obtain approval →

07 Monitor and adapt — Check compliance and outcomes; update the plan as the project changes

The mitigation hierarchy

M1 Avoid

Change location, route or design so the impact does not occur, such as moving a substation off a wetland or routing an expressway around a village.

M2 Minimise

Reduce the duration, intensity or extent of impacts that cannot be avoided, such as noise barriers, silent piling methods or restricted working hours.

M3 Restore

Repair affected areas after the works, such as reinstating farmland, replanting riverbanks or rebuilding access paths.

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M4

Offset or compensate

As a last resort, provide equivalent benefit elsewhere for residual impacts, such as habitat offsets or fair compensation for lost assets.

The ESIA is a study; the **environmental and social management plan (ESMP)** is how its commitments get done. A good ESMP lists each impact, the mitigation measure, who is responsible, when it applies, how it is monitored and the budget. It must flow into contracts: the contractor prepares a construction ESMP with topic plans for traffic, dust, noise, waste, spill response, labour camps and community engagement, and the tender must price these measures or they will quietly disappear under cost pressure. The owner's team audits compliance, reports to the regulator and lenders, and treats repeated non-compliance as seriously as a quality failure.

Practical site environmental controls

ASPECT	TYPICAL SOURCES ON SITE	CONTROLS	MONITORING
Dust and air quality	Haul roads, stockpiles, cement handling, demolition, generator exhaust	Water spraying, speed limits, covered trucks, wheel washes, enclosed silos, well-maintained engines	Daily visual checks at the boundary; periodic particulate monitoring near sensitive receptors
Noise and vibration	Piling, compaction, breakers, concrete pumps, night works	Low-noise methods such as bored instead of driven piles, acoustic screens, agreed working hours, advance notice to neighbours	Noise readings at schools, hospitals and homes; vibration monitoring and pre-condition surveys of nearby buildings
Water and drainage	Rain runoff with silt, dewatering, concrete washout, wastewater from camps	Silt fences and settlement ponds, washout pits, bunded fuel storage, treated discharge with consent	Turbidity and pH of discharges; inspection after heavy rain
Waste	Formwork offcuts, packaging, spoil, used oil, paint, ETP sludge	Segregate at source, reuse and recycle, licensed carriers, separate storage for hazardous waste	Waste register with quantities, transfer notes and disposal receipts
Spills and soil	Refuelling, transformer oil, chemical storage	Drip trays, bunds sized for the largest container, spill kits, trained responders	Weekly inspections; incident log and clean-up records

Community health and safety covers risks the project creates for people outside the fence. Construction traffic past schools and markets needs routes, timing and marshals; open excavations need fencing and lighting; labour camps need sanitation, health screening and a code of conduct so that the arrival of many workers does not bring disease, conflict or sexual exploitation and abuse. Security arrangements must be proportionate and respectful. Emergency plans for fire, flood and structural failure should reach neighbouring communities as well as the workforce. Engage early and keep engaging: much of the community's view of the project is formed by what happens on the access road every morning.

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Land acquisition and resettlement principles

- Avoid or minimise physical and economic displacement through design and route choices first.
- Compensate for lost assets at full replacement cost, without deduction for depreciation, and pay before people are moved or lose access.
- Restore and preferably improve livelihoods and living standards, including for people who lose income but not land, such as traders or fishers.
- Recognise people without formal title, such as informal settlers and tenants, and provide assistance appropriate to their situation.
- Consult affected people meaningfully, disclose entitlements clearly, and give special attention to vulnerable groups.
- Plan resettlement as a project in its own right, with a budget, schedule, monitoring and independent review.

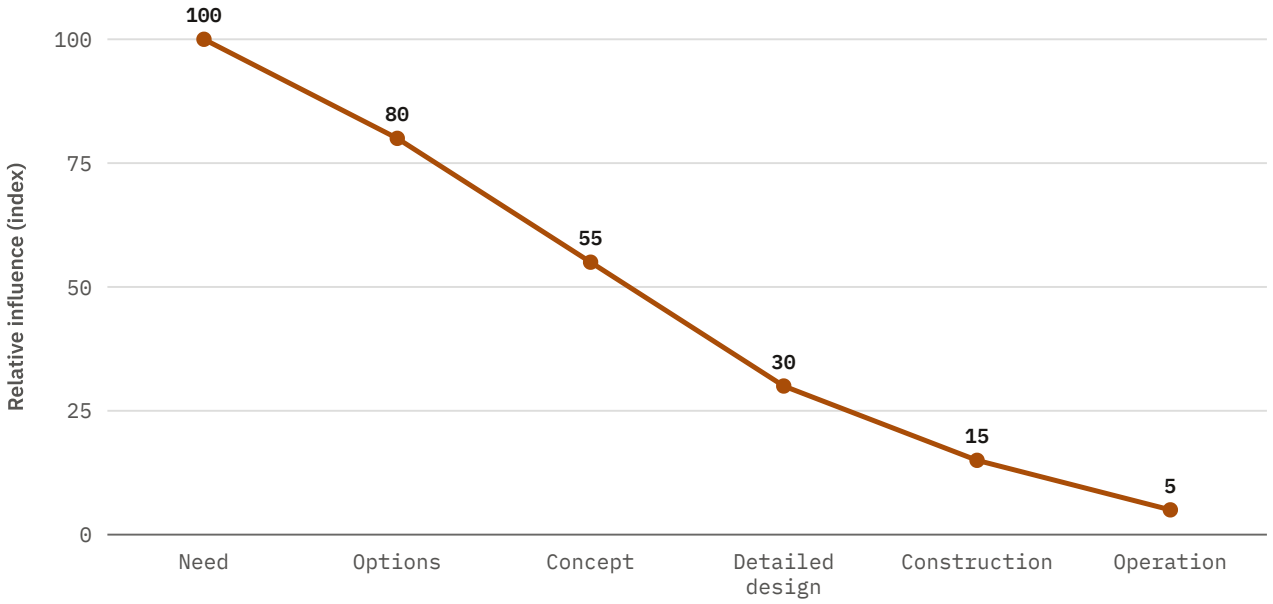
Setting up a grievance mechanism

1. **Design with the community** Agree channels that people actually use: a phone line, a box at the site office, a community liaison officer, local leaders and anonymous options.
2. **Publicise it** Explain in local languages how to raise a concern, what will happen and how long it should take, at no cost to the complainant.
3. **Register and acknowledge** Log every grievance with a reference number and acknowledge it quickly, even if the answer will take longer.
4. **Investigate and resolve** Assign an owner, visit the site, involve the contractor and agree a remedy, such as repairs, compensation or a changed working method.
5. **Close out and confirm** Tell the complainant the outcome, record whether they are satisfied and offer an independent escalation route if not.
6. **Learn from the pattern** Review trends monthly: repeated dust complaints from one village point to a control that is not working.

Carbon is now managed much like cost: many owners set a **carbon baseline** and budget alongside the cost baseline, report against it and ask for a whole-life carbon assessment. PAS 2080:2023 sets out how owners, designers, contractors and suppliers share that duty on buildings and infrastructure. Distinguish **embodied carbon**, from producing, transporting, installing, maintaining and finally removing materials such as cement, steel and aluminium, from **operational carbon** from running the asset. The largest reductions are available early, in the Define stage: build less, reuse existing structures and choose efficient structural forms. Later choices include cement replacement such as slag or fly ash, high-recycled-content steel, offsite and modular construction that cuts waste, electric site plant and grid power instead of diesel generators. Lenders and investors increasingly ask for these figures, and disclosure standards such as IFRS S1 and S2 (2023) shape what owners report. Pair carbon with climate resilience: an asset designed for yesterday's flood levels and temperatures is a risk to everyone who depends on it.

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The ability to reduce carbon falls as the project advances



An illustrative index, not measured data: the largest and cheapest savings are made while the need and options are still open; by construction most of the carbon is already locked into the design.

Whole-life carbon stages (EN 15978)

A1–A3
Product
Raw materials, transport to the factory, manufacture

A4–A5
Construction
Transport to site and the construction process, including waste

B1–B7
Use
Maintenance, repair, replacement; operational energy (B6) and water (B7)

C1–C4
End of life
Demolition, transport, waste processing and disposal

D
Beyond the life cycle
Reuse, recovery and recycling potential, reported separately

Embodied carbon covers A, B1–B5 and C; operational carbon sits in B6–B7. Module D is reported separately so that future recycling credits cannot hide today's emissions.

TEMPLATE

ESMP commitments and monitoring register

Track each environmental and social commitment from the ESIA through to evidence of compliance on site.

IMPACT OR COMMITMENT	MITIGATION MEASURE	RESPONSIBLE	MONITORING AND FREQUENCY	LIMIT OR TARGET	STATUS
Dust from haul road past a primary school	*Water bowser twice daily; 20 km/h limit; covered trucks; no deliveries at school opening and closing times*	*Contract or environment officer*	*Visual boundary check daily; particulate monitoring weekly*	*No visible dust beyond boundary; readings within national ambient standard*	*Ongoing; two complaints closed in May*

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IMPACT OR COMMITMENT	MITIGATION MEASURE	RESPONSIBLE	MONITORING AND FREQUENCY	LIMIT OR TARGET	STATUS
Vibration from compaction near older houses	*Pre-condition survey; low-amplitude rollers within 30 m; vibration monitor at nearest house*	*Site manager*	*Continuous vibration logging; weekly review*	*Below the agreed building damage threshold*	*Survey complete; monitor installed*

SCENARIO · DECIDE · VILLAGERS BLOCK THE ACCESS ROAD

On a new expressway near Narayanganj, villagers have blocked the site access road since dawn. They say vibratory compaction has cracked walls in several houses and that dust from the haul road is making children ill. The contractor's site manager says the works hold a valid environmental clearance and wants to call the police to clear the road so the concrete pour can go ahead. What should the project manager do FIRST?

(A) Support the contractor and call the police; the clearance proves the works are lawful.

(B) Go to the blockade with the community liaison officer, log the complaints in the grievance mechanism, pause compaction near the houses, and commit to an independent condition survey and visible dust controls with a date to report back.

(C) Offer cash to the village leaders to end the blockade today.

(D) Postpone the pour and wait for the villagers to go home before restarting work as normal.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Walk the boundary of a project you know, in your mind. Who lives, works or studies within a few hundred metres, which site activity affects them most, and how would they raise a complaint today?

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Key terms

Environmental and social impact assessment (ESIA)
A structured study that predicts a project's environmental and social impacts during construction and operation, rates their significance and sets mitigation, informed by baseline data and consultation.

Environmental and social management plan (ESMP)
The plan that turns ESIA commitments into specific measures with owners, timing, monitoring and budget, cascaded into contractor plans and contracts.

Mitigation hierarchy
The preferred sequence for managing impacts: avoid first, then minimise, then restore, and only then offset or compensate for what remains.

Grievance mechanism
An accessible, publicised process through which affected people can raise concerns and receive a timely, documented response, with escalation if they are not satisfied.

Involuntary resettlement
Physical or economic displacement caused by land acquisition or restrictions on land use that affected people cannot refuse, requiring compensation and livelihood support.

Livelihood restoration
Measures that restore, and ideally improve, the income and living standards of people whose land, business or access to resources is affected by a project.

Embodied carbon
Greenhouse gas emissions from producing, transporting and installing materials and products, such as cement and steel, as distinct from emissions from operating the asset.

Key takeaways

- The ESIA predicts impacts; the ESMP turns its commitments into priced, owned and monitored contract obligations.
- Apply the mitigation hierarchy in order: avoid, minimise, restore, then offset or compensate.
- Pay full replacement cost before displacement and restore livelihoods, including for people without formal title.
- Run an accessible grievance mechanism and learn from the pattern of complaints.
- Manage carbon as a baseline alongside cost, across all life-cycle stages; the biggest savings are made early in design.

KNOWLEDGE CHECK

21. A proposed 132 kV substation site sits on a seasonal wetland used by local fishers. Under the mitigation hierarchy, what should be considered first?

(A) Paying compensation to the fishers for lost catches
(B) Restoring the wetland after construction is finished
(C) Moving the substation to a site that avoids the wetland
(D) Building silt fences and settlement ponds around the site

22. Which feature is essential for a grievance mechanism to be effective?

(A) It is accessible and publicised in local languages, with logged, timely responses
(B) It is run only by the contractor so that the owner is not distracted
(C) It only accepts signed written complaints so that every case has a clear paper trail
(D) It deals only with complaints from people who hold formal land titles

23. Under good-practice resettlement principles, when should compensation for acquired homes be paid?

(A) After the asset is commissioned, from the project's operating revenue
(B) Within a year of displacement, once the final land survey is complete
(C) Only after any legal challenge to the land valuation has been resolved
(D) Before people are moved, at full replacement cost without depreciation

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24. At which point does a project usually have the greatest opportunity to reduce embodied carbon?

(A) During construction, by switching off idle plant, site lighting and site cabins

(B) Early in design, by building less and choosing efficient structural forms

(C) At handover, by buying carbon credits to cover the materials used

(D) During operation, by improving maintenance practices on the asset

25. What is the relationship between an ESIA and an ESMP?

(A) The ESMP is the regulator's approval letter for the ESIA report

(B) They are alternative names for the same document used by different lenders

(C) The ESIA assesses impacts; the ESMP turns its commitments into owned, monitored actions

(D) The ESIA covers the construction stage only, while the ESMP covers only the operation phase

26. A carbon assessment for a new bridge must report the emissions from manufacturing its reinforcing steel. In which EN 15978 stage do they belong?

(A) B6, operational energy use

(B) A1–A3, the product stage

(C) Module D, beyond the life cycle

(D) A5, the construction process

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 5 · PROTECT

Answer key

Knowledge checks

1. C The event has already happened, so it is an issue to be decided and acted on now. It may have been a risk before excavation; once it occurs, it moves to the issues log.

2. A Only the first names a present cause, an uncertain event and an effect on an objective. The others are vague concerns or judgements that give no clue where to act.

3. D An RBS is a prompt list and grouping structure. It does not quantify risks, allocate them contractually or replace the WBS.

4. B It is an uncertain event with a positive effect, so it is an opportunity. Recording and owning it lets the team act to make it more likely.

5. C Silent generation captures independent views before social pressure shapes them. It is about quality of identification, not speed or list length.

6. B Exposure = 0.25 x 800k = USD 200k. USD 600k is the impact remaining if you wrongly subtract, and the others do not match the formula.

7. A Scores of 1-5 are category labels, so multiplying them gives a ranking aid, not a measure. Quantitative exposure in money or time is needed for real comparisons.

8. D The financial consequence of exchange-rate movement is moved to a bank under the forward contract, which is transfer. The underlying purchase still happens, so it is not avoidance.

9. C Net value = (300k - 120k) - 50k = USD 130k. USD 180k ignores the response cost; the other figures do not follow the formula.

10. D Secondary risks arise directly from implementing a response. Residual risk is what remains of the original risk after the response.

11. A EMV = 0.30 x (-400k) + 0.50 x (+100k) = -120k + 50k = -USD 70k. The other answers ignore the opportunity or subtract impacts without weighting them by probability.

12. C A = 200 + 0.2 x 50 = 210; B = 190 + 0.4 x 60 = 214. Option A is lower on expected cost, even though B has the lower base price.

13. B P80 is the date that 80% of iterations meet or beat. It says nothing directly about the deterministic date, and it is a cumulative value, not a prediction of an exact finish date.

14. D The milestone needs all three paths on time: 0.6 x 0.6 x 0.6 = 0.216, about 22%. This is merge bias, which a single CPM calculation does not show.

15. A Independent sampling lets a bad draw on one activity cancel a good draw on the other, narrowing the distribution and making high-confidence values look earlier and cheaper than they are.

16. D Ground-level assembly removes most work at height, which is elimination. Better harnesses are PPE, and supervision and talks are administrative controls, all lower in the hierarchy.

17. B A permit controls high-risk work by confirming conditions and isolations before work and managing hand-back afterwards. It complements RAMS rather than replacing them.

18. C Closing near-miss reports quickly is proactive activity that prevents future harm. The other measures count harm after it has occurred, so they are lagging.

19. A Isolation must be proved at the work point with a proven instrument. Verbal confirmation, drawings and waiting do not detect back-feeds or wrong isolations.

20. B Change of use and added load require a competent structural assessment and authority approval before design is fixed. Drills, insurance and reduced hours do not make an overloaded structure safe.

21. C Avoidance comes first: if a feasible alternative site exists, the impact need not happen. Silt fences minimise, restoration repairs and compensation are later steps.

22. A People must know about it and be able to use it easily, and every grievance must be logged and answered. Restricting channels or eligibility excludes the people most affected.

23. D Compensation should be paid before displacement and at replacement cost, so that people can re-establish themselves. Paying later or with depreciation leaves them worse off.

24. B Early design decisions fix most of the material quantities and types. Site and operational measures matter, but they cannot undo a heavy design, and credits do not reduce emissions.

25. C The ESIA predicts and evaluates impacts; the ESMP is the management plan that implements the mitigation with responsibilities, monitoring and budget across construction and operation.

26. B Manufacturing materials is the product stage, A1–A3. A5 covers site construction activity, B6 the energy used to run the asset, and module D reports future reuse and recycling potential separately.

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Scenarios & case studies

Scenario · A register written for the lender, not for the site
B is best. Best. You separate issues that need decisions today from risks that need planned responses, make the most important risks actionable, and keep the sponsor and lender properly informed with a short, honest delay.

Scenario · A low score on a lethal risk
B is best. Best. Severity-5 safety risks need specific, owned controls regardless of score. Avoiding the exposure, for example by de-energising the bay or relocating the crane, is the strongest response.

Scenario · Which date goes on the plaque?
B is best. Best. The public commitment matches the owner's appetite for being wrong, while the team still works to a challenging internal target and attacks the risks that drive the spread.

Case study · The risk analysis nobody believed
The distribution was narrow because the ranges were symmetrical and tiny, the hard constraints stopped delays from propagating, discrete risks such as vessel availability and weather were absent, and independent sampling let overruns and underruns cancel. The analyst should have used a summarised but logic-sound schedule, removed or justified constraints, set skewed ranges from evidence and expert challenge, modelled key risk events with their probabilities, and linked inputs with common drivers such as weather and vessel performance. The committee should have seen the S-curve with P50 and P80, the deterministic date's own probability, the tornado drivers, and a clear statement of what was and was not modelled. A result that looks surprisingly comfortable on a first-of-a-kind project is itself a warning sign to investigate.

Case study · Lessons from Bangladesh's garment sector
The design failures were building and extending structures beyond their permitted and designed use, placing industrial loads and vibrating plant on floors not designed for them, and laying out exits and storage so that stairs could become fire traps. The management failures were ignoring visible structural distress and alarms, and ordering people back into danger. Before fixing the design for any extension or change of use, a competent structural engineer must assess the existing structure, foundations and load paths against the new loads and vibration, with permits obtained from the relevant authority; fire engineering must provide protected escape routes, fire separation and detection sized for the new occupancy. Useful leading indicators include monthly checks that exits and stairs are clear and unlocked, alarm and fire-pump test records, evacuation drill times, structural load plans displayed and respected, and the speed at which reported cracks or electrical faults are investigated. The deepest lesson is that a building's structural and fire safety is a design and governance matter, not a matter for workers to manage on the day.

Scenario · Wind rising during a girder lift
B is best. Best. The lift plan's limit is a critical control. Stopping, making safe and rebooking the closure protects people and still lets the work proceed later; the delay is a schedule problem to solve, not a reason to take the risk.

Scenario · Villagers block the access road
B is best. Best. You listen, register the grievance, stop the likely cause while it is investigated, and make specific, time-bound commitments. This addresses the real problem and is the fastest credible route to reopening the road.

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06

MODULE 6 · PROCURE

Procurement, Contracts, Claims & Variations

Buy the right capability, allocate risk deliberately and manage change and claims without losing the project to disputes

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design a procurement strategy with make-or-buy, packaging, delivery model and long-lead decisions that fit the owner's objectives and capability
- Select contract pricing mechanisms and allocate risks to the party best able to manage them, calculating payments under each
- Run a fair tender and evaluate bids with normalised prices, weighted scoring and sound integrity controls
- Apply the shared principles of international standard-form contracts: the engineer's role, notices, securities, payments, taking over and dispute boards
- Manage variations, extensions of time and claims with sound records and resolve disputes at the lowest effective level

LESSONS

1. Procurement Strategy & Delivery Models · 34 min
2. Contract Pricing Types & Risk Allocation · 35 min
3. Tendering & Bid Evaluation · 36 min
4. International Standard-Form Principles · 38 min
5. Variations, Claims & Disputes · 40 min

MODULE 6 · PROCURE · LESSON 1 · 34 MIN

Procurement Strategy & Delivery Models

Decide who designs, who builds, how the work is split and what to buy first

- Build a procurement strategy from objectives, market conditions and owner capability
- Compare design-bid-build, design-build, EPC/turnkey, management contracting, construction management and alliancing
- Make packaging and long-lead procurement decisions that protect the critical path

Long before a contract is signed, the **procurement strategy** has already fixed most of the project's commercial risk. It decides who is responsible for design, who builds, how many contracts there will be and where their boundaries fall, how each party will be paid and which items must be bought before the main contractor is even appointed. Get it right and the owner buys the capability it lacks at a fair price; get it wrong and the project spends years managing interface gaps, claims and disputes that were designed into the structure on day one. The strategy is an owner decision, approved at a stage gate (see Justify), but contractors and consultants need to understand it because it shapes their own risk.

Building a procurement strategy

1. **Confirm objectives and constraints** Rank what matters most: earliest completion, lowest cost, price certainty, quality of a complex system, local participation or owner control. You cannot maximise all of them at once.
2. **Assess the owner's own capability** Be honest about how much design management, site supervision and interface coordination your organisation can do. Many strategies fail because they assume an owner team that does not exist.
3. **Analyse the market** Find out how many capable firms exist for each package, how busy they are, typical lead times and whether they will accept the risks you plan to give them.
4. **Decide make-or-buy and packaging** Choose what to do in-house and how to split the remainder into packages with clear boundaries and manageable interfaces.
5. **Select the delivery model and pricing** Match the model to design certainty, speed and risk appetite, then select the pricing mechanism (next lesson).
6. **Plan the procurement schedule** Work back from the need-by dates on the project schedule, allowing time for tendering, evaluation, approvals, manufacture and delivery, with long-lead items first.
7. **Obtain approval and record the reasons** Document the strategy and the alternatives rejected, so that later decisions and audits can see why the structure was chosen.

> DEFINITION · MAKE OR BUY
>
> A **make-or-buy** decision asks whether the owner should perform an activity with its own resources or buy it from the market. A power utility with a strong transmission design department may design substations in-house and buy only construction; a garment group with no engineering staff will buy design, supply and installation together. Consider cost, capability, capacity at the time needed, control of critical know-how and long-term strategy. A cheaper in-house option that depends on two key engineers who are already overloaded is not really cheaper.

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Delivery models compared

MODEL	WHO DESIGNS	WHO BUILDS	OWNER'S ROLE	STRENGTHS	WATCH-OUTS
Design-bid-build (DBB)	Owner's consultant completes design before tender	Contractor builds to that design	High control; owner carries design risk	Clear competitive pricing on a full design; familiar to public owners	Slow, sequential; design errors become variations; little contractor input on buildability
Design-build (DB)	Contractor develops design from owner's requirements	Same contractor	Defines requirements and reviews	Single point of responsibility; overlaps design and construction; buildability built in	Quality depends on how well requirements are written; less owner control of detail
EPC / turnkey	Contractor: engineering, procurement and construction	Same contractor, through to commissioning	Sets performance requirements; limited intervention	Price and date certainty for performance-defined plant such as power or process	High risk premium; few bidders for large packages; owner must accept less influence
Management contracting	Owner's designers, often in stages	Works contractors employed by the management contractor	Close involvement in design and packages	Early start, phased packages, expert construction management	Cost certainty arrives late; owner still carries much of the risk
Construction management	Owner's designers	Trade contractors contracted directly to the owner	Many direct contracts; CM is the owner's adviser	Transparency, speed, owner control of each package	Owner carries interface and coordination risk across many contracts
Alliancing / collaborative	Joint team of owner, designers and contractors	Joint team	Active member of the alliance board	Shared incentives, open-book costs, fast joint problem-solving on complex or uncertain work	Needs mature, trusting parties and strong governance; weak cost discipline if targets are soft

Modern strategies often blend these models. **Early contractor involvement (ECI)** brings a contractor in during design, usually under a pre-construction agreement, to advise on buildability, sequence, offsite and DfMA options and cost. **Two-stage tendering** selects that contractor first on quality, fees and key rates, then agrees the construction price once the design has matured. **Framework agreements** appoint a pool of firms for a programme of similar works, and **alliances** go furthest by sharing pain and gain across the whole team. Whatever the model, write the owner's information requirements (under ISO 19650), carbon targets and any offsite intentions into the strategy so that bidders can price them.

Packaging splits the scope into contracts. Fewer, larger packages reduce the number of interfaces the owner must manage but narrow the market to large firms and may increase risk premiums. More, smaller packages widen competition and can involve capable local contractors, but every boundary becomes a place where responsibility can fall through: who connects the transformer to the switchgear, who provides the crane for the vendor's supervisor, who seals the cable entry through the civil contractor's wall. On a metro line, a typical split separates viaduct and station civil works, track, power supply, signalling, rolling stock and depot; the

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owner or a systems integrator then has to manage the interfaces between them deliberately, with interface matrices, joint programmes and clear battery limits (see Define).

Few large packages or many small ones?

Few large packages

- Fewer interfaces for the owner to coordinate
- Single point of responsibility for integrated systems
- Smaller owner team required
- Fewer bidders and higher risk premiums
- Harder for local firms to participate as main contractors

Many smaller packages

- Wider competition and specialist expertise in each package
- Earlier starts by tendering packages in sequence
- More opportunities for local contractors and suppliers
- Many interfaces for the owner to define and police
- Delay or failure in one package can stall several others

Six questions for choosing a delivery model

Q1

How complete is the design?

A finished design suits design-bid-build; a performance specification suits design-build or EPC; an evolving solution suits collaborative models.

Q2

How urgent is completion?

Overlapping design and construction through design-build, management or construction management saves time but delays cost certainty.

Q3

What can the owner manage?

Construction management and many packages need a strong owner team; EPC needs far less day-to-day capacity.

Q4

Who should carry which risks?

Transfer risks to parties who can control them; transferring everything buys certainty at a high premium.

Q5

What will the market accept?

If capable bidders refuse the risk profile, the tender fails or attracts the wrong firms.

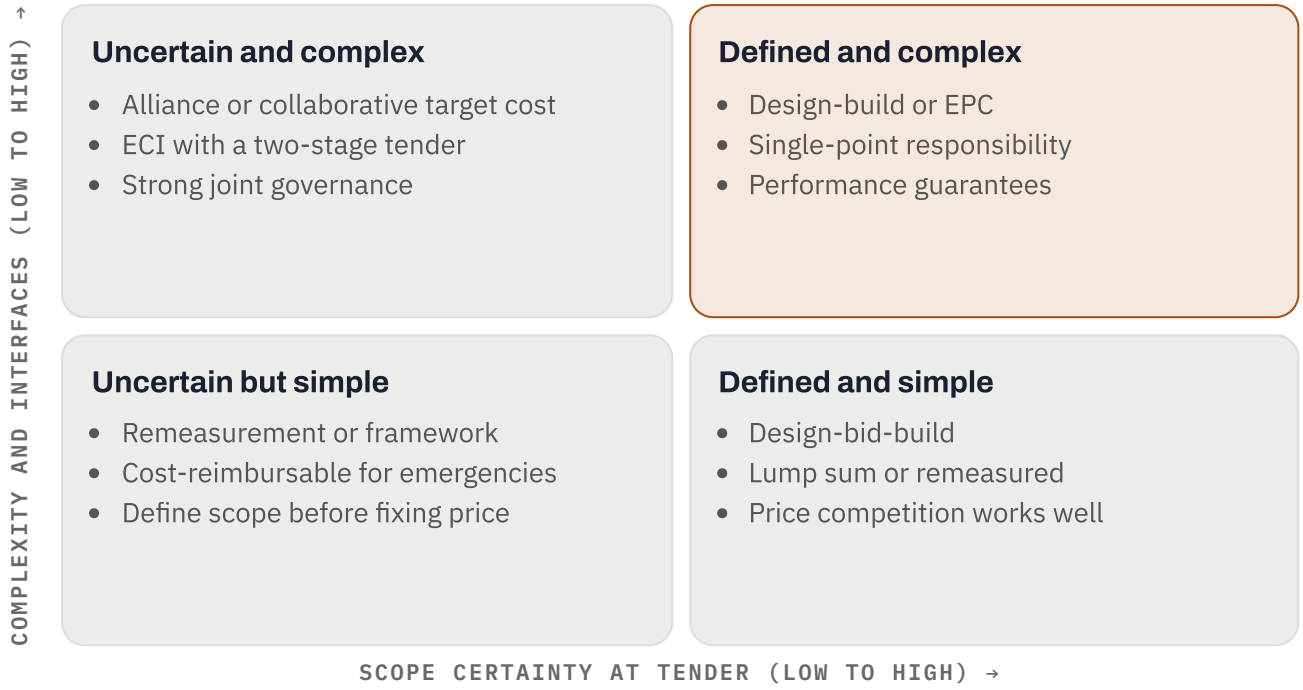
Q6

How complex are the interfaces?

Tightly integrated systems, such as process plant, favour single-point responsibility or an alliance.

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A first filter: scope certainty and complexity



Use it to shortlist, not to decide: owner capability and market appetite (Q3 and Q5) can still move the answer.

EXAMPLE · LONG-LEAD ITEMS DRIVE THE STRATEGY

Some equipment takes longer to design, manufacture and deliver than the civil works take to build: large power transformers, gas-insulated switchgear, gas and steam turbines, tunnel boring machines, large pumps and valves, HV cable and specialist process packages such as dyeing machines or ETP membranes. Owners often reserve factory slots or place these orders before the main contractor is appointed, then either **free-issue** them to the contractor or **novate** the supply contract so the contractor takes it over. Free-issue keeps control but leaves the owner liable if the item arrives late or damaged; novation gives single-point responsibility but contractors will price the risk of a supplier they did not choose.

A practical sequence on a hypothetical combined-cycle plant illustrates the logic: the owner reserves the gas turbine slot at the end of feasibility, appoints an owner's engineer to write the employer's requirements, tenders an EPC contract with the turbine supply agreement attached for novation, and keeps the grid connection works as a separate package with the transmission utility. Each choice is deliberate: the turbine protects the critical path, the EPC contract gives one party responsibility for plant performance, and the grid works stay with the only organisation that can deliver them. Record such reasoning in the strategy so the next project can learn from it.

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Contract structure for the combined-cycle example



The turbine starts as an owner contract and moves under the EPC contractor by novation, while the grid works stay outside the EPC boundary, leaving an interface the owner must manage deliberately.

SCENARIO · DECIDE · CHOOSING A MODEL FOR A FAST-TRACK FACTORY EXPANSION

A textile group in Gazipur wants a new dyeing hall, eight dyeing machines and an upgraded effluent treatment plant operating within 18 months to meet a buyer's sustainability deadline. The group has two facility engineers and no design department. The concept design is incomplete, but the required production capacity and discharge standards are clear. The chairman asks for design-bid-build 'so we keep control'. What should the project manager recommend?

(A) Design-bid-build, completing the full design with a consultant before tendering construction.

(B) Design-build against clear performance requirements, with the owner ordering the dyeing machines early, an owner's engineer reviewing design, and defined hold points for the owner's approval.

(C) Construction management, with the group contracting each trade directly to stay in control.

(D) An EPC lump-sum contract tendered immediately on the current incomplete concept.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Look at a project you know well. List its contract packages and draw the interfaces between them. Which interface caused the most trouble, and would a different packaging decision have avoided it?

Key terms

Make-or-buy decision
The choice between performing an activity with the organisation's own resources or buying it from the market, based on cost, capability, capacity, control and strategy.

Contract packaging
The division of a project's scope into separate contracts, balancing competition and specialisation against the number and difficulty of interfaces.

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Design-bid-build (DBB)
A delivery model in which the owner's consultant completes the design, contractors bid on it, and the selected contractor builds it, with the owner carrying design risk.

EPC (turnkey) contract
A contract in which one contractor is responsible for engineering, procurement, construction and commissioning, handing over a working facility that meets defined performance requirements.

Long-lead item
Equipment or material whose design, manufacture and delivery time is long enough to drive the schedule, so it must be ordered early, often before the main contractor is appointed.

Design-build (DB)
A delivery model in which one contractor is responsible for both design development and construction against the owner's requirements.

Alliancing
A collaborative delivery model in which owner, designers and contractors work as one team under a shared agreement, with open-book costs and shared incentives tied to project outcomes.

Key takeaways

- The procurement strategy fixes most commercial risk before any contract is signed; record the reasons and the options rejected.
- Match the delivery model to design certainty, urgency, owner capability, risk allocation, market appetite and interface complexity.
- Fewer packages mean fewer interfaces but a narrower market; more packages need an owner team that can police every boundary.
- ECI, two-stage tendering, frameworks and alliances bring contractor knowledge in early when scope is still developing.
- Order long-lead items early, choosing free-issue or novation with eyes open to who carries late delivery.

KNOWLEDGE CHECK

1. An owner has a complete, well-checked design for a district road bridge and wants transparent competition on price. Which delivery model fits best?
   - (A) Alliancing with a joint owner-contractor team
   - (B) Design-bid-build with the finished design
   - (C) EPC turnkey with performance requirements
   - (D) Construction management with trade packages
2. What is the main drawback of splitting a metro project into many small contract packages?
   - (A) The project will be unable to attract any international contractors
   - (B) Contract prices always rise because local firms charge more than large ones
   - (C) Designers lose responsibility for the quality of their drawings
   - (D) The owner must define and manage many more interfaces between packages
3. Why might an owner reserve a gas turbine factory slot before appointing the EPC contractor?
   - (A) Because the turbine's lead time would otherwise drive the completion date
   - (B) Because EPC contractors are not permitted to buy major rotating equipment
   - (C) Because an early order automatically transfers all supply risk to the vendor
   - (D) Because turbine prices are fixed by law once a factory slot is booked

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4. An owner free-issues transformers to a contractor instead of novating the supply contract. What risk does the owner keep?

   (A) The contractor's risk of poor workmanship during installation

   (B) The risk that the contractor's own crane breaks down on site

   (C) The risk of late or defective delivery of the transformers

   (D) The contractor's risk of low productivity on cable termination

5. Which situation most strongly favours an alliancing or collaborative model?

   (A) A simple steel-framed warehouse with a complete, checked design and many interested bidders

   (B) A complex rail upgrade under live operations with uncertain scope and many interfaces

   (C) A standard pumping station repeated from an earlier, successful project

   (D) A small fit-out where the owner wants the lowest fixed price possible

6. A hospital owner wants a contractor's buildability and offsite advice during design, but cannot fix a construction price until the design matures. Which approach fits best?

   (A) A single-stage lump-sum tender on the concept design

   (B) Cost plus a percentage fee for the whole of the works

   (C) Design-bid-build once every drawing is complete

   (D) Early contractor involvement with a two-stage tender

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 6 · PROCURE · LESSON 2 · 35 MIN

Contract Pricing Types & Risk Allocation

Lump sum, remeasurement, cost-reimbursable, target cost and GMP, and who really carries each risk

- Explain how each main pricing mechanism pays the contractor and which party carries quantity, productivity and price risk
- Calculate payments under remeasurement, cost-plus, target cost and guaranteed maximum price arrangements
- Apply the principle of allocating each risk to the party best able to manage it

The pricing mechanism decides how the contractor is paid, and therefore who wins or loses when quantities grow, productivity falls or materials become more expensive. Every tender price contains three parts: the contractor's estimate of cost, an allowance for the risks it is asked to carry (the **risk premium**) and its margin. Transferring a risk to the contractor does not make it disappear; the owner pays for it through the premium, whether or not the risk occurs. Good contract strategy therefore asks not 'how do we push all risk to the contractor?' but 'which party can manage each risk at the lowest overall cost?'

What a tender price is made of



Changing the risk allocation moves mainly the risk premium, and the owner pays that premium whether or not the risks occur.

The main pricing mechanisms

PRICING TYPE	HOW THE CONTRACTOR IS PAID	QUANTITY RISK	PRODUCTIVITY AND PRICE RISK	BEST SUITED TO
Lump sum	A fixed price for a defined scope, usually paid against milestones or activity schedules	Contractor	Contractor	Complete designs or clear performance requirements; turnkey plant
Remeasurement (unit rate)	Measured quantities multiplied by tendered rates from a bill of quantities or schedule of rates	Owner	Contractor (within each rate)	Civil and infrastructure works where quantities cannot be fixed, such as earthworks and piling
Cost-reimbursable	Actual allowable cost plus a fee, either a percentage of cost or a fixed amount	Owner	Owner	Emergency works, undefined scope, research and first-of-a-kind work

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PRICING TYPE	HOW THE CONTRACTOR IS PAID	QUANTITY RISK	PRODUCTIVITY AND PRICE RISK	BEST SUITED TO
Target cost	Actual cost plus fee, adjusted by a pain/gain share of the difference between target and actual cost	Shared	Shared	Complex work with some uncertainty where collaboration on cost pays off
Guaranteed maximum price (GMP)	Actual cost plus fee up to a ceiling; contractor bears cost above it; savings often shared	Mostly contractor above the ceiling	Mostly contractor above the ceiling	Buildings with well-developed design where the owner wants a cost ceiling with transparency

Payment formulas by pricing type

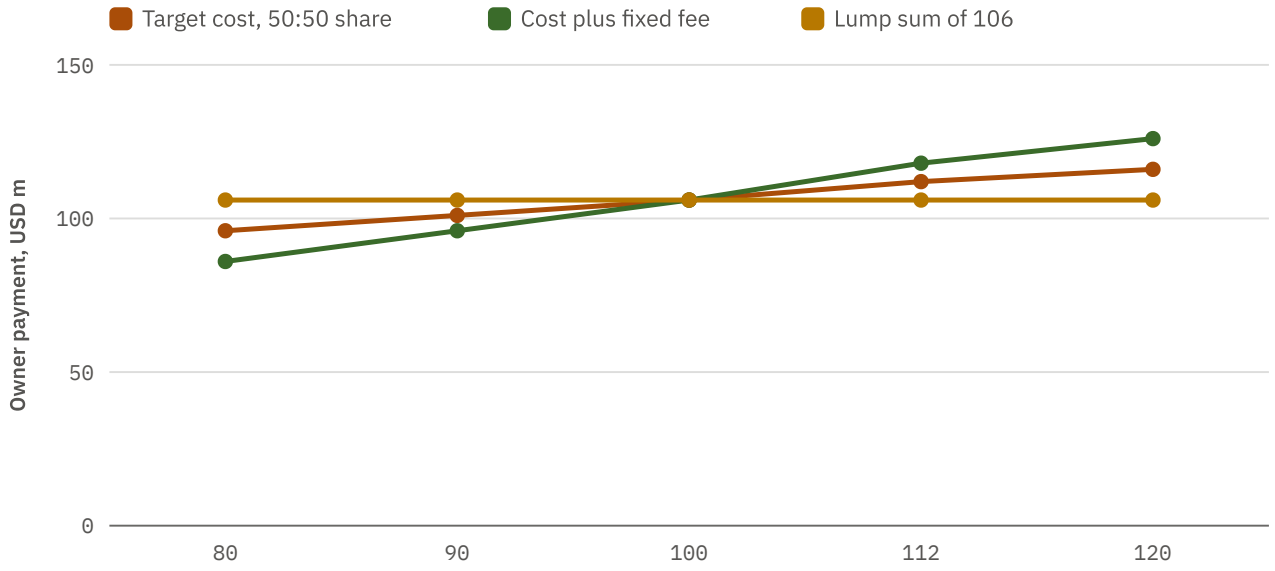
MEASURE	FORMULA	WHAT IT TELLS YOU
Remeasurement payment	`Payment = sum of (measured quantity_i x tendered rate_i) + valued variations`	If piles measure 460 m against 400 m in the bill at a rate of 5,000 per metre, the pile item is paid at 460 x 5,000 = 2,300,000.
Cost plus percentage fee	`Payment = actual cost x (1 + fee %)`	The fee grows as cost grows, so the contractor has no financial reason to save money; use only with strong controls or in emergencies.
Cost plus fixed fee	`Payment = actual cost + fixed fee`	The fee does not grow with cost, which removes the perverse incentive, but cost risk still sits with the owner.
Target cost with pain/gain share	`Payment = actual cost + fee + contractor share x (target cost - actual cost)`	Underspend gives the contractor a bonus; overspend reduces its fee. Contracts often cap the contractor's pain share, for example at the fee.
Guaranteed maximum price	`Owner pays the lower of (actual cost + fee) and the GMP, plus any agreed share of savings below it`	Costs above the GMP fall on the contractor unless the scope changes; the ceiling is adjusted only through agreed changes.
Price adjustment (indexation)	`Pn = a + b x (Ln / Lo) + c x (Mn / Mo) + ..., where a + b + c + ... = 1`	a is the fixed, non-adjustable share; each other weighting multiplies the ratio of a current index (n) to its base-date value (o), for labour, steel, cement or fuel. Payments are multiplied by Pn. Illustrative: a = 0.15, b = 0.35, c = 0.50, with indices up 10% and 20%, gives Pn = 0.15 + 0.385 + 0.60 = 1.135.

EXAMPLE · TARGET COST IN NUMBERS (ILLUSTRATIVE)

A hypothetical water treatment works has a target cost of USD 100m, a fixed fee of USD 6m and a 50:50 pain/gain share. **Underspend:** actual cost USD 90m. Gain = 100 - 90 = 10, contractor share = +5. Payment = 90 + 6 + 5 = **USD 101m**; the owner saves 5 against target plus fee (106), and the contractor earns 11 instead of 6. **Overspend:** actual cost USD 112m. Pain = 100 - 112 = -12, contractor share = -6. Payment = 112 + 6 - 6 = **USD 112m**; the owner pays 6 more than target plus fee and the contractor recovers its cost but earns no margin. Both parties feel every dollar of variance, which is the point.

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What the owner pays as actual cost varies



Illustrative, using the callout's target of 100 and fee of 6 (actual cost in USD m along the bottom). The target-cost line is half as steep as cost-plus, so each party feels half of every variance; the lump sum stays flat until change or claims move it.

WATCH OUT · FIXED PRICE IS NOT THE SAME AS CERTAIN COST

A lump sum only fixes the price of the scope as described. Every design change, late owner information, change in law or unforeseeable condition that the contract assigns to the owner becomes a variation or claim. Owners who tender lump sums on immature designs often end with a final account far above the contract price, plus disputes. Conversely, a contractor that accepts a lump sum for risks it cannot control, such as ground conditions with no site investigation, may be pushed towards insolvency, which also hurts the owner.

INSIGHT · VOLATILE PRICES CHANGED THE CONVERSATION

Sharp swings in steel, cement, fuel and freight prices in the early 2020s left contractors on long fixed-price contracts without indexation facing heavy losses, and some failed. Owners now more often share this risk through price-adjustment formulas tied to published indices, keeping a fixed portion and sometimes a cap so that budget certainty is not lost. Price the residual exposure in the owner's risk allowance rather than pretending it has gone.

Who typically carries each risk under the main pricing types

RISK	LUMP SUM	REMEASUREMENT	COST-REIMBURSABLE	TARGET COST
Quantities greater than estimated	Contractor	Owner	Owner	Shared via pain/gain
Low productivity or poor planning	Contractor	Contractor	Owner	Shared via pain/gain
Errors in the owner's design	Owner	Owner	Owner	Owner, through target adjustment

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RISK	LUMP SUM	REMEASUREMENT	COST-REIMBURSABLE	TARGET COST
Unforeseeable ground conditions	Depends on contract terms; often owner	Owner (quantities) and contract terms	Owner	Usually owner, through target adjustment
Inflation in labour and materials	Contractor unless a price-adjustment formula applies	Contractor unless a price-adjustment formula applies	Owner	Shared, or adjusted by formula

Mixing mechanisms inside one contract

- **Provisional sums** hold money for work that cannot yet be defined, such as utility diversions, and are spent only on instruction.
- **Daywork schedules** price small unforeseen tasks by labour and plant hours at tendered rates.
- **Price-adjustment formulas** link payments to published indices for steel, cement, fuel and labour on long contracts.
- **Milestone or activity schedules** turn a lump sum into payments that track real progress rather than time.
- **Reimbursable elements** such as specialist vendor supervision can sit inside an otherwise lump-sum EPC contract.
- **Incentives** for early completion, performance guarantees and liquidated damages for delay or shortfall in plant output align commercial results with project objectives.

CASE STUDY · PAIN, GAIN AND A CAP

A hypothetical municipal water treatment works is let under a target cost contract. The target is USD 80m, the fixed fee is USD 4m, the pain/gain share is 50:50, and the contractor's pain share is capped at the value of its fee. Costs are recorded on an open-book basis and audited monthly. The target can be adjusted only for owner-instructed changes and defined owner risk events.

1. What does the owner pay if the actual cost is USD 72m?
2. What does the owner pay if the actual cost is USD 92m, and who bears the overrun?
3. What must be in place for this mechanism to drive collaboration rather than arguments?

Model answer: module answer key.

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SCENARIO · DECIDE · A LUMP SUM FOR AN UNKNOWN BUILDING

A public owner wants to convert and extend a 1970s hospital block in Chattogram. Drawings of the existing structure are missing, the extent of concrete deterioration is unknown, and the new wing has a well-developed design. The finance department insists on a single lump-sum contract 'so the budget cannot move'. What should the project manager advise?

(A) Proceed with one lump sum; contractors will price the unknowns and the budget will be fixed.

(B) Commission intrusive surveys now, then use a lump sum for the defined new wing and a remeasured or target-cost arrangement with provisional sums for the refurbishment.

(C) Use cost plus percentage fee for everything, since the existing structure is uncertain.

(D) Postpone the project until a complete as-built survey of every element is available.

Circle your choice. Answer and feedback: module answer key.

REFLECT

For a contract you have worked under, list the three biggest risks the contractor carried. Which of them could the contractor genuinely control, and what do you think it cost the owner in risk premium?

Key terms

Lump sum contract
A contract in which the contractor agrees to deliver a defined scope for a fixed price, carrying quantity and productivity risk, with adjustments only for changes and risks the contract assigns to the owner.

Cost-reimbursable contract
A contract in which the contractor is paid its actual allowable costs plus a fee, leaving most cost risk with the owner.

Pain/gain share
The agreed proportions in which owner and contractor split cost overruns (pain) and underruns (gain) against a target, often with caps on the contractor's pain.

Risk premium
The amount a contractor adds to its price to cover risks it is asked to carry; the owner pays it whether or not the risks occur.

Remeasurement contract
A contract in which the work is measured as executed and paid at tendered unit rates, so the owner carries quantity risk while the contractor carries the risk within each rate.

Target cost contract
A contract in which the contractor is paid actual cost plus fee, adjusted by a pre-agreed share of the difference between an agreed target cost and the actual cost.

Guaranteed maximum price (GMP)
A pricing arrangement in which the owner reimburses cost plus fee up to an agreed ceiling, the contractor bears cost above it, and savings below it are often shared.

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Key takeaways

- Every tender price contains cost, a risk premium and margin; transferring risk is paid for whether or not the risk occurs.
- Lump sum suits defined scope; remeasurement suits uncertain quantities; cost-reimbursable suits undefined or emergency work.
- Target cost and GMP share or cap overruns; both need open-book access, clear allowable costs and prompt target adjustment.
- Index-linked price adjustment shares inflation risk on long contracts without giving up budget control.
- Allocate each risk to the party best able to manage it at the lowest overall cost.

KNOWLEDGE CHECK

7. Under a remeasurement contract, bored piles are billed at 600 m but 680 m are installed as instructed, at a rate of 4,500 per metre. What is the pile payment?

   (A) 2,700,000, the billed quantity at the tendered rate
   (B) 2,880,000, the billed quantity plus a 6.7% allowance
   (C) 3,060,000, the measured quantity at the tendered rate
   (D) 3,400,000, the measured quantity at a revised fair rate

8. Why is cost plus percentage fee generally avoided except in emergencies?

   (A) The fee rises as cost rises, so the contractor has no incentive to control cost
   (B) It requires the contractor to carry all of the quantity and productivity risk on site
   (C) It is illegal for public owners to reimburse contractor costs directly
   (D) It prevents the owner from instructing any changes to the scope

9. Under a target cost contract with a 40% contractor share, the target is 50m, fee 3m and actual cost 55m with no cap. What is the payment?

   (A) 58m, actual cost plus the full fee
   (B) 53m, the target plus the full fee
   (C) 60m, the actual cost plus fee and share
   (D) 56m, actual cost plus fee less 2m pain

10. Which principle should guide how risks are allocated between owner and contractor?

    (A) Every risk should be transferred to the contractor so that the owner's budget becomes certain
    (B) Each risk should sit with the party best able to control it and bear it at the lowest cost
    (C) Risks should be split equally between the parties so the contract is seen as fair
    (D) Risks should stay with the owner because contractors always overprice them

11. Under a guaranteed maximum price of USD 40m with no scope changes, the contractor's actual cost plus fee reaches USD 43m. What does the owner pay?

    (A) USD 40m, with the USD 3m excess carried by the contractor
    (B) USD 43m, because the actual cost was properly incurred
    (C) USD 41.5m, with the excess split equally between the parties
    (D) USD 40m plus any financing costs the contractor incurred

12. A contract adjusts payments by Pn = 0.2 + 0.3 x (Ln / Lo) + 0.5 x (Mn / Mo). Since the base date the labour index has risen 10% and the materials index 20%. What is Pn?

    (A) 1.15, the simple average of the two rises
    (B) 1.30, adding both index rises in full
    (C) 1.13, the weighted sum
    (D) 1.03, ignoring the materials index rise

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 6 · PROCURE · LESSON 3 · 36 MIN

Tendering & Bid Evaluation

Run a fair competition, evaluate on value, and award a contract you can defend

- Plan a tender process from prequalification to contract award
- Evaluate tenders technically and commercially using normalised prices and weighted scoring
- Recognise abnormally low tenders, conflicts of interest and other integrity risks, and respond correctly

Tendering has three aims: obtain a competitive price, select a contractor that can actually deliver, and do both through a process that every bidder, auditor and lender can see was fair. Public owners must follow procurement law; in Bangladesh, public procurement is governed by national legislation and rules and much of it runs through the national electronic government procurement system, while projects financed by development banks also follow the lender's procurement regulations. Private owners have more freedom but face the same practical truth: a rushed or opaque tender produces weak competition, poor contractors and contracts that are fought over for years.

The tender process

01 Prequalify — Shortlist firms with the capability and capacity to deliver

→

02 Issue documents — Complete tender package with evaluation criteria and weights

→

03 Clarify — Site visit, questions answered to all, addenda issued

→

04 Receive and open — Sealed or electronic submission, recorded opening

→

05 Evaluate — Compliance, technical scoring, commercial normalisation

→

06 Clarify and verify — Post-qualification checks; clarifications without changing price

→

07 Award — Approval, letter of acceptance, securities, debrief of unsuccessful bidders

Typical prequalification criteria

CRITERION	EVIDENCE REQUESTED	HOW IT IS USUALLY ASSESSED
Legal eligibility	Registration, licences, tax and debarment status	Pass or fail
Financial capacity	Audited accounts, turnover, liquidity, available credit lines	Pass or fail against minimum thresholds
Similar experience	Completed projects of comparable type, size and complexity, with owner references	Pass or fail, sometimes scored
Key personnel	CVs of the proposed project manager, design manager, HSE and commissioning leads	Scored or minimum requirements
Equipment and resources	Owned or committed cranes, piling rigs, batching plants, specialist tools	Pass or fail for critical items
HSE, quality and integrity record	Safety statistics, management system certificates, litigation and claims history	Pass or fail, with judgement on serious incidents

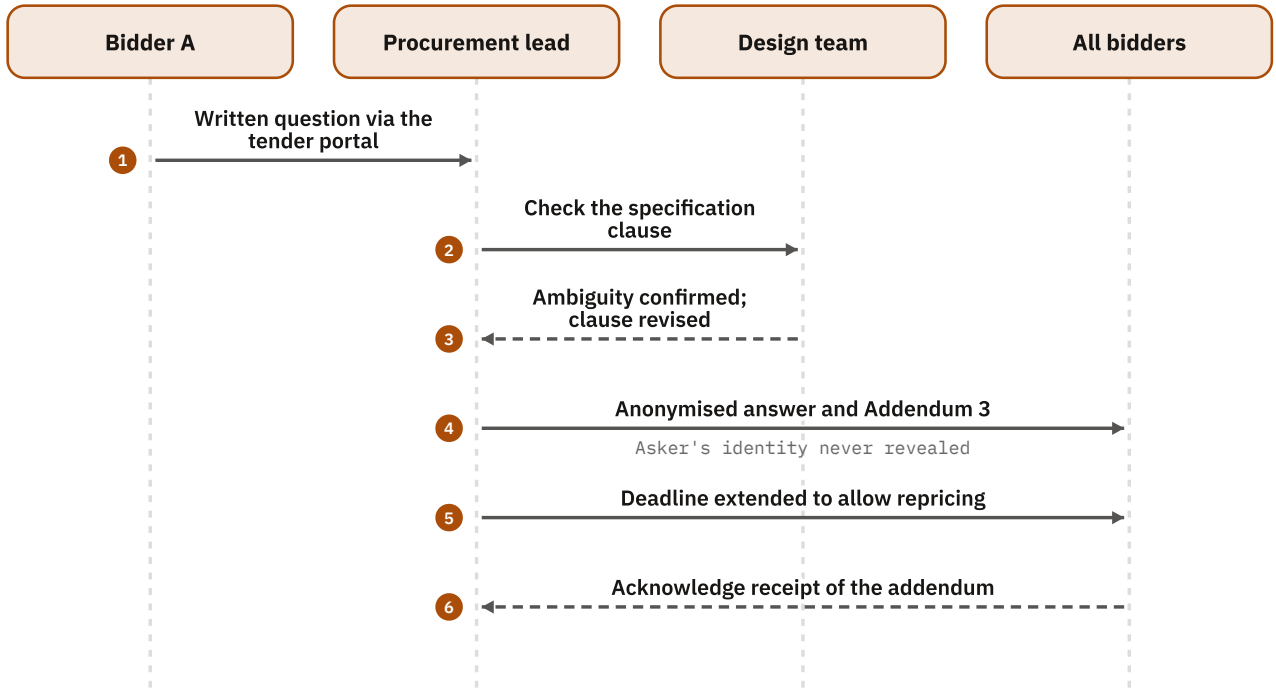
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What a complete tender package contains

- Instructions to tenderers: procedure, deadlines, submission format and the evaluation method with weights
- Conditions of contract, including the general conditions and project-specific amendments
- The owner's requirements or specifications, with drawings and the site and ground investigation information
- Pricing documents: bill of quantities, activity schedule or price breakdown, and daywork schedules
- Programme requirements: key dates, milestones, access dates and interfaces with other packages
- Forms: tender form, tender security, performance security, declarations of integrity and conflicts

During the tender period, keep every bidder on the same footing. Questions are answered in writing and the answer goes to all bidders without revealing who asked; any change to the documents is issued as a numbered **addendum**, and the deadline is extended if bidders need time to price it. Do not meet bidders privately or hint at budgets. A well-run site visit, with the ground investigation data and the existing utility records available to all, reduces the risk premiums that bidders add for things they cannot see. At the deadline, open tenders in a recorded session so that nobody can later suggest a late or altered tender was accepted.

Handling a tender clarification fairly



Notice that the answer goes to every bidder, not just the one who asked, and that acknowledgements prove each tender was priced on the same documents.

Evaluating tenders

1. **Check responsiveness** Confirm each tender is complete, signed, on time, with the required securities and without material deviations from the conditions.
2. **Evaluate technically** Score methodology, programme, key staff, design approach and HSE plan against the published criteria, by evaluators who score independently first.
3. **Normalise prices** Correct arithmetic errors under the stated rules, price any exclusions, qualifications and omissions, and convert currencies at the stated date so tenders can be compared like for like.

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4. **Score price and combine** Apply the published price formula and weights; never introduce new criteria after opening.
5. **Verify the preferred bidder** Confirm capacity and financial standing, check references and examine any abnormally low rates.
6. **Report and recommend** Write an evaluation report that explains every score and adjustment, so the award can be audited and defended.

Price scoring and normalisation

MEASURE	FORMULA	WHAT IT TELLS YOU
Normalised tender price	`Normalised price = tender price + priced exclusions and qualifications + corrected arithmetic errors`	Makes tenders comparable. A low price that excludes the crane for transformer placement is not really low.
Price score	`Price score = (lowest normalised price / tender's normalised price) x price weight`	With a 30-point price weight, the lowest tender scores 30; a tender 25% higher scores 30 / 1.25 = 24.
Combined score	`Combined score = technical score (out of its weight) + price score`	For example, technical 60 of 70 plus price 27 of 30 gives 87; the tender with the highest combined score offers the best value under the published method.

Combined scores for three tenders



Illustrative. Normalised prices of 10.0m, 9.0m and 10.4m give price scores of 27, 30 and 26. B is cheapest, but A offers the best value under the published weights.

TEMPLATE

Tender evaluation matrix

Record weighted scores for each tenderer so that the recommendation is transparent and auditable.

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CRITERION	WEIGHT	TENDERER A	TENDERER B	TENDERER C	EVALUATOR NOTES
Methodology and programme	*25*	*20*	*22*	*14*	*C has no lifting study for the girders and an unrealistic piling rate*
Price (normalised)	*30*	*27*	*30*	*26*	*B lowest after pricing A's excluded temporary works*

Evaluations increasingly look beyond price and method. Owners score carbon reduction proposals, social value such as local employment and training, offsite and DfMA approaches, and the bidder's ability to work to the owner's information requirements in a common data environment. Publish such criteria, and how they will be measured, in the tender documents. Digital and AI tools can help check compliance and compare long submissions, but evaluators must verify every finding and remain accountable for each score.

WATCH OUT · ABNORMALLY LOW TENDERS

A tender far below the estimate and the other bids is not automatically a bargain. It may reflect a misunderstanding of the scope, a missing package, a buy-in strategy that relies on claims, or a contractor in financial trouble. Ask the bidder in writing to explain and substantiate its prices with rate build-ups, supplier quotes and resources. If it can show a genuine advantage, such as owned equipment nearby or a better method, accept it; if it cannot, follow the procurement rules for rejection. Awarding to an unsustainable price often ends in slow progress, disputes or insolvency that cost far more than the saving.

Integrity in tendering

Good practice

- Every evaluator signs a conflict-of-interest declaration before seeing tenders
- Criteria and weights are published in the tender documents and not changed
- All clarifications are answered in writing to all bidders
- Tender opening is recorded and witnessed
- Unsuccessful bidders receive a debrief explaining their scores

Red flags

- Specifications that only one manufacturer can meet without justification
- Late changes to evaluation criteria after tenders are opened
- Repeated rotation of winners among the same few firms
- Evaluators meeting bidders privately during evaluation
- Consultant who drafted the specification also bidding, or advising a bidder

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SCENARIO · DECIDE · A CONFLICT SURFACES MID-EVALUATION

You chair the evaluation committee for a hypothetical 230 kV substation contract funded by a development bank. Technical scores have been entered when one evaluator tells you that her brother-in-law was recently appointed a director of Tenderer B, which is currently ranked first. She says she scored fairly and asks to stay on the committee to avoid delay. What should you do FIRST?

(A) Let her stay, since her scores are already entered and she has disclosed the relationship.

(B) Record the declaration, remove her from the committee, and have her scores set aside and the affected criteria re-scored by an independent replacement, informing the approving authority.

(C) Cancel the whole tender and start again to avoid any suggestion of bias.

(D) Ask Tenderer B to withdraw so that the conflict no longer exists.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Recall a tender you prepared or evaluated. Was any criterion written in a way that favoured one bidder or could not be scored objectively? How would you rewrite it?

Key terms

Prequalification
Assessment of potential bidders' eligibility, capability, capacity and record before tendering, so that only firms able to deliver are invited to bid.

Two-envelope procedure
A tendering method in which technical and financial proposals are submitted separately and price envelopes are opened only for technically compliant tenders.

Conflict of interest
A situation in which a person's private interests could improperly influence, or appear to influence, their duties in a procurement, requiring declaration and management.

Letter of acceptance
The owner's formal written acceptance of a tender, which typically creates the contract and starts the periods for providing securities and beginning the works.

Tender documents
The package issued to bidders, containing instructions, conditions of contract, requirements and specifications, drawings, pricing documents, forms and evaluation criteria.

Abnormally low tender
A tender priced so far below the estimate and other bids that its sustainability is doubtful, requiring the bidder to justify its prices before award.

Tender security
A bank guarantee or bond submitted with a tender that the owner can call if the bidder withdraws during the validity period or refuses to sign after award.

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Key takeaways

- Prequalify on capability and capacity, then issue complete documents with the evaluation method and weights.
- Answer every clarification in writing to all bidders and issue numbered addenda; never meet bidders privately.
- Normalise prices for exclusions and errors before scoring, and never add criteria after opening.
- Investigate abnormally low tenders before deciding; manage conflicts of interest transparently.
- Write an evaluation report that explains every score so the award can be audited and defended.

KNOWLEDGE CHECK

13. A tender is USD 9.6m but excludes temporary works for the girder launch, which the evaluators price at USD 0.9m. What is its normalised price?

(A) USD 8.7m, after deducting the excluded work
(B) USD 9.6m, as submitted by the tenderer
(C) USD 9.15m, with half of the exclusion added
(D) USD 10.5m, including the priced exclusion

14. With a 40-point price weight, the lowest normalised price is 80m. What price score does a tender of 100m receive?

(A) 40 points, the full price weight
(B) 32 points, scaled by 80 divided by 100
(C) 20 points, half the price weight
(D) 8 points, from the difference in millions

15. A bidder asks a question that reveals an ambiguity in the specification. How should the answer be handled?

(A) Answer only that bidder, since it identified the ambiguity first
(B) Ignore it and let the evaluators interpret the specification after tender opening
(C) Answer in writing to all bidders and issue an addendum if the documents change
(D) Discuss it with the bidder by phone to save time before the deadline

16. A tender is 35% below the estimate and the other bids. What should the evaluation committee do?

(A) Ask the bidder to explain and substantiate its prices before deciding
(B) Award to it immediately, since price is the most important factor
(C) Reject it automatically because it is outside the expected range
(D) Ask the other bidders whether they can now reduce their prices to match it

17. In a two-envelope procedure, when are the financial envelopes opened?

(A) Before technical evaluation, so that price can guide scoring
(B) At the same time as the technical proposals, in one session
(C) Only if the owner's cost estimate has been exceeded by every single bidder
(D) After technical evaluation, and only for technically compliant tenders

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · PROCURE · LESSON 4 · 38 MIN

International Standard-Form Principles

How widely used international forms share risk, run payments, handle notices and avoid disputes

- Describe the main families of international standard-form contracts and how they allocate design and ground risk
- Explain the role of the engineer or contract administrator, notices, time limits, securities and the payment cycle
- Trace the path from taking over through the defects period to final release, and explain how dispute boards work

Many engineering projects, especially those financed by development banks or built by international contractors, use standard-form contracts published by industry bodies. The best known internationally are those of FIDIC, the International Federation of Consulting Engineers, whose family of forms is often identified by the colours of their covers; development banks commonly use harmonised versions of them, and many countries publish their own national forms built on similar ideas. FIDIC's current Red, Yellow and Silver Books are the 2017 editions, reprinted with amendments in 2022, though many contracts still run on the 1999 editions. The NEC family, now NEC4, is built around early warnings and compensation events. This lesson describes the **principles** these forms share, in general terms. It does not reproduce any form: on a real project, read the actual edition, the particular conditions and the amendments, because projects frequently change the standard text, and the detail matters.

Three common contract families

Construction to the employer's design

- Employer's consultant designs; contractor builds
- Usually remeasured against a bill of quantities
- An independent engineer administers the contract
- Employer carries risk of its design and of unforeseeable ground

Plant and design-build

- Contractor designs to the employer's requirements
- Usually lump sum with a payment schedule
- An engineer administers the contract
- Contractor carries its design risk; unforeseeable ground generally remains with the employer

EPC / turnkey

- Contractor delivers a complete, working facility
- Lump sum with high price and date certainty
- Employer's representative instead of an independent engineer
- Contractor carries much more risk, including most ground and data risks

In the employer-designed and design-build families, the **engineer** (in some forms called the contract administrator) is appointed by the employer and wears two hats. For most purposes the engineer acts for the employer: issuing instructions, reviewing designs, inspecting work and certifying payment. But when the parties disagree over a claim or a matter that needs a decision, the engineer must first consult both parties, encourage agreement and, failing that, make a fair determination in accordance with the contract, acting neutrally rather than as the employer's advocate. The engineer's authority may be limited by the employer for certain decisions, such as variations above a value, and those limits must be stated in the contract. In turnkey forms, an employer's representative replaces the engineer and the neutral role largely disappears.

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From award to final release (illustrative 30-month contract)

MONTH 0
Letter of acceptance
Contract formed; periods for securities and commencement start

MONTH 0–1
Commencement
Performance security, site access, initial programme

MONTHS 1–30
Execution
Design, construction, variations, claims; monthly statements and payment certificates

MONTH 30
Taking over
Tests passed; care passes to the employer; delay damages stop

MONTHS 30–42
Defects period
Notified defects remedied; retention and securities reduce

MONTH 42+
Final release
Performance certificate, final statement and payment, securities released

The dates assume a 12-month defects notification period. Each certificate along the line shifts risk and money, so it must be issued on evidence.

Notices and time limits in practice

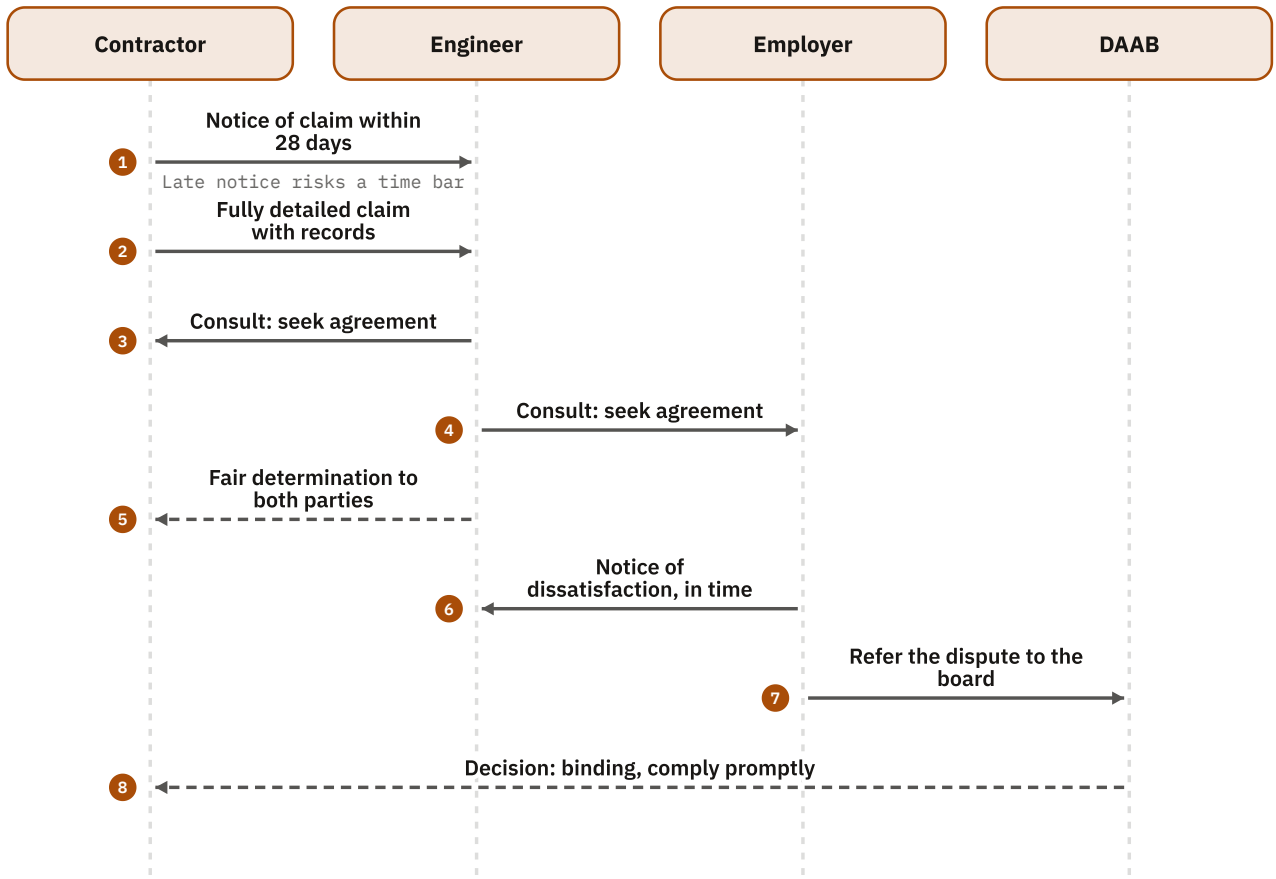
SITUATION	WHO GIVES NOTICE	WHY TIMING MATTERS	PRACTICAL HABIT
Event the contractor believes entitles it to time or money	Contractor	Forms set a short fixed period from awareness, such as 28 days under the 2017 FIDIC forms or eight weeks for an NEC4 compensation event; late notice can lose the entitlement	Weekly commercial review of diaries, RFIs and instructions for events that need notice
Event the employer believes entitles it to money or an extension of the defects period	Employer or engineer	Employer claims are usually also subject to notice requirements	Treat employer claims with the same discipline as contractor claims
Early warning of a matter that may affect cost, time or performance	Either party	Allows joint action before the problem grows	Raise early warnings in progress meetings and confirm them in writing
Disagreement with an engineer's determination	Either party	A notice of dissatisfaction within the stated period preserves the right to take the matter further	Diary the deadline the day the determination arrives
Detailed claim with supporting particulars	Claimant	A fully detailed claim must follow the initial notice within a further fixed period	Build the claim file from contemporaneous records from day one

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WATCH OUT · TIME BARS ARE REAL

A **time bar** is a contract provision under which a claim is lost, or seriously weakened, if notice is not given within the stated period. Courts and tribunals in many jurisdictions enforce them, although some legal systems soften their effect. The cure is cultural, not legal: a site team that confirms verbal instructions in writing the same day, a commercial team that scans every diary and RFI for notifiable events each week, and a notice register that nobody is allowed to ignore. A well-founded claim can be lost on a missed date; a weak claim cannot be rescued by a well-timed notice.

A claim from notice to dispute board



Generalised from the 2017 FIDIC claims procedure (Clause 20, 28-day notice); particular conditions often change periods and steps. Either party can be the claimant or the dissatisfied party.

Typical allocation of key risks in international forms (in general terms)

RISK	EMPLOYER GENERALLY CARRIES	CONTRACTOR GENERALLY CARRIES
Physical conditions on site	Conditions an experienced contractor could not reasonably have foreseen at tender, in employer-designed and design-build forms	Foreseeable conditions; under turnkey forms, most ground conditions
Design	The employer's own design and data it is responsible for	The contractor's design, and under turnkey forms the interpretation of the employer's requirements
Delay caused by the employer's side	Late access, late information, late approvals, other contractors employed by the employer	Its own planning, resources, subcontractors and suppliers

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RISK	EMPLOYER GENERALLY CARRIES	CONTRACTOR GENERALLY CARRIES
Exceptional events	Time, and in some cases cost, for events beyond either party's control such as war or major natural catastrophe	Ordinary commercial and operational risks
Change in law after the base date	Time and cost effects of new or changed laws	Laws in force at the base date, priced in the tender
Weather	Time for exceptionally adverse climatic conditions under many forms	Normal seasonal weather, including predictable monsoon conditions

Securities and the payment cycle

- **Performance security**: a bank guarantee, often payable on demand, provided after award and kept valid until the contractor has completed the works and remedied defects.
- **Advance payment guarantee**: protects the employer's advance, reducing as the advance is recovered from interim payments (see Budget for advances and retention).
- **Monthly statements**: the contractor submits a statement of the value of work done, materials and adjustments.
- **Interim payment certificates**: the engineer certifies the amount it fairly considers due within a set period, and may withhold for defective work but not refuse to certify altogether.
- **Payment and late payment**: the employer pays within a set period; late payment typically attracts financing charges and, if prolonged, rights to slow down, suspend or ultimately terminate.
- **Final statement and final payment**: after the performance certificate, the final account is settled and remaining securities are released.

Taking over marks the point at which the works, or a section of them, are complete enough to be used for their intended purpose, with only minor outstanding work and defects that do not substantially affect use, and after the tests on completion have been passed. The engineer issues a taking-over certificate; responsibility for care of the works passes to the employer, delay damages stop running and part of the retention is usually released. The **defects notification period** then begins, typically a year or as stated in the contract, during which the contractor must make good defects notified to it. When it expires and all defects are remedied, a performance certificate confirms that the contractor's obligations are complete. The Commission stage covers the engineering of handover; here the point is that each certificate shifts risk and money, so it must be issued on evidence, not on pressure.

How a standing dispute avoidance and adjudication board works

1. **Appoint at the start** One or three experienced, independent members are appointed jointly by the parties at the beginning of the contract, not when a dispute has already arisen.
2. **Visit regularly** The board visits the site at intervals, reads progress reports and gets to know the project and the people before tensions build.
3. **Help informally** If both parties agree, the board can give an informal view on an emerging disagreement, which often resolves it without a formal referral.
4. **Decide referred disputes** If a dispute is formally referred, the board hears both sides and gives a reasoned decision within a set period; the parties must comply with it promptly.
5. **Preserve or accept** A party unhappy with the decision gives notice of dissatisfaction within the stated period; otherwise the decision becomes final and binding.

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6. **Settle or arbitrate** After a period for amicable settlement, unresolved disputes go to arbitration, where the board's decision remains binding until revised.

SCENARIO · DECIDE · THE ENGINEER UNDER PRESSURE

You are the engineer on a design-build water treatment works near Dhaka. The contractor has given timely notice and a well-documented claim for an extension of time after excavation for the clarifier met a buried, unrecorded concrete structure that the site investigation could not reasonably have revealed. The employer's director calls you and says: 'You work for us. Reject it; let them fight for it.' What should you do?

(A) Reject the claim as instructed, since the employer appointed you and pays your fees.

(B) Consult both parties, assess the claim against the contract and the records, make a fair, reasoned determination, and explain to the director why this protects the employer.

(C) Avoid making any determination so that the matter can go straight to arbitration.

(D) Grant a reduced extension as a compromise so that both parties feel partly satisfied.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Contract notice register**

Track every contractual notice and its deadlines so that no entitlement is lost and no response is late.

REF	DATE SENT OR RECEIVED	FROM AND TO	EVENT OR SUBJECT	CONTRACT BASIS (GENERAL)	NEXT DEADLINE	STATUS
N-023	*12 Mar 2026*	*Contractor to engineer*	*Notice of claim: unrecorded concrete structure at clarifier excavation*	*Unforeseeable physical conditions*	*Fully detailed claim due within the contract period*	*Open: records being compiled*
N-024	*15 Mar 2026*	*Engineer to contractor*	*Instruction to remove obstruction and redesign clarifier base locally*	*Variation instruction*	*Contractor's proposal on cost and time*	*Open: proposal due*

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REFLECT

Who on your project is responsible for spotting events that need a contractual notice, and how quickly does information from the site diary reach them? What would you change?

Key terms

Contract administrator (the Engineer)
The person or firm named in the contract to administer it, issuing instructions and certificates for the employer and making fair, neutral determinations when the parties disagree.

Time bar
A contract provision under which entitlement to time or money is lost, or weakened, if the required notice is not given within the specified period.

Performance security
A bank guarantee or bond provided by the contractor after award, which the employer can call if the contractor fails to perform its obligations.

Dispute avoidance and adjudication board (DAAB)
A standing panel of independent experts appointed at the start of a contract to help the parties avoid disputes and to give decisions on referred disputes that are binding unless later revised.

Contractual notice
A formal written communication required by the contract to trigger or preserve rights, such as notice of a claim, sent to the stated address within the stated period.

Unforeseeable physical conditions
Ground, water and other physical conditions at the site that an experienced contractor could not reasonably have foreseen at tender, often an employer risk in employer-designed and design-build forms.

Interim payment certificate
The engineer's periodic certificate of the amount due to the contractor for work done, materials and adjustments, which the employer must pay within the contract period.

Key takeaways

- Read the actual edition, particular conditions and amendments; standard forms are often changed.
- The engineer acts for the employer day to day but must consult and determine claims fairly under the contract.
- Notices have deadlines that can bar a good claim; run a notice register and weekly records review.
- Securities, payment certificates, taking over and the performance certificate each shift risk and money.
- A standing dispute board appointed at the start helps avoid disputes; its decisions bind unless revised.

KNOWLEDGE CHECK

18. Under a typical EPC/turnkey form, which party usually carries most of the risk of unexpected ground conditions?

(A) The contractor, as part of a turnkey price
(B) The employer, through remeasured quantities
(C) The engineer, as the neutral administrator
(D) The dispute board, through its decisions

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19. When determining a disputed claim, how should the engineer act?

(A) As the employer's advocate, minimising every claim
(B) As the contractor's adviser, since the claim is theirs
(C) Neutrally, after consulting both parties, following the contract
(D) By splitting the claimed amount evenly between the two contracting parties

20. A contractor becomes aware of an event giving entitlement to time but gives notice ten weeks later, well after the period stated. What is the main risk?

(A) The engineer must award double the time to compensate
(B) The entitlement may be lost or weakened under a time bar
(C) The employer must pay financing charges on the claim
(D) The dispute board must hear the claim before anything else

21. What usually happens to responsibility for care of the works when a taking-over certificate is issued?

(A) It stays with the contractor until the final payment is made
(B) It passes to the engineer until the defects period expires
(C) It is split between the parties according to the retention
(D) It passes to the employer, and delay damages stop running

22. What distinguishes a standing dispute avoidance board from an ad hoc one?

(A) It is appointed only after a dispute has been formally referred
(B) Its decisions can never be challenged in later arbitration
(C) It is appointed at the start and visits regularly to help avoid disputes
(D) It is chosen by the engineer alone rather than by the two parties jointly

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · PROCURE · LESSON 5 · 40 MIN

Variations, Claims & Disputes

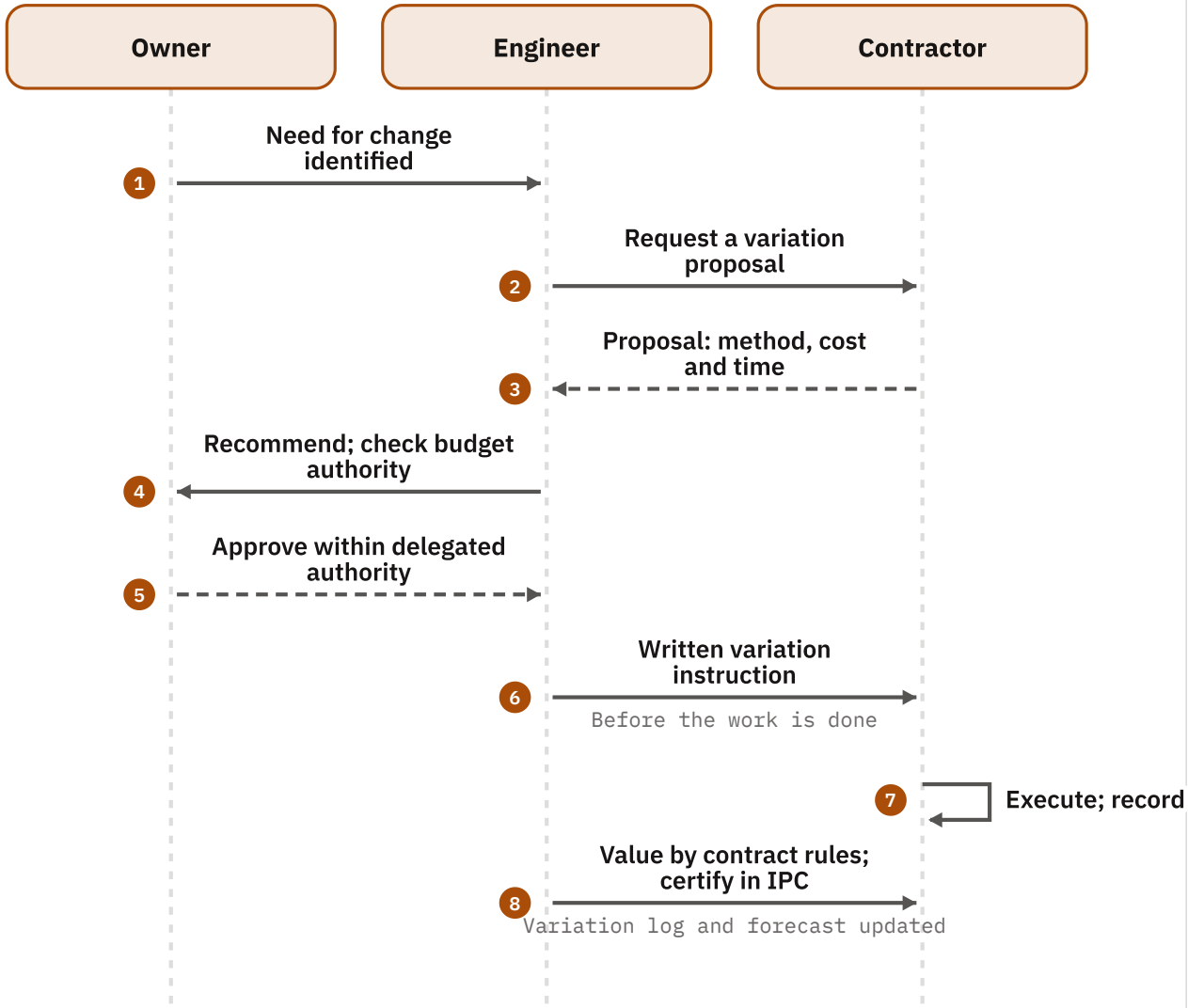
Instruct and value change properly, prove delay and disruption, and resolve disagreements at the lowest level

- Manage a variation from instruction through valuation, recording and payment
- Explain extensions of time, delay analysis methods, concurrent delay, prolongation and disruption
- Keep the records and notices that make claims provable, and choose the right route from negotiation to arbitration

Change is normal on engineering projects: designs develop, owners change their minds, ground behaves differently from the borehole logs and authorities impose new conditions. A **variation** is a change to the works instructed under the contract. A **claim** is a request by either party for an entitlement the contract provides, usually additional time, additional payment, or both. Variations and claims are not signs of failure; unmanaged ones are. Disputes arise when a claim is not agreed and the parties cannot settle it. The professional aim is to price change fairly and quickly, prove entitlement with records, and settle disagreements at the lowest, cheapest level possible.

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The life of a variation



The written instruction comes before the work, and the contractor's separate cost code makes valuation quick and credible.

How variations are usually valued

BASIS	WHEN IT APPLIES	EXAMPLE
Contract rates applied directly	The varied work is of similar character and conditions to items already priced	Extra 120 m of the same cable tray on the same level, at the tendered rate
Contract rates used as a basis	Similar work but different quantity, location or conditions that justify adjustment	Same tray installed at 18 m height from a mobile platform; rate adjusted for access and lower productivity
New fair rates	No relevant rate exists in the contract	A new chemical dosing skid: reasonable cost plus a reasonable allowance for overheads and profit
Daywork	Small, unpredictable tasks that are hard to measure	Breaking out an unexpected old foundation over two days, recorded by labour and plant hours
Agreed quotation in advance	The parties agree a price before the work starts	Owner requests a revised lobby layout and accepts the contractor's quotation before instructing

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An **extension of time** (EOT) gives the contractor more time to complete, protecting it from delay damages, when a delay is caused by an event the contract places at the employer's risk and that event actually delays completion. The second condition matters: a late drawing for a non-critical stair finish may cost money but not time if the activity has enough float. The contractor must show cause and effect with a programme analysis, give notice in time and substantiate the claim. Whether an EOT also carries money depends on the type of event: employer-caused delays usually do; neutral events such as exceptionally adverse weather often give time only. Contracts differ on who owns float, so read yours. Software, including AI-assisted tools, can process programme updates quickly, but a qualified planner must own the method and be able to explain every step to a tribunal.

Delay analysis methods (in general terms)

METHOD	HOW IT WORKS	STRENGTHS	WEAKNESSES
As-planned versus as-built	Compares planned and actual dates to show where and by how much the work slipped	Simple, intuitive, uses actual records	Shows what slipped, not why; weak on cause when several events overlap
Impacted as-planned	Adds delay events to the original programme and measures the shift in completion	Quick and cheap	Ignores what actually happened and the contractor's own delays; often criticised
Time impact analysis	Inserts a fragnet for each event into the programme updated to just before the event, and measures the effect	Contemporaneous and prospective; well suited to EOT decisions during the project	Depends on accurate programme updates; can be time-consuming for many events
Windows analysis	Divides the project into periods and analyses the critical path and delay within each period using updated programmes	Tracks the shifting critical path with actual progress	Needs regular, reliable updates; choice of windows can influence results
Collapsed as-built	Removes delay events from the as-built programme to show when work would have finished but for them	Grounded in what actually happened	Requires a reliable as-built logic; subjective when rebuilding it

INSIGHT · CONCURRENT DELAY

Concurrent delay occurs when two or more delay events, one the employer's risk and one the contractor's, affect the critical path at the same time and each would have delayed completion on its own. Treatment depends on the contract and the governing law, but a common approach gives the contractor time (so it is not penalised by delay damages) but not money (because it would have incurred the cost anyway). True concurrency is rarer than claimed: often one event is critical and the other only absorbs float. Careful, contemporaneous programme analysis is the only reliable way to tell the difference, and many contracts now state how concurrency will be handled.

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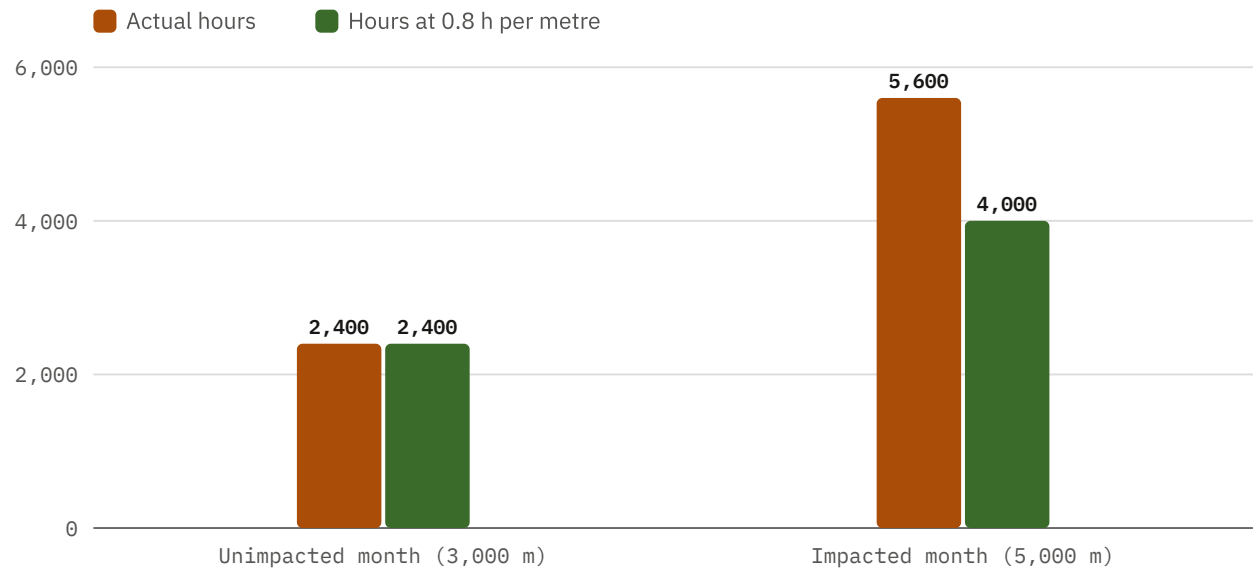
Quantifying prolongation and disruption

MEASURE	FORMULA	WHAT IT TELLS YOU
Prolongation cost	`Prolongation cost = compensable delay (days) x actual time-related cost per day`	Time-related costs include site management, supervision, offices, cranes kept on hire and site security for the extra period; they must be proven from records, not taken from the tender rate alone.
Measured-mile lost hours	`Lost hours = actual hours in the impacted period - (quantity installed in the impacted period x hours per unit in an unimpacted period)`	Compares productivity on the same work when disrupted and when not, using the project's own records.
Disruption cost	`Disruption cost = lost hours x all-in labour rate (+ associated plant and supervision)`	Converts lost productivity into money; the cause of disruption must still be linked to an employer-risk event.

EXAMPLE · A MEASURED MILE ON CABLE PULLING (FIGURES ILLUSTRATIVE)

In an unimpacted month, crews pulled 3,000 m of cable in 2,400 hours: 0.8 hours per metre. In the month when the owner's late equipment changes forced crews to work around partially installed switchboards, they pulled 5,000 m in 5,600 hours. Expected hours = 5,000 x 0.8 = 4,000. Lost hours = 5,600 - 4,000 = **1,600**. At an all-in rate of BDT 450 per hour, the disruption cost of labour is 1,600 x 450 = **BDT 720,000**. The claim is credible because both figures come from the project's own allocation sheets for the same work by the same crews.

Measured mile: actual versus expected hours



Illustrative figures from the example. In the unimpacted month the bars match; in the impacted month the 1,600-hour gap is the lost productivity claimed.

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Records that win, or settle, claims

☐ Daily site diaries signed by supervisors, with weather, resources, work done and problems
☐ Labour and plant allocation sheets by activity or cost code
☐ Monthly programme updates with actual start and finish dates and a narrative of delays
☐ Dated and located photographs, drone surveys and laser scans of progress and problems
☐ RFI and submittal registers with dates of request and response
☐ Every verbal instruction confirmed in writing the same day
☐ Notice register with deadlines and evidence of delivery
☐ Minutes of progress meetings agreed by both parties
☐ Cost records that separate the cost of each variation and disruption event
☐ Records held in the common data environment with status, revision and date (ISO 19650)

The dispute resolution ladder (lowest and cheapest first)

Site-level negotiation	`Project managers and commercial leads` `Fast, cheap, preserves relationships`
Senior management negotiation	`Executives not involved day to day` `Fresh view and authority to trade issues`
Mediation	`Independent mediator helps parties reach their own settlement` `Confidential; not binding unless a settlement is signed`
Dispute board or adjudication	`Independent experts decide quickly` `Binding in the interim; may become final if not challenged`
Arbitration or litigation	`Final and binding award or judgment` `Slow, expensive, public in court, private in arbitration`

CASE STUDY · A STATION BOX, TWO DELAYS AND A MISSING RECORD

On a hypothetical underground metro station, the owner was responsible for diverting a large water main before the contractor could excavate the east end of the station box. The diversion finished eleven weeks late. During the same period the contractor's main crawler crane was out of service for four weeks after a hydraulic failure, and the contractor's diaries for that month are incomplete. The contractor gave notice of the diversion delay on time and now claims eleven weeks' extension of time and prolongation costs. The owner's team argues that the crane breakdown means the delay was concurrent and nothing is due.

1. What must the contractor show to obtain an extension of time for the diversion delay?

2. How should the crane breakdown affect the assessment?

3. How could the contractor's missing diary entries affect the outcome, and what should both parties do now?

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Model answer: module answer key.

SCENARIO · DECIDE · A VERBAL INSTRUCTION AT THE TRENCH

You are the contractor's project manager on a hypothetical gas-insulated substation. During a site walk, the owner's engineer tells your foreman to re-route a 60 m cable trench around a planned future transformer bay, which requires breaking out a newly cast slab. The foreman wants to start tomorrow to keep the crew busy. Nothing has been written down. What should you do FIRST?

(A) Start tomorrow and claim the cost in the final account at the end of the project.

(B) Refuse to do any work until the owner agrees the full price of the change.

(C) Confirm the instruction in writing today, give notice of the expected cost and time effect within the contract period, and record the resources on a separate cost code from the start.

(D) Ask the foreman to do the work quietly at night so that the main programme is not disrupted.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Variation and claims log**

Track every change and claim from instruction to agreement so that the forecast final account is always current.

REF	DESCRIPTION	INSTRUCTED BY AND DATE	NOTICE DATE	TIME CLAIMED OR AGREED	COST CLAIMED OR AGREED	STATUS
V-031	*Re-route 60 m cable trench around future transformer bay, including slab break-out*	*Owner's engineer, verbal 04 Jun 2026, confirmed in writing 04 Jun*	*05 Jun 2026*	*6 days claimed (non-critical: 0 agreed)*	*BDT 2.4m claimed / 2.1m agreed*	*Agreed, in IPC 14*
C-007	*Prolongation due to late water main diversion at east end*	*Employer-risk event*	*18 Feb 2026*	*11 weeks claimed*	*Under assessment*	*Engineer's determination due 30 Jun*

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REFLECT

Think of the last claim you saw rejected or reduced. Was it lost on entitlement, on notice, on programme analysis or on records? What one habit on site would have changed the outcome?

Key terms

Variation
A change to the works, such as an addition, omission, or change in quality, quantity, sequence or method, instructed under the contract by a person with authority, and valued under its rules.

Concurrent delay
A situation in which an employer-risk delay and a contractor-risk delay affect the critical path at the same time, each sufficient on its own to delay completion.

Disruption
Loss of productivity caused by interference with the contractor's planned working methods, even where completion is not delayed, usually measured by comparing affected and unaffected work.

Arbitration
A private, binding method of resolving disputes in which an independent tribunal chosen under the contract hears the case and makes an award enforceable in most countries.

Extension of time (EOT)
An adjustment of the completion date granted when an event at the employer's risk delays completion, protecting the contractor from delay damages for that period.

Prolongation cost
The time-related costs a contractor incurs because the project lasts longer, such as site management, facilities and plant on hire, recoverable for compensable delay.

Time impact analysis
A prospective delay analysis method that inserts a fragnet representing a delay event into the programme updated to just before the event and measures its effect on completion.

Key takeaways

- A variation needs a written instruction from someone with authority, ideally before the work, and is valued by the contract's rules.
- An extension of time needs an employer-risk event that actually delays completion, proven by programme analysis and timely notice.
- Genuine concurrent delay commonly gives time but not money; check the contract and governing law.
- Contemporaneous records win or settle claims: diaries, allocation sheets, programme updates, photographs and notices.
- Resolve disputes at the lowest effective level, from site negotiation up to arbitration only as a last resort.

KNOWLEDGE CHECK

23. A late drawing delays a non-critical activity by eight days, but that activity has fifteen days of total float. What is the contractor most likely entitled to?

(A) An eight-day extension of time plus the associated prolongation cost
(B) No extension of time, though direct costs may be recoverable
(C) A fifteen-day extension of time to restore the float
(D) Delay damages from the employer for the lost float

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24. A varied item is similar to an existing bill item but must be installed at height from a mobile platform. How is it usually valued?

(A) At a new rate based solely on the contractor's quotation

(B) By daywork, because the conditions are different

(C) At the existing rate with no adjustment at all, as the work is broadly similar

(D) Using the existing rate as a basis, adjusted for the different conditions

25. Which delay analysis method inserts fragnets into the programme updated to just before each delay event?

(A) Time impact analysis

(B) As-planned versus as-built

(C) Impacted as-planned analysis

(D) Collapsed as-built analysis

26. Crews installed 2,000 m of duct in 1,900 hours while disrupted. In an unimpacted period they achieved 0.7 hours per metre. How many hours were lost?

(A) 1,900 hours, the total hours worked

(B) 1,330 hours, the unimpacted rate times the hours

(C) 500 hours, the actual minus the expected hours

(D) 1,400 hours, the expected hours for the quantity

27. Under a common approach to genuine concurrent delay, what does the contractor typically receive for the concurrent period?

(A) Neither time nor money, because it also caused the delay

(B) Money but not time, so delay damages still apply

(C) Time and money, as the employer's delay came first

(D) Time but not money, so no delay damages for that period

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · PROCURE

Answer key

Knowledge checks

1. B With a finished design and a priority on competitive pricing, design-bid-build is the natural fit. The other models are designed for situations with less design certainty or different priorities.

2. D More packages create more boundaries where responsibilities can be missed, and the owner must coordinate them. Competition usually widens rather than narrows, and price effects vary.

3. A Long-lead items can sit on the critical path, so reserving production early protects the schedule. The other statements are not general rules.

4. C With free-issue, the owner remains responsible to the contractor for timely and correct supply, so late or defective transformers become the owner's problem. Workmanship and productivity remain the contractor's.

5. B Alliancing suits complex, uncertain work where joint problem-solving and shared incentives add value. Simple, well-defined work is usually better served by competitive fixed pricing.

6. D Two-stage tendering appoints the contractor early on quality, fees and rates, uses its advice during design, then agrees the price once the design is mature. A lump sum on a concept invites premiums, cost plus percentage removes cost discipline, and design-bid-build loses the early input.

7. C Remeasurement pays measured quantities at tendered rates: 680 x 4,500 = 3,060,000. The billed quantity is only an estimate, and the rate changes only if the work's character or conditions change.

8. A A percentage fee rewards higher cost. A fixed fee, target cost or strong cost controls address this; the other statements are not true of cost-plus contracts.

9. D Payment = 55 + 3 + 0.4 x (50 - 55) = 55 + 3 - 2 = 56m. The contractor's pain share reduces the fee; the other answers ignore or reverse the share.

10. B Allocating risk to the party best able to manage it minimises total cost, including risk premiums and dispute costs. Blanket transfer, equal splits or blanket retention ignore that logic.

11. A Without changes to scope, the GMP caps the owner's liability, so the contractor bears the USD 3m excess. Savings below the GMP may be shared, but overruns above it are not.

12. C Pn = 0.2 + 0.3 x 1.10 + 0.5 x 1.20 = 0.2 + 0.33 + 0.60 = 1.13. The fixed portion does not move, and each index rise counts only in proportion to its weighting.

13. D Normalisation adds the cost of exclusions so tenders are compared like for like: 9.6 + 0.9 = 10.5m. Ignoring or halving the exclusion makes this tender look cheaper than it really is.

14. B Price score = (80 / 100) x 40 = 32. Only the lowest tender receives the full 40 points under this formula.

15. C Equal treatment requires the same information for every bidder. Private answers or post-opening interpretation would make the competition unfair.

16. A Abnormally low tenders must be investigated: the price may be genuine or may hide errors or a claims strategy. Automatic award or rejection skips that check, and asking others to match would be bid shopping.

17. D Keeping price sealed until technical evaluation is complete prevents price from influencing technical scores, and only compliant tenders proceed.

18. A Turnkey forms give the contractor most ground and data risk in return for a higher price. Employer-designed and design-build forms generally leave unforeseeable conditions with the employer.

19. C The determination role requires consultation and a fair decision under the contract. Advocacy for either side or arbitrary splits are not acceptable.

20. B Many forms make timely notice a condition of entitlement. Late notice risks losing the claim regardless of its merits; it does not create extra entitlements.

21. D Taking over transfers care of the works to the employer and stops delay damages. The contractor still remedies defects during the defects notification period.

22. C A standing board knows the project from the beginning and can help prevent disputes. Its decisions are binding but can be revised in arbitration if a timely notice of dissatisfaction is given.

23. B An EOT requires delay to completion. With fifteen days of float, completion is not delayed, though any direct costs of the disruption may still be claimable.

24. D Where work is similar but conditions differ, contract rates are used as the basis and adjusted. Daywork and new rates are for cases with no relevant rate or small unmeasurable work.

25. A Time impact analysis models each event on the contemporaneous programme. Impacted as-planned uses the original programme; collapsed as-built removes events from the as-built record.

26. C Expected hours = 2,000 x 0.7 = 1,400; lost hours = 1,900 - 1,400 = 500. The other figures are the actual or expected hours themselves or a misapplied calculation.

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27. D A widely used approach grants time, so the contractor is not penalised, but no prolongation cost, because it would have incurred the cost anyway. Contracts and laws vary, so always check the terms.

Scenarios & case studies

Scenario · Choosing a model for a fast-track factory expansion
B is best. Best. Performance requirements are clear, time is short and owner capacity is small. Design-build gives single-point responsibility, early ordering protects the critical path, and an owner's engineer preserves control where it matters.

Case study · Pain, gain and a cap
At an actual cost of USD 72m the gain is 8, the contractor's share is 4, and the owner pays 72 + 4 + 4 = USD 80m, saving 4 against target plus fee of 84. At USD 92m the pain is 12 and the contractor's 50% share would be 6, but it is capped at the 4m fee, so the owner pays 92 + 4 - 4 = USD 92m: the contractor recovers cost with no margin and the owner bears 8 of the 12 overrun. The mechanism needs a realistic target agreed with good information, a clear definition of which costs are allowable, open-book access and audit, prompt target adjustment for changes, and joint governance that reviews forecasts monthly. If the target is unrealistically low, the contractor quickly reaches the cap and loses any incentive; if it is padded, the owner pays gain share for no real improvement.

Scenario · A lump sum for an unknown building
B is best. Best. Reducing uncertainty before tender and matching each pricing mechanism to its level of definition gives a fair price for known work and transparent control of the unknowns.

Scenario · A conflict surfaces mid-evaluation
B is best. Best. This manages the conflict transparently, protects the integrity of the award and keeps a documented trail for the lender and auditors.

Scenario · The engineer under pressure
B is best. Best. The determination role requires neutrality. A fair, documented decision reduces dispute risk and cost, and explaining it helps the employer understand its own interest.

Case study · A station box, two delays and a missing record
The contractor must show that the late diversion was an employer-risk event, that notice was given properly, and through a time impact or windows analysis that it actually delayed the critical path to completion, and by how much. The crane breakdown is a contractor-risk event; if analysis shows it was on the critical path at the same time and would have delayed completion on its own, those four weeks may be concurrent, which under many contracts gives time but not prolongation cost for that period, whereas if the crane only consumed float it does not reduce the entitlement. The missing diaries weaken the contractor's ability to prove resources and productivity, so the tribunal may rely on other records such as crane hire invoices, photographs and progress updates, or award less. Both parties should agree the facts that can be agreed, share programme updates, and try senior negotiation or the dispute board before positions harden into arbitration.

Scenario · A verbal instruction at the trench
C is best. Best. Written confirmation establishes the variation, the notice protects entitlement, and separate records make valuation straightforward and credible.

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07

MODULE 7 · INTEGRATE

Stakeholders, Site Leadership & Digital Engineering

Bring people, information and data together so the site builds the right thing, safely, from the latest information

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Map and engage authorities, utilities, regulators, lenders and communities so that permits and consents never surprise the schedule
- Lead multi-company site teams with presence, psychological safety, fair conflict handling and sound decisions under pressure
- Run a site communication system of meetings, diaries, RFIs, submittals and document control that produces action and evidence
- Plan BIM uses, a common data environment and digital twin uses aligned with ISO 19650 information management principles
- Govern AI and data-driven project controls with data quality checks, verification and clear human accountability

LESSONS

1. Stakeholder & Community Engagement · 35 min
2. Leading Site Teams · 35 min
3. Site Communication, Meetings & Reporting · 35 min
4. BIM & Digital Twins · 40 min
5. AI & Data-Driven Project Controls · 40 min

MODULE 7 · INTEGRATE · LESSON 1 · 35 MIN

Stakeholder & Community Engagement

Authorities, utilities, regulators, lenders and neighbours – mapping who matters and earning permission to build

- Identify the stakeholders of an engineering project and analyse them with power-interest and salience tools
- Build a stakeholder register and engagement plan that connect to the schedule and the risk register
- Plan permits, consents and utility interfaces as critical-path work rather than paperwork
- Engage affected communities in ways that earn and keep a social licence to operate

Engineering projects are built in public. A **stakeholder** is any person, group or organisation that can affect the project or is affected by it, and on a capital project the list is long. A factory extension inside a fenced plot in Gazipur may involve the owner's board, the buyer's compliance auditors, the fire service, the power distribution company and a handful of neighbours. An elevated metro viaduct along a city arterial road involves the traffic police, the city corporation, half a dozen utility owners, lenders, thousands of shopkeepers and commuters, and the media. Project managers who treat stakeholders as a communications task tend to discover, too late, that one missing authority approval or one angry neighbourhood can stop a concrete pour more effectively than any technical problem.

Typical stakeholders on an engineering project

STAKEHOLDER GROUP	TYPICAL EXAMPLES	WHAT THEY CONTROL OR CARE ABOUT	ENGAGEMENT APPROACH
Permitting authorities	Planning or development authority, environment department, fire service, building regulator	Permits, approvals, inspections and the right to occupy or operate	Pre-application meetings, complete submissions, a consents register with a named owner for every approval
Utilities and infrastructure owners	Power distribution company, gas utility, water and sewerage authority, telecoms operators, railway, roads authority	Their buried and overhead assets, outage windows, crossing and diversion approvals	Joint surveys, agreed diversion designs, outages booked well ahead, a single coordination forum
Regulators and safety bodies	Electricity regulator, nuclear or aviation safety authority, port authority, labour inspectorate, boiler inspectorate	Licences to construct and operate, compliance inspections, powers to stop work	Formal correspondence, inspection readiness, early warning of anything that may need their consent
Lenders and investors	Development banks, export credit agencies, commercial lenders, the owner's board	Drawdowns, covenants, environmental and social standards, climate and sustainability disclosures, reporting	Monitoring reports, site visits by the lender's technical adviser, transparent variance reporting
Affected communities	Neighbours, shopkeepers, farmers, fishers, schools, hospitals, places of worship	Access, noise, dust, safety, livelihoods, local jobs and respect	A community liaison officer, notice before disruptive work, local hiring, visible follow-up on complaints

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STAKEHOLDER GROUP	TYPICAL EXAMPLES	WHAT THEY CONTROL OR CARE ABOUT	ENGAGEMENT APPROACH
Internal and delivery partners	Owner's operations team, designers, main contractor, subcontractors, suppliers, the workforce	Requirements, interfaces, workload, payment and recognition	Integrated planning, interface meetings, clear decision rights and honest reporting

Analysis turns a long list into priorities. The **power-interest grid** places each stakeholder by how much power they have over the project and how much interest they take in it: manage high-power, high-interest stakeholders closely; keep high-power, low-interest stakeholders satisfied; keep low-power, high-interest stakeholders informed; and monitor the rest. The **salience model** adds legitimacy and urgency: an affected village has a legitimate claim even if its formal power is low, and a utility owner that controls next week's shutdown has urgency. For each important stakeholder also record their current attitude and the engagement level you need – unaware, resistant, neutral, supportive or leading – and, critically, which activity on your schedule depends on them.

Power-interest grid for an elevated metro corridor (illustrative)

POWER OVER THE PROJECT →

Keep satisfied
- Aviation authority (crane heights)
- Lenders' boards
- Roads authority before works reach its roads

Manage closely
- Utility owners with diversions on the critical path
- Traffic police
- Environment department
- Lenders' technical adviser

Monitor
- Commodity suppliers
- Public outside the corridor

Keep informed
- Market traders near stations
- Schools and residents along the route
- Local media

INTEREST IN THE PROJECT →

Positions are a snapshot: stakeholders move between quadrants as the work front moves. Use salience as a check – a low-power group with a legitimate, urgent claim, such as traders losing access, needs active engagement, not just updates.

INSIGHT · STAKEHOLDERS MOVE AROUND THE GRID

A residents' association that had little power during design can gain a great deal once night piling starts and it complains to the environment department, the media or the courts. A lender that is quiet in year one becomes very interested when a covenant is at risk. **Re-map stakeholders at every stage gate and whenever the work front moves into a new neighbourhood**, rather than filing the analysis after kickoff.

Six steps to engaging stakeholders on an engineering project

1. **Identify** Run workshops with design, construction, operations and land staff; walk the route or site; review land, utility and permit records; and ask every stakeholder you meet who else you should be talking to.
2. **Analyse** Place each stakeholder on the power-interest grid, note salience, current and desired engagement, key concerns, and what they control on the critical path – a permit, an outage, an access road, a drawdown.

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3. **Plan** Write a **stakeholder engagement plan**: who engages whom, through which channel, how often and with what message. Put permits, outages and consultations into the schedule as real activities with durations and predecessors.
4. **Engage** Deliver the plan through meetings, site visits, written notices, a public information point, a hotline and local media. Listen as much as you inform, and record every commitment made on the project's behalf.
5. **Track commitments** Enter each promise – for example no heavy deliveries during school hours – into a commitments register and flow it into site rules and contractor obligations so it is actually kept.
6. **Review and escalate** Monitor sentiment, complaints and permit status monthly and at stage gates; escalate early when a stakeholder threatens a milestone, bringing options rather than just bad news.

TEMPLATE

Stakeholder register

Record each significant stakeholder, what they need, how much they matter and how the project will engage them, with a direct link to schedule activities that depend on them.

STAKEHOLDER	INTEREST OR CONCERN	POWER / INTEREST	CURRENT → DESIRED ENGAGEMENT	OWNER AND CHANNEL	CRITICAL-PATH LINK	NEXT ACTION AND DATE
City traffic police (zone office)	*Lane closures and diversions for pier-cap formwork and girder launching*	*High / Medium*	*Neutral → Supportive*	*Construction manager; monthly meeting plus written requests*	*Night closure permits for girder launches on spans 14–20*	*Submit traffic management plan for spans 14–20 by 5 May*
Market traders' association near Station 3	*Loss of footfall and customer access during foundation works*	*Medium / High*	*Resistant → Neutral*	*Community liaison officer; weekly walk-through*	*Hoarding layout and temporary access routes for station foundations*	*Agree temporary shopfront access plan at next weekly meeting*

Permits and consents are schedule work, not paperwork. A typical engineering project may need environmental clearance, planning or building permission, road-cutting and traffic permits, tree-felling consent, river or navigation clearance for marine works, aviation height clearance for tall cranes near an airport, fire safety approval, a grid connection agreement, boiler registration with the boiler inspectorate and, at the end, an occupancy or operating licence. Each approval has a lead time, prerequisite documents, a validity period and conditions. Keep them in a **consents register** with a named owner, show each one in the schedule with realistic durations and logic – environmental clearance before site clearance, for example – and flow every condition attached to an approval into the scope of whoever must comply with it.

Utilities deserve special attention on urban and brownfield projects because they combine high power, poor information and long lead times. Records of buried services are often incomplete or outdated, so serious projects verify them with trial pits and ground-penetrating radar before finalising foundation positions. Each

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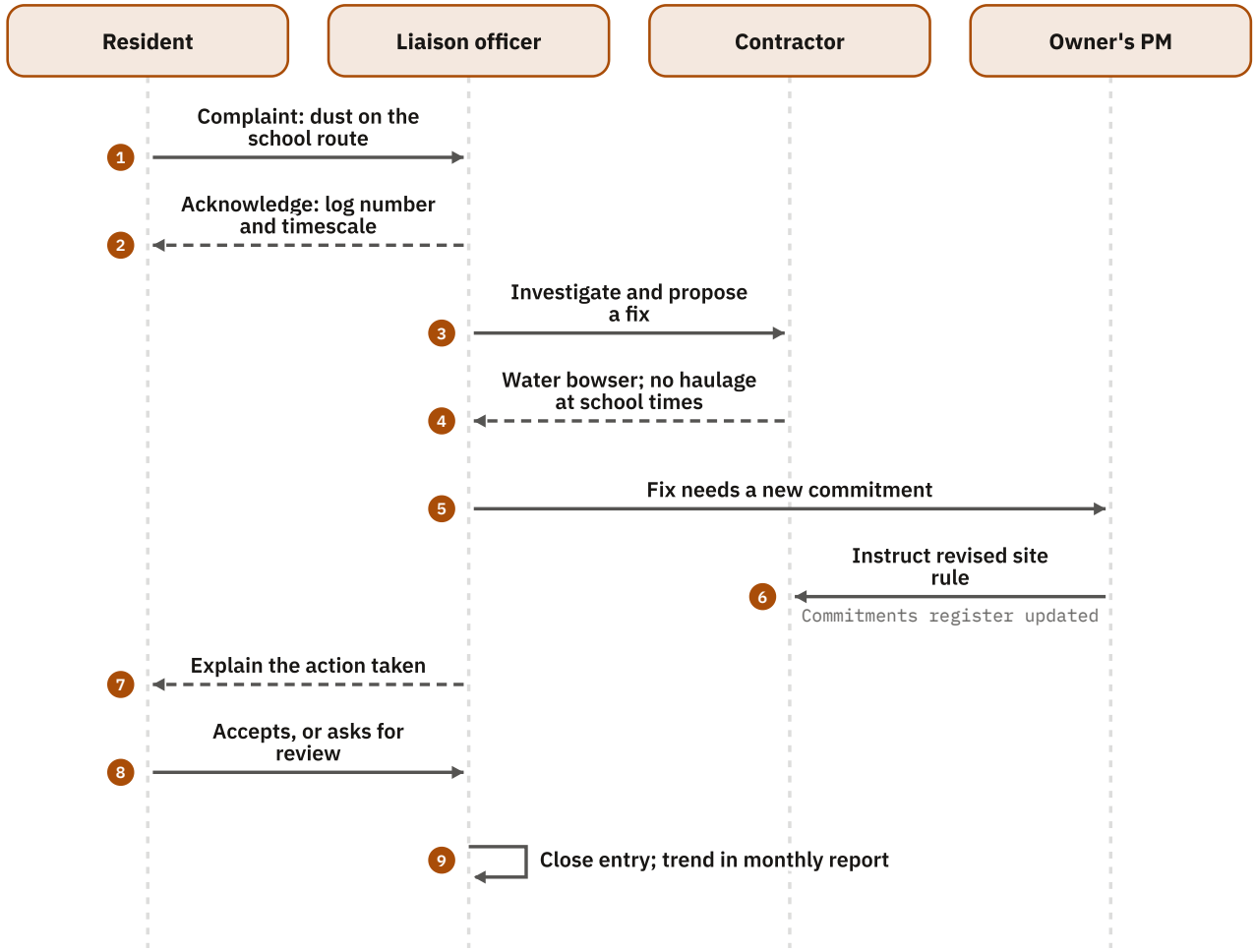
utility owner usually insists on approving the diversion design, may use its own crews, and grants shutdown windows only on its own calendar. A buried 33 kV cable or a large water main that appears on no drawing can stall a pile cap for months; a project that surveys early, agrees diversion designs jointly and books outages well ahead turns that risk into a managed sequence.

> **WATCH OUT · A PROMISE TO THE COMMUNITY IS A CONTRACT IN PRACTICE**
> Commitments made by a liaison officer at a public meeting – no piling after 10 pm, repair the access road after the monsoon, priority jobs for local workers – are binding in the community's eyes even if they appear in no contract. If the contractor's night-shift plan contradicts them, trust collapses quickly and complaints turn into stop-work orders. **Every commitment goes into the register, into site rules and into the subcontract**, with someone checking it is kept.

Lender-financed projects add formal requirements. The **IFC Performance Standards** (2012), the **Equator Principles** (EP4, 2020) applied by many commercial banks to project finance, and the **World Bank Environmental and Social Framework** (2018), which has a dedicated standard on stakeholder engagement and information disclosure, all expect meaningful consultation, disclosure in a form affected people can understand, and a **grievance mechanism** – an accessible route for complaints, free of charge and without retaliation, with recorded responses and timescales. Treat the grievance log as a leading indicator: a rising number of unresolved complaints in one neighbourhood often predicts the next stop-work order.

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Closing a community grievance



A grievance is closed only when the complainant has heard what was done and has a route to ask for review. The commitment it creates goes into the register and the site rules, not just the reply.

CASE STUDY · UTILITY DIVERSIONS ON AN ELEVATED METRO CORRIDOR (ILLUSTRATIVE)

A hypothetical elevated metro extension runs along a congested arterial road in a large South Asian city, with a long viaduct and several stations. The design assumed that utility records were reliable. Three months before piling, trial pits show that a large share of the planned pier positions conflict with services that are not on any record: a water main, medium-voltage power cables, telecom ducts and a storm drain. Each utility owner has its own approval process, crews and outage calendar. Traders object to the hoardings that diversions need, and the traffic police restrict daytime lane closures. The sponsor still expects piling to start on the baseline date. All figures and details are illustrative.

1. Which stakeholders are now on the critical path, and how would you map and prioritise them?

2. What should the project manager change in the schedule and the engagement plan?

3. How would you stop the same problem recurring on the next contract package?

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Model answer: module answer key.

SCENARIO · DECIDE · NIGHT POUR, WORRIED NEIGHBOURS

A continuous raft pour for a new hospital block in a residential area of Chattogram has to run through the night because the raft cannot have a cold joint. The building permit allows work only until 10 pm unless the environment department grants an extension, and the residents' committee complained about generator noise last month. The batching plant, pumps and labour are booked for a pour in three days, and the extension application is still unanswered. What should the project manager do FIRST?

(A) Proceed with the overnight pour as booked; once concrete is flowing nobody can stop it and the residents will understand afterwards.

(B) Meet the residents' committee and the authority now: explain why the pour must be continuous, offer specific mitigation and a hotline, seek written approval, and ask the designer for a fallback split-pour detail in case approval does not arrive.

(C) Postpone the pour indefinitely until every resident has agreed in writing to night working on the site.

(D) Leave it to the contractor, because noise control and working hours are within the contractor's scope under the contract.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Choose a project you know. Which single stakeholder outside your organisation could have stopped work for the longest time, and was their approval or goodwill shown anywhere in the schedule? What would you add to the register and the plan today?

Key terms

Stakeholder register
A controlled list of the individuals, groups and organisations that can affect or are affected by a project, recording their interests, influence, attitude, engagement owner, channel and links to project activities.

Salience model
A stakeholder analysis approach that judges priority by three attributes – power, legitimacy and urgency – so that stakeholders with legitimate or time-critical claims are not overlooked just because their formal power is low.

Social licence to operate
The informal, ongoing acceptance of a project by its local community and wider society, earned through fair treatment and kept promises; it is not issued by any authority and can be withdrawn at any time.

Power-interest grid
A two-by-two analysis tool that positions stakeholders by their power over the project and their interest in it, guiding whether to manage them closely, keep them satisfied, keep them informed or monitor them.

Stakeholder engagement plan
The document that sets out how, by whom, through which channels and how often each stakeholder group will be engaged, including consultation, disclosure and commitments, and how engagement will be monitored.

Utility diversion
The relocation, protection or temporary rerouting of existing services such as water mains, sewers, power cables, gas pipes or telecom ducts to clear space for new works, normally designed and approved with the utility owner.

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Consents register
A log of every permit, licence, clearance and approval a project needs, with the issuing body, prerequisites, lead time, validity, attached conditions, responsible owner and current status, linked to schedule activities.

Key takeaways

- Analyse stakeholders by power, interest, legitimacy and urgency, and re-map them at every stage gate and new work front
- Permits, consents and utility outages are scheduled activities with owners, durations and logic – not paperwork
- Verify buried services by trial pits and radar before foundation positions are frozen
- Log every commitment and grievance, flow commitments into site rules and subcontracts, and close the loop visibly
- Lender-financed projects must meet the lender's environmental and social standards, including a working grievance mechanism

KNOWLEDGE CHECK

1. An affected fishing community has little formal power over a coastal LNG terminal, but its livelihoods are directly harmed by dredging. Which analysis tool best explains why it must still be a priority?

   (A) The power-interest grid, because the community's interest is high
   (B) The salience model, as its claim is legitimate and urgent
   (C) A responsibility assignment matrix for the dredging package
   (D) A RACI chart that names the community as consulted

2. A planning authority approval has a lead time of four months and must be in place before site clearance. Where should the project manager show it?

   (A) In the communications plan as a monthly reporting item for the sponsor
   (B) Only in the risk register, as a threat to the start date
   (C) In the contractor's method statement for site clearance
   (D) In the schedule as an activity that is a predecessor to clearance

3. Why should trial pits and ground-penetrating radar be carried out before foundation positions are frozen on an urban project?

   (A) Records are often incomplete, and late finds force redesign and outages
   (B) They are needed to calculate the working bearing capacity of the piles
   (C) Utility owners no longer need to approve diversions once surveys exist
   (D) They replace the need for a separate traffic management plan during piling

4. A community liaison officer promised residents that heavy deliveries would avoid school hours. What is the most important next step?

   (A) Publish the promise on the project website and site notice boards for transparency
   (B) Ask residents to confirm the promise in writing within one week of the meeting
   (C) Record it in the commitments register and flow it into site rules and contracts
   (D) Mention it at the next monthly steering committee meeting for information

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5. Which stakeholder group typically sends a technical adviser to review progress, costs and environmental and social performance before releasing funds?

   (A) The building regulator's inspection team
   (B) Lenders such as development banks
   (C) The utility coordination forum
   (D) The project's community liaison team

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MODULE 7 · INTEGRATE · LESSON 2 · 35 MIN

Leading Site Teams

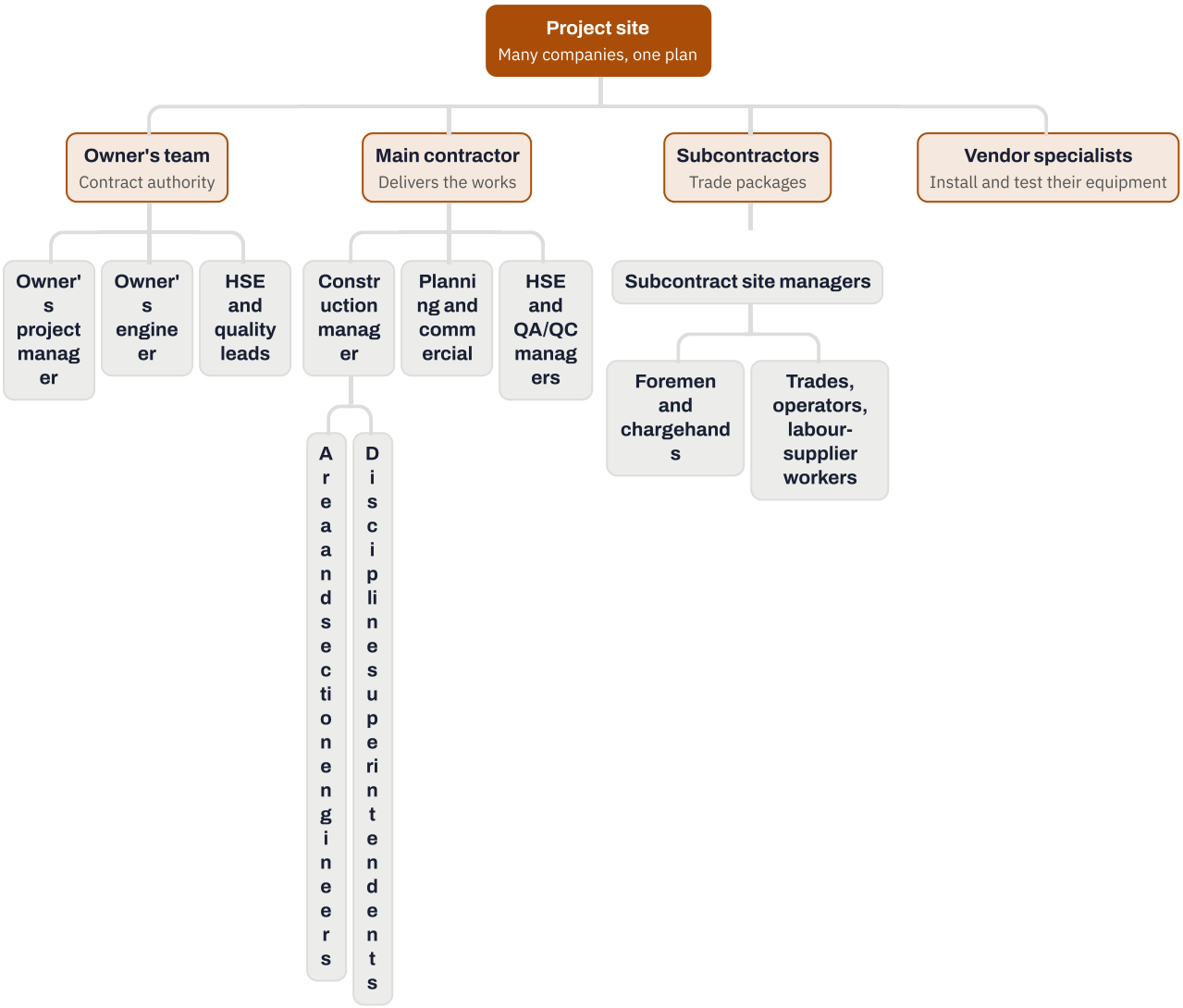
Multi-company crews, foremen, conflict, welfare and decisions under pressure – the PM as leader on site

- Describe how authority and influence flow through a multi-company site organisation
- Apply situational leadership and psychological safety with supervisors and crews
- Resolve common site conflicts and make sound decisions under time pressure
- Treat labour welfare as a leadership and productivity responsibility

A construction or plant site is a temporary organisation made of many companies: the owner and its engineer, a main contractor, a dozen or more subcontractors, labour suppliers, vendor specialists and inspectors. Most of these people do not report to the project manager. Authority comes from contracts, but performance comes from leadership – from whether supervisors understand the plan, whether crews trust that raising a concern is safe, and whether decisions are made quickly and explained. A project manager who stays in the site office reading reports sees the project weeks late; one who walks the work fronts, knows the foremen by name and listens before deciding sees it as it happens.

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A typical multi-company site organisation



Authority runs through contracts between companies, so most people on site do not report to the project manager. The foremen at the bottom of each branch are where the plan becomes work.

The foreman is the most important leader on site. Foremen translate the plan into crew instructions, decide the sequence each morning, notice hazards first and set the norms their crews follow. Many were promoted for trade skill rather than leadership, so invest in them: short training in planning, briefing and recognising fatigue; the look-ahead early enough to prepare; a voice in method statements; and public backing when they stop unsafe work. Watch **span of control** too – as a rough rule, a foreman can directly supervise about a crew on one compact work face, but not forty people spread over three floors of a building or two kilometres of pipeline.

Situational leadership matches the style to the person and the task. A supervisor who is new to a task needs clear direction; one who is learning needs coaching – direction plus explanation and encouragement; a capable but hesitant one needs support more than instruction; and a proven, confident foreman can be given the work front and trusted. The same foreman may need direction on an unfamiliar heavy lift and full delegation on routine formwork, so judge readiness task by task.

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Situational leadership styles

DIRECTIVE BEHAVIOUR ↑

Directing
- New to the task
- Clear instructions, close follow-up

Coaching
- Learning, not yet consistent
- Explain the why, invite ideas

Delegating
- Proven and confident
- Agree outcomes, then step back

Supporting
- Capable but hesitant
- Listen, encourage, share decisions

SUPPORTIVE BEHAVIOUR →

As competence and confidence grow on a given task, the style moves from directing through coaching and supporting to delegating. Step back to directing when the task is new or the risk is high.

ZEPMS site leadership habits

S1
Be present
Walk the work fronts daily and at different times, including night shifts and weekends. See ground conditions, housekeeping and morale for yourself instead of relying on reports.

S2
Make the plan visible
Keep the look-ahead, constraints and today's priorities on a board every supervisor can read. Foremen own their part of the plan and say when it is not achievable.

S3
Make it safe to speak
Thank people for bad news, ask what could go wrong today, separate learning from blame and close the loop on every concern raised.

S4
Decide and explain
Make timely decisions at the right level, explain the reasoning, record them and revisit them openly if the facts change.

S5
Look after people
Treat welfare, fair pay, rest, heat stress and recognition as leadership duties that apply equally to every company's workers on site.

Psychological safety is the shared belief that people can speak up with a concern, a question or a mistake without being punished or humiliated. On site it decides whether you hear about the missing shear studs before the deck is poured, whether a rigger questions a lift plan, and whether a subcontractor admits early that its steel delivery will slip. Leaders build it by admitting their own uncertainty, asking direct questions, responding to bad news with appreciation, and showing that reports lead to action. It is not the same as being soft: high standards and psychological safety work together, because people can only meet standards they feel free to discuss.

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Command-and-control vs engaged site leadership

Command-and-control

- Instructions flow down; information is expected to flow up unasked
- Problems are hidden until they become visible failures
- Foremen wait to be told the sequence each morning
- Safety stops are seen as lost production
- Blame follows incidents, so near misses go unreported

Engaged leadership

- Plans are built with foremen, who commit to what they can deliver
- Problems surface early because raising them is rewarded
- Foremen prepare from a visible look-ahead and raise constraints
- Safety stops are backed publicly by the project manager
- Incidents are analysed for causes, and reporting rises

EXAMPLE · A MORNING HUDDLE THAT WORKS

On a hypothetical semiconductor fab, each steel erection crew holds a ten-minute huddle at its work face before the shift. The foreman covers yesterday's progress against plan, today's lifts and exclusion zones, constraints such as missing bolts or a late delivery, the wind forecast, one safety point and an open question: does anyone see a problem with today's work? The project manager attends one huddle a day, listens more than speaks, and makes sure the constraints raised are cleared by the right person before the afternoon.

Conflict on site is normal; unmanaged conflict is expensive. Typical sources include competition for crane time, access and laydown space; ambiguous design information; payment disputes; trade interfaces such as ductwork against cable trays in a ceiling void; language and cultural differences; and simple fatigue. Conflict-handling styles range from competing and accommodating to avoiding, compromising and collaborating. None is always right: collaborating solves recurring interface conflicts, directing is necessary when safety is at stake, and a short pause can be wise when emotions are high – as long as the issue returns to the table.

Choosing a conflict approach on site

SITUATION	USEFUL APPROACH	WHY IT FITS
Two subcontractors keep fighting over the tower crane every morning	Collaborate on a weekly crane allocation meeting with shared rules	The conflict recurs, so a fair system works better than daily arbitration
A crew wants to work under a suspended load to save time	Direct and stop the work immediately	Life-safety rules are not negotiable and there is no time for discussion
Mechanical and electrical foremen argue over a ceiling-void route	Collaborate with the coordination model on screen	Both have valid constraints; a shared view finds a better route
A heated argument breaks out about a payment certificate in a site meeting	Pause, take it offline and reconvene with records	Emotion blocks problem-solving; the facts are in documents, not voices
Small disagreement on where to stack pallets for one day	Accommodate or compromise quickly	Low stakes do not justify a long negotiation

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Deciding under pressure

01 Stabilise
Make the situation safe first; stop work or isolate if needed

→

02 Sense
Gather facts fast from the people closest to the problem

→

03 Options
Frame two or three realistic options, including doing nothing

→

04 Decide
Decide at the right level with imperfect information

→

05 Communicate
Tell those affected what to do and why

→

06 Review
Check the outcome, adjust and record the decision

Labour welfare is a leadership duty

- Safe drinking water, shade and scheduled rest breaks, with heat-stress plans for hot seasons
- Clean toilets and washing facilities, with separate, secure facilities for women workers
- Safe, uncrowded accommodation where the project provides it, with working fire precautions
- Wages paid in full and on time, including by labour suppliers and small subcontractors
- Reasonable working hours, rest days and active management of fatigue on long shifts
- Free personal protective equipment that fits, and a confidential route for workers to raise concerns

INSIGHT · SAFETY IS A FUNDAMENTAL RIGHT AT WORK

In 2022 the ILO added a safe and healthy working environment to its fundamental principles and rights at work, alongside freedom of association and freedom from forced labour, child labour and discrimination. ILO Convention 167 (1988) sets the core standards for safety and health in construction, and ISO 45001:2018 – amended in 2024 so organisations consider climate change – frames the management system, with worker consultation and participation at its heart. On site this means leaders who plan for heat, consult crews before methods change and back every justified stop; buyers, lenders and regulators increasingly check that it is real.

SCENARIO · DECIDE · THE FOREMAN WHO STOPPED THE LIFT

On a combined-cycle power plant site, a rigging foreman employed by a subcontractor halts the lift of a heat recovery steam generator module because gusts are approaching the wind limit in the lift plan and one tag line is damaged. The main contractor's construction manager is furious: the heavy-lift crane costs a great deal per day and the turbine hall sequence is already tight. He tells you he intends to have the foreman removed from site. You are the owner's project manager. What should you do FIRST?

(A) Publicly support the stop, confirm the lift stays on hold until conditions are within the plan and the tag line is replaced, then discuss schedule recovery with the construction manager privately.

(B) Leave the matter to the contractor, because lifting methods and its own supervision are the contractor's responsibility under the contract.

(C) Ask the lifting supervisor whether the wind limit in the lift plan could be raised slightly so the lift can proceed today.

(D) Allow the lift to proceed with extra tag lines because the gusts are intermittent and the crane is costly to keep idle.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Think about the best site leader you have worked with. Which of the five ZEPMS site leadership habits did they practise most naturally, and which one do you find hardest? What will you do differently on your next site walk?

Key terms

Psychological safety
A shared belief within a team that people can raise concerns, questions, ideas and mistakes without fear of punishment or humiliation, enabling problems to surface early.

Span of control
The number of people or work fronts a supervisor can effectively direct, check and support; it shrinks when work is dispersed, complex or hazardous.

Gemba walk
A leader's regular visit to the place where the work is actually done to observe, ask questions and remove obstacles, rather than relying only on reports.

Labour welfare
The facilities, conditions and treatment provided for workers – water, sanitation, rest, accommodation, fair and timely pay, reasonable hours and respect – beyond the minimum needed to do the work.

Situational leadership
An approach in which the leader adjusts the balance of direction and support to the competence and commitment of the person or crew for the task in hand, from close instruction to delegation.

Team development stages
The typical progression of a new team through forming, storming, norming and performing (and later adjourning), used to anticipate early friction on multi-company sites.

OODA loop
A decision cycle of observe, orient, decide and act that helps leaders respond quickly and repeatedly to fast-changing situations such as site emergencies or disruption.

Key takeaways

- Authority comes from contracts, but performance comes from leadership – be present at the work fronts and know the foremen
- Match the style to each supervisor's readiness for the task: direct and coach new foremen, delegate to proven ones
- Psychological safety and high standards work together: thank people for bad news and act on it visibly
- Choose the conflict approach by stakes and recurrence – direct on life safety, build a fair system for recurring clashes
- Welfare and safe working are rights and productivity drivers for every company's workers on site

KNOWLEDGE CHECK

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6. A newly promoted foreman is skilled at pipe fitting but unsure how to plan a crew's day. Under situational leadership, what should the superintendent provide at first?

   (A) Full delegation, so the foreman learns quickly by making all decisions alone
   (B) More direction and coaching, then less as competence grows
   (C) A written warning to motivate faster improvement in planning
   (D) Reassignment to the trade crew until leadership skills develop

7. Which practice most directly builds psychological safety on a site?

   (A) Publishing a league table of crews ranked by incident counts
   (B) Requiring foremen to sign that every hazard has been eliminated
   (C) Rewarding the crews that report zero near misses each month with a bonus
   (D) Thanking people who report problems and acting visibly on reports

8. Two subcontractors argue every morning over tower crane time. What is usually the most effective approach?

   (A) Agree a shared crane allocation process with clear rules
   (B) Let the stronger subcontractor decide each day to save time
   (C) Avoid the issue until it settles down on its own
   (D) Split every day's crane hours exactly in half

9. During a site emergency, what is the first step in the 'deciding under pressure' sequence?

   (A) Gather all the facts before taking any action
   (B) Communicate the problem to the sponsor and lenders
   (C) Stabilise the situation and make people safe
   (D) Frame several options for the steering committee

10. Why is paying labour-supplier workers on time a concern for the project manager rather than only for the labour supplier?

    (A) It is a contractual pass-through item that the PM must certify every month
    (B) Late pay harms welfare, morale, productivity and reputation on site
    (C) It sets the critical path of every work package directly
    (D) It allows the PM to reduce retention on the main contract

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MODULE 7 · INTEGRATE · LESSON 3 · 35 MIN

Site Communication, Meetings & Reporting

Diaries, meetings, RFIs, submittals, document control and dashboards that drive action

- Design a meeting and reporting rhythm that suits the project stage
- Run RFI, submittal and transmittal processes with clear turnaround times and records
- Apply document control so the site always builds from the current revision
- Build dashboards and escalation routes that trigger decisions rather than describe status

Many site errors are really communication errors: a pile cap cast to a superseded drawing, cable trays installed before an RFI was answered, an oral instruction that nobody recorded and that later becomes a disputed variation. On an engineering project, communication is a controlled system with formal channels, agreed turnaround times and records that survive. Those records are also evidence: when a delay or claim is argued months later, the diary, the RFI log and the minutes will decide the outcome (developed in Procure).

A typical meeting rhythm during construction

MEETING	FREQUENCY	CHAIR	MAIN ATTENDEES	PURPOSE AND OUTPUTS
Start-of-shift huddle	Daily, at the work face	Foreman	Crew, sometimes the area engineer	Today's tasks, hazards, lifts and constraints
Daily coordination	Daily	Construction manager	Area supervisors, crane coordinator, HSE, QA/QC	Resolve today's and tomorrow's clashes for access, cranes, permits and inspections
Weekly progress and look-ahead	Weekly	Contractor's project manager	Subcontractors, planner, QA/QC, owner's representative	Progress against plan, look-ahead constraints, overdue RFIs and submittals
Design and interface coordination	Weekly or fortnightly	Design manager	Designers, BIM coordinator, discipline leads	Model coordination, clash resolution, RFI answers, design changes
Monthly progress review	Monthly	Owner's project manager or engineer	Contractor management, cost and planning leads	Performance, earned value, risks, changes and decisions needed
Steering committee	Monthly or quarterly	Sponsor	Senior owner, contractor and lender representatives	Strategic decisions, escalations and stage-gate approvals

PRO TIP · MEETINGS THAT PRODUCE ACTION

Issue the agenda in advance and start with safety. Review previous actions before new business. Record each **decision** and **action** with an owner and a date, not just discussion. Issue minutes within a working day, and ask that disagreement with the minutes is raised in writing promptly – silent acceptance of inaccurate minutes is a common source of later disputes.

The **site diary** is the project's memory. A useful daily entry records weather, labour by trade and company, plant on site and plant standing idle, materials delivered, work done by area, instructions received, visitors and

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inspections, incidents, and any delays with their causes. It should be written on the day, signed by a responsible person and supported by photographs with date and location. Diaries that read 'work progressing as normal' every day are worthless; diaries that note 'excavator idle 13:00–17:00 awaiting water main isolation by utility' are the raw material for progress measurement, delay analysis and fair settlement of claims. Many sites now keep the diary in a digital app with geotagged photographs, 360° captures and drone images attached; the discipline is the same, because an image without a date, a location and a note of what it shows proves very little.

RFI, submittal and transmittal – not the same thing

Request for information

- Raised by the contractor or site team
- Asks the designer to clarify, confirm or resolve a conflict
- Should quote drawing and clause references
- Often proposes a solution to speed the answer
- Answer may trigger a change if scope is affected

Submittal

- Raised by the contractor or supplier
- Submits shop drawings, method statements, samples or data sheets for review
- Required by the specification at defined points
- Reviewed and coded by the engineer or designer
- Released for procurement or construction only when accepted

Transmittal

- Used by any party
- A cover record of documents sent: what, which revision, to whom, why
- Proves when information was issued and received
- Carries submittals and RFI answers between parties
- Managed by document control in the CDE or document system

RFI life cycle

01 Raise — Site drafts the question with references, a sketch, a proposed solution and a required-by date

→

02 Log — Document control numbers it and routes it to the responsible designer

→

03 Review — Designer or engineer answers within the agreed turnaround

→

04 Respond — Formal answer issued; if scope changes, the change process starts

→

05 Close — Answer distributed to site, drawings revised and old copies withdrawn

TEMPLATE

RFI log

Track every request for information from raising to closure so that overdue answers are visible in weekly meetings and possible cost or time impacts are flagged early.

RFI NO.	SUBJECT AND LOCATION	DRAWING OR SPEC REFERENCE	RAISED / REQUIRED BY	RESPONSIBLE PARTY	STATUS	POSSIBLE COST OR TIME IMPACT
RFI-0142	*Clash between cable tray and HVAC duct, control building level 2, grid C/4*	*E-LAY-204 rev D; M-DUC-112 rev C*	*3 Mar / 10 Mar*	*MEP design lead*	*Open – overdue 2 days*	*Tray installation on level 2 on hold; possible reroute*
RFI-0143	*Anchor bolt projection for transformer plinth T2 not dimensioned*	*C-FDN-031 rev B*	*4 Mar / 7 Mar*	*Civil designer*	*Answered 6 Mar – closed*	*None*

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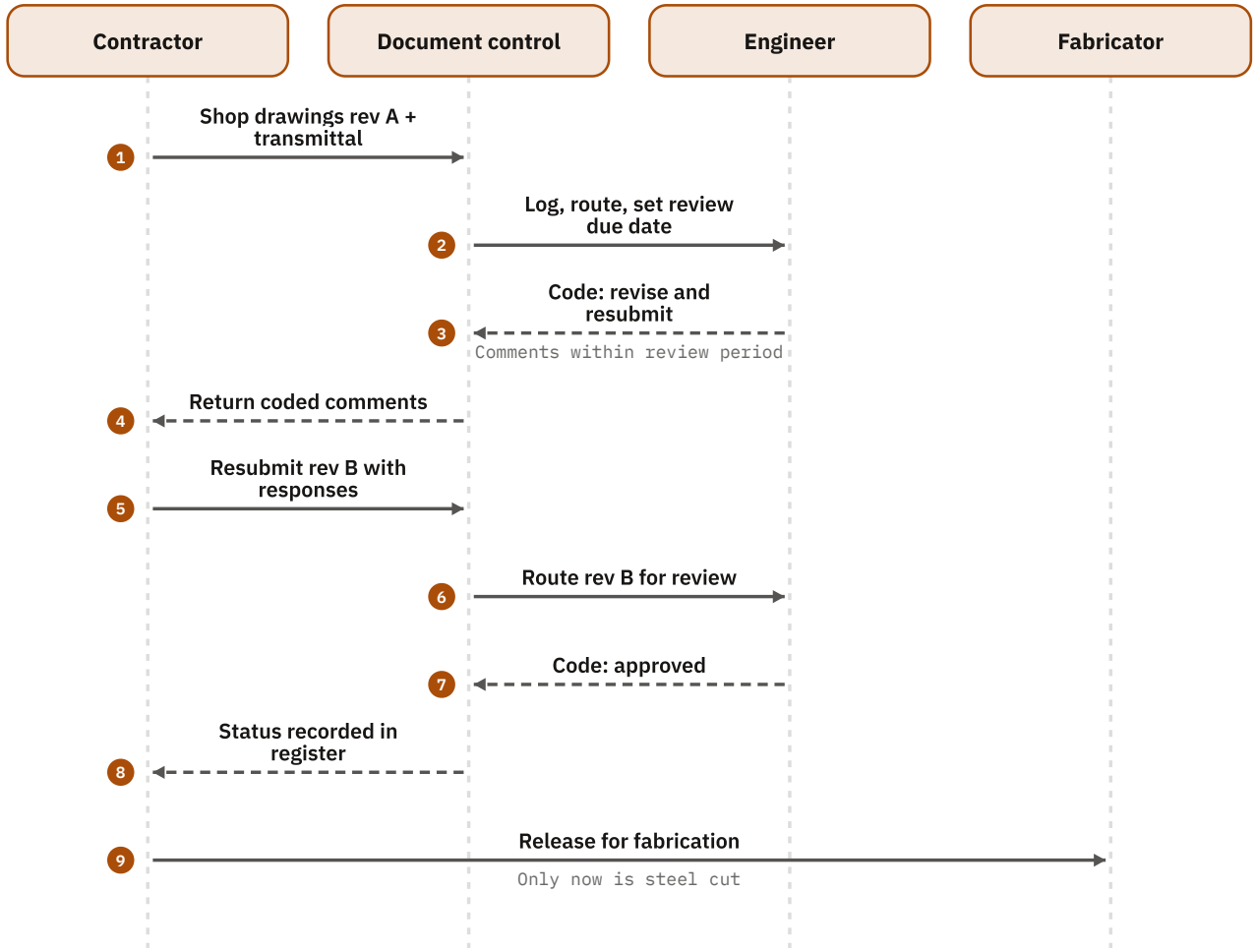
RFI NO.	SUBJECT AND LOCATION	DRAWING OR SPEC REFERENCE	RAISED / REQUIRED BY	RESPONSIBLE PARTY	STATUS	POSSIBLE COST OR TIME IMPACT

Running the submittal process

1. **Plan the register** Extract every required submittal from the specification into a register with planned submission and approval dates linked to procurement and construction activities.
2. **Prepare and self-check** The contractor checks shop drawings, method statements, samples and data sheets against the specification before submitting, so reviewers are not doing the contractor's quality control.
3. **Submit by transmittal** Issue through document control with a transmittal that states the revision, purpose and required review date.
4. **Review and code** The designer or engineer reviews within the agreed period and returns a status code – for example approved, approved with comments, revise and resubmit, or rejected. Code names vary by project, so define them in the procedures.
5. **Resubmit and close** The contractor addresses comments and resubmits. Fabrication, purchase or construction proceeds only once the status allows it.

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A submittal from first issue to release



Every hop passes through document control, leaving a dated record of what was sent and what was decided. The fabricator is released only after the status allows it – not when the first revision is sent.

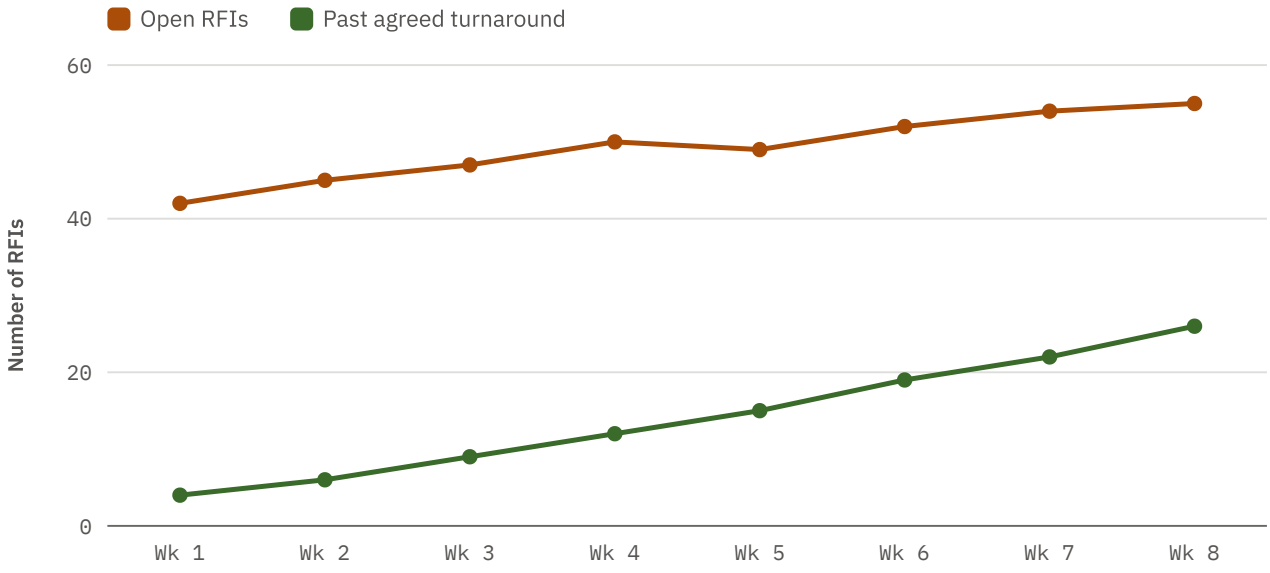
Document control ensures that everyone works from the right information. Every document has a unique number, a revision and a status such as for information, for review or for construction. Controlled documents are issued through the common data environment (see the next lesson) or a document management system, and superseded revisions are withdrawn or clearly stamped. On site, keep a drawing register that supervisors check weekly, make tablets display the current revision automatically, and allow only documents marked for construction at the work face. Most 'built to the wrong drawing' failures trace back to a printed copy that nobody withdrew. Under the ISO 19650 approach each information container also carries a status showing what it is suitable for and a revision code, so 'for construction' is a recorded state in the CDE rather than a stamp someone remembers to apply.

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Dashboards that drive decisions

- Show a few indicators that someone will act on, each with an owner and a threshold
- Prefer trends over snapshots – four weeks of rising overdue RFIs says more than one number
- Include leading indicators: open constraints, overdue submittals, RFIs older than the turnaround, inspections waiting
- Report by exception – highlight what needs a decision, not everything that is fine
- Draw from the same data as the schedule and cost systems so there is one version of the truth
- Allow drill-down from summary to the actual record, so people can check the evidence
- Link progress claims to evidence – dated photos, scans or drone surveys – so reviewers can verify them quickly

Open RFIs vs RFIs past turnaround (illustrative)



Illustrative data. The open total looks steady, but the overdue count rises every week – the trend that predicts blocked work fronts and deserves a decision now.

An **escalation matrix** tells everyone when and to whom an issue must be raised. It defines thresholds – for example a delay of more than a few days to a critical activity, a cost impact above a set value, a safety or environmental breach, or a stakeholder threatening a stop – and the level and time limit for each: foreman to area supervisor within hours, construction manager the same day, project manager within a day, sponsor or steering committee at the next meeting or sooner. Escalating early with options is professional; escalating late with excuses destroys trust. Where the contract requires formal notices, the escalation does not replace the notice.

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SCENARIO · DECIDE · TWO REVISIONS ON SITE

During cable tray installation in the control building of an LNG terminal, the electrical supervisor discovers that one crew has been working from printed revision C of the tray layout. The common data environment shows revision E, issued ten days ago, which reroutes the trays to avoid new HVAC ducts. About sixty metres of tray have already been installed to revision C. What should the project manager do FIRST?

(A) Let installation continue so the crew keeps its productivity, and raise a non-conformance report later once the area is complete.

(B) Formally discipline the supervisor whose crew used the old drawing, to send a clear message to all crews on site.

(C) Stop tray installation in the affected area, withdraw superseded prints, confirm the current revision with the designer, and assess the installed work.

(D) Ask the designer to revert the layout to revision C, since sixty metres of tray have already been installed to it.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Look at last week's minutes, diary entries or reports from a project you know. Would they help someone reconstruct what happened and why, a year from now? What two changes would make them more useful as evidence and as a basis for decisions?

Key terms

Site diary
A daily, contemporaneous record of site conditions and events – weather, labour, plant, deliveries, work done, instructions, visitors, incidents and delays – signed by a responsible person and used for progress and claims.

Submittal
A document, drawing, sample or data set that a contractor or supplier must submit for review under the specification – such as shop drawings, method statements or product data – before procurement or construction proceeds.

Document control
The function and procedures that number, register, issue, distribute, revise and archive project documents so that everyone uses the current approved revision and superseded copies are withdrawn.

RAG status
A red-amber-green rating used in reports and dashboards to show whether an item is on track, at risk or off track against defined thresholds; meaningful only when the thresholds are written down.

Request for information (RFI)
A formal written query from the construction team to the designer or engineer seeking clarification, confirmation or resolution of missing, conflicting or ambiguous design information.

Transmittal
A cover record that accompanies documents sent between parties, listing each document, its revision and purpose, and proving when it was issued and received.

Escalation matrix
A table defining which types and sizes of issue must be raised to which management level, by whom and within what time, so decisions are made at the right level early enough.

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Key takeaways

- Site communication is a controlled system: formal channels, agreed turnaround times and records that survive
- Write diaries and minutes as evidence – specific, dated and located – and challenge inaccurate minutes promptly in writing
- RFIs ask, submittals seek acceptance and transmittals prove issue; fabrication waits for the right submittal status
- Only current, published revisions reach the work face; withdraw superseded prints the day a new revision is issued
- Dashboards should trend leading indicators with owners and thresholds, and escalation should come early with options

KNOWLEDGE CHECK

11. Which diary entry is most useful for later delay analysis?

(A) Excavation work progressing normally on the eastern section today

(B) Weather fine; all crews productive; no significant issues noted

(C) Excavator idle 13:00–17:00 at pier 12 awaiting utility isolation

(D) General progress satisfactory across the whole site this week

12. A contractor sends shop drawings for steel connections for the engineer's review. What is this document package called?

(A) A submittal

(B) A request for information

(C) A non-conformance report

(D) A site instruction

13. Why should superseded drawing prints be withdrawn or stamped as soon as a new revision is issued?

(A) To reduce paper and printing costs on the project

(B) To satisfy the lender's monthly reporting format

(C) To prevent crews from building to obsolete information

(D) To speed up the designer's review of all open RFIs on the project

14. Which indicator on a weekly dashboard is a leading indicator of future delay?

(A) Number of RFIs older than their agreed turnaround time

(B) Total concrete volume poured on the project since mobilisation

(C) Cumulative cost spent to date on the project

(D) Number of workers who attended last month's induction

15. What is the main purpose of an escalation matrix?

(A) To replace formal contractual notices with quicker informal emails

(B) To list the contact details of every stakeholder

(C) To rank subcontractors by their performance each month

(D) To define which issues go to which level, and how fast

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MODULE 7 · INTEGRATE · LESSON 4 · 40 MIN

BIM & Digital Twins

Coordination, clash detection, 4D and 5D, the common data environment, reality capture and twins in operation

- Explain the main BIM uses – 3D coordination, clash detection, 4D sequencing and 5D cost – and their value on site
- Describe information management with a common data environment and information requirements, in line with the ISO 19650 series
- Plan reality capture to compare as-built conditions with the design model
- Distinguish digital twin maturity levels and choose uses that serve operations

Building information modelling (BIM) is often mistaken for 3D modelling. In practice it is the process of creating and managing structured information about a built asset across its whole life: geometry, but also properties such as equipment ratings, materials, manufacturers, maintenance intervals and links to documents. A good model lets a team coordinate trades before anything is fabricated, generate quantities, rehearse the construction sequence and hand the owner an information asset for operations. A poor one is an expensive 3D picture that site teams ignore because it is out of date.

BIM uses and what they deliver

BIM USE	WHAT IT MEANS	TYPICAL BENEFIT ON SITE	PREREQUISITE
3D design coordination	Combining discipline models into one federated view for review	Problems found on screen instead of during installation	Agreed coordinates, modelling standards and regular model exchange
Clash detection	Rule-based automated checks between models for hard and clearance clashes	Fewer site reroutes of ducts, pipes and trays; less rework	Models at the right level of information need, with tolerances defined
4D sequencing	Linking model elements to schedule activities to visualise construction over time	Tested crane positions, access, logistics and phasing before mobilising	A logic-linked schedule with codes that map to model elements
5D cost	Linking quantities from the model to cost data and the cost plan	Faster, traceable quantity take-off and cost impact of design changes	Consistent classification and a model built for measurement
Reality capture and verification	Laser scanning, photogrammetry and 360° imagery compared with the model	Early detection of deviations and accurate existing-condition surveys	Survey control, scanning plan and processing capacity
Design for manufacture and assembly (DfMA)	Model-driven fabrication of modules, precast elements, pipe spools and racks	Fewer site hours, factory-controlled quality and safer work at height	Earlier design freeze, fabrication-grade information and agreed tolerances
Asset information for operations	Structured asset data handed to the owner's maintenance systems	Faster mobilisation of maintenance and fewer missing records	Asset information requirements defined at the start

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EXAMPLE · THREE KINDS OF CLASH

A **hard clash** is a physical intersection: a chilled-water pipe through a steel beam. A **clearance or soft clash** violates required space: a valve with no room for a hand wheel, a switchboard without its front access zone, insulation thickness ignored. A **workflow or 4D clash** happens in time rather than space: two crews scheduled in the same plantroom on the same day, or a crane slewing over an area where roofing is planned. Good coordination looks for all three.

Running a model coordination cycle

1. **Federate** Collect the latest shared models from every discipline in the common data environment and combine them in a coordination tool on agreed coordinates.
2. **Test** Run clash and clearance rules zone by zone – for example structure against mechanical, mechanical against electrical – with tolerances agreed in the BIM execution plan.
3. **Group and prioritise** Group hundreds of raw clashes into real issues, discard duplicates and false positives, and prioritise by the construction sequence.
4. **Resolve** Review issues in the coordination meeting with discipline leads, assign each to an owner and due date, and record decisions.
5. **Re-test and sign off** Re-run the tests on updated models and sign off each zone as coordinated before it is released for fabrication or installation.

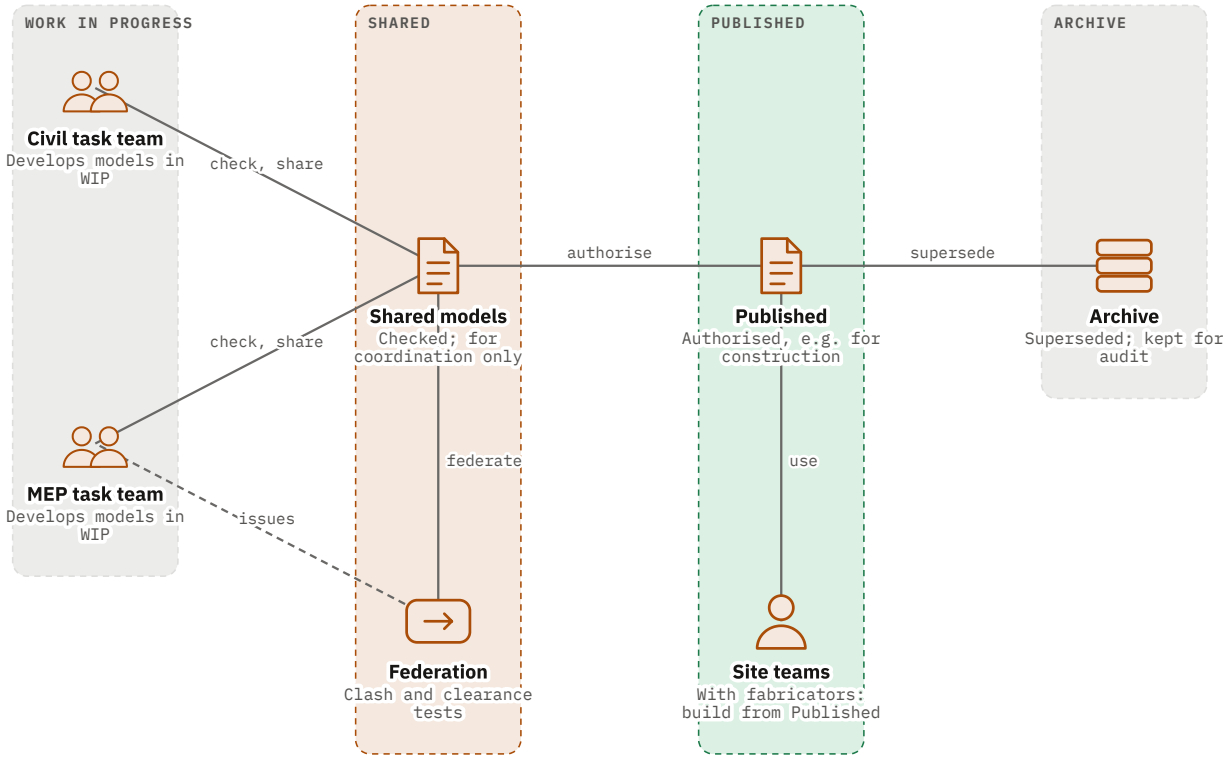
The **ISO 19650** series is the international framework for managing information over the whole life of a built asset when using BIM. Parts 1 and 2 (2018) set out the concepts and the delivery-phase process, part 3 (2020) covers the operational phase, part 4 (2022) information exchange, and part 5 (2020) a security-minded approach to sensitive asset information; together they have superseded the earlier UK PAS 1192 documents. In outline, the client – the appointing party – defines what information it needs and why; each lead appointed party responds with a BIM execution plan and information delivery plans; information is produced and exchanged in a **common data environment (CDE)** that moves it through defined states; and the level of information need limits detail to what is genuinely useful. The standards describe principles and processes rather than particular software. Check the current parts, amendments and any national annex before writing them into a contract, and state their requirements in your own project documents rather than assuming everyone has read them.

The information requirements cascade (top first)

Organisational information requirements	`What the owner organisation needs for strategic and business decisions` `Example: energy use and availability data across all plants`
Asset and project information requirements	`Asset information needed to operate and maintain` `Project information needed to make decisions during delivery`
Exchange information requirements	`What each appointed party must deliver` `When, in what format and at what level of information need`
Execution and delivery plans	`BIM execution plan` `Master and task information delivery plans`
Information models	`Project information model during delivery` `Asset information model handed over for operation`

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Information states in a common data environment



Information moves right only through a check, review or authorisation. Coordination uses Shared information, but only Published information may be built from, and superseded information goes to the Archive rather than disappearing.

WATCH OUT · MODEL STATUS MUST BE CONTRACTUAL, NOT ASSUMED

Site teams will build from whatever looks most current. State in the contract and the BIM execution plan whether the model or the drawings take precedence, what level of information need each element has reached, and what may be relied on for setting out or fabrication. A beautifully detailed model element that is only 'shared for coordination' is not an instruction to build.

Reality capture records the physical world digitally. Terrestrial and mobile laser scanners produce dense point clouds; drones with photogrammetry produce surface models and orthophotos; 360° cameras record interiors week by week. Compared with the design model, these datasets reveal deviations early: embedded plates out of position before steel erection, slab levels outside tolerance before raised floors, or earthworks volumes for monthly measurement. In brownfield work – a new pipe rack threaded through an operating fertiliser plant, or new equipment in an existing textile mill – a scan of existing conditions is often the only reliable basis for design, because record drawings rarely match reality.

Digital twin maturity – one practical ladder

LEVEL	DESCRIPTION	TYPICAL DATA	EXAMPLE USE IN OPERATION
0 – Reality capture	A static digital record of the asset as built	Point clouds, photos, as-built model	Planning a retrofit or a maintenance access route
1 – Descriptive	A model linked to asset data and documents	Asset register, O&M data, warranties	Finding a valve, its data sheet and its maintenance history quickly

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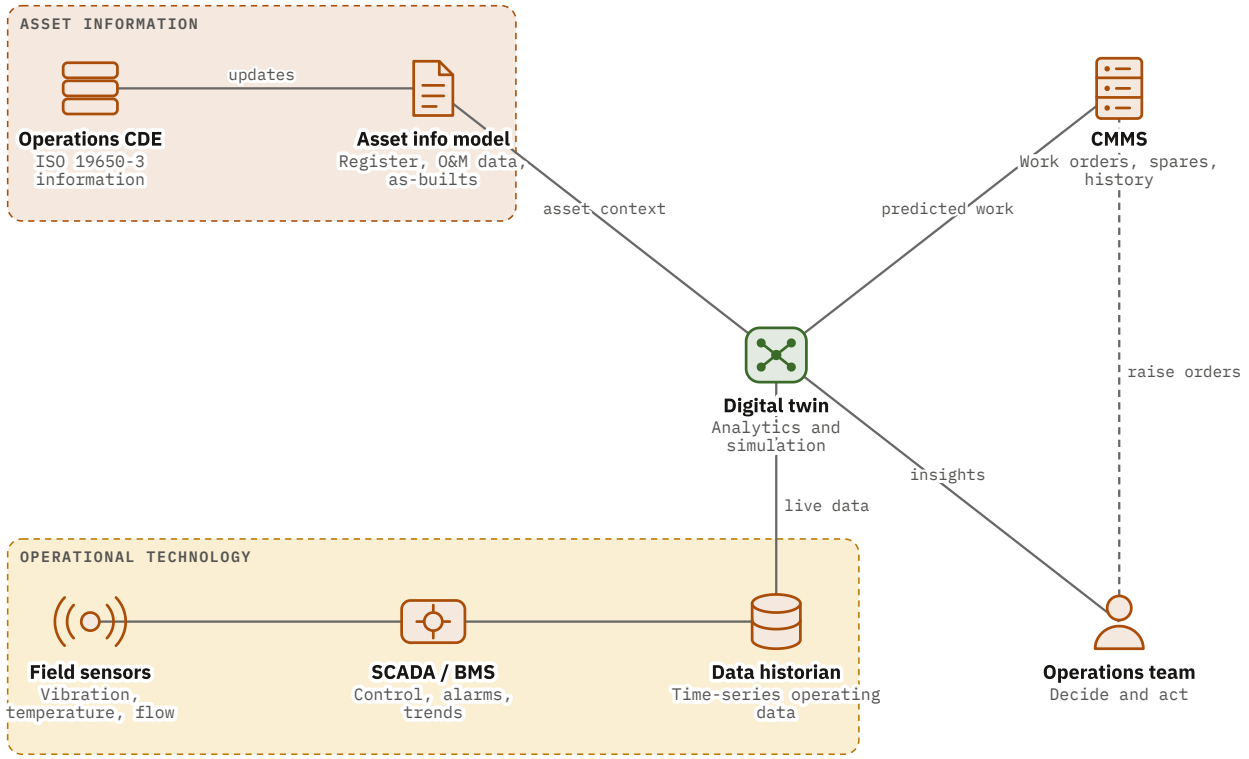
LEVEL	DESCRIPTION	TYPICAL DATA	EXAMPLE USE IN OPERATION
2 – Informative	Live or frequent operational data shown in context	Sensor feeds, building management or SCADA data	Seeing abnormal temperatures in a substation or plantroom on the model
3 – Predictive	Analytics forecast future condition or performance	Historical trends, condition monitoring	Predicting bearing failure in a pump or fouling in a heat exchanger
4 – Prescriptive	Simulation tests options and recommends actions, with people deciding	Physics or data-driven models, scenarios	Testing a new operating mode for a turbine or a ventilation strategy

INSIGHT · START FROM THE DECISION, NOT THE TECHNOLOGY

There is no single universal scale for digital twins, and organisations use different ladders. What matters is the operating decision the twin will improve – fewer unplanned outages, lower energy use, faster maintenance – and whether the owner has the people and data processes to keep it alive. A twin that nobody updates after handover becomes another out-of-date model.

A shared vocabulary helps: ISO/IEC 30173:2023 sets out digital twin concepts and terminology, operational information is managed under ISO 19650-3, and the twin works best inside an asset management system aligned with ISO 55000 and ISO 55001 (current editions 2024). The typical data flow of an informative-to-predictive twin for a plant is shown below – note that it starts with asset information defined in delivery, not with sensors bought at the end.

Data flows in an operational digital twin



The twin combines static asset information with live operating data from the operational technology network. Its value appears only when insights become work orders and operating decisions taken by people.

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CASE STUDY · BIM FOR A HOSPITAL BLOCK (ILLUSTRATIVE)

A hypothetical owner is building a multi-storey hospital block in Chattogram with operating theatres, intensive care, medical gas systems, pressurised isolation rooms and dense mechanical and electrical services in shallow ceiling voids. The owner's facilities team wants usable asset data at handover because its maintenance records for the existing hospital are poor. The design consultant and the main contractor both use BIM authoring tools, but no information requirements have yet been written, and the contractor proposes to 'model as needed'. All details are illustrative.

1. What should the owner's exchange information requirements specify, and when?

2. Where will clash detection and 4D sequencing add the most value on this project?

3. How can the owner make sure the asset information is usable at handover rather than a last-minute data dump?

Model answer: module answer key.

SCENARIO · DECIDE · THE MODEL SAID NO CLASHES

On an underground metro station, the coordination report for a concourse ceiling zone showed no remaining clashes and the zone was released for installation. A week later, site crews find cable trays clashing with sprinkler mains in several places. Investigation shows that the sprinkler subcontractor's model in the CDE was three weeks old because its latest version was still in its work-in-progress area and had never been shared. What should the project manager do FIRST?

(A) Instruct the sprinkler subcontractor to bear all rework costs and record the failure on its performance scorecard.

(B) Let site foremen route the trays and sprinkler mains around each other in the field to keep installation moving.

(C) Abandon model-based coordination for this zone and return to traditional two-dimensional overlay drawings.

(D) Pause installation in the zone, get all current models shared and re-run coordination, then tighten the CDE sharing rules in the BIM execution plan.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

On your current or last project, which single BIM use would have saved the most rework or delay, and what information requirement or CDE rule would have been needed to make it work?

Key terms

Building information modelling (BIM)
The process of creating, coordinating and managing structured graphical and non-graphical information about a built asset through design, construction and operation, using shared digital models and data.

4D BIM
The linking of model elements to schedule activities so that the planned construction sequence can be visualised, tested and communicated over time.

Common data environment (CDE)
The agreed single source of information for a project or asset, with defined workflows, permissions and states such as work in progress, shared, published and archived.

Digital twin
A digital representation of a physical asset that is connected to it by data flowing from the asset, used to monitor, understand, predict or optimise its performance in operation.

Clash detection
The automated, rule-based checking of federated discipline models to find physical intersections, clearance violations and sequencing conflicts before fabrication or installation.

5D BIM
The linking of model-derived quantities to cost data so that estimates, cost plans and the cost effect of design changes can be produced and traced from the model.

Exchange information requirements (EIR)
The client's specification of the information each appointed party must deliver, including content, format, level of information need and delivery dates.

Key takeaways

- BIM is structured information across the asset life, not a 3D picture; plan each use – coordination, 4D, 5D, DfMA – for a purpose
- ISO 19650 (parts 1–5) has superseded PAS 1192: the client states its information needs and the CDE controls information states
- Build only from Published information; Shared models are for coordination
- Reality capture finds deviations early and gives brownfield projects a reliable existing-condition baseline
- A digital twin starts from an operating decision and asset information requirements defined at the start of the project

KNOWLEDGE CHECK

16. A ventilation duct is modelled with no space left for the access panel needed to service a fire damper. What type of clash is this?

(A) A hard clash between two physical elements
(B) A workflow clash caused by the construction schedule
(C) A clearance or soft clash
(D) A coordinate system mismatch between two models

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17. In a common data environment, what must happen before information is used for construction?

(A) It must be published after authorisation for that purpose
(B) It must be shared with all other teams for coordination
(C) It must remain in the author's work-in-progress area
(D) It must be archived together with all of its previous revisions first

18. Which document tells each appointed party what information to deliver, in what format and when?

(A) The organisational information requirements document
(B) The exchange information requirements
(C) The asset information model for operations
(D) The task team's latest clash detection report

19. A twin shows live chiller temperatures from the building management system on the model, but makes no forecasts. Which maturity level does this best match?

(A) Reality capture
(B) Descriptive
(C) Prescriptive
(D) Informative

20. What is the main purpose of 4D sequencing on a congested site?

(A) To produce bills of quantities automatically from the federated model
(B) To test the construction sequence, access and crane use over time
(C) To replace the critical path schedule with a model
(D) To certify that the federated model matches as-built conditions

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · INTEGRATE · LESSON 5 · 40 MIN

AI & Data-Driven Project Controls

Scheduling assistants, prediction, computer vision and drones – with data quality, governance and human accountability

- Evaluate where AI scheduling assistants, generative schedule options and delay-risk prediction add value – and where they fail
- Apply data quality checks and verification before acting on any AI output
- Govern AI, computer vision and drone use with clear human accountability, privacy safeguards and records
- Interpret precision and recall when judging a detection or prediction tool

Project controls generate more data than any team can read: schedules with thousands of activities, daily diaries, RFI logs, photographs, sensor feeds and weekly drone surveys. AI tools can process that volume quickly – suggesting schedule logic, generating sequence options, flagging activities likely to slip, spotting missing hard hats in camera images or measuring progress from scans. What they cannot do is carry accountability. The planner who issues a baseline, the project manager who signs a report and the engineer who certifies progress remain answerable for it, whatever tool helped. Treat every AI output as draft analysis from a fast but fallible assistant: useful, worth checking, never self-certifying.

AI and data uses in project controls

USE	WHAT IT DOES	DATA IT DEPENDS ON	TYPICAL FAILURE	HUMAN CHECK
AI scheduling assistant	Answers questions about a schedule, runs health checks, drafts logic and narratives	Current schedule file, calendars, progress data	Invents logic or misreads float; confident but wrong narrative	Planner verifies against CPM rules and site knowledge
Generative schedule options	Produces many feasible sequences against stated constraints and objectives	Quantities, productivity rates, resource and space limits	Unrealistic rates or missing constraints such as curing or outage windows	Planner and construction manager test the chosen option
AI-assisted estimating	Drafts quantities from models or drawings and suggests rates from cost history	Model quantities, cost database, price indices, supplier quotes	Stale rates during price volatility; misread drawings; wrong estimate class	Estimator samples quantities and checks rates against current market prices
Delay-risk prediction	Scores activities or milestones for likelihood of delay	Historical project data, current status, supplier updates	Trained on unlike projects; stale inputs; hidden bias	Compare with the risk register and expert judgement
Computer vision for progress	Detects installed elements from photos or scans and compares with the model	Images, point clouds, current model	Occlusion, poor lighting; installed is not tested or complete	Engineer verifies before progress is certified
Computer vision for safety	Flags missing PPE, people in exclusion zones, unsafe proximity to plant	Camera feeds with consent and privacy controls	False alarms and missed events; privacy intrusion	Supervisor reviews alerts; never the only safeguard

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USE	WHAT IT DOES	DATA IT DEPENDS ON	TYPICAL FAILURE	HUMAN CHECK
Drone surveys	Measures stockpiles, earthworks, roof and façade condition, solar array installation	Flight permissions, ground control points	Poor control points; flights near restricted airspace	Surveyor checks accuracy; flight approvals confirmed

An **AI scheduling assistant** lets a planner ask plain-language questions of a schedule – which activities drive the substation energisation milestone, where are the open ends, which constraints override logic – and drafts narratives or first-cut logic from a WBS and past projects. **Generative scheduling** goes further, generating and ranking many feasible sequences against constraints such as crane capacity, access, curing times and outage windows. Both are genuinely useful for speed and for exploring options a team would never have time to draw by hand. Both also fail in recognisable ways: logic links that do not exist in reality, calendars ignored, productivity borrowed from a different context, and fluent narratives that misstate float. Every output must be checked against the critical path principles developed in Schedule and against what the site can actually do.

> **WATCH OUT · ACCOUNTABILITY DOES NOT TRANSFER TO THE TOOL**
> If an AI-drafted schedule becomes the baseline, the planner who prepared it and the project manager who approved it own it. Contracts, regulators, lenders and courts look to named people and to records of what they checked. Write down who reviewed each AI-assisted deliverable, what they verified and what they changed. 'The software said so' is not a defence.

Verifying an AI-generated schedule or forecast before use

1. **Check the inputs** Confirm the schedule version, data date, calendars, progress updates and supplier information the tool actually used. A model fed last month's data produces last month's answer.
2. **Check the logic** Run standard health checks: open ends, dangling activities, out-of-sequence progress, hard constraints, long lags and negative lags. Compare suggested links with real construction dependencies.
3. **Check reasonableness** Test durations against site productivity, crane and crew limits, curing and testing times, permit and outage windows, and weather seasons.
4. **Trace the critical path** Confirm that the critical and near-critical paths run through the drivers the team expects, and investigate any surprise rather than accepting it.
5. **Test sensitivity** Vary key assumptions – a late transformer, slower piling, a missed outage – and see whether conclusions hold.
6. **Approve and record** A named person approves the output, notes what was changed, and stores the version and review record in the CDE.

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Data quality before you trust a dashboard or model

☐ Completeness – all areas, packages and subcontractors are represented, with no missing weeks
☐ Timeliness – the data date is current and the update cycle is known
☐ Consistency – WBS codes, activity IDs, units and calendars match across systems
☐ Accuracy – a sample of records has been spot-checked against site reality
☐ Uniqueness – duplicates removed, for example the same punch or RFI logged twice
☐ Lineage – it is clear where each figure came from and who maintains it
☐ Representativeness – training data comes from projects similar in type, region and contract model
☐ Definitions – terms such as 'complete', 'installed' and 'tested' are defined the same way everywhere

Computer vision applies trained models to images and video. For progress, fixed cameras, 360° walk-throughs and scans can be compared with the model to detect installed columns, ducts or panels. For safety, cameras can flag people without helmets or harnesses, entries into exclusion zones around excavators, or proximity to live equipment. Drones extend this outdoors: stockpile volumes, earthworks, solar park row installation, roof and façade inspections, and progress on linear works. Limits are real: dust, glare, night conditions, occlusion behind scaffolding and camera angle all reduce accuracy, and an element that looks installed may not yet be connected, tested or accepted. Drone flights need aviation authority permissions and must respect no-fly zones near airports and sensitive installations.

Judging a detection or prediction tool

MEASURE	FORMULA	WHAT IT TELLS YOU
Precision	`Precision = TP / (TP + FP)`	Of everything the tool flagged, the share that was real. Low precision means many false alarms that waste supervisors' time and erode trust.
Recall	`Recall = TP / (TP + FN)`	Of all real events, the share the tool caught. Low recall means many missed events, which matters most for safety uses.
F1 score	`F1 = 2 × precision × recall / (precision + recall)`	A single figure balancing precision and recall; useful for comparing tools, but always look at both components.
False alarm share	`False alarm share = FP / (TP + FP) = 1 − precision`	The proportion of alerts that turn out to be wrong; a direct measure of review workload.

EXAMPLE · READING THE NUMBERS (ILLUSTRATIVE)
In a trial week, a PPE-detection system raises 200 alerts. Review shows 150 were real (TP) and 50 were false (FP), and a manual audit finds 30 real violations the system missed (FN). Precision = 150 / 200 = **0.75**; recall = 150 / (150 + 30) = 150 / 180 ≈ **0.83**. A quarter of alerts waste a supervisor's time, and about one real violation in six goes undetected. The tool can support supervision, but it cannot replace it – and its performance must be re-checked when conditions change, such as monsoon light or a new work area.

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Where precision and recall come from

TOOL RAISED AN ALERT →

False positive (FP)
- 50 in the trial week
- A wasted supervisor check

True positive (TP)
- 150 in the trial week
- Real violation caught

True negative (TN)
- Normal work, no alert
- Not used in precision or recall

False negative (FN)
- 30 found by manual audit
- Missed event – the safety risk

REAL EVENT OCCURRED →

Precision uses the top row: 150 / (150 + 50) = 0.75. Recall uses the right-hand column: 150 / (150 + 30) ≈ 0.83. The counts are the illustrative trial week above.

ZEPMS rules for governing AI on projects

G1
Purpose and owner
Every AI use has a stated purpose and a named accountable person who approves its outputs for use. Many organisations frame this in an AI management system such as ISO/IEC 42001:2023.

G2
Data quality
Inputs are checked for completeness, timeliness, consistency and representativeness before results are relied on.

G3
Human verification
Outputs that affect baselines, payments, safety or contractual positions are verified by a competent person before use.

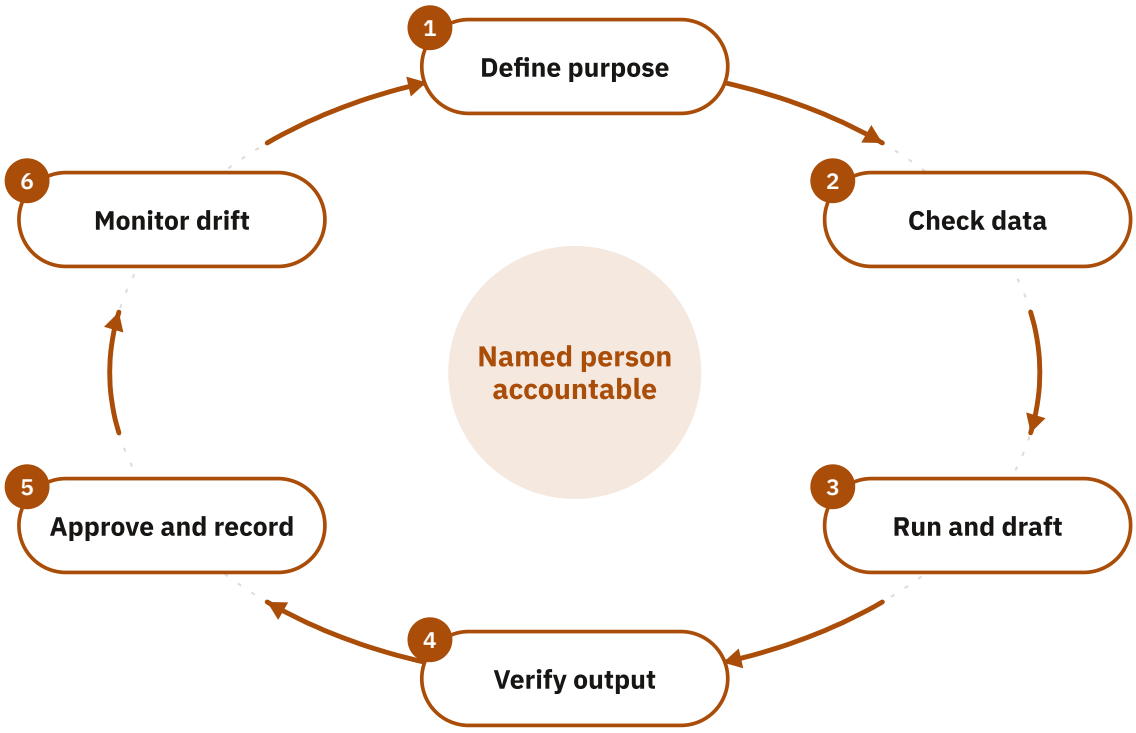
G4
Privacy and consent
People are informed, purposes are limited, retention is short and footage is not repurposed without policy and consultation. Check applicable data protection law and any AI-specific rules where the project operates.

G5
Security and confidentiality
Contract documents, designs and personal data go only into tools approved by the organisation for that class of information.

G6
Record and review
Uses, versions, reviewers and decisions are logged, and tools are re-tested periodically for drift and bias.

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Governing an AI tool in use



1. **Define purpose** — Use case, owner and limits entered in the AI use register
2. **Check data** — Completeness, timeliness, consistency and representativeness
3. **Run and draft** — The tool produces draft analysis, never a final answer
4. **Verify output** — A competent person checks against rules and site reality
5. **Approve and record** — Named approver; version, changes and review stored in the CDE
6. **Monitor drift** — Re-test accuracy, precision and recall as work or conditions change

Governance is a loop, not a one-off approval: when the work, season or site changes, monitoring sends the tool back to the data checks before its outputs are trusted again.

PRO TIP · PRIVACY ON SITE IS PART OF THE DESIGN

Cameras and drones capture workers' faces, vehicle plates and neighbouring homes. Tell workers and neighbours what is recorded and why, blur faces and plates where identification is unnecessary, limit retention, restrict access, and follow applicable data protection law. A camera installed for safety detection should not quietly become a disciplinary surveillance tool – trust lost this way also damages psychological safety on site.

TEMPLATE **AI use register**

Record each AI or advanced analytics use on the project so accountability, verification, privacy controls and review dates are explicit and auditable.

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USE CASE	TOOL TYPE (GENERIC)	DATA INPUTS AND OWNER	ACCOUNTABLE PERSON	VERIFICATION METHOD	PRIVACY AND SECURITY CONTROLS	REVIEW DATE
Weekly schedule health check and narrative draft	*AI scheduling assistant*	*Updated schedule file – lead planner*	*Lead planner; PM approves report*	*Standard logic checks plus planner review of critical path*	*Approved internal tool only; no contract documents uploaded externally*	*Monthly*
PPE and exclusion-zone alerts at turbine hall	*Computer vision on fixed cameras*	*Camera feeds – HSE manager*	*HSE manager*	*Daily supervisor review of alerts; monthly precision and recall audit*	*Worker notice, face blurring, 30-day retention, restricted access*	*Quarterly*

SCENARIO · DECIDE · THE FORECAST SAYS WE WILL MAKE IT

A delay-risk model in your project controls dashboard gives an 82% probability of meeting the energisation milestone for a 132/33 kV substation. Your planner's own review disagrees: the power transformer factory acceptance test has slipped and delivery of the gas-insulated switchgear is uncertain. You find that the model was trained mostly on past projects where equipment deliveries were rarely late, and this month's supplier updates were never fed into it. The sponsor wants to quote the 82% figure in tomorrow's board report. What should the project manager do FIRST?

(A) Let the sponsor quote the 82% figure, since the dashboard's model is more objective than the planner's judgement.

(B) Hold back the figure, update the model inputs, reconcile its result with the planner's analysis, and report a verified forecast with its assumptions and reviewer named.

(C) Report the average of the model's probability and the planner's estimate so that both views are fairly represented.

(D) Switch off the delay-risk model permanently and rely only on the planner's judgement from now on.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Pick one AI or data tool your organisation uses or is considering. Who is accountable for its outputs today, how are they verified, and what would you add to the AI use register to make that explicit?

Key terms

AI scheduling assistant
A software tool that uses AI to query, check, draft or explain project schedules in response to a planner's requests; its outputs are drafts that a competent planner must verify and approve.

Computer vision
AI techniques that interpret images, video or point clouds to detect and classify objects or events, such as installed elements, missing PPE or people inside exclusion zones.

Human-in-the-loop
A control design in which a competent person reviews, approves or overrides an automated output before it is acted on, keeping human accountability for the decision.

Data governance
The policies, roles and controls that decide who owns project data, how its quality is maintained, who may use it, for what purposes and how long it is kept.

Generative scheduling
The automated creation and ranking of many feasible construction sequences from defined activities, constraints, resources and objectives, from which planners select and refine an option.

Photogrammetry
The measurement of positions, shapes and volumes from overlapping photographs, often taken by drones, to produce surface models, orthophotos and point clouds.

Model drift
The decline in an AI model's accuracy over time as real conditions, data or work practices move away from those in its training data.

Key takeaways

- AI outputs are fast draft analysis; named people stay accountable for baselines, estimates, reports and certified progress
- Before use, verify inputs, logic, reasonableness, the critical path and sensitivity – and record who checked what
- Judge detection tools on both precision and recall; low recall rules a tool out as a primary safeguard
- Check data quality and representativeness first – stale or unlike data produces confident wrong answers
- Govern every use: register it, protect privacy and confidential information, and re-test for drift

KNOWLEDGE CHECK

21. A computer-vision system raised 80 alerts, of which 60 were real; a manual audit found 20 real events it missed. What are its precision and recall?

(A) Precision 0.60, recall 0.75
(B) Precision 0.75, recall 0.80
(C) Precision 0.80, recall 0.75
(D) Precision 0.75, recall 0.75

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22. An AI scheduling assistant drafts a revised schedule that becomes the new baseline. Who is accountable for it?

(A) The vendor that supplied and licensed the AI scheduling software

(B) The planner and project manager who reviewed and approved it

(C) No one, because the logic was generated automatically

(D) The client, because it accepted the baseline submission

23. A delay-risk model performs well for a year, then its predictions worsen after the project shifts from civil works to mechanical installation. What is the most likely explanation?

(A) Model drift as conditions move away from the training data

(B) A precision problem caused by too few camera alerts being raised

(C) Over-reliance on the human-in-the-loop review step

(D) A data governance policy that limits retention periods

24. Which practice best protects workers' privacy when safety cameras are installed on a site?

(A) Using the footage freely for any management purpose once it is recorded

(B) Keeping all footage indefinitely in case it is ever needed

(C) Informing workers, limiting purpose, blurring and short retention

(D) Placing cameras covertly so that worker behaviour is not affected

25. A dashboard's progress figures come from three subcontractors who each define 'installed' differently. Which data quality dimension is mainly at risk?

(A) Timeliness of the weekly updates

(B) Uniqueness of the progress records

(C) Lineage of the source systems

(D) Consistency of definitions

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · INTEGRATE

Answer key

Knowledge checks

1. B The salience model looks beyond power to legitimacy and urgency. The grid alone might place a low-power group in 'keep informed', which undervalues a community whose livelihood is directly affected.

2. D Permits that gate physical work belong in the schedule with realistic durations and logic, and in a consents register. Listing them only as risks or reports hides their effect on the critical path.

3. A Verifying buried services early lets designers move foundations or plan diversions before the critical path depends on them. The surveys do not set pile capacity or remove utility approvals.

4. C A commitment is only kept if it reaches the people who schedule deliveries. Publishing or reporting it is useful, but without site rules and contract obligations it will be broken.

5. B Lenders commonly appoint a technical adviser to monitor the project and confirm that drawdown conditions and environmental and social commitments are being met.

6. B Situational leadership matches style to readiness. A capable tradesperson new to supervision needs more direction and coaching at first, then increasing delegation.

7. D People speak up when doing so is rewarded and leads to action. League tables and rewards for zero reports encourage hiding problems.

8. A A recurring conflict is best solved collaboratively with a fair, transparent system. Forcing, avoiding or a rigid split ignore actual workload and priorities.

9. C Safety comes first: stop work or isolate hazards, then gather facts, frame options and decide. Waiting for all facts can cost lives.

10. B Unpaid workers are less safe, less productive and more likely to leave or protest; lenders and buyers also expect fair labour treatment across the supply chain.

11. C Specific, time-stamped entries with location and cause support delay analysis and claims. Generic statements prove nothing.

12. A Shop drawings, method statements and product data submitted for review under the specification are submittals. An RFI asks a question; it does not submit work for acceptance.

13. C Printed copies that remain at the work face are a frequent cause of rework. Withdrawal is a core document control practice.

14. A Overdue RFIs signal design information that will soon block work. The other measures describe what has already happened.

15. D An escalation matrix sets thresholds, levels and time limits so issues reach the right decision maker early. It does not replace formal notices.

16. C The duct does not intersect another object, but it violates a required maintenance clearance – a soft or clearance clash.

17. A Shared information is for coordination; published information has been authorised for uses such as construction.

18. B Exchange information requirements specify deliverables from appointed parties. Organisational requirements sit higher; the asset information model is a deliverable, not a specification.

19. D Live operational data shown in context is informative. Predictive and prescriptive levels add forecasting and simulation.

20. B 4D links model elements to schedule activities so the team can rehearse the sequence and logistics. It relies on, rather than replaces, a logic-linked schedule.

21. D Precision = 60 / 80 = 0.75. Recall = 60 / (60 + 20) = 0.75. Both happen to be equal here.

22. B Accountability stays with the named people who prepared and approved the baseline. Tools do not carry professional or contractual responsibility.

23. A When the work and data change, a model trained on earlier conditions loses accuracy – model drift. It needs re-testing or retraining.

24. C Transparency, purpose limitation, minimisation and short retention are core privacy safeguards. Covert or open-ended use damages trust and may break the law.

25. D Different definitions make figures non-comparable even if each is accurate and up to date; consistent definitions are needed before aggregation.

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Scenarios & case studies

Case study · Utility diversions on an elevated metro corridor (illustrative)
The utility owners have moved into the high-power, high-urgency corner because their approvals and outages now gate the piling sequence; the traffic police and the traders' association follow closely because they control working space and the social licence. The project manager should re-baseline the foundation sequence around confirmed clear pier positions, add diversion design, approval and outage activities with realistic durations, and consider relocating some piers or changing pile-cap geometry with the designer. A joint utility coordination forum with named decision makers, plus a traffic and access plan agreed with traders, should replace one-off letters. The sponsor needs an honest forecast with options rather than a promise to hold the date. For the next package, utility verification by trial pits and radar should be a design deliverable before foundation design freeze, and utility owners should be engaged during design rather than at mobilisation.

Scenario · Night pour, worried neighbours
B is best. Best. This engages the two stakeholders who control the outcome, makes the trade-off visible, offers real mitigation such as acoustic screens, low-noise pumps and directed lighting, and protects the structure with a contingency plan if consent is refused.

Scenario · The foreman who stopped the lift
A is best. Best. Backing a correct stop protects lives and shows every crew that speaking up is safe. Schedule recovery is a real concern, but it is solved by planning around the weather window, not by punishing the person who followed the lift plan.

Scenario · Two revisions on site
C is best. Best. Contain first, then make sure every crew is working from revision E, and evaluate what can be kept. Root cause and document control improvements follow.

Case study · BIM for a hospital block (illustrative)
The owner should define its asset information requirements first – which assets matter for maintenance, which attributes it needs, the classification system and the format its maintenance system can import – and translate these into exchange information requirements in the tender documents, with delivery milestones aligned to design stages and commissioning. Clash detection pays most in the ceiling voids, plantrooms and risers where ducts, medical gas pipes, cable trays and sprinkler mains compete for space, and 4D sequencing is valuable for phasing work next to any operational hospital areas and for crane and logistics planning. Asset data should be collected progressively, checked at each information exchange against the requirements, and test-loaded into the maintenance system before handover, with the facilities team involved in the checks. A CDE with clear states and a BIM execution plan agreed before construction would replace 'model as needed' with an accountable process.

Scenario · The model said no clashes
D is best. Best. Contain the problem, restore a trustworthy federated model, and fix the root cause – a release decision based on stale information – so other zones are not affected.

Scenario · The forecast says we will make it
B is best. Best. This fixes the data quality problem, keeps a competent person accountable for the number, and gives the board a forecast it can rely on, with caveats made explicit.

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08

MODULE 8 · COMMISSION

Quality, Commissioning, Handover & Closeout

Prove the asset works, hand it over ready to operate, close the project cleanly and feed the lessons back

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Build a quality management approach – QA, QC, project quality plan and cost of quality – that prevents defects rather than finding them late
- Control inspection and acceptance with ITPs, hold and witness points, FAT and SAT, NCRs and punch lists
- Plan and lead systems-based commissioning from mechanical completion to performance tests with a safe start-up
- Hand over an operable, documented asset with operations readiness, spares, training and usable asset information
- Close contracts and projects cleanly and turn lessons into changed practice and verified benefits

LESSONS

1. Quality Management on Engineering Projects · 35 min
2. Inspection & Test Plans, NCRs & Acceptance · 40 min
3. Commissioning · 40 min
4. Handover & Operations Readiness · 35 min
5. Closeout & Lessons Learned · 35 min

MODULE 8 · COMMISSION · LESSON 1 · 35 MIN

Quality Management on Engineering Projects

QA and QC, the quality management system, the project quality plan, cost of quality, audits and continual improvement

- Distinguish quality assurance from quality control and place both inside a quality management system
- Structure a project quality plan for an engineering project
- Calculate and interpret the cost of quality
- Use audits, supplier quality surveillance and the PDCA cycle to drive continual improvement

On an engineering project, quality means two things at once: **conformance** to the specified requirements and **fitness for purpose** in operation. The pile must reach its design capacity, the weld must pass radiography, the switchgear must perform its protection functions, and the finished plant must be operable and maintainable by the people who will run it. Defects in engineering are unusually expensive because work is quickly buried, covered, insulated or energised: honeycombing in a column found after cladding, a misaligned anchor bolt found when the turbine arrives, or a leaking roof found after the dyeing machines are installed. The economic logic of quality management is therefore simple – prevent defects rather than detect them late. ISO 21502:2020 treats quality management as one of the integrated practices of project management, planned alongside scope, schedule, cost and risk rather than left to an inspection department.

Quality assurance vs quality control

Quality assurance (QA)

- Focuses on the process
- Preventive – planned before and during work
- Examples: approved procedures, welder and procedure qualification, method statements, training, audits
- Asks: will our way of working reliably produce conforming results?
- Led by management and the QA function

Quality control (QC)

- Focuses on the product
- Detective – carried out during and after work
- Examples: slump and cube tests, dimensional surveys, radiography and other NDT, insulation resistance and pressure tests
- Asks: does this particular item conform to the requirement?
- Performed by inspectors, engineers and test laboratories

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The quality management hierarchy (top first)

Quality policy and objectives	`Organisation's commitment to quality` `Measurable objectives set by top management`
Quality management system	`Organisation-wide processes and responsibilities` `Often certified to ISO 9001 – a revision is in progress, so check the current edition`
Project quality plan	`How the QMS is applied to this project` `Project-specific responsibilities, standards and controls`
Procedures, method statements and ITPs	`How each activity is done and checked` `Acceptance criteria and inspection points`
Quality records	`Inspection reports and test certificates` `NCRs, audit reports and calibration records`

What a project quality plan should contain

SECTION	CONTENT	EXAMPLE DETAIL
Scope and quality objectives	What the plan covers and the measurable targets for the project	First-pass acceptance target for weld radiography; zero repeat NCRs of the same cause
Organisation and responsibilities	Who does what in quality, including independence of inspection	QA/QC manager reports to the project director, not to the construction manager
Standards and specifications	Codes, standards and specifications that apply, with order of precedence	Concrete, structural steel, pressure piping and electrical installation codes named in the specification
Control of documents and records	How quality documents are numbered, approved, issued and retained	Inspection records stored in the CDE against the system and tag numbers
Inspection and testing	ITPs, test laboratories, calibration of instruments and inspection resources	Accredited laboratory for cube testing; calibrated torque wrenches for bolting
Supplier and subcontractor quality	Prequalification, quality requirements in orders, surveillance and release	Vendor inspection at the transformer factory before release for shipment
Non-conformance and corrective action	How NCRs are raised, dispositioned, analysed and closed	Root cause analysis required for any repeated or major NCR

The **cost of quality** groups quality-related spending into four categories. **Prevention** costs stop defects happening: quality planning, training, procedure and welder qualification, design reviews, mock-ups and supplier prequalification. **Appraisal** costs find defects: inspection, testing, laboratory fees, third-party inspection and audits. **Internal failure** costs arise from defects found before handover: rework, scrap, re-testing, delay and the extra supervision they need. **External failure** costs arise after handover: warranty repairs, call-backs, lost production for the owner, claims and damaged reputation. Prevention and appraisal together are the cost of conformance; internal and external failure together are the cost of poor quality. Spending a little more on prevention usually removes much more in failure costs.

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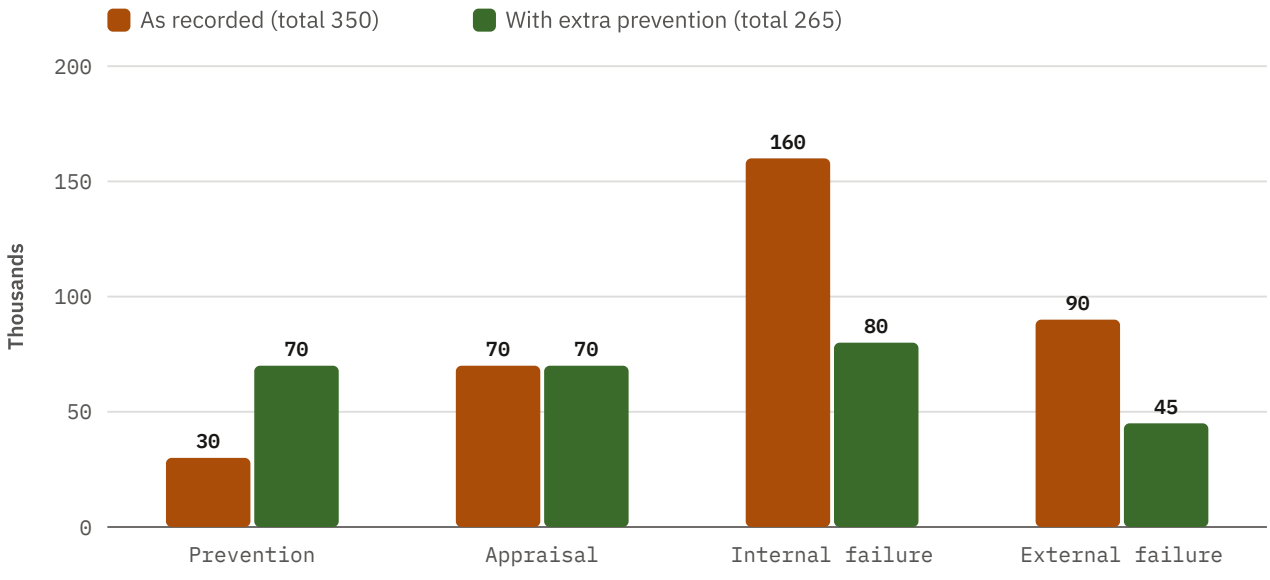
Cost of quality

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost of quality	`COQ = prevention + appraisal + internal failure + external failure`	The total cost of achieving quality and of failing to achieve it, for a project, package or period.
Cost of conformance	`Cost of conformance = prevention + appraisal`	What is spent to prevent and detect defects – the cost of good quality.
Cost of poor quality	`COPQ = internal failure + external failure`	What defects actually cost once they occur, before or after handover.
COPQ share	`COPQ share = COPQ / COQ × 100%`	The proportion of quality cost that is wasted on failure; a high share signals under-investment in prevention.

EXAMPLE · COST OF QUALITY ON A MILL EXTENSION (ILLUSTRATIVE)

A hypothetical spinning mill extension near Narayanganj records, in thousands: prevention 30, appraisal 70, internal failure 160 (slab rework, rejected pipe spools) and external failure 90 (roof leak repairs and lost production after handover). **COQ = 30 + 70 + 160 + 90 = 350**; COPQ = 160 + 90 = 250, so the COPQ share is 250 / 350 ≈ **71%**. If an extra 40 on prevention – pre-pour checklists, a roof mock-up, welder retesting – had halved both failure categories, the figures would have been 70 + 70 + 80 + 45 = **265**: a lower total cost with better quality. All figures are illustrative.

Cost of quality on the mill extension (illustrative)



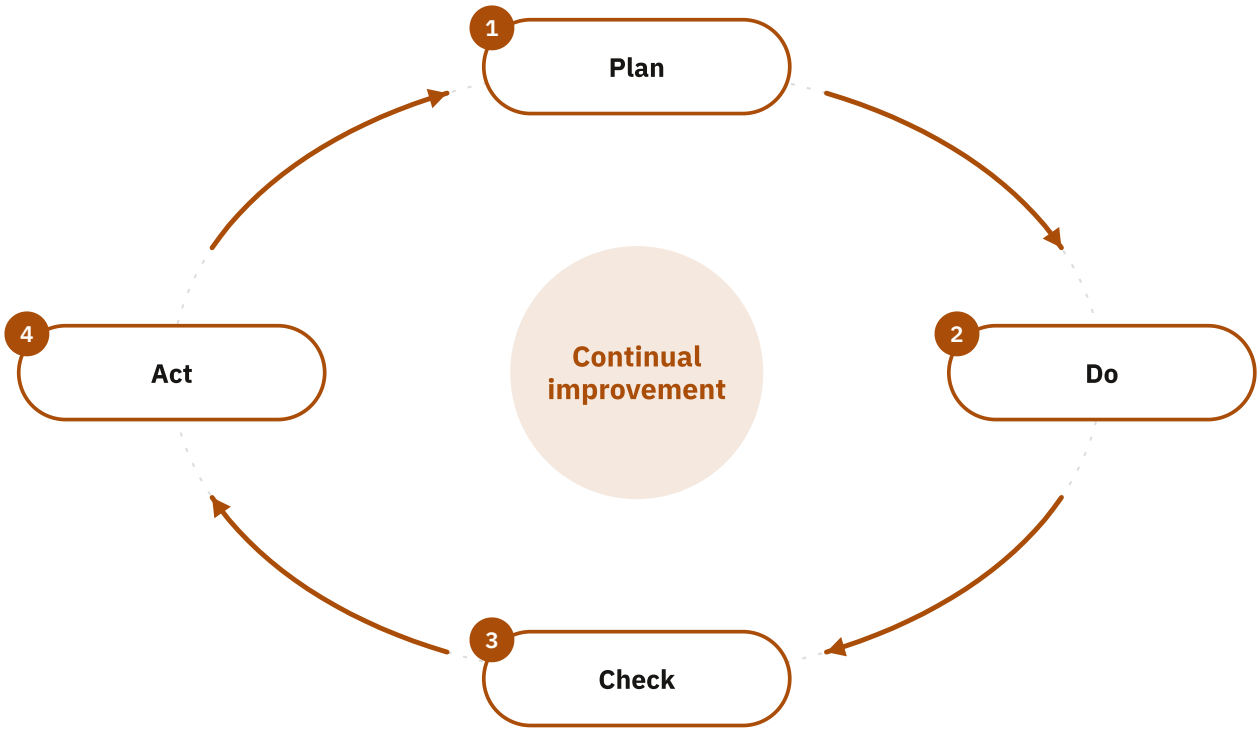
Illustrative figures from the example above. Adding 40 to prevention halves both failure bars, so the total cost of quality falls from 350 to 265 while quality improves.

Quality audits check whether the system is being followed and is effective. A system audit examines the quality management system as a whole; a process audit looks at one activity, such as concrete production or welding; a product audit re-examines a sample of finished work. Findings are graded – typically major non-

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conformity, minor non-conformity and observation – and each needs a corrective action with an owner and a date. **Supplier quality** extends the same discipline into the supply chain: prequalify vendors, write quality requirements into purchase orders, hold a pre-inspection meeting to agree the vendor's ITP, carry out surveillance at the works, and issue a formal release before shipment. A switchgear panel rejected at the factory costs days; the same panel rejected on site costs weeks. Offsite manufacture and DfMA – precast elements, volumetric modules, pipe spools, prefabricated plant rooms – move more of the quality story into the factory, so agree the factory ITP, inspect the first article and check dimensions and interfaces before dispatch, when correction is still cheap.

The PDCA cycle for continual improvement



1. **Plan** — Analyse the problem and its causes; plan a change with a measure of success
2. **Do** — Try the change on a limited scale, such as one floor or one crew
3. **Check** — Measure the results against the target and look for side effects
4. **Act** — Standardise what worked in procedures and ITPs, or start the cycle again

The cycle never ends: each standardised improvement becomes the new baseline for the next Plan step. Trials stay small until Check shows that the change works.

INSIGHT · QUALITY AND SCHEDULE ARE ALLIES

Teams under schedule pressure often see inspection as delay. In practice rework is one of the biggest destroyers of schedule: a failed pressure test, a rejected pour or a repeated NCR consumes far more time than doing it right first time. Measuring first-pass acceptance and making it visible on the site dashboard changes the conversation from 'quality slows us down' to 'rework is what slows us down'.

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SCENARIO · DECIDE · WELD REJECTIONS FROM A SPOOL FABRICATOR

Pipe spools for a hypothetical effluent treatment plant at a dyeing factory in Gazipur are arriving from an off-site fabricator. Radiography on site is rejecting an unusually high share of the shop welds, and the site team is cutting out and re-welding joints, which is delaying hydrotesting. The fabricator says the defects are caused by handling damage on site. What should the project manager do FIRST?

(A) Terminate the fabrication order immediately and move the remaining spools to a new fabricator to protect the schedule.

(B) Accept the situation, repair the welds on site and backcharge the fabricator for the cost of the rework at the end of the job.

(C) Raise an NCR, hold further shipments pending investigation, analyse the defect types with the fabricator and audit its welding process, qualifications and consumables.

(D) Increase on-site radiography to every weld so that every defective joint is found before hydrotesting.

Circle your choice. Answer and feedback: module answer key.

REFLECT

On a project you know, estimate roughly how quality costs split between prevention, appraisal, internal failure and external failure. Which single prevention measure would have removed the largest failure cost?

Key terms

Quality assurance (QA)
The planned, systematic, process-focused activities that give confidence that work will meet requirements, such as approved procedures, qualifications, training and audits.

Quality management system (QMS)
An organisation's documented and implemented set of policies, processes, responsibilities and records for directing and controlling quality, often certified against ISO 9001.

Cost of quality (COQ)
The total of prevention, appraisal, internal failure and external failure costs, used to show the economic case for preventing defects rather than detecting and repairing them.

PDCA cycle
The plan-do-check-act improvement loop in which a change is planned, trialled, measured and then standardised or revised, used for continual improvement.

Quality control (QC)
The product-focused inspection, measurement and testing activities that check whether specific work or items conform to their requirements.

Project quality plan (PQP)
The project-specific document that sets out how the quality management system will be applied to a project: objectives, organisation, standards, inspection and testing, supplier control, NCRs and audits.

Quality audit
A systematic, independent and documented examination of whether quality arrangements are followed and effective, leading to graded findings and corrective actions.

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Key takeaways

- Quality means conformance and fitness for purpose; prevent defects, because engineering work is soon buried, covered or energised
- QA builds a reliable process, QC checks the product, and the project quality plan applies the QMS to this project
- COQ = prevention + appraisal + internal failure + external failure; a high failure share signals too little prevention
- With offsite manufacture, quality control moves into the factory: agree vendor ITPs and release before dispatch
- Use audits and PDCA to standardise what works, and make first-pass acceptance visible on the dashboard

KNOWLEDGE CHECK

1. Which of these is a quality assurance activity rather than a quality control activity?

   (A) Taking concrete cubes from each truck mixer for testing
   (B) Qualifying welders before they start production welding
   (C) Measuring the cover to reinforcement before a pour
   (D) Radiographing a sample of butt welds on a pipeline

2. A package records prevention 20, appraisal 50, internal failure 110 and external failure 40 (thousands). What is the cost of poor quality?

   (A) 70 thousand
   (B) 130 thousand
   (C) 220 thousand
   (D) 150 thousand

3. Which document explains how an organisation's QMS will be applied to a specific power plant project?

   (A) The project quality plan
   (B) The quality policy statement
   (C) The inspection and test plan for piling
   (D) The supplier's mill test certificate

4. An audit finds that calibration records for torque wrenches are missing for two months. What should follow?

   (A) A note in the audit report with no further action
   (B) Immediate dismissal of the tool store supervisor
   (C) A graded finding with corrective action, owner and date
   (D) A decision to stop all bolting work for the whole project

5. A crew trials a new pre-pour checklist on one floor and measures rework before rolling it out. Which PDCA step is the trial itself?

   (A) Plan
   (B) Do
   (C) Check
   (D) Act

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · COMMISSION · LESSON 2 · 40 MIN

Inspection & Test Plans, NCRs & Acceptance

Hold, witness and review points, factory and site acceptance tests, traceability, non-conformances and punch lists

- Build an inspection and test plan with hold, witness and review points
- Plan factory and site acceptance tests and maintain material traceability
- Manage non-conformances from report through root cause to verified close-out
- Control punch lists with measures that forecast when they will be cleared

The **inspection and test plan (ITP)** is the backbone of quality control. For each work activity it states what will be checked, against which criteria, how often, by whom, what record will prove it, and which parties must be notified or present. ITPs are derived from the specification and the method statement, agreed before work starts, and used every day by inspectors and supervisors. A good ITP is short enough to be used at the work face and specific enough that two inspectors would make the same decision.

Extract from an ITP for a reinforced concrete pile cap (illustrative)

STEP	ACTIVITY OR CHECK	REFERENCE AND ACCEPTANCE CRITERIA	FREQUENCY	RECORD	CONTRACTOR	ENGINEER
1	Excavation level and pile head trimming	Drawings; pile cut-off level within specified tolerance	Each pile cap	Survey record	H	W
2	Pile integrity test results	Specification; no anomalies outside acceptance criteria	Each pile	Test report	R	R
3	Reinforcement: bar size, spacing, laps and cover	Bar bending schedule and drawings; cover within tolerance	Each pile cap	Rebar inspection checklist	H	H
4	Formwork, embedded items and cleanliness	Drawings; dimensions, holding-down bolts and sleeves in position	Each pile cap	Pre-pour checklist	H	H
5	Fresh concrete: slump, temperature and test cubes	Mix design and specification limits	Per truck or per set volume	Delivery tickets and sampling log	W	S
6	Hardened concrete strength	Cube results against specified characteristic strength	Per sample set	Laboratory certificate	R	R

Types of inspection point

H

Hold point

Work must not proceed beyond this point until the named party has inspected or reviewed it and given written release. Covering reinforcement before release breaches the ITP.

W

Witness point

The named party must be notified with agreed notice and may attend. If it does not attend after proper notice, work may proceed and the record stands.

R

Review point

Documents such as certificates, test reports or calibration records are reviewed, without attending the activity itself.

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S

Surveillance

Random or periodic monitoring at the inspector's discretion, without a notification requirement, to check that the process is under control.

WATCH OUT · NEVER BYPASS A HOLD POINT

Pouring concrete over reinforcement that has not been released, or closing a vessel before internal inspection, may force the work to be broken out or opened up at the contractor's cost, and destroys the evidence that would have shown it was right. **Verbal releases are not releases**: require a signed record or an entry in the inspection system, and plan inspections into the look-ahead so that hold points do not become delays.

Major equipment is tested twice. A **factory acceptance test (FAT)** is carried out at the manufacturer's works before shipment, where faults are cheapest to fix and test facilities are best. A power transformer is put through routine tests such as ratio, winding resistance, impedance and losses, and dielectric tests, plus any type or special tests the specification calls for. Pumps are run on a test bed against their performance curve; switchgear is checked for wiring, interlocks and insulation; a control system is tested by simulating inputs and checking logic and graphics. A **site acceptance test (SAT)** follows installation and proves that the equipment works correctly in its real environment, connected to real power, process and control interfaces. Remote or video-witnessed FATs are now common when travel is impractical; agree in advance which tests will be witnessed live, the camera coverage, access to raw test data and instrument calibration, and how the record will be signed.

Factory acceptance test vs site acceptance test

Factory acceptance test

- At the manufacturer's works, before shipment
- Proves the item meets its specification and data sheet
- Uses factory test bays, simulated loads and simulated I/O
- Defects corrected cheaply by the maker's own staff
- Release for shipment follows a successful FAT

Site acceptance test

- On site, after installation and pre-commissioning checks
- Proves the item works in its real installed environment
- Uses actual power, process connections and control interfaces
- Defects cost more and can delay commissioning
- Results feed system commissioning and handover records

Material traceability links each installed item back to its origin and test evidence. Structural steel and pressure parts carry heat numbers that match their mill test certificates; marking is transferred when plates or pipes are cut; alloy materials may be checked with positive material identification; concrete is traced by delivery tickets and cube sets; cable drums and their test reports are linked to circuits. Traceability matters most when something goes wrong: if one heat of steel or one batch of bolts is found to be defective, the project can find every location where it was used instead of opening up everything. Many projects now link barcodes or RFID tags on materials and spools to the completions database, so the search takes minutes rather than days.

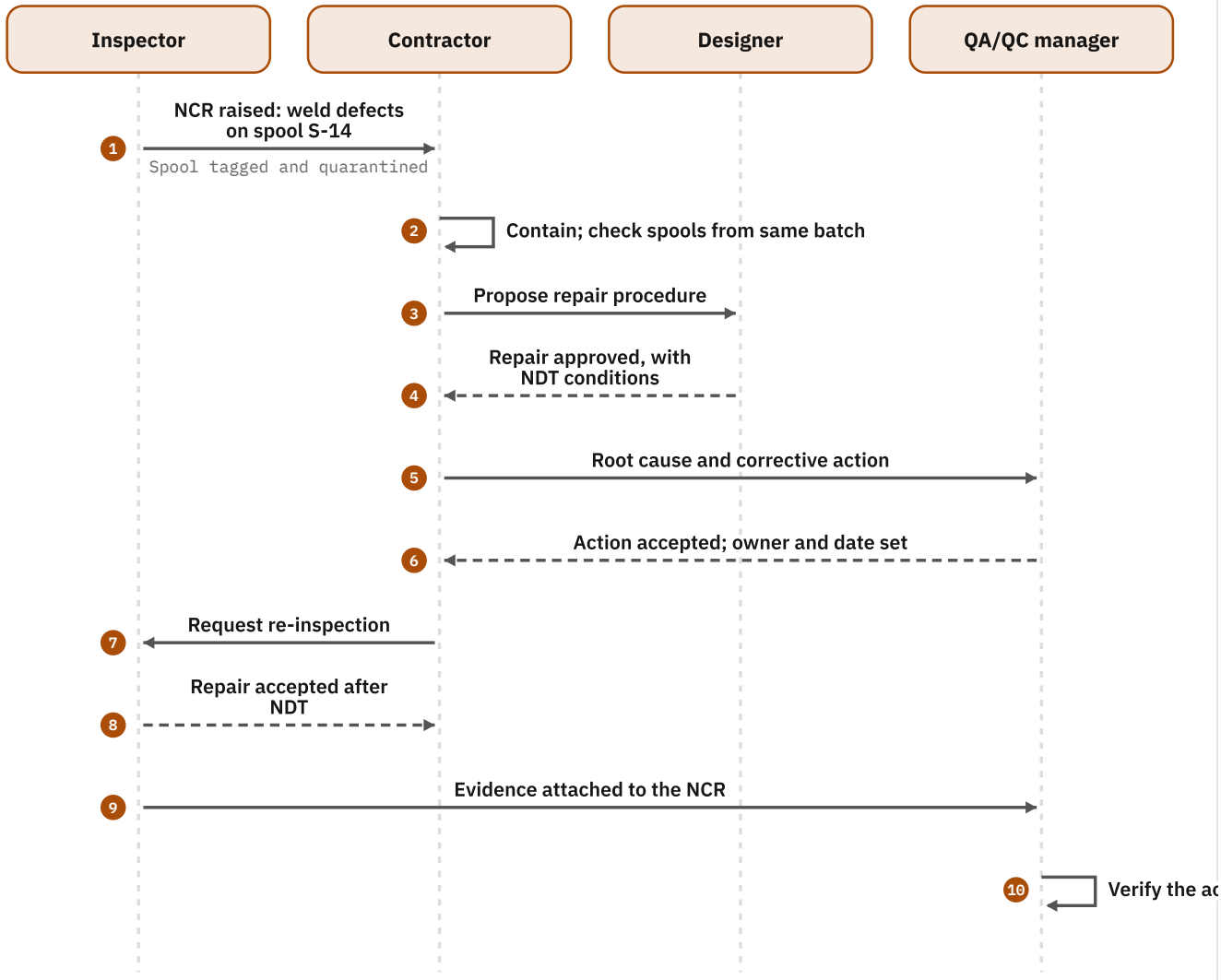
From non-conformance to verified close-out

1. **Identify and contain** Stop the affected work, tag and segregate the item, and notify the responsible parties so that nothing more is built on top of the problem.

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2. **Raise the NCR** Describe what is wrong, the requirement it breaches, the evidence, the location and the quantity affected.
3. **Disposition** Decide the treatment: rework to meet the requirement, repair to an approved procedure, reject and replace, or use as-is with a formal concession from the designer. Repair and use-as-is need engineering approval.
4. **Find the root cause** Use structured methods such as five whys or a cause-and-effect diagram covering people, method, materials, machines, measurement and environment.
5. **Correct and prevent** Fix the cause, not just the item – change a procedure, retrain, change a supplier, add an ITP check – and check other locations with the same risk.
6. **Verify and close** Re-inspect, confirm the corrective action works, attach records, close the NCR and review NCR trends monthly.

Closing an NCR on a pipe spool



The repaired spool is accepted at step 8, but the NCR closes only when the QA/QC manager has evidence that the corrective action stops the defect recurring. Repairs need the designer's approval; rework to the original requirement does not.

TEMPLATE **NCR log**

Track each non-conformance from discovery to verified close-out, so that open NCRs, ageing and repeated causes are visible to the project team.

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NCR NO.	DATE AND LOCATION	DESCRIPTION AND REQUIREMENT BREACHED	DISPOSITION	ROOT CAUSE	CORRECTIVE ACTION AND OWNER	STATUS / CLOSED DATE
NCR-031	*12 Jun – pile cap PC-14, grid B/7*	*Bottom cover 30 mm against 50 mm specified; drawing and specification*	*Rework – re-fix bottom mat before pour*	*Wrong spacer blocks issued from store*	*Spacer blocks colour-coded by size; store check added – stores supervisor*	*Closed 13 Jun*
NCR-032	*15 Jun – pipe rack PR-2*	*Bolt grade marking does not match approved submittal*	*Reject and replace bolts*	*Supplier substituted grade without approval*	*Supplier corrective action; receiving inspection to check markings – procurement lead*	*Open – awaiting replacement bolts*

A **punch list** records minor incomplete or defective items found during walkdowns near completion. Many projects categorise items: category A items must be cleared before a defined milestone such as mechanical completion or start-up because they affect safety or function; category B items can be cleared later but before handover or within an agreed period; some projects add a category for cosmetic items. Category names and rules vary, so define them in the completion procedure. Each item needs a location, system and tag, discipline, category, responsible party, target date and a photograph, and it should live in one database rather than in separate spreadsheets kept by each party.

Quality and punch-list metrics

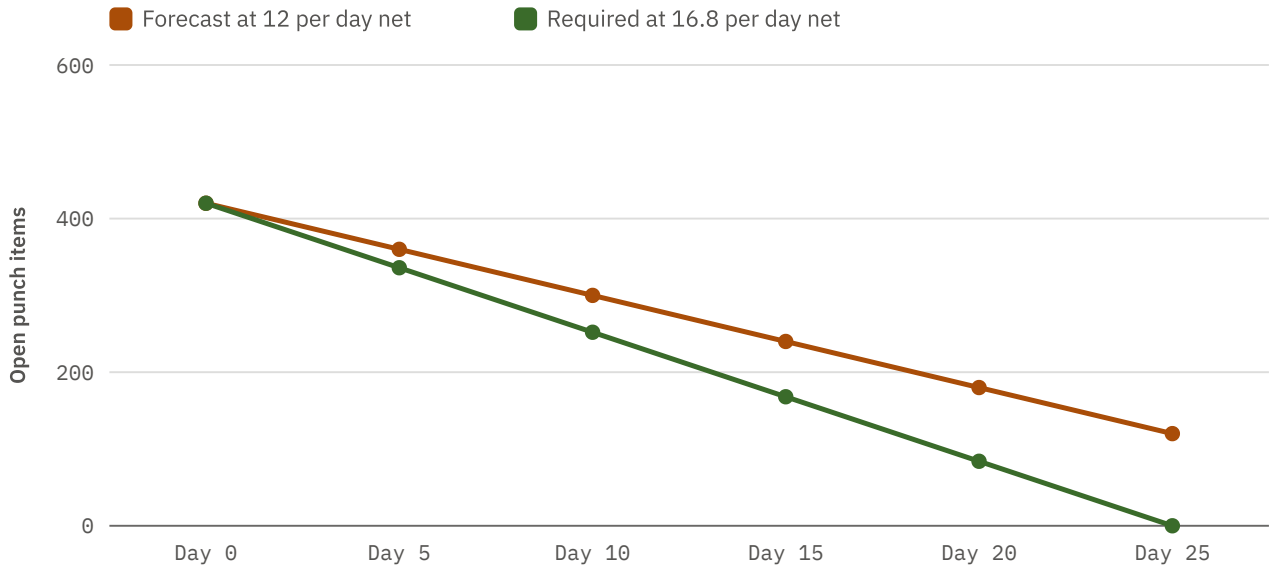
MEASURE	FORMULA	WHAT IT TELLS YOU
First-pass yield	`FPY = inspections accepted at first presentation / total inspections presented × 100%`	The share of work accepted first time without rework; a direct measure of whether quality is built in.
Rolled throughput yield	`RTY = FPY_1 × FPY_2 × … × FPY_n`	The probability that an item passes every step first time, where FPY_i is the first-pass yield of step i expressed as a fraction.
Net burn rate	`Net burn rate = items closed per day − new items raised per day`	How fast the open punch list is actually shrinking once newly found items are counted.
Forecast clearance time	`Days to clear = open items / net burn rate (valid only when net burn rate > 0)`	When the punch list will be empty at the current rates; if the net burn rate is zero or negative, the list will never clear without action.
Required net burn rate	`Required net burn rate = open items / working days remaining to the target date`	The net rate needed to clear the list by the target date; compare with the actual net burn rate.

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EXAMPLE · PUNCH-LIST BURN-DOWN (ILLUSTRATIVE)

A hypothetical substation has **420** open punch items and **25** working days to the energisation target. Crews close 18 items a day while walkdowns raise 6 new ones, so the net burn rate is 18 − 6 = **12 per day**. Forecast clearance = 420 / 12 = **35 days** – ten days late. The required net burn rate is 420 / 25 = **16.8 per day**. Options include adding a dedicated punch crew, clearing category A items first so energisation is not blocked, and joint daily walkdowns so that items are raised once and correctly rather than repeatedly.

Substation punch-list burn-down (illustrative)



Illustrative figures from the example above. At the current net rate about 120 items are still open on the target day, so the list would clear only around day 35 unless the rate rises.

SCENARIO · DECIDE · WITNESS POINT MISSED AT THE FACTORY

The ITP for 33 kV switchgear for a hypothetical solar park makes the FAT a witness point for the owner's inspector, with ten days' notice. The vendor gave only two days' notice, the inspector could not travel, and the vendor ran the FAT anyway. The test reports all show passes and the vendor asks for release for shipment so that it can meet its delivery date. What should the project manager do FIRST?

(A) Release the switchgear for shipment, since the reports show that every test was passed and delivery is on time.

(B) Reject the switchgear outright and require the vendor to manufacture new panels with a properly witnessed FAT.

(C) Accept the reports now and plan to repeat the full FAT on site after installation, where it can be witnessed.

(D) Review the FAT records against the ITP and require key tests to be repeated with the owner witnessing, in person or remotely, before release.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Pick an ITP you have used or seen. Which hold points actually protected the project, and which were treated as formalities? How would you redesign one line of it to be clearer at the work face?

Key terms

Inspection and test plan (ITP)
A document listing, for a work activity or item, each inspection and test with its reference, acceptance criteria, frequency, record and the hold, witness, review or surveillance role of each party.

Witness point
An inspection point at which a designated party must be given agreed notice and may attend; if it does not attend after proper notice, work may proceed.

Site acceptance test (SAT)
Testing of equipment or a system after installation on site to demonstrate correct performance in its actual environment and with its real interfaces.

Punch list
A list of minor incomplete or defective items identified at walkdowns near completion, each categorised, assigned and tracked until cleared.

Hold point
An inspection point beyond which work must not proceed without the written release of the designated party, typically before work is covered or made inaccessible.

Factory acceptance test (FAT)
Testing of equipment at the manufacturer's works before shipment to demonstrate compliance with its specification, often witnessed by the purchaser.

Non-conformance report (NCR)
A formal record that work, material or equipment does not meet a specified requirement, used to contain, disposition, investigate and close the non-conformance.

Key takeaways

- An ITP states what is checked, against which criteria, by whom and with what record; hold points need written release
- FAT at the works, SAT on site – remote witnessing works only when agreed in advance and properly recorded
- Traceability lets one bad heat or batch be found without opening up everything
- Close NCRs by fixing the cause, not just the item, and verify that the corrective action worked
- Forecast punch-list clearance from the net burn rate and compare it with the rate the target date requires

KNOWLEDGE CHECK

6. Reinforcement in a raft has been fixed, but the engineer has not yet inspected it, although the ITP shows a hold point. The concrete trucks are arriving. What should happen?

(A) Pour the raft and send the engineer photographs of the rebar

(B) Hold the pour until the engineer inspects and releases it in writing

(C) Pour the raft if the contractor's own inspector has approved the rebar

(D) Pour half the raft and keep the remainder for the engineer's inspection

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7. A transformer passed all routine tests at the factory. Why is a site acceptance test still needed?

   (A) To repeat the factory tests in case the vendor's results were falsified

   (B) To satisfy the insurers that the transformer was delivered on the agreed date

   (C) To confirm correct performance after transport, installation and connection

   (D) To replace the need for any commissioning of the substation system

8. A batch of bolts is found to be the wrong grade. Which quality practice allows the team to find every location where that batch was installed?

   (A) Material traceability linking batches to locations

   (B) A higher frequency of witness points in the bolting ITP

   (C) First-pass yield measurement for bolting work

   (D) Rolled throughput yield for the steel erection

9. A punch list has 300 open items. Crews close 20 per day and walkdowns raise 5 new items per day. How many days will it take to clear at these rates?

   (A) 15 days

   (B) 12 days

   (C) 25 days

   (D) 20 days

10. Of 200 weld inspections presented in a month, 170 were accepted first time. What is the first-pass yield?

    (A) 80%

    (B) 85%

    (C) 90%

    (D) 75%

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · COMMISSION · LESSON 3 · 40 MIN

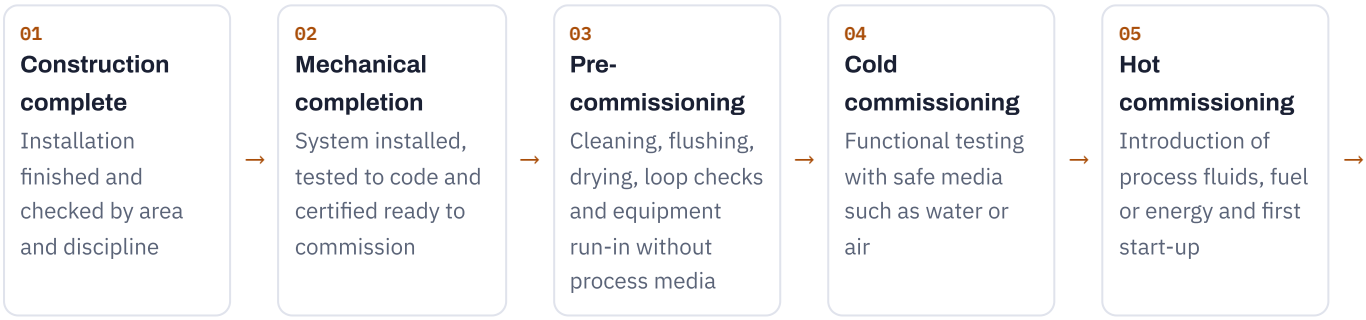
Commissioning

Mechanical completion, pre-commissioning, cold and hot commissioning, systems, performance tests and safe start-up

- Sequence the transition from mechanical completion through pre-commissioning, cold and hot commissioning to performance testing
- Organise commissioning by systems and subsystems and track readiness with certificates
- Apply start-up safety controls, including a pre-start-up safety review, as systems become live
- Structure a commissioning team and its interfaces with construction and operations

Construction builds by area and discipline; commissioning proves by system. **Commissioning** is the structured process of verifying that installed systems are complete, function as designed, and can be operated safely and reliably by the owner. It begins in design – with isolation valves, test points, flushing connections, temporary strainers and a commissioning philosophy – not in the last month of construction. On a boiler, a substation, an effluent treatment plant or a metro signalling system, commissioning is where hidden design, installation and integration problems finally surface, so it deserves its own plan, team and schedule. On modern plants the scope also covers digital systems – control networks and their cybersecurity settings, the data historian and any digital twin connections – and verifies energy performance, because operational energy and carbon are increasingly managed targets.

From construction to performance



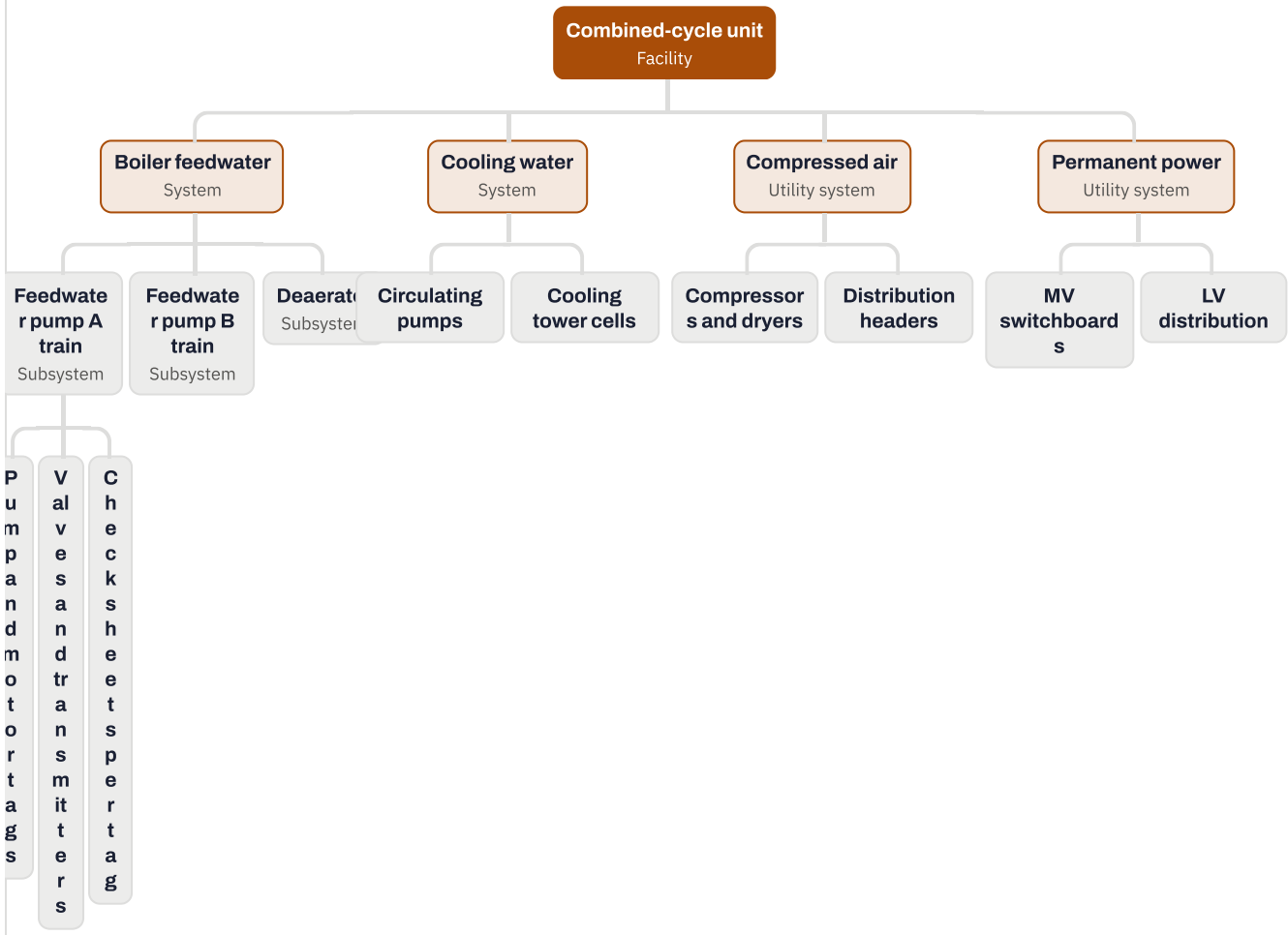
What happens at each commissioning stage

STAGE	TYPICAL ACTIVITIES	MEDIA AND ENERGY	COMPLETION EVIDENCE
Mechanical completion	Installation verified against drawings; pressure and leak tests; electrical installation tests; category A punch items cleared	None, or test media only	Mechanical completion certificate per system or subsystem
Pre-commissioning	Flushing, air or steam blowing preparation, chemical cleaning, drying; instrument loop checks; motor solo runs; insulation resistance and alignment checks	Temporary water, air or nitrogen; temporary power	Pre-commissioning check sheets complete; system ready for commissioning

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STAGE	TYPICAL ACTIVITIES	MEDIA AND ENERGY	COMPLETION EVIDENCE
Cold commissioning	Functional tests of interlocks, trips and control loops; pump and fan run-in on water or air; HVAC balancing	Water, air or inert gas; permanent power	Functional test records signed by commissioning and operations
Hot commissioning and start-up	First firing of a boiler, steam blows, introduction of gas, synchronisation of a generator, first effluent through an ETP	Process fluids, fuel, steam, high voltage	Ready-for-start-up approval and start-up logs
Performance and reliability tests	Capacity, efficiency, emissions or effluent quality guarantees; continuous running for the contractually agreed period	Normal operating conditions	Signed test reports and calculations against guarantees

Systemisation of a combined-cycle unit (illustrative)



Every tag belongs to exactly one subsystem; check sheets attach to tags, while certificates and turnover packages are issued per subsystem. Utility systems are completed first because the others depend on them.

Systemisation draws commissioning boundaries on marked-up process and instrumentation diagrams and electrical single-line diagrams, then links every tag to a system and subsystem. Construction progress by area does not tell you whether a system can be commissioned: the cooling water system may be 95% built, but if

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one valve in a remote corner is missing the system cannot be flushed. Completion teams therefore track every check sheet and punch item by subsystem and hand over formal turnover packages from construction to commissioning. Utilities come first – permanent power, compressed air, water, fire water and control room HVAC – because every other system depends on them.

> **INSIGHT · BUILD IN COMMISSIONING SEQUENCE**
> In the last third of construction, the schedule should be driven by the system completion sequence, not by what is convenient for each area or discipline. A planner who keeps scheduling by area until the end will produce a site that is 'nearly complete everywhere' and a commissioning team that has nothing it can start.

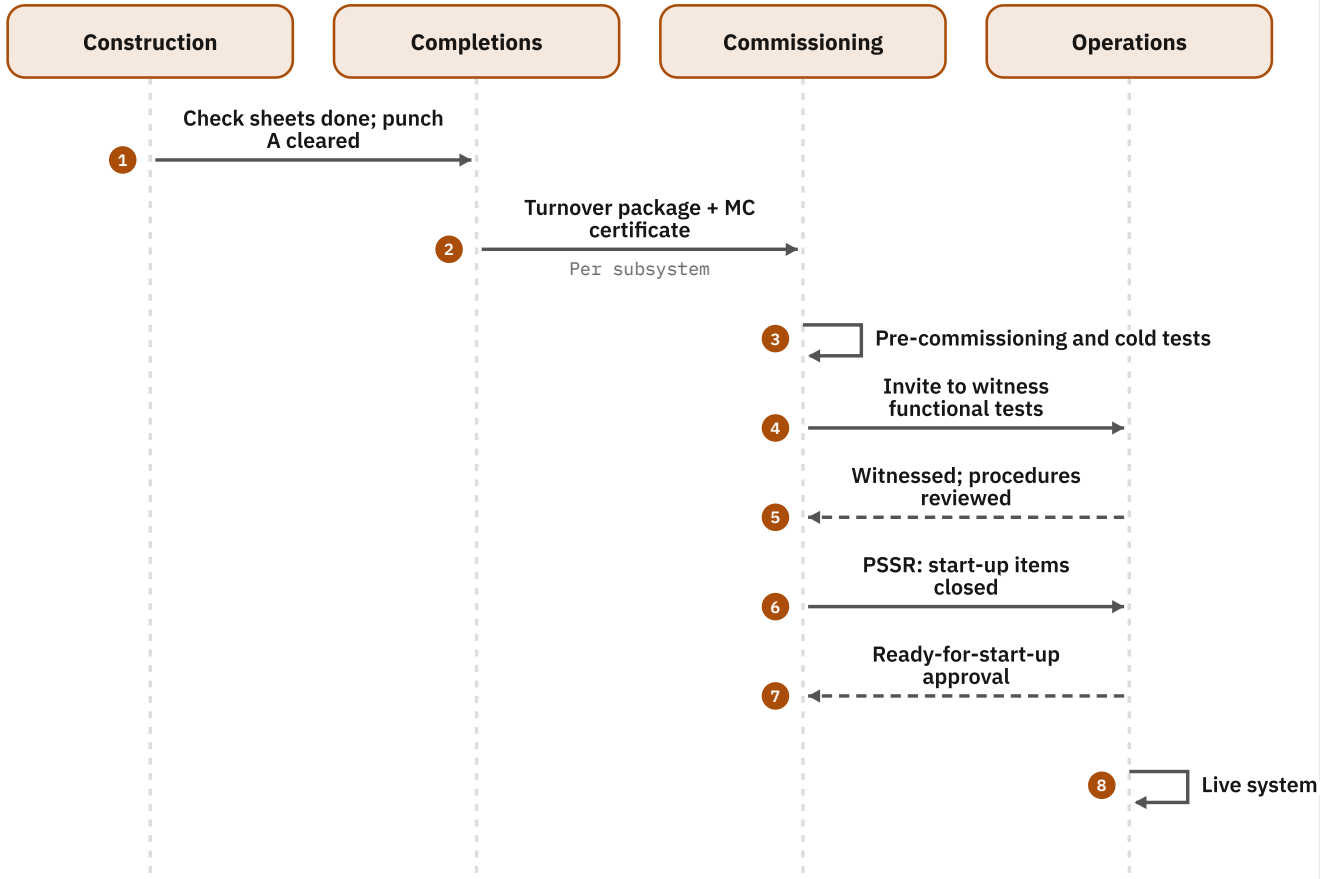
As systems are energised, pressurised or filled with process fluids, the site changes from a construction site into an operating plant – often with construction still going on next door. Start-up safety needs deliberate control: the permit-to-work system transfers to operations for live systems (see Protect), lock-out and tag-out is applied at every isolation, live areas are clearly demarcated, energisation notices are issued and briefed, and emergency response is ready before the first hazardous energy is introduced. A **pre-start-up safety review** confirms, before start-up, that the plant is built as designed, procedures and training are in place, hazard study actions are closed, and safety systems have been tested.

Pre-start-up safety review – before introducing hazardous energy or process fluid

☐ Construction and installation verified against the design, with all relevant category A punch items closed
☐ Pressure and leak tests complete and recorded for every line in the start-up boundary
☐ Safety-critical trips, interlocks and alarms function-tested and signed off
☐ Relief valves installed, set and certified; fire and gas detection operational
☐ Hazard study and design review actions affecting start-up closed or formally accepted
☐ Operating procedures issued and operators trained and assessed on them
☐ Permit-to-work, isolation and energisation arrangements in force and briefed
☐ Emergency response, first aid and fire-fighting arrangements ready and tested

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From mechanical completion to ready for start-up



Each hand-off is a signed record per subsystem, not a verbal agreement. Operations witnesses and approves before hazardous energy is introduced, then controls the live system through its own permit-to-work.

The commissioning team

ROLE	MAIN RESPONSIBILITIES
Commissioning manager	Owns the commissioning plan, sequence and schedule; chairs readiness reviews; interfaces with the construction manager and operations
System commissioning engineers	Lead the commissioning of assigned systems; write and execute procedures; sign test records
Completions team	Maintains the completions database of tags, check sheets, punch items and certificates; prepares turnover packages
Vendor specialists	Commission packaged equipment such as turbines, transformers, chillers or control systems to their own procedures
Operations staff	Shadow, then operate under supervision; review procedures; accept systems; build familiarity before taking over
HSE and owner's representatives	Manage start-up permits and isolations; witness tests and verify guarantees on the owner's behalf

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Common performance test calculations

MEASURE	FORMULA	WHAT IT TELLS YOU
Net heat rate	`Net heat rate (kJ/kWh) = fuel heat input (kJ) / net electrical energy exported (kWh)`	Fuel energy needed per unit of net electricity; lower is better. Results are normally corrected to the reference conditions in the contract.
Net thermal efficiency	`Efficiency (%) = 3600 / net heat rate (kJ/kWh) × 100`	Because 1 kWh = 3600 kJ, this converts heat rate to efficiency.
Capacity achieved	`Capacity achieved (%) = corrected measured net output / guaranteed net output × 100`	Whether the plant delivers its guaranteed output once measurements are corrected to reference conditions.
Availability in a reliability run	`Availability (%) = hours available / total hours in the test period × 100`	The share of the test period during which the plant was able to operate as required.

CASE STUDY · COMMISSIONING AN EFFLUENT TREATMENT PLANT FOR A DYEING FACTORY (ILLUSTRATIVE)

A hypothetical dyeing and finishing factory near Gazipur is completing a new biological effluent treatment plant: equalisation tank, chemical dosing, aeration basins with blowers, clarifiers, sludge handling and a treated-water discharge point monitored by the environment department. Civil and mechanical works are nearly complete. The factory wants to start production on a new line within weeks, and the factory manager suggests starting effluent flow as soon as the pumps run. The biological process needs time to establish a healthy biomass, and the discharge must meet the environmental standards that apply. All details are illustrative.

1. How would you systemise the plant and order its commissioning?

2. What should be proven in cold commissioning before any real effluent is introduced?

3. How would you handle the pressure to start production before the biological process is stable?

Model answer: module answer key.

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SCENARIO · DECIDE · PRESSURE TO SKIP THE REVIEW

A new gas-fired boiler at a textile mill is scheduled for first fire tomorrow. The pre-start-up safety review has found that leak-test records for one section of the gas line are incomplete and that two burner management system trips have not yet been function-tested. The mill's production director wants first fire to go ahead because a large export order depends on steam next week, and says the gas line 'was obviously tested'. What should the project manager do FIRST?

(A) Allow first fire tomorrow while the leak-test paperwork is completed in parallel, since the gas line was almost certainly tested.

(B) Hold first fire until the gas line section is tested and recorded and both trips are function-tested, and give the director a recovery plan with realistic timings.

(C) Proceed with first fire but bypass the two untested trips and post an operator at the burner to watch the flame continuously.

(D) Escalate the decision to the boiler vendor and let its commissioning engineer decide whether to proceed tomorrow.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a start-up you have seen or read about. Which item on the pre-start-up safety review checklist was most at risk of being skipped under schedule pressure, and what would have made it easier to hold the line?

Key terms

Mechanical completion
The milestone at which a system or subsystem is installed, tested and documented in accordance with the drawings, specifications and codes, ready for pre-commissioning, usually confirmed by a certificate.

Cold commissioning
Functional testing of systems with safe media such as water, air or inert gas to prove controls, interlocks and equipment operation before process fluids or fuel are introduced.

Systemisation
The division of a facility into commissioning systems and subsystems with defined boundaries, so that completion, testing and handover can be planned and tracked by function.

Pre-start-up safety review (PSSR)
A structured check carried out before hazardous energy or process fluids are introduced, confirming that the plant, procedures, training and safety systems are ready for safe start-up.

Pre-commissioning
Non-operational checks and preparation of installed systems before process media are introduced, such as flushing, cleaning, drying, loop checks, insulation tests and motor solo runs.

Hot commissioning
The commissioning stage in which process fluids, fuel or full energy are introduced and the plant is started up and brought towards normal operating conditions.

Performance guarantee test
A formal test, run to an agreed procedure, that demonstrates whether a plant meets its contractual guarantees such as output, efficiency, emissions or effluent quality.

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Key takeaways

- Construction builds by area; commissioning proves by system – switch the late-stage schedule to system completion
- Move from mechanical completion through pre-commissioning, cold and hot commissioning to performance tests, with evidence at each step
- Systemise early and commission utilities first; turnover packages pass from construction to commissioning per subsystem
- A pre-start-up safety review must close every safety-critical gap before hazardous energy or process fluid is introduced
- Correct performance test results to the contract's reference conditions before judging guarantees

KNOWLEDGE CHECK

11. Flushing a lube-oil system, drying a line with nitrogen and checking instrument loops normally belong to which stage?

(A) Hot commissioning
(B) Performance testing
(C) Pre-commissioning
(D) Reliability running

12. Why are utility systems such as permanent power, compressed air and fire water usually commissioned first?

(A) They are the cheapest systems to commission on a plant
(B) Other systems depend on them for their own commissioning
(C) Their vendors usually arrive on site before all the others
(D) They do not need any mechanical completion certificate

13. A gas-fired plant burns fuel with a heat input of 90,000,000 kJ in one hour while exporting 10,000 kWh net. What is its net thermal efficiency?

(A) 30%
(B) 36%
(C) 40%
(D) 45%

14. What is the main purpose of a pre-start-up safety review?

(A) To confirm that the plant is ready to start safely before hazards are introduced
(B) To agree the final account for the commissioning contractor's work
(C) To measure whether the completed plant meets its contractual efficiency and output guarantees
(D) To record the punch items left over for the defects period

15. Construction reports a cooling water system as 95% complete, but commissioning cannot flush it. What is the most likely explanation?

(A) The performance test has not yet been witnessed by the owner's team
(B) The progress figure is measured by area, while one missing item blocks the system
(C) The cooling water system has been incorrectly systemised as a utility system by the completions team
(D) The system was mechanically completed before its pre-commissioning had started

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MODULE 8 · COMMISSION · LESSON 4 · 35 MIN

Handover & Operations Readiness

Taking over, as-built records, O&M manuals, spares, training, asset information, warranties and soft landings

- Plan taking over and handover so the owner receives a safe, documented and operable asset
- Specify as-built records, O&M manuals, spares, training and structured asset information
- Prepare operations readiness from early in the project, using soft landings principles
- Manage warranties and the defects period after handover

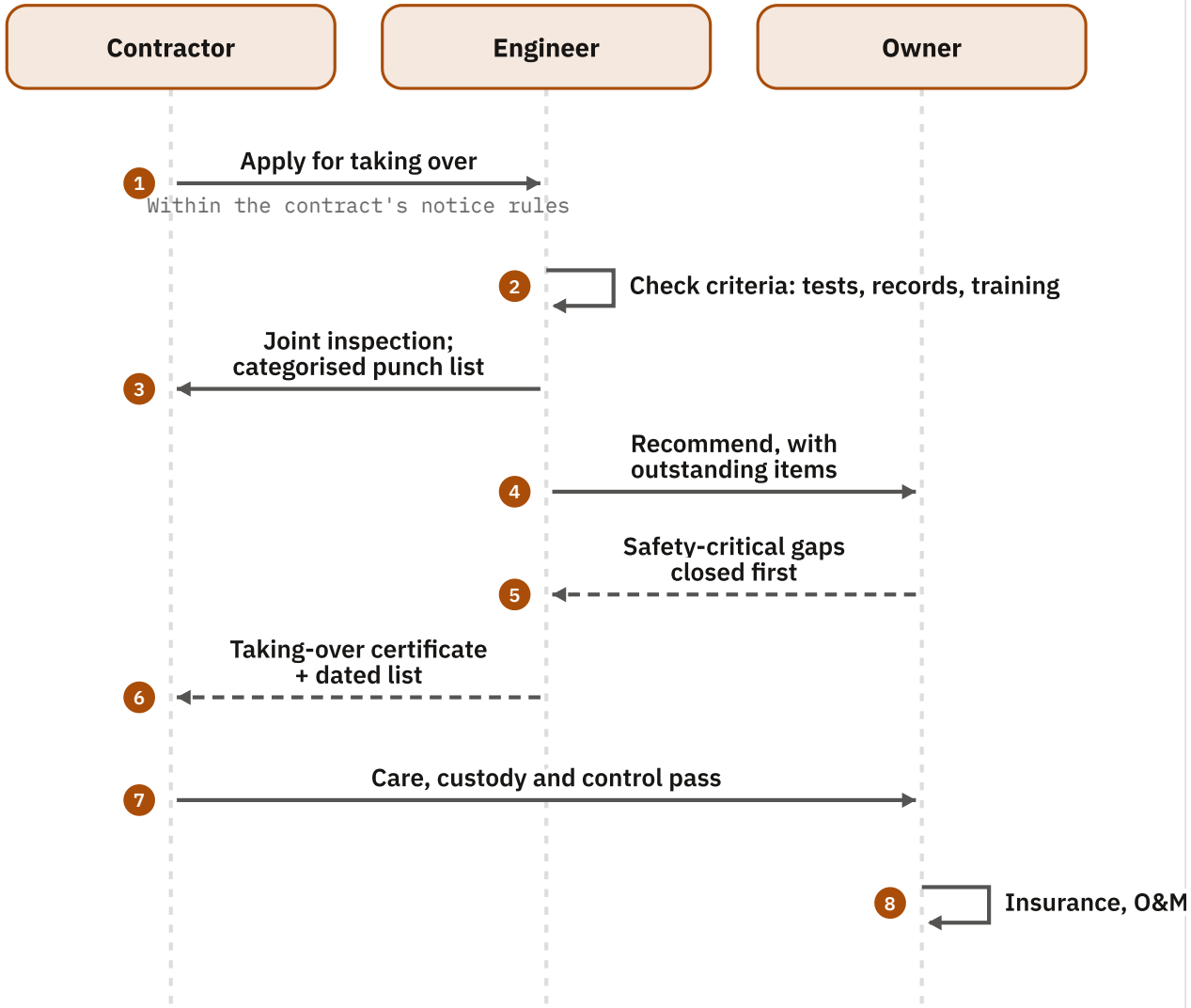
The project ends but the asset begins. **Handover** transfers care, custody and control from the contractor to the owner: responsibility for security, insurance, operation, maintenance and most risks changes hands. Too many owners discover at handover that they have a working plant but unusable drawings, an unclear spares position and operators who have never run the equipment without the vendor beside them. The cure is to treat operations readiness as a workstream that starts in design, with its own deliverables, owner and schedule, not as a pile of documents gathered in the final month.

A handover sequence that works

1. **Agree the handover strategy early** Decide whether the asset will be taken over as a whole, by section or by system, and align the contract, commissioning plan and operations plan with that choice.
2. **Hold readiness reviews** Check progressively that systems, documents, people, spares and permits will be ready, starting months before the planned date.
3. **Inspect and punch** Carry out joint completion inspections and agree a categorised punch list, distinguishing items that prevent taking over from those that do not.
4. **Deliver and check information** Receive as-built drawings, O&M manuals, test records, certificates and asset data, and check them against the requirements rather than just counting files.
5. **Train and familiarise** Complete operator and maintainer training, including hands-on sessions and assessment, before the owner's staff take responsibility.
6. **Take over formally** Issue the taking-over certificate or equivalent under the contract, recording the date, any outstanding items and the start of the defects period (see Procure).
7. **Support after handover** Provide aftercare during the defects period – visits, performance monitoring and quick remedy of defects – as agreed in the contract and soft landings plan.

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Taking over: a recorded decision



Taking over is judged against the criteria in the contract. Minor outstanding items can be listed with dates, but gaps that affect safe operation must be closed before the owner takes control.

The handover dossier – what the owner should receive

ITEM	CONTENT	COMMON PROBLEM	GOOD PRACTICE
As-built drawings	Drawings showing the works as actually constructed	Design drawings re-titled without recording site changes	Site redlines maintained continuously and verified by survey or scan before handover
O&M manuals	Operating procedures, maintenance instructions, data sheets, spares lists, vendor documents	Generic catalogues; delivered after the staff needed them	Project-specific manuals reviewed in draft by operators before training starts
Test and commissioning records	ITP records, FAT and SAT reports, commissioning sheets, performance test results	Records scattered across parties and formats	One completions database exported per system with all certificates attached

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ITEM	CONTENT	COMMON PROBLEM	GOOD PRACTICE
Spares and special tools	Commissioning, start-up and operational spares; special tools; consumables	Spares delivered without tags or storage plan	Spares checked against the agreed list, tagged, recorded and stored before handover
Warranties and certificates	Equipment warranties, guarantees, material certificates, statutory approvals such as fire, lift and boiler	Warranties started at delivery and expired before operation	Warranty register with start and end dates, contacts and conditions
Asset information	Asset register and structured data for the maintenance management system	Data not in a format the owner's systems can import	Asset information requirements defined early and test-loaded before handover; sensor tags and data mappings verified where a digital twin is planned

WATCH OUT · AS-BUILT IS NOT AS-DESIGNED WITH A NEW TITLE BLOCK
Real as-built drawings record what was actually installed: the cable route moved around a column, the pipe that went over the beam instead of under it, the buried valve chamber that shifted two metres. If site teams do not mark up changes as they happen, the information is lost for ever. Make redline maintenance a weekly supervisory duty, check it at audits, and verify critical as-built information by survey or laser scan.

Operations readiness prepares the owner's organisation to run the asset from day one. It covers recruiting and training operators and maintainers in time to take part in commissioning; writing operating, maintenance and emergency procedures; setting up the maintenance management system with the asset register, maintenance routines and spares records; securing consumables such as chemicals, lubricants and filters; arranging specialist support contracts; and obtaining the permits and licences needed to operate. On a hypothetical new hospital it also means clinical commissioning – moving equipment, staff and patients – which can take months after technical handover. Owners with an asset management system aligned to ISO 55001 (current edition 2024) can derive their asset information requirements from it, and ISO 19650-3 covers how that information is then managed in operation.

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Operations readiness from design to aftercare

DESIGN
Operators join design reviews
Operability and maintainability tested; asset information requirements set

CONTRACT AWARD
Readiness plan and owner
Handover strategy, staffing plan, maintenance system set-up

T – 12 MONTHS
Recruit and start training
Key operators and maintainers in post; vendor courses booked

T – 6 MONTHS
Draft O&M manuals reviewed
Operators check procedures before training relies on them

COMMISSIONING
Operators work under supervision
Witness tests, run systems, assessed hands-on

T
Taking over
Care, custody and control pass to the owner

DEFECTS PERIOD
Aftercare and review
Soft landings visits; performance in use measured

T is the planned taking-over date; the lead times are illustrative and grow with complexity. The point is that readiness starts in design, not in the final month.

Traditional handover vs soft landings

Traditional handover

- Operations staff appear shortly before completion
- Design decisions made with little input from operators
- Handover is a single event; the contractor leaves
- Problems after handover become defects disputes
- Performance in use is rarely measured against design intent

Soft landings

- Operators and maintainers involved from design onwards
- Design reviews test operability and maintainability
- A graduated handover with familiarisation and aftercare
- Designers and contractors stay engaged for an agreed period after handover
- Performance in use – including energy and operational carbon – is monitored and fine-tuned against design intent

Spares deserve explicit decisions. **Commissioning spares** are consumed during commissioning; **start-up spares** cover the early operating period; operational spares support routine maintenance for an agreed period; and **critical spares** – sometimes called insurance or capital spares – are long-lead items whose failure would stop the plant for months, such as a spare transformer, a turbine rotor component or a large pump casing. Vendors propose spares lists, but the owner should review them against its maintenance strategy, equipment

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standardisation and supplier lead times, and the spares should be delivered, inspected, tagged, recorded and stored before handover.

TEMPLATE **Handover readiness register**

Track each handover deliverable by system or area so that the owner can see in advance what will not be ready and agree actions before taking over.

SYSTEM OR AREA	DELIVERABLE	REQUIRED BY	RESPONSIBLE	STATUS	VERIFIED BY	NOTES
33 kV substation	*Protection relay settings and test records*	*4 weeks before energisation*	*Electrical commissioning lead*	*Received – under review*	*Owner's protection engineer*	*Two relays still show factory default settings*
HVAC – operating theatres	*Operator training with assessment on pressure regimes*	*Before clinical commissioning*	*Mechanical contractor and FM manager*	*Scheduled*	*Hospital estates manager*	*Include night-shift staff in second session*

After taking over, the **defects period** – called the defects liability or defects notification period in different contracts – begins. During it the contractor must remedy defects that it is notified of, and retention money and performance security are usually released in stages as obligations are met (see Procure for the contractual mechanisms). Equipment warranties often run separately: a vendor warranty that starts on delivery may expire before a delayed plant even starts operating, so negotiate warranty start dates linked to commissioning or taking over, and keep a register. Defects that only become apparent later may still be claimable under the general law, subject to limitation periods that vary by jurisdiction.

CASE STUDY · HANDING OVER A HOSPITAL BLOCK (ILLUSTRATIVE)

A hypothetical multi-storey hospital extension in Dhaka includes wards, intensive care, operating theatres with controlled pressure regimes, medical gas systems, standby generators and a building management system. The hospital wants to open the wards in phases, with theatres last. The contractor plans one handover at the end, and the O&M manuals are still in draft. The hospital's estates team is small and has never maintained medical gas or theatre HVAC systems. All details are illustrative.

1. What handover strategy would you recommend, and what does it require in the contract?

2. What operations readiness actions should start immediately?

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3. Which records and tests must be complete before patients occupy any area?

Model answer: module answer key.

SCENARIO · DECIDE · READY TO TAKE OVER?

A hypothetical solar park has passed its performance test and is exporting power. The contractor applies for the taking-over certificate. Outstanding items include SCADA historian data not yet transferred, O&M manuals still in draft, and the owner's operators trained on the inverters but not on the substation protection and switching. The owner's finance team wants taking over now because revenue recognition depends on it. What should the owner's project manager do FIRST?

(A) Refuse taking over until every document and every training session is complete, regardless of what the contract says.

(B) Grant taking over immediately, since the plant is generating power and the performance test has passed.

(C) Let the contractor's staff continue operating the whole plant indefinitely after taking over until the owner's team feels ready.

(D) Check the contract's taking-over criteria with the engineer, agree a dated list of outstanding deliverables, and require substation switching and protection training before owner staff operate the substation.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of an asset you have helped hand over or now operate. What one piece of information or training did the operators most need but not receive, and when in the project should it have been planned?

Key terms

As-built drawings
Drawings that record the works as actually constructed, including all changes made during construction, verified and handed to the owner for operation, maintenance and future modification.

Operations readiness
The planned preparation of the owner's people, procedures, systems, spares, contracts and permits so that the asset can be operated and maintained safely and effectively from handover.

O&M manual
The project-specific set of operating and maintenance information for an asset – procedures, maintenance instructions, data sheets, drawings, spares lists and vendor documents.

Critical spares
Long-lead or high-consequence spare parts, such as a spare transformer or turbine component, held because their failure would otherwise cause a long and costly outage.

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Asset information model (AIM)
The structured collection of graphical, non-graphical and documentary information needed to operate and maintain an asset, handed over from the project and kept updated in operation.

Soft landings
An approach in which designers and contractors involve operators from early in the project and remain engaged for a period after handover to support, monitor and fine-tune the asset in use.

Handover dossier
The complete, organised package of records delivered to the owner at handover, including as-builts, O&M manuals, test and commissioning records, certificates, warranties and asset data.

Key takeaways

- Handover transfers care, custody and control; run operations readiness as a workstream that starts in design
- Judge the dossier on usability – verified as-builts, project-specific manuals, test-loaded asset data – not file counts
- Train and assess operators on the specific systems before they take control, especially high-voltage and safety-critical ones
- Make deliberate spares decisions and link warranty start dates to commissioning or taking over
- Soft landings keeps designers and builders engaged after handover to tune performance in use, including energy

KNOWLEDGE CHECK

16. A vendor warranty for chillers starts on delivery to site. The building is delayed by a year. What is the main risk, and the best way to avoid it?

(A) The warranty may expire before operation; link its start to commissioning
(B) The chillers may be over-specified; reduce their capacity before installation
(C) The O&M manual may be outdated; ask the vendor for a revised edition
(D) The spares may be lost; store them in the owner's warehouse at delivery

17. Which practice best ensures that as-built drawings reflect reality?

(A) Re-issuing the final design drawings with an as-built title block
(B) Asking the designer to update drawings from memory after handover
(C) Maintaining site redline mark-ups continuously and verifying critical items
(D) Photographing the completed site and attaching the images to the O&M manual

18. A spare generator step-up transformer with a long manufacturing lead time is purchased and stored at a power plant. What kind of spare is this?

(A) A commissioning spare
(B) A critical spare
(C) A start-up spare
(D) A consumable item

19. What distinguishes a soft landings approach from a traditional handover?

(A) The contractor is released from all of its defects obligations at taking over
(B) Operators are involved early and designers and builders provide aftercare
(C) Handover takes place in a single event with all documents on one day
(D) The owner's staff are excluded from commissioning to avoid confusion

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20. Which is the best evidence that asset data will be usable by the owner's maintenance team?

(A) The contractor confirms that it has delivered every file required

(B) The data folder size matches the volume expected for the project

(C) The asset data has been test-loaded into the owner's maintenance system

(D) The owner's finance team has approved the final asset valuation for the balance sheet

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MODULE 8 · COMMISSION · LESSON 5 · 35 MIN

Closeout & Lessons Learned

Contract closeout, final account, demobilisation, records, team release, reviews and lessons that change practice

- Close contracts and agree final accounts on the basis of clean records
- Plan demobilisation, records archiving and the release of the team
- Run a post-project review and a benefits realisation review
- Turn lessons into changed standards, templates and decisions

Closeout is not an administrative tail. It settles money, releases securities, retires risks, preserves knowledge and gives people their next role. Projects that close badly keep leaking for years: final accounts that are never agreed, bonds that are never released, insurance that lapses at the wrong moment, and records that cannot be found when a latent defect or a claim appears. Good closeout is planned from the start, executed deliberately and owned by a named person, often while the most experienced people are already being pulled onto the next project. ISO 21502 guidance treats closing as a managed process in its own right, with the same planning and ownership as any other.

The closeout sequence

01 Taking over
Asset handed over with outstanding items listed

→

02 Defects period
Defects remedied and performance monitored

→

03 Final account
Final contract sum agreed and certified

→

04 Securities released
Retention and performance security released as obligations are met

→

05 Records archived
Project and asset records stored and retention set

→

06 Reviews
Post-project review now; benefits realisation review later

Six closeout workstreams

WORKSTREAM	KEY ACTIVITIES	EVIDENCE OF CLOSURE	TYPICAL OWNER
Contractual	Completion of all obligations, close-out of subcontracts and purchase orders, release of securities, settlement of claims	Final certificates, signed closure of each contract, released bonds	Contracts manager
Financial	Final accounts, final payments, reconciliation of budgets and commitments, closure of cost codes	Agreed final accounts, closed cost codes, final cost report	Project controls or commercial lead
Physical	Demobilisation of site facilities, removal of temporary works, reinstatement of land and access roads	Reinstatement sign-off by landowners and authorities	Construction manager
Information	Archiving of project records, transfer of asset information, retention schedule set	Archive index and confirmation of transfer to operations	Document controller

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WORKSTREAM	KEY ACTIVITIES	EVIDENCE OF CLOSURE	TYPICAL OWNER
People	Release and redeployment of staff, performance feedback, recognition	Release plan completed; appraisals and references given	Project manager and HR
Benefits and learning	Post-project review, lessons register, benefits realisation review plan	Review report, actions assigned, benefits measures agreed	Sponsor and PMO

The **final account** is the agreed final contract sum: the original contract price adjusted for valued variations, remeasured quantities, adjustments of provisional sums, price fluctuation where the contract allows it, settled claims and any deductions such as liquidated damages or contra-charges. The contractor typically submits a statement of final account, the engineer or quantity surveyor reviews it, the parties negotiate unresolved items, and a final payment certificate follows under the contract's procedure (see Procure for the mechanisms). The final account is only as good as the records behind it, which is why variations should be valued and agreed as the work proceeds rather than stockpiled for the end.

> **PRO TIP · CLOSE AS YOU GO**
> Agree variation valuations monthly while the people and records are fresh. Close each subcontract or purchase order when its scope ends rather than all together at the end. Archive each work package's records as it completes. A project that has done this arrives at the final account with a short list of genuinely difficult items instead of hundreds of forgotten ones.

Demobilisation is a planned activity with its own hazards and commitments: dismantling site offices and batching plants, removing temporary works and haul roads, disconnecting temporary utilities, returning hired plant, reconciling materials and disposing of surplus responsibly, and reinstating borrowed land, access roads and drainage as promised to landowners and communities (see Integrate). **Team release** deserves equal care. Plan it with HR months ahead, capture knowledge before key people leave, give honest performance feedback and references, and recognise contributions publicly – including those of subcontractors and the workforce. People remember how a project ended long after they forget how it started.

Project closeout checklist

☐ All contracts and purchase orders formally closed, with no open commitments in the cost system

☐ Final accounts agreed and final payments made, or unresolved items documented with positions and evidence

☐ Retention and performance securities released or scheduled for release with dates

☐ Insurances, permits and site licences transferred to operations or formally closed

☐ Site demobilised, temporary works removed and land reinstated with sign-off from owners and authorities

☐ Project records archived with an index, retention periods and access rights

☐ Asset information transferred and confirmed usable by operations

☐ As-built carbon and environmental performance recorded against the carbon baseline and handed to the owner

☐ Team released with feedback and recognition; knowledge captured from key people

☐ Post-project review held and lessons register published with owners and dates

☐ Benefits realisation review scheduled, with measures and data sources agreed

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Running a post-project review

1. **Prepare the evidence** Gather baseline and actual cost, schedule, quality, safety and change data, the risk register and key decisions, so that the review is based on facts rather than memory.
2. **Invite a cross-section** Include the owner, designers, contractors, key subcontractors, operations staff and, where possible, people from the front line – not only senior managers.
3. **Set a blame-free tone** Make clear that the purpose is learning, not fault-finding. Use a neutral facilitator if relationships are strained or claims are still open.
4. **Analyse causes** For each major variance or success, ask why repeatedly until reaching causes that a future project could control.
5. **Agree lessons and actions** Turn each lesson into a specific change with an owner and a date – a revised standard, specification, template, checklist or contract clause.
6. **Publish and track** Publish the lessons register, report actions to the PMO or leadership, and check that the next projects actually use them.

Lessons that change practice

L1

Specific

Describe what happened, where and in what context, with enough detail that a stranger would understand it.

L2

Caused

Identify the root cause, not the symptom – the late design input, not the delayed pour.

L3

Actionable

Name the change that would prevent or repeat the outcome: a standard, template, specification clause, checklist or decision rule.

L4

Owned

Give the change an owner with the authority to make it and a date by which it will be done.

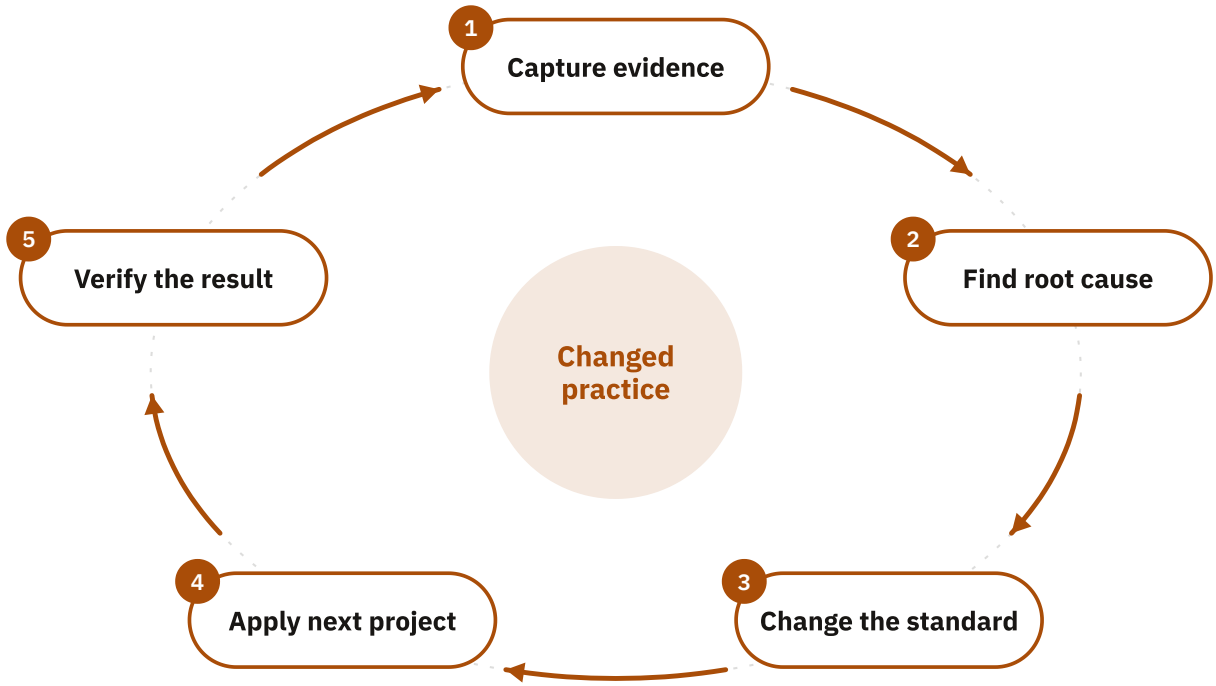
L5

Verified

Check later that the change was adopted and made a difference on a subsequent project.

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The lessons-learned loop



1. **Capture evidence** — Facts from records and front-line people, not memory
2. **Find root cause** — Ask why until reaching a cause a future project can control
3. **Change the standard** — Specification, template, checklist or contract clause, with an owner and date
4. **Apply next project** — The change is used at the right stage of the next project
5. **Verify the result** — Check that it was adopted and made a difference; if not, go round again

A lessons register that stops after the first step is only a diary. The loop closes when a later project shows the changed standard working.

TEMPLATE Lessons learned register

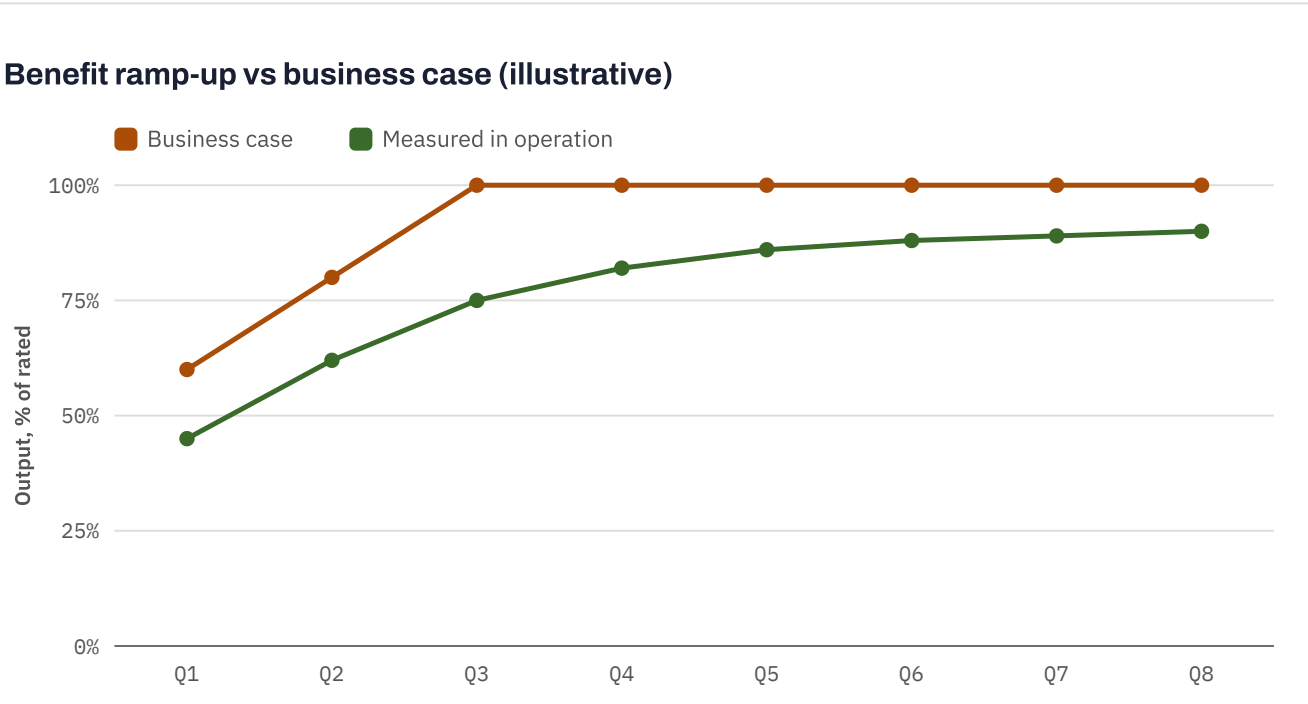
Capture lessons in a form that leads to a specific change in practice, with an owner and evidence that the change was adopted.

LESSON (WHAT HAPPENED)	ROOT CAUSE	CHANGE TO PRACTICE	OWNER	DUE DATE	APPLIED ON	VERIFIED RESULT
Pile caps delayed by unrecorded utilities on two metro stations	*Utility verification not required before foundation design freeze*	*Trial pits and radar survey added as design-stage deliverables in the design brief template*	*Head of design management*	*Next design brief revision*	*Line extension package 2*	*No unrecorded utility conflicts at pier positions*
Boiler first fire delayed by incomplete gas leak-test records	*Leak-test records not linked to the completions database*	*Leak-test certificates made a mandatory field for mechanical completion of fuel gas subsystems*	*Commissioning manager*	*Before next start-up*	*Mill boiler 2*	*Records complete at review; first fire on plan*

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LESSON (WHAT HAPPENED)	ROOT CAUSE	CHANGE TO PRACTICE	OWNER	DUE DATE	APPLIED ON	VERIFIED RESULT

A **benefits realisation review** returns, months or years after handover, to the benefits promised in the business case (developed in Justify) and measures whether they arrived: the effluent treatment plant meeting discharge standards consistently, the new production line reaching its rated output, the substation reducing outages, the hospital wards reducing waiting times. The review needs measures and data sources agreed at closeout, an owner in the operating organisation, and an honest comparison with the original assumptions. Its findings are among the most valuable lessons an organisation has, because they test whether its investment decisions were sound. Owners that publish sustainability disclosures – increasingly aligned with IFRS S1 and S2 (2023) – also need reliable as-built carbon and energy data, so secure it at closeout while the evidence is available, ideally against a carbon baseline managed in the PAS 2080 way.



Illustrative data for a new production line, by quarter after handover. Output plateaus about 10 points below the business case – the review must find out whether the assumption, the asset or its operation explains the gap.

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INSIGHT · CLOSE THE LOOP

The ZEPMS Delivery Loop is a loop, not a line. Lessons from commissioning, handover and benefits reviews should change the next business case, scope definition, schedule norms, estimates, risk registers and contract strategies. An organisation that captures lessons but never changes its standards is only keeping a diary.

SCENARIO · DECIDE · THE FINAL ACCOUNT STANDOFF

A hypothetical expressway interchange has been taken over. Of forty variations, twelve remain unagreed, and the contractor is demobilising. The owner's quantity surveyor, who knows the history of every variation, is due to move to another project next month. The owner's finance team wants the project closed in the accounts this quarter. What should the project manager do FIRST?

(A) Pay the contractor's figure for the twelve variations so that the project can be closed in the accounts this quarter.

(B) Withhold all remaining payments until the contractor withdraws its claims for the twelve disputed variations.

(C) Before the surveyor leaves, secure the records, set up a time-bound final account plan with joint valuation meetings, and document positions on each item.

(D) Close the project now and pass the twelve variations to the legal department to resolve later if the contractor pursues them.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a lesson your organisation has 'learned' more than once. Why did it not change practice the first time, and which of the five criteria – specific, caused, actionable, owned, verified – was missing?

Key terms

Final account
The agreed final contract sum, being the original price adjusted for valued variations, remeasurement, provisional sums, fluctuations, settled claims and deductions, leading to the final payment certificate.

Demobilisation
The planned removal of the contractor's people, plant, temporary facilities and materials from site, including reinstatement of land and disconnection of temporary services.

Benefits realisation review
A review carried out after the asset has been in operation for some time to measure whether the benefits promised in the business case have been achieved, and why or why not.

Contract closeout
The process of confirming that all obligations under a contract have been met, settling outstanding payments and claims, releasing securities and formally closing the contract.

Post-project review
A structured, evidence-based review held at or soon after project completion to understand what went well and badly, why, and what should change in future practice.

Lessons learned register
A controlled record of lessons from projects, each with its root cause, the change to practice it requires, an owner, a due date and evidence of adoption.

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Records retention schedule
A defined list of project and asset record types with how long each must be kept, where, in what format and who may access it, reflecting contractual, legal and operational needs.

Key takeaways

- Plan closeout from the start across six workstreams: contractual, financial, physical, information, people, benefits and learning
- Close as you go: agree variations monthly and close each subcontract when its scope ends
- Secure records and key people's knowledge before they leave, and keep records under a retention schedule
- A lesson counts only when it changes a standard, template or decision – with an owner and later verification
- Measure benefits in operation against the business case and feed the gaps into future investment standards

KNOWLEDGE CHECK

21. Why should variations be valued and agreed as the work proceeds rather than at the end of the project?

(A) Contracts never allow variations to be valued after taking over
(B) Records and people are available, so agreement is faster and fairer
(C) It allows the owner to avoid paying for most of the variations
(D) It removes the need for any final account at the very end of the project

22. Which is the best example of a lesson that will change practice?

(A) Communication between design and site should be improved next time
(B) The project team should try much harder next time to avoid delays to the critical path
(C) Utility surveys will be a design deliverable before foundation design freeze
(D) The project experienced many problems with buried services in year one

23. When is a benefits realisation review normally carried out?

(A) After the asset has operated for long enough to measure its benefits
(B) At the stage gate before the investment decision is approved
(C) During commissioning, alongside the performance guarantee tests
(D) On the day that the taking-over certificate is issued by the engineer

24. Who should normally take part in a post-project review?

(A) Only the sponsor and the project manager, to keep it confidential
(B) Only the contractor, because it carried out most of the physical work
(C) Only the PMO, which can compare results across the organisation
(D) A cross-section of owner, designers, contractors and operations staff

25. A contractor demobilises and leaves a borrowed field with compacted soil and broken drainage, contrary to a promise to the landowner. Which closeout workstream has failed?

(A) Financial
(B) Information
(C) Physical
(D) People

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · COMMISSION

Answer key

Knowledge checks

1. B Welder qualification is preventive and process-focused – QA. The other three inspect or test particular products, which is QC.

2. D COPQ = internal failure + external failure = 110 + 40 = 150. The total COQ is 220, and the cost of conformance is 70.

3. A The project quality plan applies the QMS to the project. An ITP covers one activity, and the policy sits at organisation level.

4. C Audit findings are graded and closed through corrective action with an owner and date. Doing nothing, blaming a person or stopping everything are not proportionate first responses.

5. B Doing the change on a limited scale is the 'Do' step. Measuring results is 'Check', and standardising is 'Act'.

6. B Work must not proceed past a hold point without written release from the named party. Photographs or the contractor's own approval are not substitutes.

7. C Transport and installation can damage equipment, and real connections and interfaces must be proven. The SAT feeds commissioning; it does not replace it.

8. A Traceability links material batches and certificates to where they are used, so a defective batch can be located without opening up everything.

9. D Net burn rate = 20 – 5 = 15 per day; 300 / 15 = 20 days. Ignoring new items would give an optimistic 15 days.

10. B FPY = 170 / 200 × 100% = 85%. The other 15% needed rework and re-inspection.

11. C These preparation activities are carried out before process media are introduced – pre-commissioning.

12. B Most process systems need power, air, water and fire protection to be tested and operated safely, so utilities come first.

13. C Net heat rate = 90,000,000 / 10,000 = 9,000 kJ/kWh. Efficiency = 3600 / 9,000 × 100 = 40%. The other values come from using the wrong conversion factor or confusing gross and net output.

14. A The review checks construction, procedures, training, safety systems and hazard study actions before start-up. Guarantees and punch lists are handled elsewhere.

15. B Area or quantity progress does not show whether a whole system is complete; one missing valve can block flushing. That is why commissioning tracks completion by system and subsystem.

16. A Warranties that start at delivery can run out before equipment operates; negotiating a start linked to commissioning or taking over protects the owner.

17. C As-built information must be captured as changes happen and checked, for example by survey or scan. Re-titling design drawings or relying on memory loses changes.

18. B Long-lead, high-consequence items held to avoid long outages are critical, insurance or capital spares.

19. B Soft landings involves operators from design onwards and keeps the delivery team engaged after handover to fine-tune performance.

20. C Test-loading proves the data is structured and complete enough for the maintenance system. Counting files or folder size proves nothing about usability.

21. B Contemporaneous valuation uses fresh records and the people who were there. It does not remove the final account or avoid legitimate payment.

22. C A useful lesson names a specific change that can be owned and verified. The others describe symptoms or wishes.

23. A Benefits appear in operation, so they can only be measured after a period of use; the measures should be agreed at closeout.

24. D Different parties saw different parts of the story; a cross-section produces a fuller and fairer picture.

25. C Reinstatement of land and access is part of physical closeout and is often a community commitment as well.

Scenarios & case studies

> **Scenario · Weld rejections from a spool fabricator**
> **C** is best. Best. Containment plus a data-led root cause investigation at the source addresses the real problem. Defect types such as porosity or lack of fusion point to process causes, not handling damage.

> **Scenario · Witness point missed at the factory**
> **D** is best. Best. This restores the verification the ITP intended at modest cost, keeps release in the owner's hands, and records the vendor's notice failure for performance management.

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Case study · Commissioning an effluent treatment plant for a dyeing factory (illustrative)
Systemise by process function: power and controls, chemical dosing, equalisation and transfer pumps, aeration blowers and diffusers, clarifiers and sludge handling, with utilities first. Cold commissioning with clean water should prove pumps, blowers, valves, level and dissolved-oxygen control loops, dosing pumps, alarms and interlocks, and the hydraulic path, including leak checks on tanks. Hot commissioning then introduces effluent at a controlled rate while the biomass is seeded and built up, with laboratory monitoring of the key discharge parameters. The project manager should give the factory a realistic, evidence-based ramp-up plan, explain the regulatory and reputational risk of non-compliant discharge, and agree production ramp-up limits tied to measured treatment performance rather than to the construction finish date.

Scenario · Pressure to skip the review
B is best. Best. These are safety-critical gaps that the review exists to catch. Holding the start-up while offering a clear plan treats the production need seriously without gambling with safety.

Case study · Handing over a hospital block (illustrative)
A sectional handover, by floor or department, would let the hospital open wards early, but it needs the contract to define sections, their completion requirements, separate taking-over dates, defects periods and insurance arrangements, and safe segregation between occupied and construction areas. Operations readiness should start now: recruit or contract medical gas and HVAC maintenance competence, involve the estates team in commissioning and witness testing, review the draft O&M manuals, load the asset register into the maintenance system, and agree a training plan with assessment. Before occupation, the medical gas systems need full testing and certification by competent persons, theatre and isolation room pressure regimes and air changes need to be verified, fire safety systems and approvals must be in place, and standby power must be proven on load. A soft landings approach with the contractor and designers on call during the first months would help the hospital fine-tune systems safely.

Scenario · Ready to take over?
D is best. Best. This respects the contract, protects revenue where justified and closes the safety-critical gap before the owner takes operational control of high-voltage equipment.

Scenario · The final account standoff
C is best. Best. This protects the knowledge and evidence that will decide the outcome, creates a structured path to agreement, and gives finance a realistic closure date.

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REFERENCE

Glossary

1

100% rule The WBS principle that each level must include all of the work of its parent element and no work outside it, so that no scope is missed or double-counted. (Define)

4

4D BIM The linking of model elements to schedule activities so that the planned construction sequence can be visualised, tested and communicated over time. (Integrate)

5

5D BIM The linking of model-derived quantities to cost data so that estimates, cost plans and the cost effect of design changes can be produced and traced from the model. (Integrate)

A

Abnormally low tender A tender priced so far below the estimate and other bids that its sustainability is doubtful, requiring the bidder to justify its prices before award. (Procure)

Accrual A cost recognised for work done or goods received before the invoice is booked, included in actual cost so that cost performance is not overstated. (Budget)

Actual cost (AC) The cost actually incurred, including accruals, for the work performed up to a given date, recorded on the same basis as the budget. (Budget)

Advance payment A payment made to the contractor early in the contract to fund mobilisation or long-lead purchases, usually secured by a guarantee and recovered through deductions from later certificates. (Budget)

AI scheduling assistant A software tool that uses AI to query, check, draft or explain project schedules in response to a planner's requests; its outputs are drafts that a competent planner must verify and approve. (Integrate)

Alliancing A collaborative delivery model in which owner, designers and contractors work as one team under a shared agreement, with open-book costs and shared incentives tied to project outcomes. (Procure)

Analogous estimate An estimate based on the actual cost of a similar past project, adjusted for differences in size, date, location and scope; quick but only as reliable as the comparison. (Budget)

Arbitration A private, binding method of resolving disputes in which an independent tribunal chosen under the contract hears the case and makes an award enforceable in most countries. (Procure)

As-built drawings Drawings that record the works as actually constructed, including all changes made during construction, verified and handed to the owner for operation, maintenance and future modification. (Commission)

Asset information model (AIM) The structured collection of graphical, non-graphical and documentary information needed to operate and maintain an asset, handed over from the project and kept updated in operation. (Commission)

Asset life cycle The full life of a physical asset from concept through design, construction, operation, maintenance and renewal to decommissioning; the project is only its first part. (Justify)

B

Backward pass The calculation from project finish back to the start that gives each activity's late finish and late start, taking the smallest successor late start wherever paths branch. (Schedule)

Basis of design The designers' controlled record of the technical criteria, site data, codes and standards, assumptions and key decisions used to meet the owner's requirements; the reference for all subsequent design. (Define)

Battery limit A defined physical or contractual boundary, such as a tie-in flange, cable termination or plot line, at which one party's scope ends and another's begins. (Define)

Benefit-cost ratio The present value of benefits divided by the present value of costs; a ratio above 1 indicates that benefits exceed costs at the chosen discount rate. (Justify)

Benefits realisation review A review carried out after the asset has been in operation for some time to measure whether the benefits promised in the business case have been achieved, and why or why not. (Commission)

Bill of quantities (BOQ) A list of measured quantities of every work item, prepared to an agreed method of measurement and priced with unit rates to produce an estimate or tender price. (Budget)

Bottom-up estimate An estimate built from the detailed cost of each work package or activity, using quantities, productivity and prices, then rolled up to the project total. (Budget)

Budget at completion (BAC) The total approved budget for the work in the performance measurement baseline; the value that EV will reach when all work is complete. (Budget)

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Building information modelling (BIM) The process of creating, coordinating and managing structured graphical and non-graphical information about a built asset through design, construction and operation, using shared digital models and data. (Integrate)

Business case The documented justification for an investment, setting out need, options, value for money, affordability, deliverability and expected benefits; developed in stages and revisited at each gate. (Justify)

C

Capital project A project that creates or substantially upgrades a long-lived physical asset, such as a plant, building or piece of infrastructure, funded from capital rather than operating budgets. (Justify)

Cause-risk-effect statement A three-part way of writing a risk: the existing cause or condition, the uncertain event that may occur, and its effect on named objectives. (Protect)

Change control board A formally constituted group with authority to review, approve, reject or defer change requests within defined limits, and to record its decisions and reasons. (Define)

Change request A formal proposal to modify an approved baseline, deliverable or document, recorded and assessed before a decision is taken at the authorised level. (Define)

Clash detection The automated, rule-based checking of federated discipline models to find physical intersections, clearance violations and sequencing conflicts before fabrication or installation. (Integrate)

Cold commissioning Functional testing of systems with safe media such as water, air or inert gas to prove controls, interlocks and equipment operation before process fluids or fuel are introduced. (Commission)

Commitment An obligation to spend money created by a placed order or subcontract that has not yet been invoiced, tracked so that forecasts include money already spoken for. (Budget)

Common data environment (CDE) The agreed single source of information for a project or asset, with defined workflows, permissions and states such as work in progress, shared, published and archived. (Integrate)

Computer vision AI techniques that interpret images, video or point clouds to detect and classify objects or events, such as installed elements, missing PPE or people inside exclusion zones. (Integrate)

Concurrent delay A situation in which an employer-risk delay and a contractor-risk delay affect the critical path at the same time, each sufficient on its own to delay completion. (Procure)

Confidence level (P50, P80) A point on the simulated outcome distribution; P80 is the value that 80% of iterations meet or beat, used to set funded budgets and committed dates. (Protect)

Configuration management The discipline of identifying, controlling and auditing the requirements, design documents and physical configuration of an asset so they remain consistent as changes are made. (Define)

Conflict of interest A situation in which a person's private interests could improperly influence, or appear to influence, their duties in a procurement, requiring declaration and management. (Procure)

Consents register A log of every permit, licence, clearance and approval a project needs, with the issuing body, prerequisites, lead time, validity, attached conditions, responsible owner and current status, linked to schedule activities. (Integrate)

Constructability The extent to which a design allows safe and efficient construction, considering sequence, access, temporary works, standardisation and site conditions. (Define)

Contingency drawdown The controlled release of risk contingency into a control account when an identified risk occurs, approved within delegated authority and recorded in the register and forecast. (Protect)

Contingency reserve Budget within the cost baseline set aside for identified risks and estimating uncertainty, drawn down by the project manager under documented rules. (Budget)

Contract administrator (the Engineer) The person or firm named in the contract to administer it, issuing instructions and certificates for the employer and making fair, neutral determinations when the parties disagree. (Procure)

Contract closeout The process of confirming that all obligations under a contract have been met, settling outstanding payments and claims, releasing securities and formally closing the contract. (Commission)

Contract packaging The division of a project's scope into separate contracts, balancing competition and specialisation against the number and difficulty of interfaces. (Procure)

Contractual notice A formal written communication required by the contract to trigger or preserve rights, such as notice of a claim, sent to the stated address within the stated period. (Procure)

Control account A management control point where scope, budget, schedule and actual cost are integrated for a WBS element under one responsible manager; it contains one or more work packages. (Define)

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Cost baseline The approved, time-phased budget, made up of control account budgets plus contingency reserve, against which cost and earned value performance are measured. (Budget)

Cost escalation The expected increase in costs over time due to inflation and market conditions, applied from the estimate base date to when the money is spent. (Budget)

Cost of quality (COQ) The total of prevention, appraisal, internal failure and external failure costs, used to show the economic case for preventing defects rather than detecting and repairing them. (Commission)

Cost performance index (CPI) The ratio EV / AC, showing the budgeted value earned for each unit of money spent; values below 1.0 indicate cost overrun. (Budget)

Cost variance (CV) The difference between earned value and actual cost (EV - AC); a negative value means the completed work has cost more than its budget. (Budget)

Cost-loaded schedule A schedule in which each activity carries its budgeted cost, so that planned spend can be summed by period and plotted over time. (Budget)

Cost-reimbursable contract A contract in which the contractor is paid its actual allowable costs plus a fee, leaving most cost risk with the owner. (Procure)

Crash cost slope The extra direct cost per unit of time saved when an activity is crashed: (crash cost - normal cost) / (normal duration - crash duration). (Schedule)

Crashing Shortening the duration of critical activities by adding resources such as extra shifts, crews or equipment, accepting additional direct cost to save time. (Schedule)

Critical path The longest sequence of logically linked activities through the network, which sets the shortest possible project duration. Its activities have the lowest total float. (Schedule)

Critical spares Long-lead or high-consequence spare parts, such as a spare transformer or turbine component, held because their failure would otherwise cause a long and costly outage. (Commission)

D

Data date The date up to which progress has been recorded in a schedule update; everything before it is actual, everything after it is forecast. (Schedule)

Data governance The policies, roles and controls that decide who owns project data, how its quality is maintained, who may use it, for what purposes and how long it is kept. (Integrate)

Decision rights The explicit assignment of each class of project decision to a named role or body, so that everyone knows who recommends, who decides and who is informed. (Justify)

Decision tree A diagram of decisions, chance events and outcomes that is rolled back from right to left to compare options on expected value while also showing the spread of outcomes. (Protect)

Deliverables register A controlled list of every design deliverable, such as drawings, calculations, specifications and models, with discipline, revision, planned and actual dates and status, used to plan and measure design progress. (Define)

Demobilisation The planned removal of the contractor's people, plant, temporary facilities and materials from site, including reinstatement of land and disconnection of temporary services. (Commission)

Design freeze The agreed point after which a design, or a defined part of it, may be changed only through formal change control with impact assessment. (Define)

Design review A planned, structured examination of design deliverables at a defined maturity by designers, owner, operator or contractor, with comments logged, resolved and closed. (Define)

Design-bid-build (DBB) A delivery model in which the owner's consultant completes the design, contractors bid on it, and the selected contractor builds it, with the owner carrying design risk. (Procure)

Design-build (DB) A delivery model in which one contractor is responsible for both design development and construction against the owner's requirements. (Procure)

Detailed design The design stage that completes the calculations, drawings, models, schedules and data sheets needed to procure, fabricate, construct and test, ending with documents issued for construction. (Define)

Digital twin A digital representation of a physical asset that is connected to it by data flowing from the asset, used to monitor, understand, predict or optimise its performance in operation. (Integrate)

Discount rate The annual rate used to convert future cash flows into present values, typically the owner's cost of capital or hurdle rate for financial appraisal, or a social discount rate for public economic appraisal. (Justify)

Discounted payback period The time needed for cumulative discounted cash flows to recover the initial investment; always longer than simple payback when the discount rate is positive. (Justify)

Dispute avoidance and adjudication board (DAAB) A standing panel of independent experts appointed at the start of a contract to help the parties avoid disputes and to give decisions on referred disputes that are binding unless later revised. (Procure)

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Disruption Loss of productivity caused by interference with the contractor's planned working methods, even where completion is not delayed, usually measured by comparing affected and unaffected work. (Procure)

Do-nothing option The baseline option of making no new investment, or only the minimum needed to keep operating, against which the costs and benefits of all other options are measured. (Justify)

Document control The function and procedures that number, register, issue, distribute, revise and archive project documents so that everyone uses the current approved revision and superseded copies are withdrawn. (Integrate)

E

Earned schedule An extension of earned value that finds the time at which the planned value equalled the current earned value, giving schedule variance and performance in time units. (Budget)

Earned value (EV) The budgeted value of the work actually completed by a given date, measured using agreed rules of credit. (Budget)

Embodied carbon Greenhouse gas emissions from producing, transporting and installing materials and products, such as cement and steel, as distinct from emissions from operating the asset. (Protect)

Environmental and social impact assessment (ESIA) A structured study that predicts a project's environmental and social impacts during construction and operation, rates their significance and sets mitigation, informed by baseline data and consultation. (Protect)

Environmental and social management plan (ESMP) The plan that turns ESIA commitments into specific measures with owners, timing, monitoring and budget, cascaded into contractor plans and contracts. (Protect)

EPC (turnkey) contract A contract in which one contractor is responsible for engineering, procurement, construction and commissioning, handing over a working facility that meets defined performance requirements. (Procure)

Escalation matrix A table defining which types and sizes of issue must be raised to which management level, by whom and within what time, so decisions are made at the right level early enough. (Integrate)

Estimate at completion (EAC) The forecast total cost of the project or control account at completion, based on actual cost to date plus the expected cost of the remaining work. (Budget)

Estimate class A category describing an estimate's maturity, purpose, method and expected accuracy range, from early screening estimates to definitive estimates based on complete design. (Budget)

Estimate to complete (ETC) The forecast cost of the remaining work from the status date to completion, equal to EAC minus AC. (Budget)

Exchange information requirements (EIR) The client's specification of the information each appointed party must deliver, including content, format, level of information need and delivery dates. (Integrate)

Expected monetary value (EMV) The probability-weighted value of an uncertain outcome, calculated as probability multiplied by impact and summed across outcomes; negative for threats and positive for opportunities. (Protect)

Extension of time (EOT) An adjustment of the completion date granted when an event at the employer's risk delays completion, protecting the contractor from delay damages for that period. (Procure)

F

Factored estimate An estimate that prices the major equipment and applies factors for installation, bulk materials and indirect costs to reach total installed cost, typical of process plant at feasibility stage. (Budget)

Factory acceptance test (FAT) Testing of equipment at the manufacturer's works before shipment to demonstrate compliance with its specification, often witnessed by the purchaser. (Commission)

Fast-tracking Compressing a schedule by overlapping activities that were planned in sequence, which saves time but increases the risk of rework and interface problems. (Schedule)

Fatal flaw A characteristic that makes an option unacceptable regardless of its other merits, such as an unobtainable permit, unproven technology or unaffordable cost, used to screen options early. (Justify)

Feasibility study A multidisciplinary study that develops short-listed options in enough technical, market, economic, environmental, social and legal depth to support an investment decision or financing. (Justify)

Final account The agreed final contract sum, being the original price adjusted for valued variations, remeasurement, provisional sums, fluctuations, settled claims and deductions, leading to the final payment certificate. (Commission)

Finish-to-start (FS) relationship The default logic link: the successor cannot start until the predecessor has finished, for example setting a transformer only after its plinth is complete. (Schedule)

Five-case model A widely used business case structure that tests a proposal against strategic, economic, commercial, financial and management dimensions. (Justify)

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Fixed-formula method A rule of credit that credits a fixed percentage of budget when an activity starts and the remainder when it finishes, such as 0/100, 50/50 or 20/80; suitable for short activities. (Budget)

Forward pass The calculation from project start to finish that gives each activity's early start and early finish, taking the largest predecessor early finish wherever paths merge. (Schedule)

Free float The time an activity can be delayed without delaying the early start of any immediate successor, calculated as the earliest successor ES minus the activity's EF. (Schedule)

Front-end engineering design The design stage, also called basic design, that defines process, layouts, main equipment and specifications in enough detail to support the investment decision, a reliable estimate and an EPC or construction tender. (Define)

Front-end loading Deliberately investing effort in definition, design basis, investigations and execution planning before sanctioning execution, when influence over outcomes is highest and changes are cheapest. (Justify)

G

Gemba walk A leader's regular visit to the place where the work is actually done to observe, ask questions and remove obstacles, rather than relying only on reports. (Integrate)

Generative scheduling The automated creation and ranking of many feasible construction sequences from defined activities, constraints, resources and objectives, from which planners select and refine an option. (Integrate)

Grievance mechanism An accessible, publicised process through which affected people can raise concerns and receive a timely, documented response, with escalation if they are not satisfied. (Protect)

Guaranteed maximum price (GMP) A pricing arrangement in which the owner reimburses cost plus fee up to an agreed ceiling, the contractor bears cost above it, and savings below it are often shared. (Procure)

H

Handover dossier The complete, organised package of records delivered to the owner at handover, including as-builts, O&M manuals, test and commissioning records, certificates, warranties and asset data. (Commission)

Hard constraint An imposed date, such as 'must finish on', that overrides network logic. It can hide the true critical path and should be used only for genuine fixed dates with a recorded reason. (Schedule)

HAZOP Hazard and operability study: a systematic, team-based design review that applies guide words to process parameters for each node of a design to identify deviations, their causes, consequences and safeguards. (Define)

Hierarchy of controls The ranking of risk controls from most to least effective: elimination, substitution, engineering controls, administrative controls and personal protective equipment. (Protect)

Hold point An inspection point beyond which work must not proceed without the written release of the designated party, typically before work is covered or made inaccessible. (Commission)

Hot commissioning The commissioning stage in which process fluids, fuel or full energy are introduced and the plant is started up and brought towards normal operating conditions. (Commission)

Human-in-the-loop A control design in which a competent person reviews, approves or overrides an automated output before it is acted on, keeping human accountability for the decision. (Integrate)

I

Impact assessment The multidisciplinary evaluation of a proposed change's effects on scope, cost, schedule, risk, safety, quality, interfaces and operations, carried out before a decision is taken. (Define)

Inspection and test plan (ITP) A document listing, for a work activity or item, each inspection and test with its reference, acceptance criteria, frequency, record and the hold, witness, review or surveillance role of each party. (Commission)

Interim payment certificate The engineer's periodic certificate of the amount due to the contractor for work done, materials and adjustments, which the employer must pay within the contract period. (Procure)

Internal rate of return The discount rate at which a project's NPV equals zero; for conventional cash flows the project is acceptable when its IRR exceeds the hurdle rate. (Justify)

Involuntary resettlement Physical or economic displacement caused by land acquisition or restrictions on land use that affected people cannot refuse, requiring compensation and livelihood support. (Protect)

Issue A problem or condition that has already occurred or is certain to occur and now requires a decision or action; a risk becomes an issue when it materialises. (Protect)

K

Kickoff meeting The meeting at the start of execution or of a phase that aligns owner, consultants and contractors on objectives, roles, safety, interfaces, communication and immediate priorities. (Justify)

L

Labour welfare The facilities, conditions and treatment provided for workers – water, sanitation, rest, accommodation, fair and timely pay, reasonable hours and respect – beyond the minimum needed to do the work. (Integrate)

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Lag A time delay applied to a logic relationship, such as FS + 7 days for curing. It consumes time but carries no resources or cost, so long lags should be justified or replaced by real activities. (Schedule)

Lagging safety indicator A measure of harm that has already happened, such as lost-time injuries or dangerous occurrences, useful for trends but too late to prevent those events. (Protect)

Lead (negative lag) A negative lag that allows a successor to start before its predecessor finishes, such as starting cable tray installation in a zone before ceiling framing there is complete. (Schedule)

Leading safety indicator A measure of proactive safety activity or conditions, such as near-miss reports closed or RAMS briefed, that can predict and prevent harm. (Protect)

Lessons learned register A controlled record of lessons from projects, each with its root cause, the change to practice it requires, an owner, a due date and evidence of adoption. (Commission)

Letter of acceptance The owner's formal written acceptance of a tender, which typically creates the contract and starts the periods for providing securities and beginning the works. (Procure)

Level of effort Support work, such as site management or security, that has no measurable output and earns value in step with the passage of time, so its EV always equals its PV. (Budget)

Livelihood restoration Measures that restore, and ideally improve, the income and living standards of people whose land, business or access to resources is affected by a project. (Protect)

Lock-out tag-out (LOTO) An isolation practice in which energy sources are isolated, secured with personal locks and tags, and proved dead before work begins. (Protect)

Long-lead item Equipment or material whose design, manufacture and delivery time is long enough to drive the schedule, so it must be ordered early, often before the main contractor is appointed. (Procure)

Look-ahead schedule A short-term plan, typically covering the next four to six weeks, that breaks the control schedule into crew-level tasks and identifies constraints to remove. (Schedule)

Lump sum contract A contract in which the contractor agrees to deliver a defined scope for a fixed price, carrying quantity and productivity risk, with adjustments only for changes and risks the contract assigns to the owner. (Procure)

M

Maintainability The ease with which an asset can be inspected, maintained, repaired and replaced, including access, lifting provisions, isolation and spares. (Define)

Make-or-buy decision The choice between performing an activity with the organisation's own resources or buying it from the market, based on cost, capability, capacity, control and strategy. (Procure)

Management reserve Budget outside the cost baseline but within the project budget, held by the sponsor for unforeseen work within scope and released through change control. (Budget)

Mechanical completion The milestone at which a system or subsystem is installed, tested and documented in accordance with the drawings, specifications and codes, ready for pre-commissioning, usually confirmed by a certificate. (Commission)

Merge bias The tendency for a milestone fed by several parallel paths to finish later than any single path suggests, because every incoming path must finish before the milestone is reached. (Schedule)

Milestone A zero-duration event marking a significant point such as a contract key date, permit approval, equipment delivery or sectional completion, linked into the logic rather than left floating. (Schedule)

Mitigation hierarchy The preferred sequence for managing impacts: avoid first, then minimise, then restore, and only then offset or compensate for what remains. (Protect)

Model drift The decline in an AI model's accuracy over time as real conditions, data or work practices move away from those in its training data. (Integrate)

Monte Carlo simulation A method that recalculates a cost or schedule model thousands of times, each time sampling uncertain inputs and risk events, to build a distribution of possible outcomes. (Protect)

Multi-criteria analysis A transparent method of scoring options against weighted financial and non-financial criteria, used alongside discounted cash flow when not all effects can be valued in money. (Justify)

N

Near-critical path A path whose total float is below a threshold set by the project, so that a modest delay or a realised risk could make it critical. (Schedule)

Negative float Float below zero, which appears when an imposed completion or milestone date is earlier than the date the network can achieve. It shows how much time must be recovered. (Schedule)

Net present value The sum of all of a project's cash flows discounted to today at the chosen discount rate; a positive NPV means the project creates value above the required return. (Justify)

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Non-conformance report (NCR) A formal record that work, material or equipment does not meet a specified requirement, used to contain, disposition, investigate and close the non-conformance. (Commission)

O

O&M manual The project-specific set of operating and maintenance information for an asset – procedures, maintenance instructions, data sheets, drawings, spares lists and vendor documents. (Commission)

O&M operator The organisation, in-house or contracted, that will operate and maintain the asset after handover and should influence its design and commissioning. (Justify)

OODA loop A decision cycle of observe, orient, decide and act that helps leaders respond quickly and repeatedly to fast-changing situations such as site emergencies or disruption. (Integrate)

Operability The ease with which operators can run an asset safely and effectively across its operating modes, including start-up, normal operation, upsets and shutdown. (Define)

Operations readiness The planned preparation of the owner's people, procedures, systems, spares, contracts and permits so that the asset can be operated and maintained safely and effectively from handover. (Commission)

Opportunity A risk whose occurrence would benefit one or more project objectives, for example an earlier factory slot for a turbine or early grid energisation. (Protect)

Optimism bias The systematic tendency to underestimate costs and durations and overestimate benefits at the decision stage, corrected by reference class forecasting, uplifts and independent review. (Justify)

Options analysis The structured generation, screening and comparison of alternative ways to meet a defined need, leading to a recommended preferred option with recorded reasons. (Justify)

Owner's engineer An engineering consultant appointed by the owner to advise on technical matters, prepare or review designs and monitor contractors on the owner's behalf. (Justify)

Owner's project requirements A document that sets out, in the owner's and users' terms, what the asset must achieve, including functions, capacities, performance, quality, budget and operational expectations, against which the design is checked. (Define)

P

Pain/gain share The agreed proportions in which owner and contractor split cost overruns (pain) and underruns (gain) against a target, often with caps on the contractor's pain. (Procure)

Parametric estimate An estimate that multiplies a key project parameter, such as MW, m2 or km, by a rate derived from historical data, sometimes with a scaling exponent. (Budget)

Path variance The sum of the variances of the activities on a path, from which the path standard deviation is the square root; it assumes the activities are independent. (Schedule)

Payback period The time needed for cumulative undiscounted cash inflows to recover the initial investment; simple to compute but ignores time value and cash after recovery. (Justify)

PDCA cycle The plan-do-check-act improvement loop in which a change is planned, trialled, measured and then standardised or revised, used for continual improvement. (Commission)

Percent plan complete (PPC) The number of tasks completed as planned in a week divided by the number of tasks planned, expressed as a percentage; a measure of planning reliability. (Schedule)

Performance guarantee test A formal test, run to an agreed procedure, that demonstrates whether a plant meets its contractual guarantees such as output, efficiency, emissions or effluent quality. (Commission)

Performance security A bank guarantee or bond provided by the contractor after award, which the employer can call if the contractor fails to perform its obligations. (Procure)

Performance specification A specification that states the required outcome and acceptance tests, leaving the means of achieving it to the designer, contractor or supplier. (Define)

Permit-to-work A formal written system that authorises specific high-risk work at a set place and time, after hazards are checked and isolations made, and controls its hand-back. (Protect)

PERT (beta) mean The expected activity duration (O + 4M + P) / 6, which gives the most likely value four times the weight of each extreme. (Schedule)

PERT standard deviation The approximate spread of an activity duration, (P - O) / 6, treating the optimistic-to-pessimistic range as about six standard deviations. (Schedule)

Photogrammetry The measurement of positions, shapes and volumes from overlapping photographs, often taken by drones, to produce surface models, orthophotos and point clouds. (Integrate)

Planned value (PV) The authorised budget for the work scheduled to be completed by a given date, read from the time-phased cost baseline. (Budget)

Post-project review A structured, evidence-based review held at or soon after project completion to understand what went well and badly, why, and what should change in future practice. (Commission)

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Power-interest grid A two-by-two analysis tool that positions stakeholders by their power over the project and their interest in it, guiding whether to manage them closely, keep them satisfied, keep them informed or monitor them. (Integrate)

Pre-commissioning Non-operational checks and preparation of installed systems before process media are introduced, such as flushing, cleaning, drying, loop checks, insulation tests and motor solo runs. (Commission)

Pre-feasibility study An early, lower-cost study that screens a long list of options using desk data and limited site work, eliminating those with fatal flaws and selecting a short list for detailed study. (Justify)

Pre-start-up safety review (PSSR) A structured check carried out before hazardous energy or process fluids are introduced, confirming that the plant, procedures, training and safety systems are ready for safe start-up. (Commission)

Precedence diagramming method (PDM) A way of drawing a schedule network in which activities are boxes (nodes) and arrows show the logic between them, allowing FS, SS, FF and SF relationships with lags. It is the method used by virtually all modern scheduling software. (Schedule)

Prequalification Assessment of potential bidders' eligibility, capability, capacity and record before tendering, so that only firms able to deliver are invited to bid. (Procure)

Prescriptive specification A specification that defines the materials, equipment types, methods or workmanship to be used, so that compliance is checked by conformity to the description. (Define)

Probability-impact matrix A grid that combines defined probability and impact scales to rank risks into priority bands, often shown as a red-amber-green heat map. (Protect)

Program Evaluation and Review Technique (PERT) A network scheduling technique that uses optimistic, most likely and pessimistic durations, a beta-distribution approximation and path variances to estimate the probability of meeting a target date. (Schedule)

Project charter The document, signed by the sponsor, that formally authorises a project, appoints the project manager and records purpose, objectives, high-level scope, budget, milestones and delegated authority. (Justify)

Project life cycle The sequence of phases a project passes through from identifying the need to handing over the finished asset, typically identify, select, define, execute and commission. (Justify)

Project management office A function that defines standards, provides project controls services and assures reporting for a project or portfolio; it may be supportive, controlling or directive. (Justify)

Project quality plan (PQP) The project-specific document that sets out how the quality management system will be applied to a project: objectives, organisation, standards, inspection and testing, supplier control, NCRs and audits. (Commission)

Project risk An uncertain event or condition that, if it occurs, would have a positive or negative effect on at least one project objective such as safety, cost, time, quality or scope. (Protect)

Project sponsor The senior individual accountable for the business case and benefits, who champions the project, secures resources and takes or escalates decisions beyond the project manager's authority. (Justify)

Prolongation cost The time-related costs a contractor incurs because the project lasts longer, such as site management, facilities and plant on hire, recoverable for compensable delay. (Procure)

Psychological safety A shared belief within a team that people can raise concerns, questions, ideas and mistakes without fear of punishment or humiliation, enabling problems to surface early. (Integrate)

Public-private partnership A long-term contract in which a private party finances, builds and often operates a public asset or service, repaid through user charges or government payments, with risks shared by agreement. (Justify)

Punch list A list of minor incomplete or defective items identified at walkdowns near completion, each categorised, assigned and tracked until cleared. (Commission)

Q

Quality assurance (QA) The planned, systematic, process-focused activities that give confidence that work will meet requirements, such as approved procedures, qualifications, training and audits. (Commission)

Quality audit A systematic, independent and documented examination of whether quality arrangements are followed and effective, leading to graded findings and corrective actions. (Commission)

Quality control (QC) The product-focused inspection, measurement and testing activities that check whether specific work or items conform to their requirements. (Commission)

Quality management system (QMS) An organisation's documented and implemented set of policies, processes, responsibilities and records for directing and controlling quality, often certified against ISO 9001. (Commission)

R

RAG status A red-amber-green rating used in reports and dashboards to show whether an item is on track, at risk or off track against defined thresholds; meaningful only when the thresholds are written down. (Integrate)

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Records retention schedule A defined list of project and asset record types with how long each must be kept, where, in what format and who may access it, reflecting contractual, legal and operational needs. (Commission)

Remeasurement contract A contract in which the work is measured as executed and paid at tendered unit rates, so the owner carries quantity risk while the contractor carries the risk within each rate. (Procure)

Request for information (RFI) A formal written query from the construction team to the designer or engineer seeking clarification, confirmation or resolution of missing, conflicting or ambiguous design information. (Integrate)

Residual risk The risk that remains after responses have been implemented; residual exposure is what contingency and management attention must still cover. (Protect)

Resource calendar A calendar showing when a specific resource, such as a heavy-lift crane or a commissioning team, is available to the project. (Schedule)

Resource histogram A bar chart showing the quantity of a resource required in each time period, used to spot peaks, valleys and over-allocation. (Schedule)

Resource levelling Adjusting activity dates so that resource limits are never exceeded, accepting that the project end date may move later. (Schedule)

Resource loading Assigning the quantities of labour, plant, equipment and specialist crews required by each activity in the schedule so that demand can be totalled over time. (Schedule)

Resource over-allocation A period in which the schedule requires more of a resource than is available, meaning the programme cannot be worked as drawn. (Schedule)

Resource smoothing Adjusting activity dates within their float to reduce peaks and valleys in resource demand without changing the project end date. (Schedule)

Resource-constrained critical path The sequence of activities that drives completion once resource limits are applied, including links created by waiting for a scarce resource rather than by logic alone. (Schedule)

Retention money A percentage withheld from interim payments to the contractor as security for completion and defects, usually released in stages at taking-over and the end of the defects period. (Budget)

Risk appetite The amount and type of risk an organisation is willing to accept in pursuit of its objectives, expressed in practice through thresholds and escalation rules. (Protect)

Risk assessment and method statement (RAMS) A task-specific document set that identifies hazards and controls and describes the safe sequence, plant, competence, supervision and emergency arrangements for the work. (Protect)

Risk breakdown structure (RBS) A hierarchy of risk categories, such as technical, ground, supply chain, commercial, external and organisational, used to prompt identification and to group and report risks. (Protect)

Risk correlation The tendency of uncertain inputs that share a common cause to move together; ignoring it narrows the simulated spread and understates high-confidence values such as P80. (Protect)

Risk exposure The expected effect of a risk, calculated as its probability multiplied by its impact on cost or time; used to compare risks and size responses. (Protect)

Risk owner The one named person accountable for monitoring a risk and making sure its agreed responses are carried out, even when actions are delegated. (Protect)

Risk premium The amount a contractor adds to its price to cover risks it is asked to carry; the owner pays it whether or not the risks occur. (Procure)

Risk register The living record of identified risks with their statements, owners, scores, responses, triggers and status, reviewed regularly to drive action. (Protect)

Risk trigger An observable event or condition that warns that a risk is about to occur or that a planned response or fallback must now be activated. (Protect)

Rule of credit A pre-agreed method stating how a type of work earns value as it progresses, such as units completed, weighted milestones or a fixed formula. (Budget)

S

S-curve A graph of cumulative planned (or actual) cost or progress over time, typically shaped like an S because spending is slow at the start, fast in the middle and slow at the end. (Budget)

Salience model A stakeholder analysis approach that judges priority by three attributes – power, legitimacy and urgency – so that stakeholders with legitimate or time-critical claims are not overlooked just because their formal power is low. (Integrate)

Scenario analysis Assessing a project under a small number of internally consistent futures in which several related inputs change together, such as low demand combined with high fuel prices. (Justify)

Schedule baseline The approved version of the schedule against which progress and variances are measured; it changes only through formal change control. (Schedule)

Schedule level The degree of detail in a schedule, from Level 1 (one-page milestone summary) to Level 5 (day-by-day crew tasks), with each level serving a different audience while staying consistent with the others. (Schedule)

Schedule performance index (SPI) The ratio EV / PV, showing the rate of progress against plan; values below 1.0 indicate work behind schedule. (Budget)

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Schedule risk analysis (SRA) A Monte Carlo analysis of a logic-linked schedule with duration ranges and risk events, producing finish-date confidence levels, drivers and criticality for each activity. (Protect)

Schedule variance (SV) The difference between earned value and planned value (EV - PV), measured in money; a negative value means less work has been done than planned. (Budget)

Scope baseline The approved version of the scope statement, WBS and WBS dictionary, which can be changed only through formal change control and is the reference for measuring scope performance. (Define)

Scope creep The uncontrolled growth of project scope through additions that are not assessed, approved or funded, often through many small informal requests. (Define)

Scope statement The documented description of a project's deliverables, boundaries, exclusions, assumptions, constraints, interfaces and acceptance criteria, used as the basis for detailed planning and control. (Define)

Sensitivity analysis Testing how a result such as NPV changes when one input is varied at a time, to identify the variables that matter most. (Justify)

Site acceptance test (SAT) Testing of equipment or a system after installation on site to demonstrate correct performance in its actual environment and with its real interfaces. (Commission)

Site diary A daily, contemporaneous record of site conditions and events – weather, labour, plant, deliveries, work done, instructions, visitors, incidents and delays – signed by a responsible person and used for progress and claims. (Integrate)

Site selection study An assessment of candidate locations against criteria such as land, flood risk, ground conditions, access, utilities, neighbours and permits to choose the most suitable site. (Justify)

Situational leadership An approach in which the leader adjusts the balance of direction and support to the competence and commitment of the person or crew for the task in hand, from close instruction to delegation. (Integrate)

Social licence to operate The informal, ongoing acceptance of a project by its local community and wider society, earned through fair treatment and kept promises; it is not issued by any authority and can be withdrawn at any time. (Integrate)

Soft landings An approach in which designers and contractors involve operators from early in the project and remain engaged for a period after handover to support, monitor and fine-tune the asset in use. (Commission)

Span of control The number of people or work fronts a supervisor can effectively direct, check and support; it shrinks when work is dispersed, complex or hazardous. (Integrate)

Stage gate A planned decision point at the end of a life-cycle phase where a named authority reviews defined deliverables and decides to proceed, recycle, hold or stop. (Justify)

Stakeholder engagement plan The document that sets out how, by whom, through which channels and how often each stakeholder group will be engaged, including consultation, disclosure and commitments, and how engagement will be monitored. (Integrate)

Stakeholder register A controlled list of the individuals, groups and organisations that can affect or are affected by a project, recording their interests, influence, attitude, engagement owner, channel and links to project activities. (Integrate)

Steering committee A group of senior stakeholders, usually chaired by the sponsor, that gives direction, approves gates and major changes within delegated limits and resolves conflicts the project cannot. (Justify)

Submittal A document, drawing, sample or data set that a contractor or supplier must submit for review under the specification – such as shop drawings, method statements or product data – before procurement or construction proceeds. (Integrate)

Switching value The change in a single input that would make NPV zero or reverse the preferred option; a direct measure of how much error a business case can tolerate. (Justify)

Systemisation The division of a facility into commissioning systems and subsystems with defined boundaries, so that completion, testing and handover can be planned and tracked by function. (Commission)

T

Target cost contract A contract in which the contractor is paid actual cost plus fee, adjusted by a pre-agreed share of the difference between an agreed target cost and the actual cost. (Procure)

Team development stages The typical progression of a new team through forming, storming, norming and performing (and later adjourning), used to anticipate early friction on multi-company sites. (Integrate)

Tender documents The package issued to bidders, containing instructions, conditions of contract, requirements and specifications, drawings, pricing documents, forms and evaluation criteria. (Procure)

Tender security A bank guarantee or bond submitted with a tender that the owner can call if the bidder withdraws during the validity period or refuses to sign after award. (Procure)

Threat A risk whose occurrence would harm one or more project objectives, for example late delivery of a power transformer. (Protect)

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Three-point estimate A duration or cost estimate expressed as optimistic, most likely and pessimistic values, so that both the expected value and the uncertainty can be calculated. (Schedule)

Time bar A contract provision under which entitlement to time or money is lost, or weakened, if the required notice is not given within the specified period. (Procure)

Time impact analysis A prospective delay analysis method that inserts a fragnet representing a delay event into the programme updated to just before the event and measures its effect on completion. (Procure)

Time value of money The principle that money is worth more now than the same amount in the future, because of earning potential, inflation and risk; the basis of discounting. (Justify)

To-complete performance index (TCPI) The cost efficiency that must be achieved on the remaining work to meet a target, either BAC or an approved EAC: remaining work divided by remaining funds. (Budget)

Tolerance The permitted deviation in cost, time, scope, quality or other measures within which a management level may act without escalating to the level above. (Justify)

Toolbox talk A short, task-focused safety briefing given to a crew at the work face, usually at the start of a shift, in the workers' own language. (Protect)

Tornado chart A bar chart that ranks model inputs by how strongly they influence the result, with the largest driver at the top, directing attention to the risks that matter most. (Protect)

Total float The time an activity can be delayed without delaying project completion, calculated as LS - ES or LF - EF. It is shared by the activities on the same path. (Schedule)

Transmittal A cover record that accompanies documents sent between parties, listing each document, its revision and purpose, and proving when it was issued and received. (Integrate)

Trend notice An early-warning record raised when a potential change or deviation is first identified, prompting assessment and forecast updates before a formal change request is needed. (Define)

Triangular mean The expected duration (O + M + P) / 3 under a triangular distribution; it weights the extremes more heavily than the PERT mean. (Schedule)

Two-envelope procedure A tendering method in which technical and financial proposals are submitted separately and price envelopes are opened only for technically compliant tenders. (Procure)

U

Unforeseeable physical conditions Ground, water and other physical conditions at the site that an experienced contractor could not reasonably have foreseen at tender, often an employer risk in employer-designed and design-build forms. (Procure)

Units-completed method A rule of credit in which earned value is the budget multiplied by the proportion of units completed, such as metres of cable pulled or piles installed. (Budget)

Utility diversion The relocation, protection or temporary rerouting of existing services such as water mains, sewers, power cables, gas pipes or telecom ducts to clear space for new works, normally designed and approved with the utility owner. (Integrate)

V

Value engineering A function-based, structured review that seeks to deliver the required functions at the lowest whole-life cost without compromising performance, safety or quality. (Define)

Variance at completion (VAC) The difference between the budget at completion and the estimate at completion (BAC - EAC); a negative value is a forecast overrun. (Budget)

Variance threshold A pre-agreed limit on cost or schedule variance, or on CPI and SPI, beyond which a variance must be explained, escalated or acted on. (Budget)

Variation A change to the works, such as an addition, omission, or change in quality, quantity, sequence or method, instructed under the contract by a person with authority, and valued under its rules. (Procure)

W

WBS dictionary A companion document giving, for each WBS element, its description of work, deliverables, acceptance criteria, responsible party, estimate and schedule information. (Define)

Weighted-milestone method A rule of credit that divides a discrete item of work into observable steps, each carrying an agreed percentage of the budget that is earned when the step is reached. (Budget)

Witness point An inspection point at which a designated party must be given agreed notice and may attend; if it does not attend after proper notice, work may proceed. (Commission)

Work breakdown structure A hierarchical, deliverable-oriented decomposition of the total project scope into progressively smaller elements, down to work packages. (Define)

Work package The lowest level of the WBS: a defined piece of work small enough to be estimated, scheduled, assigned to one responsible party and measured for progress. (Define)

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CAPSTONE

Capstone: Take an engineering project round the ZEPMS Delivery Loop

Choose a real capital project from your work, or a realistic one (for example: a new effluent treatment plant for a garment factory near Gazipur, a 100 MW solar park, a four-storey hospital block in Chattogram, or an elevated metro station). Take it through all eight stages of the ZEPMS Delivery Loop and produce a project execution pack that an investment committee could approve, a contractor could price and a site team could build from.

Deliverables

☐ Business case with options analysis, NPV, IRR and payback, plus a sensitivity table

☐ Project charter with governance, stage gates and decision rights

☐ Scope statement with battery limits, exclusions, WBS to work-package level and WBS dictionary

☐ Critical path schedule with float analysis, PERT estimates for key activities and a resource histogram

☐ Cost estimate with estimate class, contingency, management reserve and a cost-loaded S-curve

☐ Earned value status report at a simulated mid-point with CPI, SPI, EAC and corrective actions

☐ Risk register with a quantitative analysis of the top risks (P50/P80) and an HSE plan

☐ Procurement and contract strategy with contract types, tender evaluation criteria and a variation procedure

☐ Stakeholder engagement plan, site communication rhythm and a BIM information-management (ISO 19650) and digital-twin use plan

☐ Inspection and test plan, commissioning plan, handover checklist and closeout report

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Investment logic	○ No financial case	○ NPV, IRR and payback calculated	○ Options compared with sensitivities, benefits and funding risks
Scope & design definition	○ Scope vague	○ Scope statement and WBS	○ Battery limits, interfaces, WBS dictionary and change control
Scheduling	○ Bar chart without logic	○ Logic-linked CPM with float	○ PERT ranges, levelled resources and compression options
Cost & earned value	○ Single-figure budget	○ Estimate with contingency and S-curve	○ EVM baseline, forecasts and thresholds for action
Risk & HSE	○ Few risks listed	○ Register with owners and responses	○ Quantified exposure, hierarchy of controls and leading indicators
Procurement & contracts	○ Contract type not justified	○ Strategy with reasons	○ Risk allocation, evaluation criteria and claims procedure
People & digital	○ Roles unclear	○ Stakeholder and communication plans	○ Engagement strategy, site leadership and governed BIM/AI use

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CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Quality, commissioning & closeout	○ No completion plan	○ ITP and handover checklist	○ Systems-based commissioning, operations readiness and lessons that change practice

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 40 questions at random from a separate, secret bank that tests the same skills, with a 60-minute limit and a 80% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Justify — Initiation, Feasibility & the Business Case

1. A hypothetical rooftop solar retrofit for a Gazipur garment factory costs 400 (USD thousand) now and saves 150 at the end of each of the next three years. At a 10% discount rate, what is the NPV to the nearest whole number?

   (A) +50
   (B) -27
   (C) +27
   (D) -50

2. At a gate review for a hypothetical LNG import jetty, the estimate still rests on a benchmark unit rate, no marine geotechnical survey has been done and the permit pathway is unknown. The sponsor wants approval to sign the main construction contract. What should the gate reviewers decide?

   (A) Recycle the phase, funding the survey, permit plan and a firmer estimate before deciding again
   (B) Approve the award, because delaying the jetty would defer the revenue it was built to earn
   (C) Approve the award but add a large contingency to cover the missing survey and permit work
   (D) Stop the project permanently, because any gap in definition at this stage shows the case is unsound

3. A hypothetical pump replacement costs 200 now and returns 80 at the end of each year for four years. Discount factors at 12% are 0.893, 0.797, 0.712 and 0.636. What is the discounted payback, to one decimal place?

   (A) 2.5 years
   (B) 2.8 years
   (C) 3.0 years
   (D) 3.2 years

4. A public flood embankment in a hypothetical river town has a present value of benefits of 540 and a present value of costs, capital plus maintenance, of 450. What is the benefit-cost ratio and what does it indicate?

   (A) 0.83, showing that costs exceed benefits in present value
   (B) 90, which is the benefit-cost ratio expressed as a money value
   (C) 1.20, showing that benefits exceed costs in present value
   (D) 1.20, showing that the IRR is exactly 20% above the discount rate

5. In the business case for a hypothetical textile mill's new boiler house, NPV becomes zero if gas savings fall by 6%, while the vendor guarantees only 90% of the claimed savings. What is the most useful conclusion for the investment committee?

   (A) The case is robust, because the vendor has guaranteed most of the savings
   (B) The case is fragile, because guaranteed savings sit below the switching point
   (C) The case is robust, because a 6% change in savings is only a small variation
   (D) The case cannot be judged until an IRR has been calculated for the whole project

6. During execution of a hypothetical 132/33 kV substation, the project manager forecasts that a cable route diversion will add cost beyond her approved tolerance but within the sponsor's limit. What should she do FIRST?

   (A) Escalate to the sponsor with the impact assessment and options for a decision
   (B) Absorb the cost in the contractor's next payment without recording it
   (C) Approve the diversion herself, since the steering committee meets only quarterly
   (D) Wait for the board to consider it at the annual capital planning meeting

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7. A hospital trust compares three options to expand diagnostic imaging but has left out a do-nothing option, arguing that 'doing nothing is not acceptable'. Why should the project manager insist on including it?

   (A) Because funders always pick the cheapest option once it is shown

   (B) Because do-nothing removes the need for any sensitivity analysis

   (C) Because it allows the trust to skip a full options analysis

   (D) Because it is the baseline for measuring every option's benefits

8. The owner of a hypothetical combined-cycle gas plant is deciding when to involve its future operations team. The EPC contract will be signed in six months. When does the operations team's input create the most value?

   (A) During commissioning, when they can witness the performance tests

   (B) After taking over, when they become responsible for the asset

   (C) During front-end definition, before the design basis is frozen

   (D) During the defects period, when warranty claims are raised

Define — Scope, Requirements & Design Definition

9. During FEED for a hypothetical LNG regasification terminal, the owner's operations team asks for the unloading arms to remain operable at a higher wind speed than the basis of design assumes. Where should this requirement be captured FIRST so that all designers work to it?

   (A) In the EPC contractor's method statement for arm installation

   (B) In the minutes of the next site coordination meeting only

   (C) In a revised basis of design, approved through change control

   (D) In a verbal instruction from the operations manager to the designer

10. The scope for a hypothetical pharmaceutical warehouse says the contractor will provide 'all HVAC works' but is silent on the chilled water supply from the owner's central plant 200 metres away. Which addition to the scope statement best prevents a future dispute?

    (A) A longer list of HVAC equipment with brand-neutral performance data sheets

    (B) A battery limit stating where the contractor's chilled water scope begins

    (C) A requirement that the contractor follows all applicable HVAC codes

    (D) A clause asking the two parties to cooperate in good faith on HVAC

11. A contractor's WBS for a hypothetical cement plant upgrade is organised as civil, mechanical, electrical and instrumentation, each spanning all plant areas. The owner wants to measure progress area by area to plan a phased start-up. What is the best response?

    (A) Restructure the upper WBS levels by plant area, with disciplines below

    (B) Replace the WBS with a cost breakdown structure organised by cost type

    (C) Keep the discipline WBS and ask for a separate monthly narrative for each area

    (D) Let each discipline lead choose its own WBS structure and coding

12. In a HAZOP of a hypothetical caustic dosing system at an effluent treatment plant, the team identifies 'more flow' of caustic if the dosing pump controller fails high, with only a manual valve as a safeguard. What is the correct outcome for the study?

    (A) Note the deviation only, since a HAZOP records hazards but never actions

    (B) Ask the vendor to guarantee the controller will never fail in service

    (C) Remove the node from the study because the pump is vendor-supplied

    (D) Raise an action with an owner to provide an adequate added safeguard

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13. On a hypothetical university laboratory block, structural design was frozen two months ago. The faculty now asks for a heavier fume-cupboard extraction fan on the roof, which will increase roof loads. What should the project manager do?

(A) Instruct the contractor to install the fan and check the roof afterwards

(B) Reject the request outright, since a frozen design cannot be changed at all

(C) Process it as a change request with structural and MEP impact assessment

(D) Ask the faculty to buy the fan separately so it falls outside the project

14. A hypothetical substation project has one very large control account, '4.2 Transformers and switchgear', covering design, supply, installation and testing over two years under one manager. Reports show the account on track, yet site teams report late switchgear deliveries. What is the most likely weakness?

(A) The 100% rule has been applied far too strictly to the electrical scope

(B) The control account is too large, so problems hide inside its totals

(C) The RAM has too few parties marked as informed for the account

(D) The basis of design does not list the correct code editions

15. On a hypothetical metro depot, a value engineering exercise at detailed design replaced the specified roof system with a cheaper one without checking its fire rating or maintenance access. Which statement best describes the problem?

(A) It was cost cutting, since the required functions were not verified

(B) Value engineering is never appropriate once FEED has been completed

(C) Value engineering should be done only by the contractor after award

(D) The roof change should have been logged as a trend notice and closed

Schedule — Scheduling: CPM, PERT, Levelling & Crashing

16. During a monthly update of a hypothetical cement mill upgrade, the planner reports that kiln burner replacement has ES = day 20, EF = day 26, LS = day 24 and LF = day 30. The mechanical contractor asks how long this activity can slip before completion is threatened. What is the correct answer?

(A) 4 days, which is its total float (LS - ES or LF - EF)

(B) 6 days, which is the duration of the activity itself

(C) 10 days, measured from its early start to its late finish

(D) 0 days, because only free float may be used by a contractor

17. A hypothetical offshore wind substation topside has a critical path with a PERT mean of 120 working days and a path variance of 64 days squared. The developer asks for the chance of completing the topside within 128 working days. What is the best estimate?

(A) About 50%, because the target is close to the PERT mean

(B) About 55%, using z = 8 / 64 = 0.125

(C) About 98%, treating the 8-day gap as two standard deviations

(D) About 84%, using z = (128 - 120) / 8 = 1.0

18. A planner on a hypothetical airport terminal expansion reports a 70% chance of meeting the baggage-system handover milestone, calculated with PERT on the critical path. Two other near-critical paths, systems integration and airside works, feed the same milestone. What should the project manager tell the sponsor?

(A) The chance is above 70%, because parallel paths give the project more ways to recover lost time

(B) The chance is exactly 70%, because PERT already accounts for the variance of every path in the project

(C) The true chance is probably below 70% because of merge bias, so all paths should be analysed together

(D) The chance cannot be estimated at all until the critical path work has been completed on site

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19. A hypothetical hospital block forecasts completion 3 days late against a contract date carrying delay damages of 8k per day; site indirect costs are 3k per day. On the critical path, activity X can be crashed 1 day at 5k per day and activity Y 2 days at 9k per day; activity Z, which has float, can be crashed at 2k per day. Assuming no other path becomes critical, what is the most economical plan?

(A) Crash Z by 3 days, because it has the cheapest cost slope available

(B) Crash X by 1 day and Y by 2 days, recovering all 3 days of the delay

(C) Crash nothing, because every critical slope exceeds the 3k daily indirect cost

(D) Crash X by 1 day only and accept 2 days of delay damages at 8k per day

20. An owner reviewing a contractor's baseline for a hypothetical container yard expansion at Chattogram port finds that over a third of the activities carry 'must finish on' constraints. What is the main risk for the owner if this baseline is accepted as submitted?

(A) Float and the critical path are distorted, so delays cannot be analysed reliably

(B) The schedule will be far too detailed for monthly reporting to the port authority

(C) The contractor will be unable to load resources onto the constrained activities

(D) The scheduling software will refuse to calculate early and late dates at all

21. A hypothetical semiconductor fab cleanroom fit-out has a fixed handover date. The early-start schedule shows electricians peaking at 220 for three weeks, while welfare facilities and supervision suit about 160. Several electrical activities have 10-20 days of float. Which technique should the planner apply first?

(A) Resource levelling, accepting a later handover date to hold the peak at 160

(B) Crashing the electrical activities so that the peak period becomes shorter

(C) Fast-tracking the electrical work so that it overlaps more with mechanical work

(D) Resource smoothing, moving electrical work within its float to cut the peak

22. At a schedule update on a hypothetical elevated expressway ramp, the deck waterproofing crew has started a span before parapet installation, its predecessor, has finished. The planner must choose how the software will treat the remaining waterproofing work. Which approach is usually more realistic?

(A) Deleting the logic link permanently, because the crew has shown it is unnecessary

(B) Progress override, so the remaining waterproofing ignores the parapet link from now on

(C) Retained logic, so the remaining waterproofing still waits for the parapets to finish

(D) Adding a hard 'finish on' constraint to the waterproofing at its baseline date

23. On a hypothetical factory roof replacement, 'install skylights' has an early finish of day 18 and a total float of 9 days. Its only successor, 'fix roof closures', has an early start of day 22. The skylight subcontractor will now finish 3 days late. What is the effect on the schedule?

(A) Completion moves 3 days later, because the activity itself has been delayed

(B) No effect on the successor or on completion, as 3 days is within free float of 4

(C) The successor starts 3 days late, but completion is protected by total float

(D) Completion moves 6 days later, because total float has now been reduced to 6 days

Budget — Cost Estimation, Budgeting & Earned Value

24. A hypothetical 132 kV transmission line project reports PV = 48, EV = 42 and AC = 40 (BDT million) at month 7. Tower foundations in a flood-prone section are on the critical path. What does the data show, and what should the project manager look at first?

(A) Over budget but ahead of schedule, so look for overtime that is buying progress at a premium

(B) Within budget for the work done but behind schedule, so check resourcing on critical towers

(C) Over budget and behind schedule, so prepare a full recovery plan for the sponsor straight away

(D) On budget and on schedule, so no action is needed until the next monthly progress report

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25. A hypothetical port crane-rail extension has BAC = 800, PV = 400, EV = 300 and AC = 375 (BDT million). Late work will need acceleration, so the project manager wants an EAC that reflects both cost and schedule performance. What is that EAC?

(A) 1,208.3
(B) 875.0
(C) 1,000.0
(D) 1,066.7

26. Midway through a hypothetical effluent treatment plant project, the government introduces a new zero-liquid-discharge requirement that nobody had foreseen, adding a water recycling stage. The risk register did not include it. How should the additional cost be funded?

(A) From contingency reserve, because any extra cost during construction is contingency
(B) From the contractor's profit margin, because the scope is part of the lump sum
(C) By cutting other work package budgets proportionally so that the baseline total stays unchanged
(D) Through change control, releasing management reserve or new funding and updating the baseline

27. On a hypothetical substation project, the electrical subcontractor proposes to report progress on installing four large power transformers by monthly percent-complete estimates from its supervisor. The owner wants objective earned value. Which approach should the project manager agree before work starts?

(A) Supervisor estimates of percent complete, checked only once at year end against the final account
(B) Level of effort, so that earned value follows the planned value each month without disputes
(C) Weighted milestones for each transformer, such as delivered, set, dressed, oil-filled and tested
(D) A 0/100 rule across the whole four-transformer package, which lasts about five months

28. A hypothetical university library building has BAC = 500, EV = 200 and AC = 260 (BDT million). The sponsor insists that the original budget must still be met. What TCPI does this require, and what does it suggest about the target?

(A) 0.77, matching current performance, so the budget is achievable without any change
(B) 1.25, far above the 0.77 achieved so far, so the target is very unlikely to be met
(C) 1.30, the inverse of the current CPI, so a modest improvement would be sufficient
(D) 0.60, the remaining work divided by BAC, so efficiency can fall and still meet it

29. An energy utility is screening a hypothetical 200 MW solar park in northern Bangladesh. Only the capacity, a candidate site and cost-per-MW data from recent similar parks are available. The board will decide whether to fund a feasibility study. What estimate should the project manager present?

(A) A parametric estimate from cost per MW, with location and date adjustments and a wide range
(B) A definitive bottom-up estimate accurate enough to fix the construction budget at this stage
(C) A bill of quantities priced with unit rates for modules, mounting structures, cabling and civil works
(D) A single figure from the most recent park, scaled by capacity, with no range stated

30. At month-end on a hypothetical cement plant upgrade, the cost team has not accrued for refractory bricks delivered and installed in the kiln because the supplier's invoice has not arrived. The work is credited in EV. What is the effect on this month's earned value report?

(A) CPI is understated, because EV has been reduced by the missing invoice amount
(B) SPI is overstated, because the installed bricks are counted twice in planned value
(C) No effect, because earned value ignores all costs until the invoices have actually been booked
(D) AC is understated, so CPI looks better than it really is until the accrual is posted

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Protect — Risk, Health, Safety & Environment

31. On a Chattogram container terminal project, the crane supplier formally notifies a ten-week shipping delay for two ship-to-shore cranes. The register lists 'crane delivery may slip' as an open risk. What should the project manager do FIRST?

(A) Run a new identification workshop on supplier risks before taking any action on the cranes

(B) Raise the probability score of the crane risk to 5 and continue monitoring it closely at the next monthly risk review

(C) Draw down the full contingency allocated to the crane package before the impact is known

(D) Treat it as an issue: assess the critical-path impact, decide recovery actions and record that the risk has occurred

32. A river intake cofferdam risk has a 30% probability and a USD 1.2m impact. A sheet-pile upgrade costing USD 150k would reduce the probability to 10%. On expected value, what is the net value of the upgrade?

(A) USD 240k, so it is clearly worth funding at once

(B) -USD 30k, so it should be rejected as uneconomic

(C) USD 90k, so it is worth funding on expected value

(D) USD 210k, so it should be funded from contingency

33. A hypothetical hospital project's register contains a threat with a 20% chance of costing USD 500k, a threat with a 50% chance of costing USD 120k, and an opportunity with a 40% chance of saving USD 150k. What is the net EMV?

(A) -USD 220k, an expected net cost to the project

(B) -USD 100k, an expected net cost to the project

(C) -USD 160k, an expected net cost to the project

(D) +USD 40k, an expected net saving to the project

34. A cost risk analysis for a hypothetical LNG jetty gives a base estimate of USD 92m, P50 of USD 98m, P80 of USD 104m and P90 of USD 108m. The owner's policy is to fund to P80. How much risk contingency should be added to the base estimate?

(A) USD 12m, the gap between the base and the P80 value

(B) USD 16m, the gap between the base and the P90 value

(C) USD 104m, which is the full P80 value held as contingency

(D) USD 6m, the gap between the base and the P50 value

35. A planner presents a commissioning date built from the most likely duration of every activity. The schedule risk analysis says this date has only a 10% chance of being met, and the sponsor asks how that is possible. What is the best explanation?

(A) The simulation software adds a standard safety margin to every activity duration before it runs the model

(B) Most-likely durations are always the pessimistic estimates provided by the contractor's planners

(C) Monte Carlo results are random, so a second run would probably show a much higher probability

(D) Right-skewed ranges, merge bias and risk events push most outcomes later than the single-point plan

36. A subcontractor will start pulling cables in a basement substation tomorrow. Its RAMS was copied from another project, has been signed by the foreman but has not been briefed to the crew, and does not mention the live switchboard in the next room. What should the project manager require FIRST?

(A) That work goes ahead, with the HSE officer checking compliance very closely during the first shift

(B) That the subcontractor is replaced by a firm with a better recorded accident history

(C) That the RAMS is revised for this task and location and briefed to the crew before work starts

(D) That each worker signs the existing generic document again to confirm it has been read

37. A metro station's monthly safety report shows six months with no lost-time injuries. Over the same period near-miss reports fell sharply, and a walk-round audit found missing edge protection on two levels. What is the best interpretation?

(A) Performance is excellent, so safety inspection resources can be reduced for the next quarter

(B) Reporting may have collapsed, so good injury figures could be hiding rising risk that needs investigating

(C) The fall in near-miss reports shows that most hazards on the site have now been eliminated

(D) Lagging indicators are reliable on their own, so the leading indicators can now be dropped from the report

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38. A lender's monitoring consultant finds that the contractor's price for a hypothetical 220 MW power plant excludes the cost of ESMP measures and that the site has no grievance log, although villagers live next to the haul road. What should the owner's project manager do FIRST?

(A) Require a priced construction ESMP and a working grievance mechanism, managed as contract obligations

(B) Ask the regulator to waive the requirements because the environmental clearance has already been issued

(C) Wait for the first formal community complaint before investing in a grievance process on site

(D) Transfer all environmental responsibility to the contractor and stop the owner's compliance audits

Procure — Procurement, Contracts, Claims & Variations

39. An owner is planning a hypothetical 400 MW combined-cycle plant. The gas turbine has a long manufacturing lead time, but the EPC tender will not be issued for another nine months. What should the project manager recommend to protect the schedule?

(A) Reserve or order the turbine now under an owner contract designed to be novated to the EPC contractor

(B) Wait for the EPC contractor, since any early turbine order would remove the single-point responsibility

(C) Ask the turbine vendor to hold its current price until the EPC contractor is finally appointed next year

(D) Shorten the EPC tender period to three weeks so that the contractor can place the turbine order sooner

40. A target cost contract has a target of USD 50m, a fixed fee of USD 3m and a 50:50 pain/gain share. The actual cost is USD 46m. How much does the owner pay the contractor?

(A) USD 49m, which is the actual cost plus the fee less the shared gain

(B) USD 46m, which is the actual cost alone because the target was not reached

(C) USD 53m, which is the target cost plus the full fixed fee for the contract

(D) USD 51m, which is the actual cost plus the fee plus a USD 2m gain share

41. After a cyclone damages a coastal embankment, an owner needs urgent repairs to start within days, before the extent of the damage can be surveyed. Which pricing mechanism is most appropriate for the first phase?

(A) A lump sum, so that the owner has full price certainty from the very first day

(B) A remeasurement contract based on a bill of quantities prepared after the survey

(C) A cost-reimbursable contract with a fixed fee and tight cost verification controls

(D) A guaranteed maximum price agreed after the contractor's design is fully complete and checked

42. In a tender with a 30-point price weight, the lowest normalised price is USD 12m. Another tender has a normalised price of USD 15m. What price score does it receive?

(A) 30 points, because it is a fully compliant tender

(B) 24 points, from 12 divided by 15, times 30

(C) 15 points, as the price weight less 15 points

(D) 22.5 points, as three quarters of the price weight

43. The lowest tender for a hypothetical river protection package is 30% below the engineer's estimate and well below the other bids. The procurement officer wants to award immediately to save money. What should the project manager do FIRST?

(A) Ask the bidder in writing to explain and substantiate its prices before any award decision

(B) Award immediately, because the procurement rules require the lowest evaluated price to be accepted

(C) Reject the tender at once, since any bid this far below the estimate must be unrealistic

(D) Invite the other bidders to revise their prices downwards to match the lowest tender

44. On a hypothetical design-build pumping station, the contractor's excavation hits strong artesian water that the site investigation did not reveal and an experienced contractor could not have foreseen. What should the contractor do FIRST?

(A) Continue the work and include the extra cost in the final account at the end of the contract

(B) Refer the matter directly to arbitration so that the entitlement is decided quickly and finally

(C) Stop all work on site until the employer agrees in writing to pay for the dewatering costs

(D) Give written notice of the event within the contract period and start keeping detailed records

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45. A programme analysis shows that a late owner-supplied transformer and the contractor's own late switchgear installation both delayed completion by the same three weeks, each on its own. Under a common approach to concurrent delay, what is the contractor likely to receive?

(A) Three weeks' extension of time with full prolongation costs for the whole three weeks

(B) No extension of time, so delay damages apply for the full three-week period

(C) Three weeks' extension of time but no prolongation costs for the concurrent period

(D) Prolongation costs for the three weeks but no extension of the completion date

Integrate — Stakeholders, Site Leadership & Digital Engineering

46. A hypothetical 220 kV transmission line crosses farmland, a river and a highway. Six weeks before stringing, the roads authority has not approved the highway crossing and nobody on the team owns the application. What should the project manager do FIRST?

(A) Report the missing approval to the steering committee as a new risk and wait for it to decide on the next step

(B) Instruct the contractor to string the highway span at night when traffic is light, then apply retrospectively for the crossing approval

(C) Assign a named owner, add the approval and its prerequisites to the schedule and consents register, and engage the authority directly

(D) Redesign the line route to avoid the highway entirely so that no roads authority approval is needed at all

47. During a weekly meeting on a hypothetical garment factory expansion near Gazipur, a junior site engineer hesitates and then says quietly that some column rebar looked congested. The construction manager dismisses the comment. What is the best leadership response from the project manager?

(A) Support the construction manager publicly to protect his authority, then check the columns quietly later in the week

(B) Thank the engineer, ask for specifics, arrange an immediate check before the next pour, and report back what was found

(C) Ask the engineer to submit a written report through the formal quality channel so it can be logged and reviewed in time

(D) Note the comment in the minutes and move on, since the construction manager has more experience of column details

48. A contractor installing ducts in a hospital plantroom finds that the drawing shows a duct passing through a beam. It needs an answer within three days to avoid stopping work. Which document should it raise?

(A) A request for information citing both drawings with a proposed solution

(B) A transmittal listing all of the current mechanical and structural drawings on site

(C) A non-conformance report against the structural designer's drawings

(D) A submittal containing shop drawings for the engineer's formal review

49. Clash detection on a federated model for a hypothetical semiconductor fab returns 4,000 raw clashes in one zone. What should the BIM coordinator do next?

(A) Send the full clash list to every discipline lead and ask each team to resolve its own clashes independently

(B) Release the zone for installation and let site supervisors resolve the clashes in the field as they appear

(C) Increase the clash tolerance until the count falls to a level that the coordination team can comfortably review in one meeting

(D) Group the raw clashes into real issues, remove false positives, prioritise by construction sequence, and assign owners

50. An owner wants its new port terminal to have a digital twin that predicts crane and conveyor failures. At what point should the asset information requirements be defined?

(A) During commissioning, when sensors are first connected to the control system for the first time

(B) After taking over, once operations staff have experience of the new equipment and its failures

(C) At the start of the project, so that design and handover deliver the data the twin needs

(D) During the defects period, when the contractor is still available to answer questions about the data

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51. A computer-vision tool for detecting people inside excavator exclusion zones achieves high precision but low recall in trials. What does this mean for its use on a hypothetical expressway project?

(A) It raises too many false alarms, so supervisors will stop trusting its alerts
(B) It misses many real intrusions, so it cannot be relied on as a primary safeguard
(C) It is accurate enough to replace banksmen and physical barriers around plant
(D) It needs larger cameras with wider lenses to reduce the number of alerts that it raises

52. A project manager on a hypothetical offshore wind project wants to use a public AI chatbot to summarise the full EPC contract, including its pricing schedules. What should the project manager do?

(A) Use only tools approved for confidential information, and have a person verify the summary
(B) Upload only the pricing schedules, since those are the hardest sections to read through
(C) Upload the contract, because summarising text is a low-risk task for any AI chatbot
(D) Ask the contractor to upload the contract instead, so that the confidentiality risk is shared

53. On a hypothetical elevated expressway, the monthly report shows every package as green, yet the critical-path girder erection is four weeks behind. The RAG thresholds were never written down. What should the project manager change first?

(A) Add more indicators to the report so that every package has at least ten measures
(B) Replace the RAG ratings with a written narrative for each package written by its lead
(C) Ask package managers to rate their own packages more strictly from the next month
(D) Define written thresholds for each RAG status, linked to schedule and cost tolerances

Commission — Quality, Commissioning, Handover & Closeout

54. On a hypothetical LNG terminal, the QA/QC manager reports to the construction manager, who is measured mainly on progress. Inspectors say they are pressured to release hold points quickly. What should the project manager change FIRST?

(A) Increase the number of inspectors so that every hold point can be released faster without delaying the work
(B) Ask the construction manager to sign a declaration that inspectors will not be pressured in future
(C) Remove hold points from the ITPs for low-risk activities so that inspectors face fewer release decisions
(D) Make the QA/QC manager report independently of construction, as set out in the project quality plan

55. A package on a hypothetical cement plant records prevention 25, appraisal 45, internal failure 120 and external failure 60 (thousands). What are the cost of quality and the COPQ share?

(A) COQ 250 thousand; COPQ share 28%
(B) COQ 180 thousand; COPQ share 72%
(C) COQ 250 thousand; COPQ share 72%
(D) COQ 250 thousand; COPQ share 48%

56. During a pre-pour inspection on a hypothetical viaduct pier, the engineer's inspector is delayed by two hours. The contractor's foreman says the rebar is fine and wants to pour before the concrete trucks queue. The ITP shows a hold point. What should happen?

(A) The pour proceeds because the contractor's foreman has the authority to release his own work
(B) The pour waits for the inspector's written release, and the delay is recorded for later review
(C) The pour proceeds after photographs of the rebar are taken and sent to the inspector by phone
(D) The pour proceeds with an NCR raised in advance so that any defect can be dealt with afterwards

57. Construction on a hypothetical combined-cycle plant reports 92% overall progress, but the commissioning team has accepted only two of thirty systems. What is the most likely cause, and the right response?

(A) Progress is measured by area and quantity; switch the remaining construction schedule to system completion priorities
(B) The commissioning team is too slow and should be enlarged with additional vendor specialists from the turbine supplier
(C) The progress percentage is wrong and should be recalculated by the planner using weighted milestones for each area
(D) The systemisation boundaries are too small and should be merged into fewer, larger systems to speed up acceptance

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58. A hypothetical biomass boiler fails its performance test because measured steam output is below the guarantee. The contractor says the fuel was wetter than the design fuel. What should the owner's project manager check FIRST?

(A) Whether the boiler's warranty period has already started under the terms of the supply contract

(B) Whether the operators who ran the test had completed all of their training on the new boiler

(C) Whether the boiler vendor has delivered the complete set of critical spares on time

(D) Whether the test procedure and corrections to reference fuel conditions were applied as agreed

59. Six months after handover of a hypothetical metro depot, maintenance staff cannot find the isolation points for several HVAC units because the as-built drawings show the design layout. What does this most clearly indicate?

(A) The depot was handed over too early, before the HVAC performance tests were complete

(B) The O&M manuals were too long for the maintenance staff to read in the time available

(C) Site changes were not captured continuously as redlines and verified before handover

(D) The maintenance staff did not receive enough classroom training on the building systems

60. At the closeout of a hypothetical hospital project, the team produces a lessons list that says 'improve coordination' and 'plan earlier'. The PMO notes the same lessons appeared on the last three projects. What should the project manager do?

(A) Circulate the list more widely across the organisation so that more people read the lessons this time

(B) Rework each lesson into a specific change with a root cause, owner, due date and later verification

(C) Add the lessons to the risk register of the next project as generic risks with a low rating

(D) Remove the repeated lessons from the register, since they are already well known to everybody

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EXAM PREPARATION

Practice exam answers & explanations

1. B PV of savings = 150 x (0.9091 + 0.8264 + 0.7513) = 150 x 2.4869 = 373, so NPV = 373 - 400 = -27. +50 ignores discounting (450 - 400), +27 has the sign reversed and -50 is not supported by any correct discounting.

2. A The gate exists to prevent irreversible commitment before key information is available, so recycling to complete the front end is right. Approving with or without extra contingency commits to a contract on unknown ground and permits, while stopping permanently overreacts to gaps that can be closed.

3. D Discounted inflows are 71.4, 63.8, 57.0 and 50.9; cumulative PV is 192.2 after year 3, leaving 7.8 to recover from 50.9 in year 4, so 3 + 7.8 / 50.9 = 3.2 years. 2.5 years is the simple payback, and the other values misplace the recovery point.

4. C BCR = 540 / 450 = 1.20, so present-value benefits exceed costs. 0.83 inverts the ratio, 90 is the NPV rather than a ratio, and a BCR says nothing directly about the IRR.

5. B The switching value is a 6% fall, but the guarantee protects only down to a 10% shortfall, so a guaranteed outcome could still give a negative NPV. Calling the case robust ignores that gap, and an IRR would not change the conclusion that the case is sensitive to savings.

6. A Tolerances define who decides: a change beyond the PM's limit but within the sponsor's goes to the sponsor with the evidence. Approving it herself exceeds her authority, hiding it undermines control, and the board is the wrong level and far too slow.

7. D Even an unacceptable do-nothing case is the reference against which benefits are measured, so leaving it out distorts every option's value. Funders do not always pick the cheapest option, and do-nothing neither replaces sensitivity analysis nor the options analysis itself.

8. C Operator input is cheapest and most influential before the design basis is frozen and the EPC contract signed. Later involvement is still useful, but at commissioning, takeover or the defects period changes are expensive or impossible.

9. C The basis of design is the single reference every discipline works to, and changing it needs approval. Meeting minutes and method statements are not design references, and a verbal instruction bypasses control and reaches only one designer.

10. B The dispute risk lies at the boundary between the owner's plant and the contractor's works, so a clearly defined battery limit resolves it. More equipment detail, code compliance or a goodwill clause leave the boundary just as unclear.

11. A A deliverable-oriented WBS with plant areas at the upper levels lets progress be measured by area for phased start-up. A narrative adds no structured data, a CBS shows money rather than deliverables, and separate discipline structures fragment control.

12. D When safeguards are inadequate, a HAZOP raises an owned action, for example a high-flow trip or a flow-limiting device. HAZOPs do record actions, no vendor can guarantee zero failures, and vendor packages remain within the study.

13. C After a freeze, changes are still possible but must go through change control with impact assessment. Outright rejection misreads the freeze, installing first is unsafe, and buying the fan separately hides a structural change from control.

14. B Very large control accounts average out problems, so a late delivery can be masked by progress elsewhere. The 100% rule, the number of informed parties and code editions do not explain why a delay is invisible in the reports.

15. A Value engineering must keep required functions such as fire performance and maintainability; without that check it is cost cutting. VE is not limited to contractors, is still possible after FEED with care, and a trend notice does not replace assessment.

16. A Total float = LS - ES = 24 - 20 = 4 days, and LF - EF = 30 - 26 gives the same answer. The duration (6) and the span from ES to LF (10) are not float, and whether float may be used is a contract question, not a calculation.

17. D The path standard deviation is the square root of the variance, 8 days, so z = (128 - 120) / 8 = 1.0, about 84%. Dividing by the variance gives the wrong z of 0.125, the gap is one standard deviation rather than two, and 50% applies only at the mean itself.

18. C When several paths merge, all must finish on time, so the joint probability is lower than the critical path alone suggests. PERT uses only the critical path's variance, parallel paths add risk rather than recovery options, and useful estimates are possible long before completion.

19. B Each day recovered saves 8k of damages plus 3k of indirect cost, 11k in all. X (5k) and Y (9k) are both below 11k, so crashing both costs 23k and saves 33k. Z is not critical, comparing slopes with indirect cost alone ignores the damages, and stopping after X leaves savings unclaimed.

20. A Hard constraints override logic, so float and the critical path reflect typed dates rather than the network, which undermines progress analysis and later delay assessment. Constraints do not change the level of detail, prevent resource loading or stop the software calculating.

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21. D With a fixed end date and activities that have float, smoothing reduces the peak without moving handover. Levelling would accept a later date unnecessarily, while crashing and fast-tracking concentrate work further instead of spreading it.

22. C Retained logic assumes the rest of the work still depends on the predecessor, which is usually true when only part of an area has been released. Progress override and deleting the link can hide a real dependency, and a hard constraint would distort float.

23. B Free float = successor ES - EF = 22 - 18 = 4 days, so a 3-day slip delays neither the successor nor completion; total float falls from 9 to 6 days. The other options ignore float or confuse total float with free float.

24. B CPI = 42 / 40 = 1.05 (work done for less than its budget) and SPI = 42 / 48 = 0.875 (behind plan). Economical but slow often means too few resources, so the critical tower foundations should be checked first. The other options misread one or both indices.

25. A CPI = 300 / 375 = 0.80 and SPI = 300 / 400 = 0.75, so CPI x SPI = 0.60 and EAC = 375 + (800 - 300) / 0.60 = 375 + 833.3 = 1,208.3. 1,000.0 is BAC / CPI, 875.0 assumes the rest is done at budget, and 1,066.7 divides BAC by SPI, which is not a recognised EAC.

26. D An unforeseen requirement is not an identified risk, so it falls outside contingency. It needs a change request: the sponsor may release management reserve or seek new funding, and the baseline is then updated. Contractors are not normally expected to absorb employer-driven scope from profit, and cutting other budgets hides the true cost.

27. C Transformers are discrete items with distinct, observable stages, which suits weighted milestones agreed in advance. Level of effort ignores output, checking only at year end leaves EV unreliable all year, and 0/100 on a five-month package would show no progress until the end.

28. B TCPI = (BAC - EV) / (BAC - AC) = 300 / 240 = 1.25, while CPI to date is 200 / 260 = about 0.77, and such a jump in efficiency is very unlikely without a major change. 0.77 is past performance, 1.30 is merely the inverse of CPI, and 0.60 divides by the wrong quantity.

29. A With only capacity and benchmark data, a parametric estimate adjusted for location and date, shown with a wide screening-class range, fits the decision. Definitive and BOQ estimates need far more design, and a single unqualified figure invites false precision.

30. D Work done but not invoiced is a real cost; without the accrual AC is too low, so CPI = EV / AC is overstated. EV and PV are not affected by invoice timing, and EVM relies on actual cost including accruals.

31. D The event has happened, so it is now an issue requiring decisions. Rescoring it as a risk delays action, drawing contingency before quantifying the impact skips control, and a workshop does not address the delay.

32. C Exposure falls from 0.30 x 1.2m = 360k to 0.10 x 1.2m = 120k, a reduction of 240k; subtracting the 150k cost leaves +90k. The other figures omit the response cost or misapply the formula.

33. B EMV = -100k - 60k + 60k = -USD 100k. Ignoring the opportunity gives -160k; -220k adds the opportunity as a cost; +40k does not weight the impacts by probability.

34. A Risk-based contingency is the chosen confidence value minus the base estimate: 104m - 92m = 12m, so the funded total is the P80 of 104m. The P50 and P90 gaps reflect different appetites, and the full P80 value is the funded total, not the contingency.

35. D Skew puts each mean above its most likely value, merge bias makes joint on-time completion unlikely, and risk events add delay. Simulations do not add hidden margins, and with enough iterations results are stable between runs.

36. C RAMS protect people only when task-specific and briefed. Supervising a generic plan does not close the gap, replacing the firm is disproportionate as a first step, and extra signatures do not add any control.

37. B Falling near-miss reports alongside visible hazards suggest under-reporting, not safety. Lagging measures can look good by chance; leading indicators and audits reveal the real conditions.

38. A ESMP measures must be priced and contractually required, and a grievance mechanism must exist before complaints escalate. A clearance does not remove obligations, and the owner cannot hand away its oversight role.

39. A Early reservation with planned novation protects the critical path while preserving single-point responsibility. Waiting lets the lead time drive completion, price holds do not secure a slot, and a rushed tender raises risk premiums.

40. D Payment = 46 + 3 + 0.5 x (50 - 46) = 46 + 3 + 2 = USD 51m. The contractor earns a bonus for underspending; the other answers reverse the share, ignore it or drop the fee.

41. C When work must start before the scope can be defined, cost-reimbursable with a fixed fee is practical, with cost verification controls; once the damage is surveyed, later phases can move to a target cost or remeasured contract. Lump sum, remeasurement and GMP all need definition that does not yet exist.

42. B Price score = (12 / 15) x 30 = 24. Only the lowest tender gets the full 30; the other answers apply invented rules rather than the published formula.

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43. A An abnormally low tender must be investigated before award: it may be genuine or may hide errors or a claims strategy. Automatic award or rejection skips that check, and inviting others to match is bid shopping.

44. D Timely notice preserves entitlement and contemporaneous records prove it. Waiting risks a time bar, stopping work may be a breach, and arbitration is the last step, not the first.

45. C A common approach to genuine concurrency gives time, so no delay damages, but not money, because the contractor would have incurred the cost anyway. Contracts and governing law can change this, so the terms must be checked.

46. C The gap is ownership and visibility of a critical-path consent: a named owner, schedule logic and direct engagement solve it. Working without approval is unlawful and unsafe, escalation without action wastes time, and a reroute is a disproportionate first response.

47. B Acting visibly on a concern builds psychological safety and may prevent a defect before concrete covers it. Quiet checks or formal delays signal that speaking up is not valued, and deferring to seniority ignores evidence.

48. A An RFI asks the designer to resolve conflicting design information, ideally with references and a proposed solution. A submittal seeks acceptance of the contractor's own documents, a transmittal only records issue, and an NCR records work that does not conform.

49. D Raw clash counts include duplicates and trivial items; grouping, prioritising and assigning owners turns them into manageable issues. Blanket distribution creates confusion, loosening tolerances hides real problems, and field resolution causes rework.

50. C Asset information requirements – in ISO 19650 terms, derived from what the organisation needs to operate the asset – shape what is modelled, tagged, sensed and handed over, so they must be defined at the start. Defining them later means retrofitting data that was never collected, which is expensive or impossible.

51. B Low recall means many real events go undetected, which is dangerous for a life-safety use. High precision means alerts are mostly real, so false alarms are not the problem; the tool supports but cannot replace physical controls and banksmen.

52. A Confidential contract information should go only into tools the organisation has approved for that class of information under its AI and information-security policy, and any summary must be verified by a competent person before use. Uploading to an unapproved tool risks disclosure regardless of who does it.

53. D RAG ratings mean nothing without written thresholds tied to tolerances; defining them makes status objective. More indicators add noise, narratives lose comparability, and asking for stricter self-rating keeps the judgement subjective.

54. D Independence of inspection is a structural safeguard; reporting to the person measured on progress creates a conflict of interest. More inspectors or fewer hold points do not remove the pressure, and a declaration changes nothing about the incentives.

55. C COQ = 25 + 45 + 120 + 60 = 250. COPQ = 120 + 60 = 180, and 180 / 250 = 72%. The 28% figure is the cost of conformance share, 180 is COPQ itself rather than COQ, and 48% uses internal failure alone.

56. B A hold point means no further work without written release from the named party. Photos, self-release or a pre-emptive NCR all bypass the control the ITP created; recording the delay supports any later discussion of cost or time.

57. A High area progress with few complete systems is the classic sign that construction is not working in commissioning sequence. More commissioning staff, recalculated progress or larger systems do not complete the missing items that block each system.

58. D Performance results are only meaningful when the agreed procedure and corrections to reference conditions, such as fuel moisture, have been applied. Warranty dates, training and spares matter elsewhere but do not decide whether the guarantee was met.

59. C As-builts that show the design layout mean field changes were never recorded. Manual length, early handover or training do not explain drawings that fail to match what was built.

60. B Generic lessons do not change practice; specific, owned and verified changes do. Wider circulation, deletion or adding vague risks all leave the underlying practice unchanged.

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NOTES

My notes

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